

Paula Alejandra Barco Alzate

A comprehensive recipe calculator for evaluating multiple environmental impact categories – A first prototype

Graduate thesis in Master in Industrial Ecology

Supervisor: Konstantin Stadler

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NTNU

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Abstract

This study presents the development of the DELI Calculator, the first prototype for a food recipe environmental impacts calculator. Leveraging automated Python scripts, users can input specific recipe(s) information that they wish to assess. Building upon the comprehensive data of Poore and Nemecek (2018) database, the DELI Calculator includes GHG emissions, land use, acidification, eutrophication, freshwater withdrawals and scarcity-weighted freshwater withdrawals. The creation of this first prototype encompassed critical aspects like usability, scalability, maintainability and universality. The primary objective was to empower consumers with a versatile tool capable of assessing multiple impact categories, thereby facilitating informed, sustainable food choices. The DELI Calculator underwent testing through various test cases to evaluate its effectiveness across different dietary patterns.

By employing an adaptation of the Waterfall Software Development Life Cycle (SDLC), the process endured both the fulfillment of the thesis's objectives and the establishment of a structured methodological approach. Subsequent test sessions involving eight users provided valuable insights, with user experience evaluated through surveys. These findings informed the refinement of the DELI Calculator, emphasizing its potential to promote environmental consciousness and support sustainable dietary practices.

Sammendrag

Dette studiet presenterer utviklingen av DELI Calculator, den første prototypen for utregning av miljøpåvirkningen til matoppskrifter. Gjennom automatiserte programsnutter i Python kan brukeren oppgi ingrediensene til matoppskrifter som de ønsker å undersøke. Ved å bygge på den omfattende databasen til Poore og Nemecek (2018), inkluderer DELI Calculatoren både GHG utslipp, arealbruk, forsurening, eutrofiering, uttak av ferskvann og knapphetsvektet uttak av ferskvann. Byggingen av prototypen bestod av kritiske aspekter som brukervennlighet, skalerbarhet, vedlikeholdbarhet og universalitet. Hovedmålet var å styrke forbrukere med et allsidig verktøy som kan vurdere flere påvirkningskategorier, og dermed fasilitere for informerte og bærekraftige matvalg. DELI Calculator gjennomgikk testing via flere typer tester for å kunne evaluere effektiviteten på tvers av ulike kostholdsvaner.

Ved å anvende en adaptasjon av metoden Waterfall Software Development Life Cycle (SDLC), tilfredsstilte prosessen både målene til avhandlingen og etableringen av en strukturert og metodisk tilnærming. Påfølgende testrunder som involverte åtte brukere gav verdifull innsikt. I tillegg ble brukeropplevelsen evaluert gjennom et spørreskjema. Disse funnene gav innspill til forbedringer av DELI Calculator og forsterket potensialet til kalkulatoren for å kunne promotere miljøbevissthet og støtte bærekraftige matvaner.

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To my family and friends, who have always believed in me and believed in my dream of studying Industrial Ecology. A dream that started building up in my mind almost 5 years ago and that now is coming to an end. I am profoundly grateful for their love and encouragement, which have sustained me every step of the way.

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As I embark on the next chapter of my journey, I am committed to upholding the legacy of knowledge sharing and contributing to overcoming the environmental challenges that confront our world.

Table of content

Abstract.....	1
Acknowledgments	2
Table of content	3
1. Introduction	5
1.1. Project background	5
1.2. Motivation	5
1.3. Goal and scope for this research.....	8
2. Theoretical background.....	8
2.1. Poore and Nemecek (2018) database	8
3. Methods.....	12
3.1. Methodological overview	12
3.2. Steps in the development process	13
3.2.1. Requirement analysis.....	14
3.2.2. Design phase.....	14
3.2.2.1. Service blueprinting.....	16
3.2.3. Implementation and repository structure.....	18
3.2.3.1. Source (./src).....	18
3.2.3.2. Data (./data)	19
3.2.3.3. Code.....	20
3.2.4. Testing and validation	21
3.2.4.1. Sensitivity analysis	23
3.2.4.2. Uncertainty analysis	23
3.2.5. User experience evaluation.....	23
4. Results	25
4.1. Comparative LCA for different dietary patterns	25
4.1.1. Relative environmental performance of recipes	25
4.1.2. Environmental impacts distribution by Ingredient for each recipe	27
4.1.3. Results per impact category.....	33
4.1.4. Sensitivity analysis	36
4.1.4.1. Sensitivity analysis for the amounts of the main ingredients	37
4.1.4.2. Sensitivity analysis for reducing the quantity of the main ingredients by 10% in meat-based Lomo Saltado recipes and substituting with a plant-based option	

4.1.4.3.	Sensitivity for the impact values	41
4.1.5.	Uncertainty analysis	42
4.2.	Test sessions	45
4.2.1.	Results from test sessions	45
4.2.2.	User experience results	48
5.	Discussion	51
5.1.	Environmental impacts of diets	52
5.1.1.	GHG emissions	52
5.1.2.	Land use	53
5.1.3.	Acidification	53
5.1.4.	Eutrophication	54
5.1.5.	Freshwater withdrawals	54
5.1.6.	Scarcity-weighted freshwater withdrawals	55
5.2.	Evaluation of the prototype	57
5.3.	Limitations and future research	59
6.	Conclusions	60
7.	References	61
8.	Supplementary Information	65
8.1.	DELI Calculator – TUTORIAL	65
8.2.	Project work results – Autumn 2023	75
8.2.1.	Comparison methodology for open-access databases	75
8.2.2.	Adjustment of the Pedigree matrix components	77
8.2.3.	Pedigree Matrix evaluation	81
8.2.4.	Database selection	82
8.3.	AGRIBALYSE 3.0	83
8.4.	Instructions for completing the Recipe_template	84
8.5.	Instructions to replicate the results from the sensitivity analysis	84
8.6.	Impact values per ingredient for the Lomo Saltado recipe with the highest impact on each impact category	85
8.7.	Results from the test sessions	89
8.8.	Results from the user experience survey	150

1. Introduction

1.1. *Project background*

The work presented in this Master's thesis builds upon my specialization project "*Life Cycle Databases for Food Sustainability Analysis*". This specialization project established the foundation for the thesis, resulting in some overlap in the theoretical background and methodological sections. Although the specialization project is not published, it provided essential insights that have been further developed and expanded in this thesis. Information about this project is included to ensure completeness and coherence for the reader, particularly in the methodological section.

The main goal of the specialization project was to conduct a comprehensive review of available open-access Life Cycle Assessment (LCA) databases for food. As a result, the database by Poore and Nemecek (2018) was selected after quantifying the overall data quality level using the Pedigree Matrix method (JRC-IES, 2010). This database met the study's objective of improving the range of environmental impacts included in the environmental analysis of food. Further details of this process are provided later in this document.

The Poore and Nemecek (2018) database serves as a data source for the development of a prototype of a robust and reliable tool to assess the environmental impacts of food across multiple impact categories.

1.2. *Motivation*

Humanity is currently facing a triple planetary crisis, consisting of climate change, pollution, and biodiversity loss (United Nations Climate, 2022). These interconnected challenges pose significant threats to global ecosystems and humans, increasing the risk of jeopardizing processes fundamental to safeguarding future generations.

Global and local food production systems are essential for providing society with healthy, safe and nutritious food in sustainable and equitable ways (Fanzo et al., 2021). However, these systems are among the most severely affected by the planetary crisis and face challenges related to food availability and security (Passarelli, 2021). Furthermore, food production systems are major contributors to greenhouse gas (GHG) emissions, biodiversity loss, and resource depletion (Poore & Nemecek, 2018; Ritchie et al., 2020). Global food production systems are estimated to account for 34% of global anthropogenic GHG emissions (Crippa et al., 2021), 32% of global terrestrial acidification, and approximately 78% of eutrophication (Poore & Nemecek, 2018).

Agriculture is a major driver of biodiversity loss, threatening 24,000 of the 28,000 (86%) species at risk of extinction due to habitat destruction, pesticide use, and monoculture practices (United Nations, 2021). Agriculture also consumes half of the planet's habitable

land and accounts for 70% of global freshwater withdrawals, further stressing vital water resources and impacting aquatic ecosystems (Ritchie et al., 2020). Unsustainable agricultural practices lead to soil nutrient depletion, desertification, and genetic erosion, undermining the long-term viability of food production systems (Food and Agriculture Organization of the United Nations (FAO), 2019).

Mitigating the negative impacts of food production systems is imperative, especially in light of the projected global population growth (United Nations, 2022) and the resulting increase in food demand. Household consumption plays a substantial role in this regard, contributing to 72% of global GHG emissions (Dubois et al., 2019), with 13% of these emissions attributable to food consumption (Ivanova et al., 2016). Dietary choices and food preferences therefore have a significant impact on environmental footprints (Carvalho et al., 2023; Dubois et al., 2019), demonstrating that the implementation of individual dietary changes is essential to address these negative impacts.

Food choices are influenced by various factors. This includes cultural customs and food availability, in addition to philosophical, religious, and medical needs, all of which affect people's lifestyles (Chen & Antonelli, 2020; Monterrosa et al., 2020). Consumers may also find it difficult to assess the environmental impact of the food they consume. The first step in promoting public engagement in sustainable diets is to provide knowledge and tools that facilitate informed action (Petersson et al., 2021).

Environmental assessment methodologies such as LCA, provide a formal framework for assessing the environmental impacts of products and services (Heller et al., 2013). Nevertheless, despite the existence of international standards and endorsements, there are still shortcomings in accurately reflecting biodiversity impacts and meeting the specific needs of businesses and value chains (Crenna et al., 2020). The prominence of a limited number of indicators is evident in many studies that examine the environmental impacts of food. These studies often focus on GHG emissions, water footprint, and energy consumption, as seen in the works of Gualtieri et al. (2023); O'Malley et al. (2023); Pairotti et al. (2015). As a result, interpretation of the results of these studies often neglects a significant portion of the total environmental impact. Moreover, there are also limitations in terms of impact drivers and geographical scope (Crenna et al., 2020).

To address these limitations and provide valuable data for informed decision-making, openly accessible data is indispensable. Open science and FAIR data offer promising approaches for addressing these limitations by improving data transparency, accessibility, interoperability, and reusability (Vicente-Saez & Martinez-Fuentes, 2018; Wilkinson et al., 2016). By making environmental impact data on food more accessible and comprehensive, stakeholders in the food industry can make more informed sustainability decisions, potentially facilitating the development of tools that calculate a holistic environmental assessment of food (Cucurachi et al., 2019).

The selection process and comparative analysis of open-access of 13 databases conducted in the previous specialization project, highlighted several limitations that affect the assessment of the environmental impacts of food products (Barco, 2023). These findings revealed that most databases tend to focus on a limited number of impact categories, such

as GHG emissions, and land use or water footprint. In addition, the number of studies included in these databases varies considerably, affecting their comprehensiveness and the range of food items covered. Methodological appropriateness and consistency also vary depending on the scope and documentation of the studies, leading to discrepancies in assessments. Temporal representativeness is also an issue, with 70% of the databases containing data between six and ten years old at the time of writing, potentially misrepresenting current practices. Geographical representativeness is limited, with a bias towards Europe, reducing applicability in other regions. Technological representativeness is also crucial, with most databases detailing individual food items, while a few use broader categories, leading to inaccuracies. Few include environmental impacts for complete meals, but even these lack the supporting material to understand the reference recipe behind the meal. Additionally, some databases rely on well-known, but commercially licensed LCA databases that restrict public access (Barco, 2023).

The environmental impacts of individual products alone are insufficient for understanding the real impact of cooking, as people typically follow recipes involving various ingredients in differing quantities. Additionally, people often create new recipes based on available ingredients, resulting in meals that may not have specific names. Assessing the environmental impacts of such flexible and dynamic cooking practices requires tools that can account for these variations. Therefore, environmental impact assessment tools must adapt to the flexible nature of cooking, considering the combined effects of multiple ingredients, varying quantities, and different preparation methods to provide a comprehensive understanding of the environmental footprint of everyday meals.

With this vision in mind, the project work concluded by selecting the Poore and Nemecek (2018) database as the data source for the future tool to calculate environmental impacts of food recipes. This database was selected for its broad coverage of environmental impact categories, despite its limitations in food data resolution. Although lacking specificity at the food level, this database offers recent and comprehensive data reflecting global production methods and supply chain dynamics across 119 countries (Poore & Nemecek, 2018).

Moreover, the role of digital tools and publicly accessible information is paramount in monitoring and mitigating the environmental impacts of societal dietary patterns (Principato et al., 2023). While some other existing databases and online tools provide comprehensive coverage of impact categories, geographical regions, and food items, their utility is often limited by licensing and budgetary constraints (Barco, 2023). For instance, innovative applications such as the Eaternity App demonstrate promising approaches to engaging consumer and promoting sustainable food consumption habits (Eaternity, 2017). However, these applications still typically provide information on the environmental impacts of specific, pre-selected recipes or individual products, overlooking the dynamic nature of actual cooking practices (Barco, 2023). The absence of a publicly accessible application that enables users to input detailed information about the recipes they prepare themselves highlights the need for more flexible tools. Addressing this gap is essential for accommodating the ever-changing landscape of food consumption habits, which is the primary objective of this thesis.

1.3. Goal and scope for this research

The primary objective of this thesis is to develop a prototype for a comprehensive food recipe calculator, capable of evaluating multiple impact categories, thereby empowering consumers to make more sustainable food choices. This prototype places emphasis on the automation of retrieval of LCA scoring, and not the development of the user interface. The main research question of this thesis is: “What are the environmental impacts of selected recipes from a broader impact category perspective?”

To achieve the primary objective, the following secondary objectives were identified:

- Assess the feasibility of combining databases in order to address limitations in data resolution for the Poore and Nemecek (2018) database, which were identified during the specialization project work.
- Select a diverse range of recipes that represent various dietary types.
- Perform a comparative LCA for the selected recipes.
- Automate Python scripts to perform the environmental analysis process by submitting recipe information.

2. Theoretical background

2.1. Poore and Nemecek (2018) database

Poore and Nemecek developed a comprehensive database containing six environmental indicators: GHG emissions, land use, freshwater withdrawals, freshwater withdrawals weighted by local water scarcity, acidifying emissions and eutrophying emissions (Table 1).

Table 1. Indicators and characterizations used in Poore & Nemecek 2018 database.

Indicator	Description	Characterization	Emissions/ Uses Characterized	Unit
Land Use - Occupation time	Refers to the continuous use of some area for a certain period of time for specified land use type.	None	Seed, on- and off-farm arable and permanent crops, fallow land, temporary pasture, permanent pasture	m ²
GHG Emissions	Refers to the quantification of the total amount of GHG emitted over the life cycle of a product.	IPCC 2013 AR5 100-year factors with climate-carbon feedbacks	CO ₂ , CH ₄ , N ₂ O to air	Kg CO ₂ eq

Acidification	Refers to the potential contribution of various sources to increased acidity in the environment.	CML2 Baseline	SO ₂ , NH ₃ , NO _x to air	Kg SO ₂ eq
Eutrophication	Refers to an increase in plant growth brought on by an increase in the availability of nutrients.	CML2 Baseline	NH ₃ , NO _x to air, NO ₃ ⁻ , NH ₄ ⁺ , P, N to water	Kg PO ₄ ³ eq
Freshwater Withdrawals	Refers to the total amount of freshwater removed from a source.	None	Irrigation, drinking, pond, and processing water	Liters
Scarcity-Weighted Freshwater Withdrawals	It takes into account both the volume of water withdrawn and the scarcity of water in the region from which it is drawn.	AWARE* (Boulay et al., 2017)	Irrigation, drinking, pond, and processing water	Liters eq

* Assessing impacts of water consumption based on available water remaining (AWARE).

These impacts are assessed at midpoint level, meaning that the characterization factors are along the cause-effect pathway before the endpoint is reached (Huijbregts et al., 2016). For example, a midpoint level would look at the increased concentration of a chemical in water or land used for a particular activity, which shows a stronger relationship with environmental flows. On the other hand, the endpoint level is usually shown as an impact on human health, ecosystem quality and resource depletion, such as species extinction (Huijbregts et al., 2016). Endpoint indicators are often easier to interpret in terms of the relevance of environmental flows (Hauschild & Huijbregts, 2015).

The freshwater withdrawal impact in Poore and Nemecek (2018), includes irrigation withdrawals, irrigation withdrawals incorporated into feed, animal drinking water, aquaculture pond water and processing water. For scarcity-weighted freshwater withdrawals, the authors assumed that all irrigation water is either evapotranspired or embedded in the product and does not return to the watershed by percolation, which could lead to an overestimation. Therefore, the authors directly multiplied freshwater withdrawals by the spatially explicit AWARE characterization factors by basin and/or country, differentiating the location of each feed crop and the location of post-farm water use (Poore & Nemecek, 2018).

Figure 1 illustrates the emissions included and excluded from the analysis for all the different impact categories. The database provides a detailed breakdown of GHG emissions and land use impacts. For GHG emissions, the data is categorized by life cycle stages: Land Use Change (LUC), Feed, Farm, Processing, Transport, Packaging and Retail. For land use, values are shown for three different land types: arable land, fallow land and permanent pasture. Arable land refers to land used directly for crop production, including fields where crops are grown for harvest. Fallow land is intentionally left uncultivated for a period of time to allow it to recover its fertility or to control weeds and pests. Permanent pasture is land intended for the long-term grazing of livestock, and not used for arable crop production.

The Poore and Nemecek (2018) database was compiled to offer a comprehensive overview of global agricultural practices. The database contains information on more than 38,700 commercially viable farms across 119 countries and includes data on 40 aggregated food groups. These food groups collectively represent about 90% of global protein and calorie consumption, providing a robust foundation for understanding the environmental impacts of food production. The defined system boundaries include primary production, processing, transport, packaging and retail stages. For each study, complete records were kept of both outputs and inputs, including elements such as the amount and type of fertilizer used, irrigation use, soil characteristics and weather conditions. Where data were missing, such as climate data, study coordinates and spatial datasets were used to fill the gaps (Poore & Nemecek, 2018).

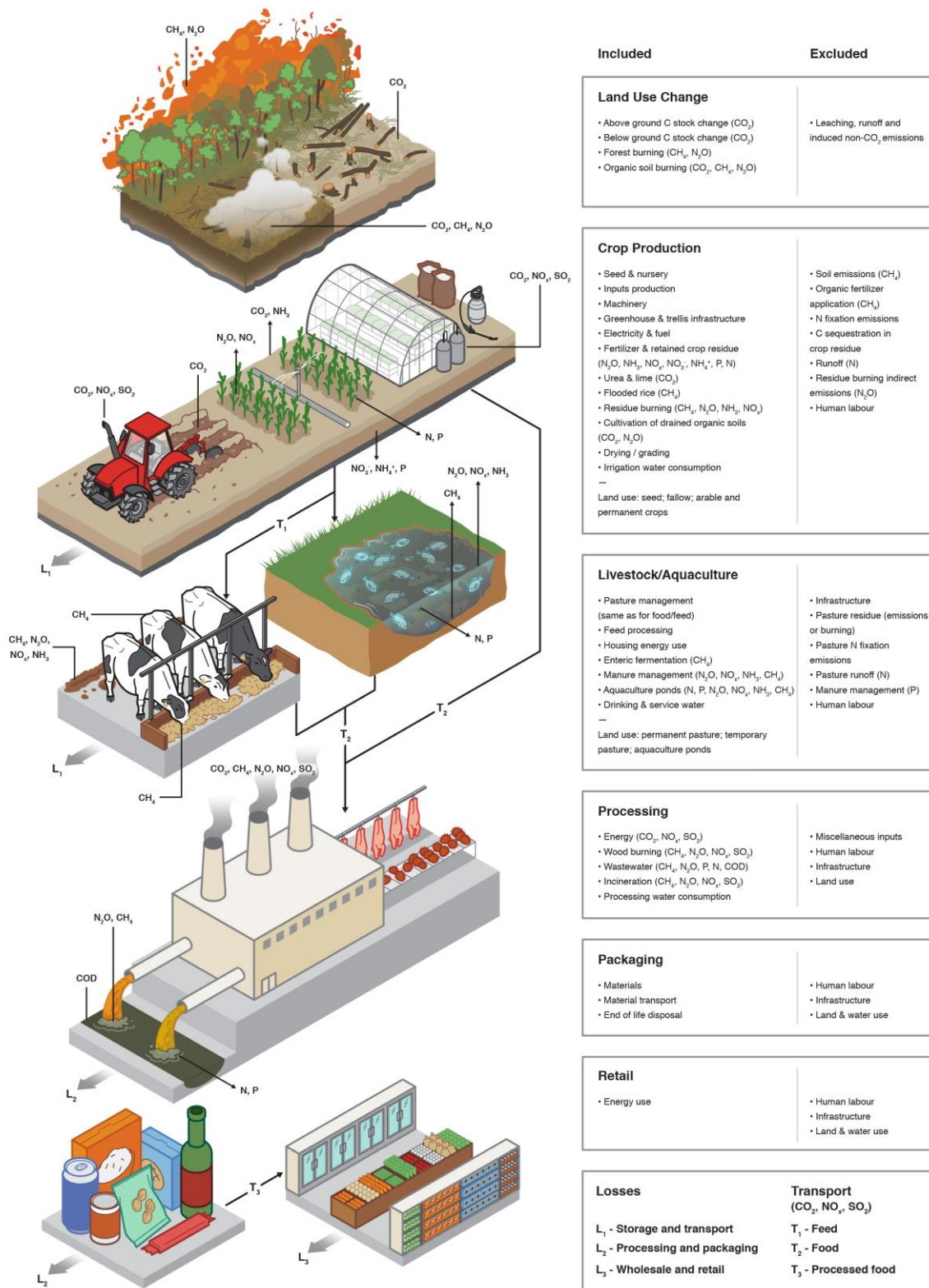


Figure 1. Emissions and resource uses included or excluded by supply chain stage from Poore & Nemecek (2018).

3. Methods

As previously stated, the objective of the prototype is to obtain the calculated environmental impacts from different impact categories for the recipe that the user is submitting to the tool. This is achieved by the submission of the specific quantities of ingredients used during the cooking process, thereby allowing for flexibility. Moreover, the prototype enables the user to compare the environmental impacts of different recipes and ingredients.

The prototype is called DELI Calculator (see Figure 2). The abbreviation DELI stands for “foD rEcipe environmentaL Impact”. DELI Calculator comprises a number of Python files and an Excel template, which the user employs to input the requisite information about the recipe. Once the template has been completed and saved to the appropriate folder, the user can run the Python script. The resulting visualizations will be displayed within the tool. The only task that the user needs to perform is to input the recipe into the Excel template; there is no need to modify the source code. For detailed instructions on how to use the DELI Calculator, please refer to the tutorial in section 8.1. of the Supplementary Information (SI). The DELI Calculator package, with all the necessary content to run the tool, can be found in the attached documents or at the GitHub repository: https://github.com/PaulaBarco/DELI_Calculator.

In this document, the terms “Food Recipe Environmental Impact Calculator” and “DELI Calculator” are used interchangeably.



Figure 2. DELI Calculator logo and slogan.

3.1. Methodological overview

In order to develop the DELI Calculator, an adaptation of the Waterfall Software Development Life Cycle (SDLC) model defined by Saravanos and Curinga (2023) was employed. This approach ensured a structured and efficient approach process, as shown in Figure 3. The SDLC follows a series of well-defined steps, including requirements gathering, design, implementation, testing, and deployment. However, in this thesis, the deployment stage was adjusted to “User experience evaluation”, since deployment would imply that the software is available to the public and this is still the first prototype.

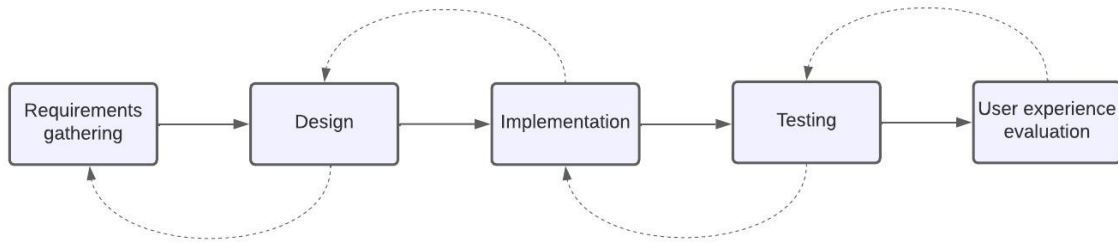


Figure 3. Methodology for developing the DELI Calculator. The figure outlines the iterative and collaborative process, from requirements gathering and design to implementation, testing and user experience evaluation.

The methodology commenced with the requirements gathering phase, during which meetings with supervisors were held to identify the functional and non-functional requirements of the DELI Calculator. Subsequent to this, the design phase focused on conceptualizing and specifying the architecture of the DELI Calculator. Design decisions were made with a focus on usability, scalability, maintainability and universality. Consequently, the code was made generic to accommodate various recipes, allowing different data sources to be integrated without altering the source code. This ensures that the DELI Calculator can handle as many recipes as the user wishes to input. However, the code will still require updates, maintenance, and refactoring for future enhancements.

Once the design phase was finalized, the implementation began with developing the DELI Calculator by using the Python programming language. This phase involved writing code to implement the functionality specified in the design. Coding standards and best practices were adhered in order to guarantee the quality and maintainability of the code.

Following implementation, the DELI Calculator underwent testing to validate its functionality and performance. This included unit testing, integration testing and system testing with the objective of identifying and addressing any defects or issues. Test cases were developed based on the objectives of the thesis regarding testing different dietary patterns. Finally, a user experience evaluation was carried out in which a tutorial was developed so that the DELI Calculator could be tested by selected end users, with the intention of getting feedback from them and allowing adjustments for the final delivery.

The methodology employed in developing the DELI Calculator was iterative and collaborative, allowing for continuous feedback throughout the development process. Python was selected as the primary tool due to its versatility, simplicity and robustness. Additionally, Python is compatible with a diverse array of systems, which expands the potential user base and facilitates the realization of open access information.

3.2. Steps in the development process

This section explains the details of each step and gives more information about the DELI Calculator's features.

3.2.1. Requirement analysis

Some of the key requirements for the DELI Calculator were related to the database selection process carried out in the previous project work. This selection process included criteria outlined by the International Reference Life Cycle Data System (ILCD) handbook, utilizing the Pedigree Matrix Data Quality Indicators (DQI): completeness, precision/uncertainty, methodological appropriateness and consistency, temporal representativeness and geographical representativeness (JRC-IES, 2010). Some of these criteria were in turn adjusted to check the resolution of the food item inventory, the system boundary, the impact categories, the quality of the sources and the number of studies involved in the databases. Details of the criteria used, the Pedigree Matrix and the results from the project work can be found in section 8.2. of the SI.

Selecting the Poore and Nemecek (2018) database enabled the DELI Calculator to provide a wide range of impact categories in its results, fulfilling the primary objective of this thesis.

Requirements were then defined for the DELI Calculator itself, in terms of how the information would be collected from the users and how the results would be presented to them were defined. It was said that the users should enter the recipe information regardless of the type of recipe they were cooking, and that the DELI Calculator should generally offer the most conventional ways of using the ingredient units in the recipes. In addition, the users should be able to use the DELI Calculator with as little effort as possible and receive the results as PNG files and also displayed within the tool. It was also said that the DELI Calculator should be intuitive, simple and easy to use.

3.2.2. Design phase

In order to meet the requirements defined in the previous section, it was crucial to draw a sketch of how the DELI Calculator would work from start to finish. To do this, it was important to consider the user experience when entering data into the DELI Calculator. These involved asking questions such as what the most common ingredient units in recipes in general are and what kind of ingredients the user would expect to find. Two main components emerged from this brainstorming: the need to convert from traditional ingredient units in the kitchen to the units that follow the functional unit in the database, and the need for the database to have a better resolution of food items inventory.

For the first component, the environmental impact values in the Poore and Nemecek (2018) database have a functional unit of kilograms (kg) of food. However, ingredients in recipes are not always measured in kilograms or grams. For instance, a recipe might call for three carrots, which makes it difficult to determine the exact weight in grams. Therefore, in such cases, ingredients need to be converted into kilograms to accurately scale the environmental impacts to the precise amount needed. This conversion process must account for various units such as tablespoons (tbsp), teaspoons (tsp), cups, slices, cloves, bunches and deciliters (dl), as well as whole units like one potato or three carrots.

Different conversion factors for selected food items from food groups such as oils, vegetables, fruit and cereals were identified through various sources found via Google search. The conversion table, which includes references to the sources, is available in an Excel file titled “conversion_factors” included in the attached documents at “./data/interim” or at the GitHub repository.

For the second component, the resolution limitations of the Poore and Nemecek (2018) database for food items were addressed by increasing the number of food items using AGRIBALYSE 3.0 (Asselin-Balençon et al., 2022). Details of AGRIBALYSE 3.0 can be found in Section 8.3. of the SI. Following the same food groups as in the Poore and Nemecek (2018) database, new food items were added, specifying “PA based on AGRIBALYSE”, where PA refers to the first two letters of the author's name of this thesis (Paula Alejandra). This expansion increased the resolution from 195 to 528 food items. It is important to note that the new database assigned the impact values from the 40 aggregated food groups from Poore and Nemecek (2018) database to the corresponding new food items. Thus, food items within the same group share identical impact values, regardless of the specific item. Figure 4 shows the first rows of the food item list. The full table is available in an Excel file titled “ingredient_foodgroup” included in the attached documents at “./data/interim” or at the GitHub repository.

Ingredient	Food group	Datasource
Bread wheat	Wheat & Rye (Bread)	Poore&Nemecek
Rye	Wheat & Rye (Bread)	Poore&Nemecek
Soft wheat	Wheat & Rye (Bread)	Poore&Nemecek
Wheat	Wheat & Rye (Bread)	Poore&Nemecek
Wheat and rye bread	Wheat & Rye (Bread)	Poore&Nemecek
Wheat bread	Wheat & Rye (Bread)	Poore&Nemecek
Wheat flour	Wheat & Rye (Bread)	Poore&Nemecek
Bagel	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Bran grain bread	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Bread, French bread, baguette	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Bread, gluten free	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Brioche	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Chocolate croissant	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Corn tortilla wrap, to be filled	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Crispbread, extruded and grilled	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Croissant	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Croutons	Wheat & Rye (Bread)	PA based on AGRIBALYSE
English muffin, prepacked	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Grissini or bread stick	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Milk roll, prepacked	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Panini bread	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Pita bread	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Rolls for hamburger/hotdog (buns), prepacked	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Rusk	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Rye bread, and wheat	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Sandwich loaf	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Wheat tortilla wrap, to be filled	Wheat & Rye (Bread)	PA based on AGRIBALYSE
Maize	Maize (Meal)	Poore&Nemecek

Figure 4. Illustration of the first rows of the Food item list.

3.2.2.1. *Service blueprinting*

To better describe how the DELI Calculator works, the Service Blueprinting (SB) approach was used (Bitner et al., 2008). This is a customer-centric approach to service innovation and improvement. SB is useful because it clearly shows the steps in a service process, making it easier to design a new process or document and improve an existing one (Ryu et al., 2020).

In the economic system, a service consists of someone using the service (customer) and someone offering/facilitating the service (employee). According to Bitner et al. (2008), SB consists of five categories that illustrate the main components of the service being mapped out. These categories are physical evidence, customer actions, on-stage (visible) employee actions, back-stage (invisible) employee actions, and support processes.

Firstly, **physical evidence** includes elements encountered by both customers and employees, such as a website, locations, and emails. Secondly, **customer actions** describe the various activities customers perform throughout the service experience, such as browsing websites and using a digital app. Thirdly, **frontstage or visible employee actions** encompass the interactions customers have with employees. Fourthly, **backstage or invisible contact employee actions** involve behind-the-scenes activities that are critical to service delivery, such as content creation and order packaging. Fifthly, **support processes** include internal activities that support service delivery, such as those performed by third-party vendors and payment systems. In addition, SB include lines to delineate each category, facilitating a comprehensive understanding of how service components interact and enabling the identification of potential sources of customer dissatisfaction.

In the context of this thesis, the customers are referred to as users, and the employee is referred to as the developer, who is the author of this thesis. In this context, the developer conducted test sessions, interacting with the users to demonstrate how the DELI Calculator works.

Figure 5 shows the SB for the DELI Calculator. Since the DELI Calculator does not have a user interface (at this point), users will receive a package containing the Python source code and an Excel template. Users begin by opening the provided Excel file named “Recipe_template” and entering the recipes they wish to evaluate. After saving the completed template, users run the Python script using a Python interpreter to process the data. The resulting visualizations and analyses will then be available for users to review.

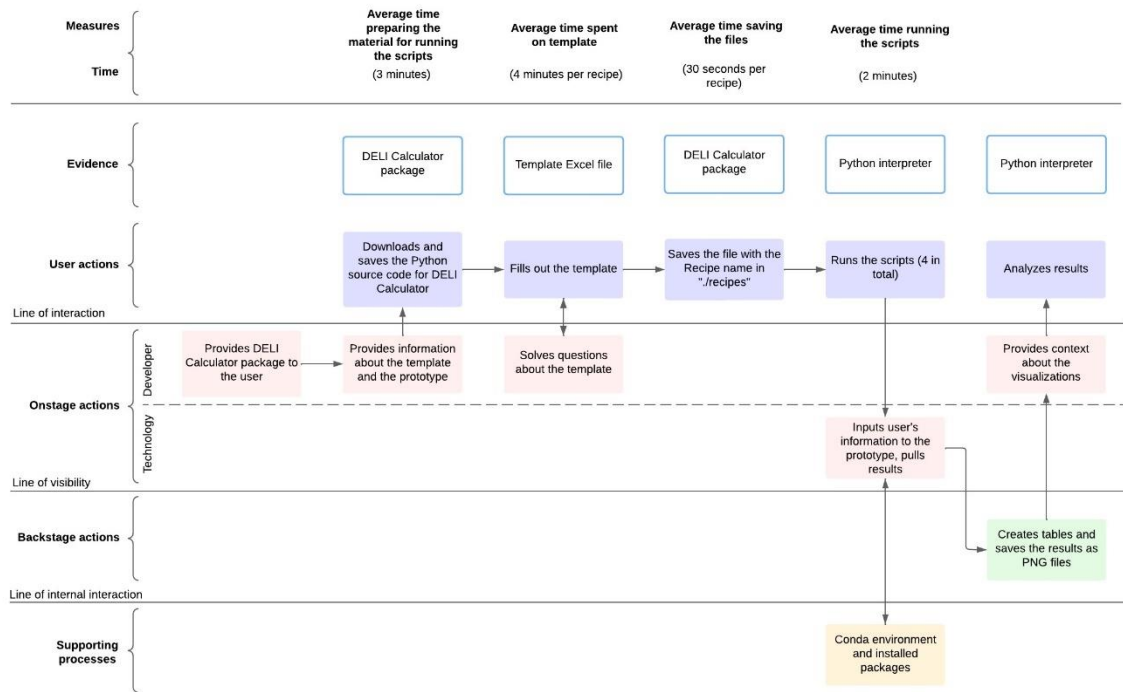


Figure 5. Designed Service Blueprint for the Food Recipe Environmental Impact Calculator.

The Python source code is the initial version and may be developed further into a full Python package available on PyPI (Python Package Index) or SaaS (Software as a Service - cloud/web-based) solution in the future.

The template has three columns: Ingredient, Amount and Unit (see Figure 6). The Ingredient column contains the list of 528 food items. The Amount column is blank because the user will be entering the exact amounts, and the Unit column shows the different typical units for each food. In this template, the column labels and the Ingredients column were locked so that the user could not change them. For the Amount column, the user can enter the amounts for each food item as desired. For the Unit column, if the unit is different from grams and the user has the exact amount in grams, the user can type “grams” instead of the original unit, as the code currently only converts other units to grams, so no conversion is necessary in this case. See the instructions for completing the template in section 8.4. of the SI.

Ingredient	Amount	Unit
Bread wheat		cup
Rye		cup
Soft wheat		cup
Wheat		cup
Wheat and rye bread		cup
Wheat bread		cup
Wheat flour		cup

Figure 6. Illustration of the first rows of the Recipe template file.

Once the template is completed, the user saves the file (with a name that is recognizable for the results) in the “./recipes” folder within the received source package. A new file must be established for each recipe. Thus, the user ends up with as many files as there are recipes to evaluate. Once this is done, the user can open the DELI_Calculator folder in any tool that allows the code to be executed, such as Visual Studio Code (VS Code). The user must then run the Python scripts in the following order: `cleaning`, `harmonized_datasources`, `analyze_data` and `visualizations`. The `visualizations` script will retrieve the graphs with the results for the recipes evaluated.

Throughout the process, the developer provided information about the different steps of the procedure to run the DELI Calculator and answered questions.

As part of the design phase, a number of time metrics were specified in order to test the procedure during the test phase. These metrics are specified in the upper part of Figure 5.

- Average time preparing the material for running the scripts: this metric quantifies the duration it takes for users to download the Python source code and save it on their computers.
- Average time spent on template: this metric captures the time users spend completing the `Recipe_template` for each recipe. It includes the time from opening the file to saving it with the corresponding recipe name.
- Average time saving the files: given that each recipe corresponds to a separate file, users must save all completed templates in the “./recipes” folder. This metric measures the time spent on this saving process.
- Average time running the scripts: this metric accounts for the time users dedicated to running the scripts. This metric is broken down into user time and computation time. Notably, users employing personal computers may experience variations in computation time.

3.2.3. Implementation and repository structure

In this phase, the goal was to develop the Python code to meet the specifications of the designed DELI Calculator. The implementation was carried out using VS Code and real-time collaboration with supervisors was facilitated through GitHub. The following subsections provide a detailed explanation of the Python code and the structure of the repository.

3.2.3.1. Source (./src)

The code can be explained in the following logical sections, which also shows the correct order to read the different Python files in order to use the DELI Calculator. In section 3.2.3.3., the detailed explanation of the last four sections of the code is provided.

1. constants.py: defines constants used throughout the different scripts, mainly related to defining paths to different data directories.
2. project_functions.py: contains generic functions mostly for reading data sets.
3. cleaning.py: focuses on cleaning the Poore and Nemecek (2018) database, which contains environmental impact data for various food groups.
4. harmonize_datasources.py: improves the resolution of the cleaned dataset by merging it with additional details.
5. analyze_data.py: includes the conversion to grams, the calculation of the recipe's impacts and the generation of all the different tables needed for the visualizations.
6. visualizations.py: focuses on creating the different charts to compare recipes and provide the breakdown into ingredients.

In this folder, there is another Python file called “sensitivity_uncertainty.py”. While this part is not part of the DELI Calculator package itself, it plays an important role in the results of this thesis. Running this code is necessary for performing sensitivity and uncertainty analyses on certain values. More details are provided in section 3.2.4.1. and 3.2.4.2. respectively.

3.2.3.2. *Data (./data)*

There are four folders indicating the level of transformation of the data.

1. raw - it contains the original Poore and Nemecek (2018) database downloaded from the academic article website. The database can be downloaded by following this link: <https://www.science.org/doi/10.1126/science.aag0216> ("Resources" section).
2. interim - it contains all the other databases that are either saved from the code or created to be merged with other databases.
 - a. clean_poore_and_nemecek: this is the database obtained from the cleaning.py file.
 - b. conversion_factors: this file contains the conversion factors from units like tbsp, tsp, cup, and so on to grams.
 - c. food_item_poore_and_nemecek: this file is obtained after merging the ingredient_foodgroup file and clean_poore_and_nemecek in order to add the food items to the original impact database.
 - d. recipe_template: the user has to download this template and fill out the amounts for each ingredient that is part of the recipe, and specify the unit, if it is unit, grams, tbsp, tsp, cup, bunch, etc. Then, save the document in this folder again without changing the name of the file.
 - e. ingredient_foodgroup: this file allows to increase the resolution of the Poore and Nemecek (2018) database in terms of food detail.
3. recipes – it contains the Recipe_template files filled in by the user.
4. impacts – it contains all the results obtained by the DELI Calculator.

3.2.3.3. Code

The specific Python version and Python functions or libraries dependencies for running the DELI Calculator are:

- Python (version 3.8.13)
- Pandas (version 1.4.2 or later), for data manipulation and analysis.
- Pathlib (part of the standard library in Python 3.4 and later), for file system path manipulation.
- OS (part of the standard library), for interacting with the operating system, in particular for listing files in a directory.
- Numpy (version 1.22.4 or later), for numerical operations on arrays.
- Matplotlib.pyplot (version 3.5.3 or later), for creating visualizations.

As mentioned above, the code for the DELI Calculator is made up of six logical sections. Here are the details of the last four sections.

- `cleaning.py`: In the cleaning process for the Poore and Nemecek (2018) database, the main steps performed were to subset the necessary data, rename columns for clarity, drop rows with Not a Number (NaN) values and save the cleaned dataset.
- `harmonize_datasources.py`: To improve the resolution of the cleaned dataset, a merge was performed between the “fooditem_foodgroup” database, which contains the 528 food items and their corresponding food group based on the Poore and Nemecek (2018) database. Some cleaning was then required, essentially moving the Ingredient column to the beginning of the table. The improved dataset was then saved.
- `analyze_data.py`: In order to accurately multiply the amounts entered by the user by their corresponding impacts, a recipe conversion to kilograms was necessary. The code iterates through each Excel file contained in the “./recipes” folder to read the data, filtering and converting units to grams using predefined conversion factors from another database. Once converted, the code merges and cleans the data to ensure unit consistency.

Next, the cleaned recipe data is merged with the impact data from the Poore and Nemecek (2018) database to perform the impact calculations. This involves multiplying the ingredient quantities by their respective impact factors. Finally, the values are converted from grams to kilograms to ensure that the impact values are appropriately scaled to match the amounts provided in grams. The code then stores the impact data for each ingredient and the total impact for each recipe in separate Excel files.

After completing the previous step and saving all relevant impacts, the code creates and saves the impact labels. It then aggregates the data from all the impact files into a single DataFrame, which is also saved in an Excel file. Additionally, the code concatenates individual ingredient impact files into a single DataFrame and saves it. Finally, the code calculates the total impacts for specific categories, such as GHG emissions and land use, and saves the results. These categories are further detailed in the database, broken down into life cycle stages and land types respectively.

- `visualizations.py`: Here the code creates four main graphs.
 - Spider web chart: This section creates a spider web chart to visualize the relative environmental performance of recipes. It reads data and processes it to scale values relative to the recipe with the highest impact for each category.
 - Stacked bar chart (per recipe and adding the ingredient breakdown): This section creates stacked bar charts to visualize the distribution of impacts per ingredient for each recipe. It processes the data for each recipe and calculates the percentage contribution of each ingredient.
 - Bar chart and Stacked bar chart (per impact category): This section filters the data for a specific impact category, and a bar chart is created to compare the recipes within that category. Next, the recipe with the highest overall impact is selected, and a stacked bar chart is generated to show the breakdown by ingredients for that recipe. This process is repeated for each of the six impact categories.

3.2.4. *Testing and validation*

The aim of this phase was to validate the functionality and performance of the DELI Calculator. Unit testing, integration testing and system testing were carried out to identify and resolve any issues.

In terms of unit testing, each individual component of the code was tested by displaying the results in the interactive window in VS Code. This was to ensure that it performed as expected and to verify its correctness and reliability. After the unit testing, the next step was to ensure that different modules and components worked together seamlessly. Integration testing was performed to validate the interaction between scripts and to detect any errors when the results of one script were used in the following script. This was done by comparing the actual values obtained from the DELI Calculator with the values manually calculated in Excel. The Excel file containing the integration testing can be found in the attached documents.

Once the integration testing was complete, system testing was carried out to evaluate the DELI Calculator as a whole. The aim of this testing step was to verify that the overall system met the specified requirements and performed well under various conditions. For this purpose, test cases were developed based on the objectives of this thesis, with a particular focus on testing different dietary patterns. These test cases helped to simulate real world scenarios and validated the DELI Calculator's ability to accurately handle different dietary inputs.

The test cases were based on a recipe and tried different substitutions for the main ingredient to represent different dietary patterns. The recipe used for the test cases was *Lomo Saltado* (Figure 7), a dish that originated in Peru but is well known throughout Latin America. The recipe is shown below. In addition, the recipe amounts were given for 4 servings as this is the standard format for recipes in general.



Figure 7. Lomo saltado with beef. Retrieved from [HOLA.com](hola.com)([Lomo saltado peruano \(hola.com\)](http://hola.com))

Lomo Saltado

Ingredients:

- 455 grams (1 lb) beef sirloin or tenderloin
- 3 tbsp soy sauce
- 2 tbsp red wine vinegar
- 3 cloves garlic
- 1 large red onion
- 2 medium tomatoes
- 1 yellow chili pepper (ají amarillo)
- 1/4 cup fresh cilantro
- 4 tbsp vegetable oil
- Salt and pepper to taste
- 4 servings of cooked white rice
- 455 grams French fries

The different variations of this recipe included different types of meat to replace the beef, and also plant based options. The substitutions chosen were chicken, salmon, tofu, mushrooms and chickpeas. The amounts for the substitutions were not based on a calorie and protein analysis, but simply transferred the same amount of 455 grams to the other foods. Each recipe has its own Excel file input.

As mentioned in section 3.2.2., the Poore and Nemecek (2018) database assigns impact values to food group, and these values are applied uniformly to all food items within each group. The ingredients selected for this analysis adhere to the following impact distribution.

- Beef uses the impact values for Bovine Meat (beef herd).
- Chicken uses the impact values for Poultry Meat.

- Salmon uses the impacts values for Fish (farmed).
- Tofu uses the impact values for tofu.
- Mushrooms uses the impact values for the category “Other vegetables”, which also includes vegetable such as artichokes, chicory, cucumber and similar items.
- Chickpeas uses the impact values for “Other pulses”, a category that includes various types of beans such as lentils, brown beans, adzuki beans, faba beans, and others.

3.2.4.1. *Sensitivity analysis*

The results of an LCA study can be influenced by various factors, making it essential to assess the robustness and reliability of the findings (Goldstein et al., 2016). In this study, after evaluating the distribution of impacts for each recipe, a sensitivity analysis was conducted to determine the sensitivity of the results to changes in the amount of the primary ingredient and the impact values. These two parameters significantly contribute to the final result as explained in section 3.2.3.1.2.

The max/min method was applied, wherein the amount in grams of the main ingredient and the impact values were altered to assess the impact of a $\pm 10\%$ change on the results. To replicate these values, please refer to the instructions in section 8.5 in the SI.

3.2.4.2. *Uncertainty analysis*

An uncertainty analysis was performed to understand the variability of the data for the ingredient with the highest impact on the results. This analysis was motivated by the availability of values from the databases reviewed in the previous project work, making it interesting to examine how variable this input can be, as this variability could impact the outcome of the DELI Calculator. The impact of GHG emissions was chosen for this analysis because it had the highest number of values, as 9 databases were involved and 100% of them included this category. For the remaining categories – land use, acidification, eutrophication, freshwater withdrawals and scarcity-weighted freshwater withdrawals – no databases were found that covered these impact categories with the same units or methodologies as those used in the Poore and Nemecek (2018) database. The uncertainty analysis included a statistical analysis of the mean, standard deviation and variance and an estimation of the confidence interval. In addition, histograms and box plots were used to visualize the distribution of the values. To replicate these values, the code can be found in the Python file called “sensitivity_uncertainty.py”.

3.2.5. *User experience evaluation*

This phase included five key aspects: user testing sessions, user preparation, contextual introduction, self-guided testing and feedback collection.

During this phase, test sessions were conducted with eight selected users to evaluate the DELI Calculator in real time. These users were chosen based on their good computer skills and knowledge of Python. Seven of these users are part of the same master's program as this thesis, while one user is external. The group comprised individuals from various nationalities, providing an international perspective on the DELI Calculator. The group included four Norwegians, two Germans, one Brazilian, and one New Zealander.

These sessions were structured to provide users with context about the DELI Calculator, helping them understand its purpose and functionality. Following the procedure defined in the BP approach, the defined time metrics were also tested, and users were asked to bring at least two recipes to evaluate to ensure efficient use of time during the sessions.

The sessions were designed to last approximately 20 minutes, during which the users interacted with the DELI Calculator on their own computers. This hands-on approach allowed for a more authentic testing experience and enabled users to provide practical feedback based on their direct interaction with the tool. After each session, a short survey was sent to the test users to collect formal feedback and directly measure the user experience.

The survey consisted of seven questions, the first five of which followed a Likert scale format, and the remaining two questions were open. The Likert scale went from 1 to 5 and the purpose of these questions was to evaluate the user experience. The last two questions were designed to gather comments on things that the user would like to stand out or highlight, as well as suggestions for improvement. The questions are shown below.

1. From 1 to 5, how well do you understand the purpose of this Food recipe environmental impact calculator? (1. not at all, 5. very well)
2. From 1 to 5, how useful do you think the results from the Food recipe environmental impact calculator are? (1. not useful at all, 5. very useful)
3. From 1 to 5, how easy/clear was it to enter the recipe details into the Excel template? (1. not easy/clear at all, 5. very easy/clear)
4. From 1 to 5, how clear was the procedure to use the Food recipe environmental impact calculator? (1. not clear at all, 5. very clear)
5. From 1 to 5, how would you rate the overall experience using the Food recipe environmental impact calculator? (1. not pleasant, 5. very pleasant)
6. Is there anything that you want to highlight from this prototype?
7. Are there any comments or suggestions that can help us improve the user experience and prototype?

The survey format can also be viewed at: <https://forms.gle/zMoggUVJTa44GBhj9>.

4. Results

4.1. Comparative LCA for different dietary patterns

4.1.1. Relative environmental performance of recipes

After saving the Recipe_template files with each variation of *Lomo Saltado*, the DELI Calculator was used to analyze the environmental impacts of different dietary patterns. Table 2 shows the composition of these recipes and the quantities of ingredients used in each of them.

Table 2. Composition of the recipes for Lomo Saltado and the quantities of ingredients used in each of them.

Ingredients (g/meal)	<i>Lomo Saltado</i> with beef	<i>Lomo Saltado</i> with chicken	<i>Lomo Saltado</i> with salmon	<i>Lomo Saltado</i> with chickpeas	<i>Lomo Saltado</i> with mushrooms	<i>Lomo Saltado</i> with tofu
Beef	455	-	-	-	-	-
Chicken	-	455	-	-	-	-
Salmon	-	-	455	-	-	-
Chickpeas	-	-	-	455	-	-
Mushrooms	-	-	-	-	455	-
Tofu	-	-	-	-	-	455
Rice	400	400	400	400	400	400
French fries	455	455	455	455	455	455
Olive oil	55.2	55.2	55.2	55.2	55.2	55.2
Tomatoes	300	300	300	300	300	300
Onions	110	110	110	110	110	110
Chilli pepper	30	30	30	30	30	30
Vinegar	30	30	30	30	30	30
Soy sauce	60	60	60	60	60	60
Coriander	4	4	4	4	4	4
Garlic	15	15	15	15	15	15
Salt	20	20	20	20	20	20
Black pepper	40	40	40	40	40	40

Figure 8 shows the first result of the DELI Calculator, which produces a spider web chart comparing the relative environmental performance of the six different versions of the *Lomo Saltado* recipe: beef, chicken, chickpeas, mushrooms, salmon and tofu.

Relative environmental performance of recipes

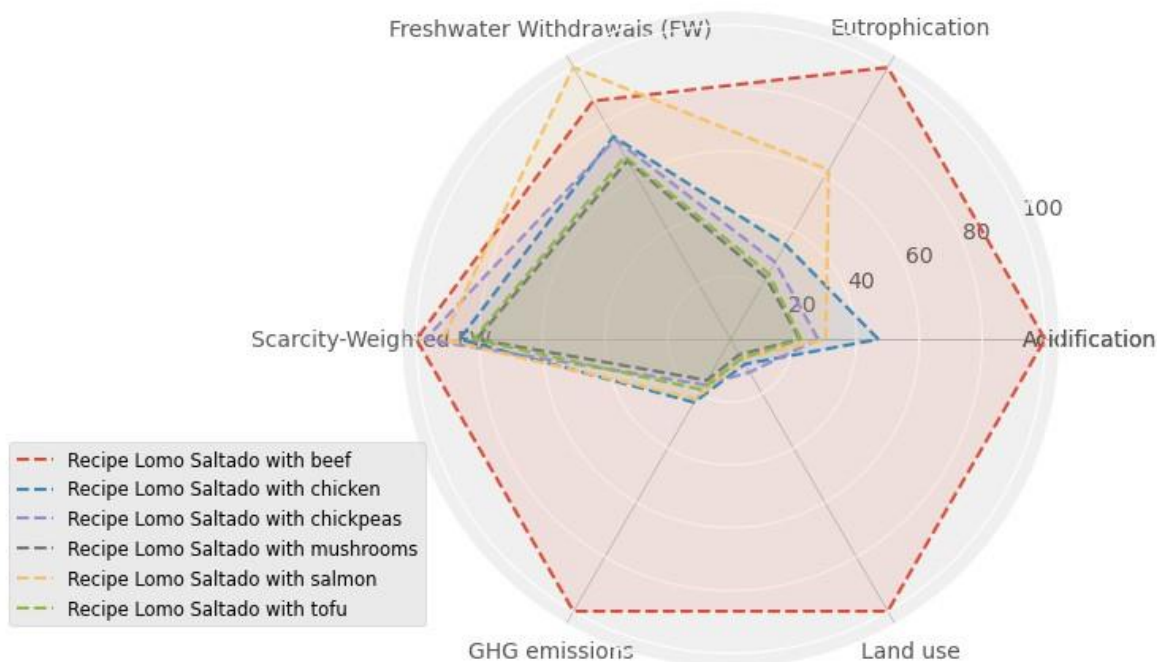


Figure 8. Spider web chart showing the relative environmental performance of different Lomo Saltado recipes. The highest impact in each category is set to 100%, with other values scaled accordingly.

The recipe with beef has the highest GHG emissions. Chicken and salmon versions follow, with each producing significantly lower GHG emissions, approximately 23% and 22% respectively of those generated by the beef version. The recipes with tofu, chickpeas and mushrooms show even lower GHG emissions, at respectively 19%, 17%, and 15% compared to beef.

In terms of land use, the *Lomo Saltado* with beef also has the highest impact. The other versions have much lower impacts, ranging between 6% and 12% of the land use required for the beef version. Chickpeas have the highest impact among these alternatives at 12%, while mushrooms have the lowest at 6%.

In terms of acidification potential, the beef version is again the highest. Chicken and salmon versions follow, representing 47% and 30% of the acidification caused by beef. The remaining versions – chickpeas, tofu, and mushrooms- have acidification impacts under 28%, with mushrooms having the lowest impact.

For eutrophication, the *Lomo Saltado* with beef is the highest, followed by the recipe with salmon at 62% of beef's impact. Chicken, chickpeas, tofu and mushrooms versions follow with 35%, 28%, 24% and 23% respectively.

The results for freshwater withdrawals show a slightly different pattern, with the salmon recipe having the highest freshwater withdrawal, and with the beef version accounting for

88% of this impact. This is followed by chicken and chickpeas recipes with 75% and 74% respectively. *Lomo Saltado* with tofu and mushrooms have the lowest impact at 67% and 66% respectively.

In terms of scarcity-weighted freshwater withdrawals, the results are more even across the recipes. The recipe with beef again has the highest impact, closely followed by the chickpeas recipe at 97%. Salmon, chicken, tofu, and mushrooms versions follow with 92%, 86%, 82% and 81% respectively.

Overall, the beef recipe has the highest environmental impact in most of the impact categories, except for Freshwater Withdrawals where the recipe with salmon leads. The salmon recipe has a high environmental impact, especially in freshwater withdrawals, scarcity-weighted freshwater withdrawals and acidification. The recipe with chicken has a moderate impact on all the indicators. The recipe with chickpeas has a high freshwater withdrawal, but a low impact in the other categories. Finally, the recipes with tofu and mushrooms consistently have the lowest environmental impacts across all categories, making them the most environmentally friendly options for this recipe.

4.1.2. Environmental impacts distribution by Ingredient for each recipe

As a second result of the DELI Calculator, Figures 9 – 14 present the results for the *Lomo Saltado* recipe variations, highlighting the contribution of the ingredient to each impact category. As previously mentioned, the land use category is divided into different land types and the GHG emissions are broken down into life cycle stages.

When analyzing the results for the *Lomo Saltado* with beef, it can be seen that beef is the main contributor to land use with an average contribution of 82%, including arable, fallow and permanent pasture. Beef also significantly contributes to GHG emissions across all life cycle stages with an average contribution of 69%. Additionally, it also leads in acidification and eutrophication impacts with 80% and 78% respectively (see Figure 9). In contrast, the picture changes when analyzing freshwater withdrawals and scarcity-weighted freshwater withdrawals, where rice has the highest impact, contributing 55% and 42% respectively, followed by beef and olive oil. While French fries, olive oil, and other minor ingredients contribute to different impact categories, their impacts are minor compared to those of beef and rice.

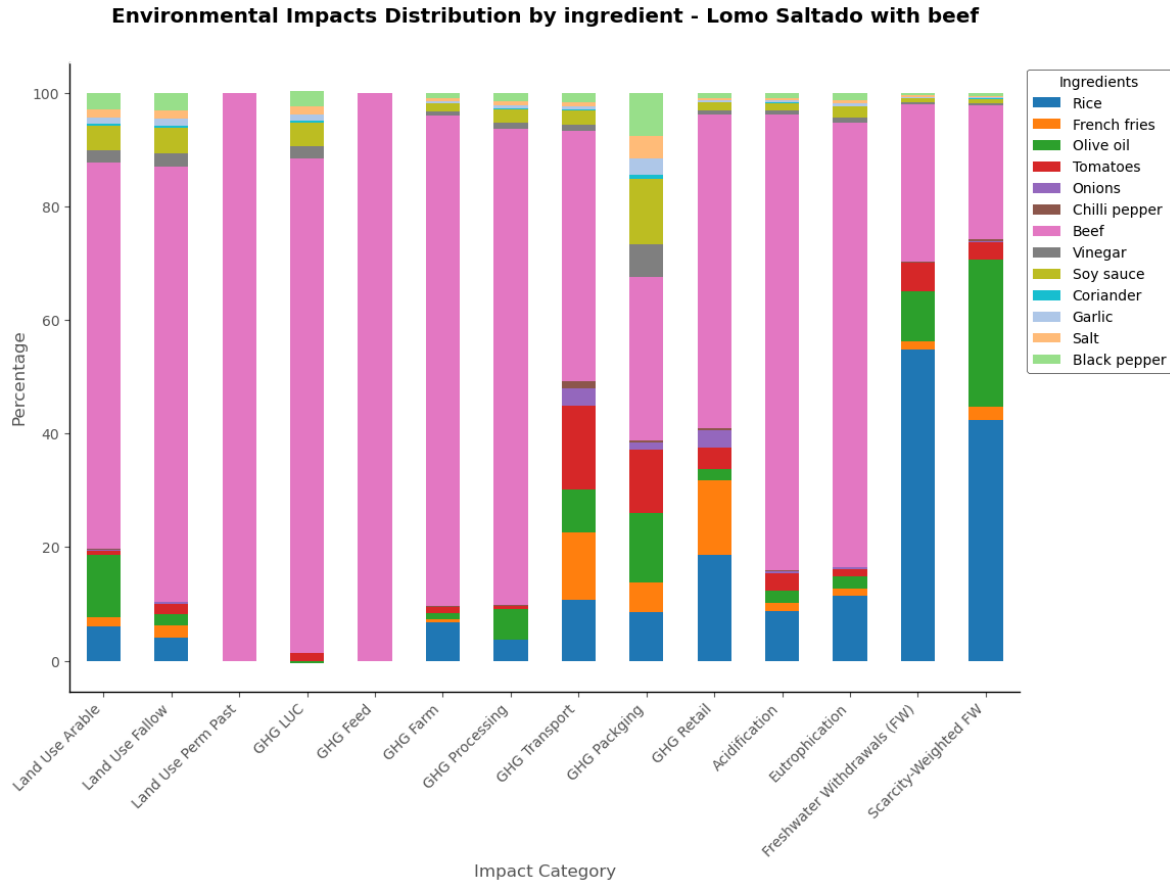


Figure 9. Stacked bar showing the environmental impacts distribution by ingredient for the *Lomo Saltado* with beef recipe

Following the pattern of earlier results, the *Lomo Saltado* with chicken reveals that chicken has the highest impact in GHG emissions, particularly in the feed stage with a contribution of 100%, followed by land use change, feed, processing and retailing stages with an average of 57%. It also significantly impacts land use with 45%, acidification with 58% and eutrophication with 37%, though its overall contribution to these impacts is 30% lower in average compared to the contribution of beef (see Figure 10). There are no impacts for land use with permanent pasture, which makes sense as this category is primarily associated with livestock. Additionally, rice surpasses chicken in terms of freshwater withdrawals and scarcity-weighted freshwater withdrawals. Moreover, rice and olive oil have a higher contribution across all impact categories compared to the *Lomo Saltado* with beef. Notably, olive oil is also contributing with negative emissions in the GHG emissions for the land use change.

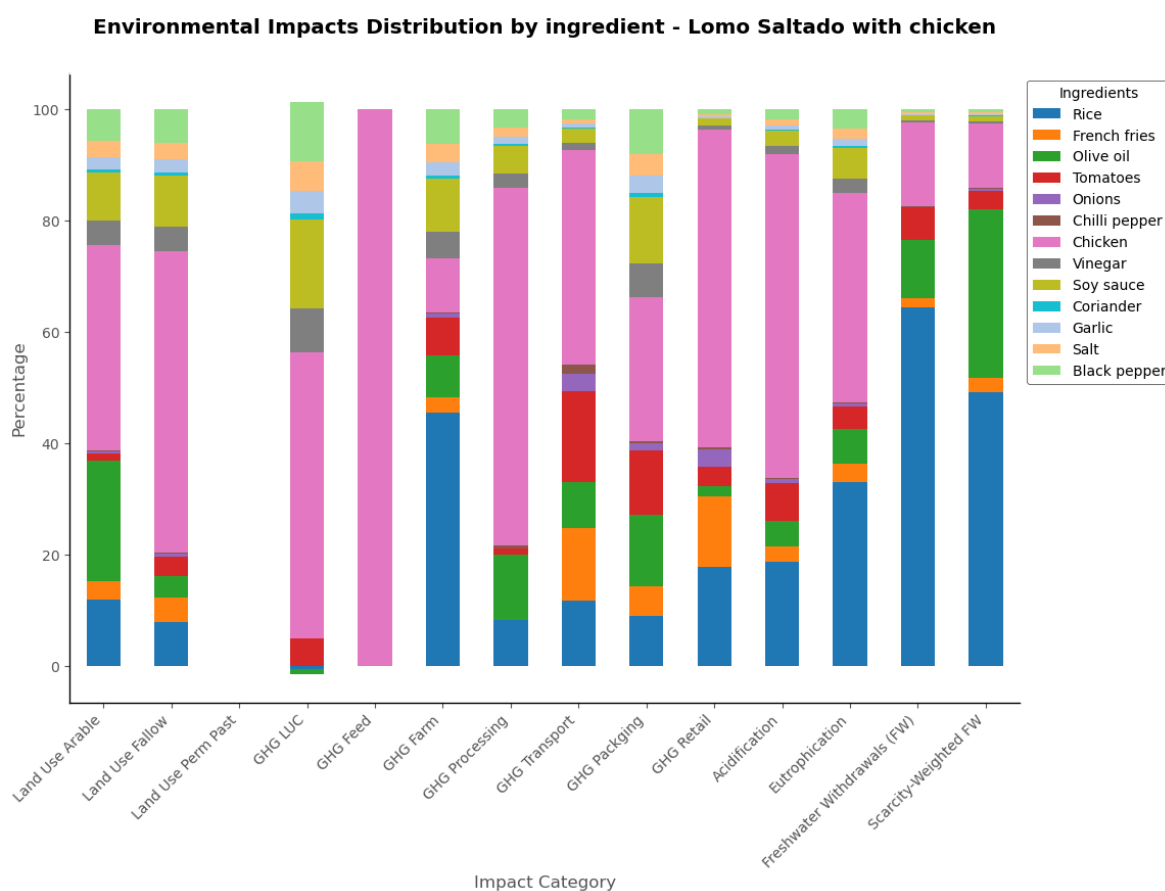


Figure 10. Stacked bar showing the environmental impacts distribution by ingredient for the *Lomo Saltado* with chicken recipe

The results for the recipe with salmon exhibit slightly different behavior compared to the previous recipes. Salmon has the highest overall impact; however, its contributions are more evenly distributed with the other ingredients (see Figure 11). For example, in land use (arable), olive oil has a slightly higher contribution than salmon with 26% versus 22%, and salmon contributes 31% to fallow land use. In terms of GHG emissions, salmon has the highest impact in the feed and farm stages, with 100% and 37%, respectively. Ingredients like soy sauce, rice, olive oil, tomatoes and French fries increasingly contribute to the GHG emissions for this recipe. For example, olive oil and rice have higher GHG emissions contributions during processing, while tomatoes, olive oil and soy sauce have higher contributions to GHG emissions during packaging. For acidification, salmon leads with 35%, followed by rice at 29%. In eutrophication, salmon has the highest impact by far, with 65%. Although salmon has the highest overall impact, its overall contribution is 46% lower than beef and 20% lower than chicken, except for freshwater withdrawals and scarcity-weighted freshwater withdrawals. Again, there are no impacts for land use with permanent pasture, which makes sense as this category is primarily associated with livestock. Additionally, rice continues to show a high contribution to freshwater withdrawals and scarcity-weighted freshwater withdrawals.

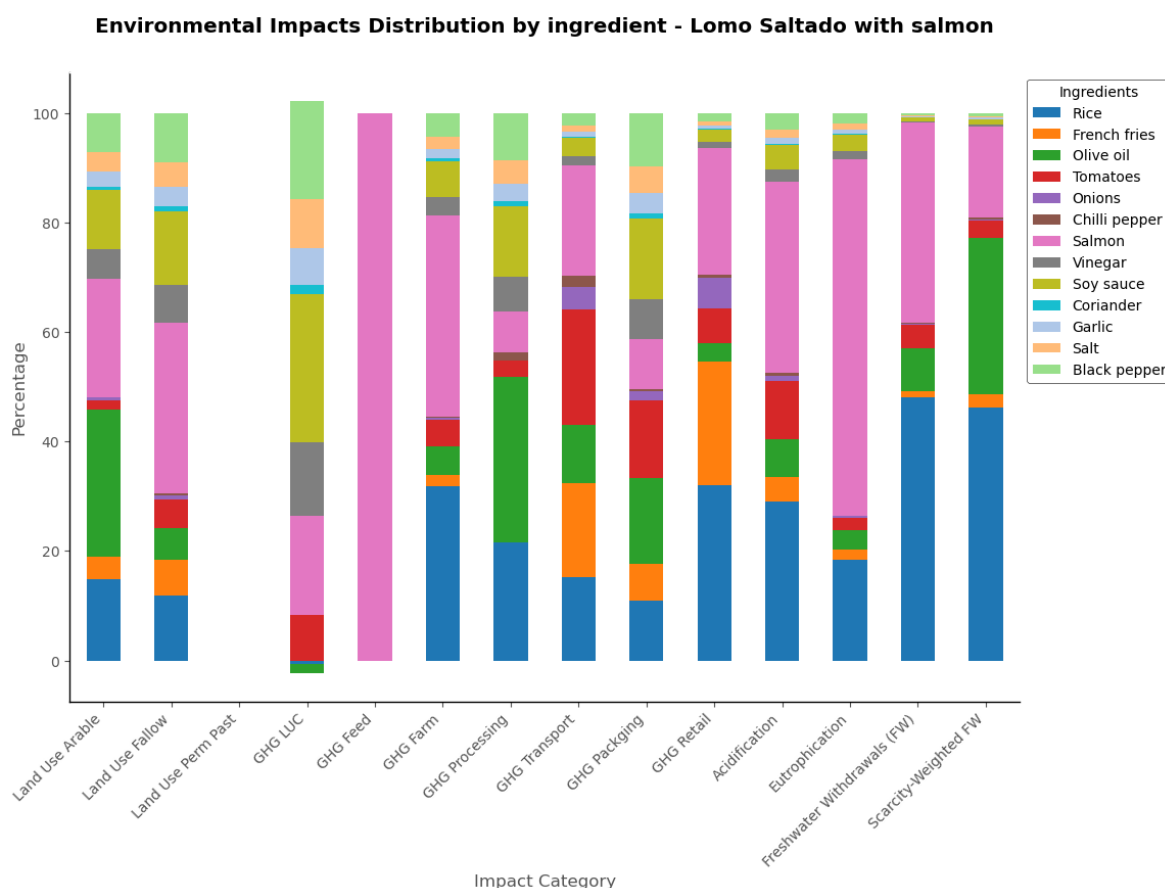


Figure 11. Stacked bar showing the environmental impacts distribution by ingredient for the Lomo Saltado with salmon recipe

Similar to the previous recipe, the *Lomo Saltado* with chickpeas exhibits a more even distribution of impacts among the ingredients (see Figure 12). Chickpeas have the highest impact on land use with an average contribution of 61% and a moderate average contribution of 20% to all other categories. Rice emerges as the ingredient with the highest overall impact in all the impact categories except land use, significantly contributing to GHG emissions (20%), acidification (32%), eutrophication (41%), freshwater withdrawals (65%) and scarcity-weighted freshwater withdrawals (44%). There are no impacts associated with land for permanent pasture or GHG emissions from feed, as this recipe does not involve livestock. Additionally, olive oil's contribution is more pronounced compared to previous recipes, with a surprising 33% for GHG emissions in the processing stage and 26% for scarcity-weighted freshwater withdrawals. However, olive oil also contributes negative emissions in the GHG emissions for land use change. Moreover, ingredients such as tomatoes, soy sauce, vinegar, and black pepper start to have a noticeable impact on GHG emissions from land use change.

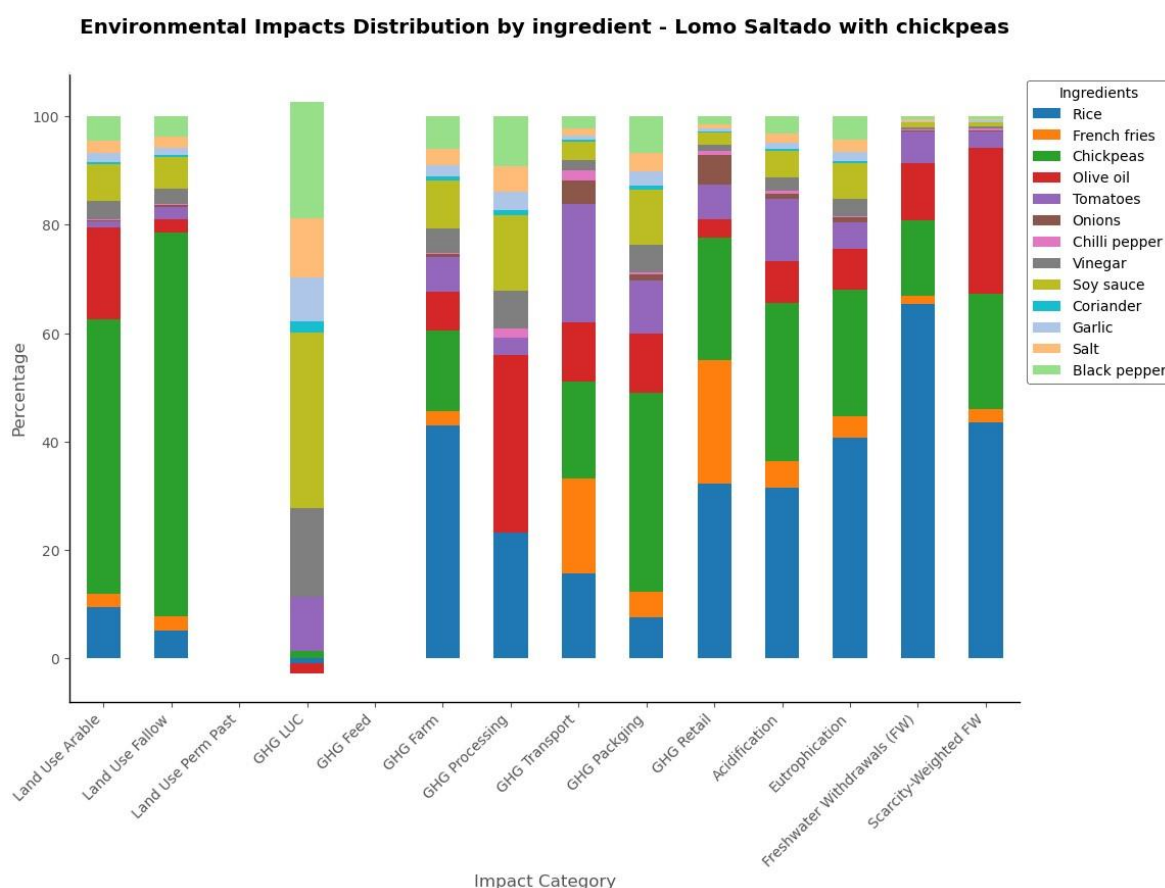


Figure 12. Stacked bar showing the environmental impacts distribution by ingredient for the *Lomo Saltado* with chickpeas recipe

In line with the results for the recipe with chickpeas, in the recipe with mushrooms, rice has the highest overall impact, contributing an average of 18% to land use. It also contributes 50% and 40% to GHG emissions in the farming, and retailing stages, respectively, 40% to acidification, 51% to eutrophication, 73% to freshwater withdrawals, and 52% to scarcity-weighted freshwater withdrawals (see Figure 13). As with the chickpeas version, there are no impacts associated with land for permanent pasture or GHG emissions for feed since this recipe does not involve livestock. Unlike other recipe versions, mushrooms, the main ingredient, only have the highest contribution in GHG emissions for transportation at 27%. Olive oil's contribution is more pronounced compared to previous recipes, with 33% in land use (arable), and 26% and 16% in GHG emissions for the processing and packaging stage, respectively. Olive oil also continues to contribute negative emissions in GHG emissions for land use change. Additionally, ingredients such as tomatoes, soy sauce, vinegar, and black pepper still have a noticeable impact on GHG emissions from land use change.

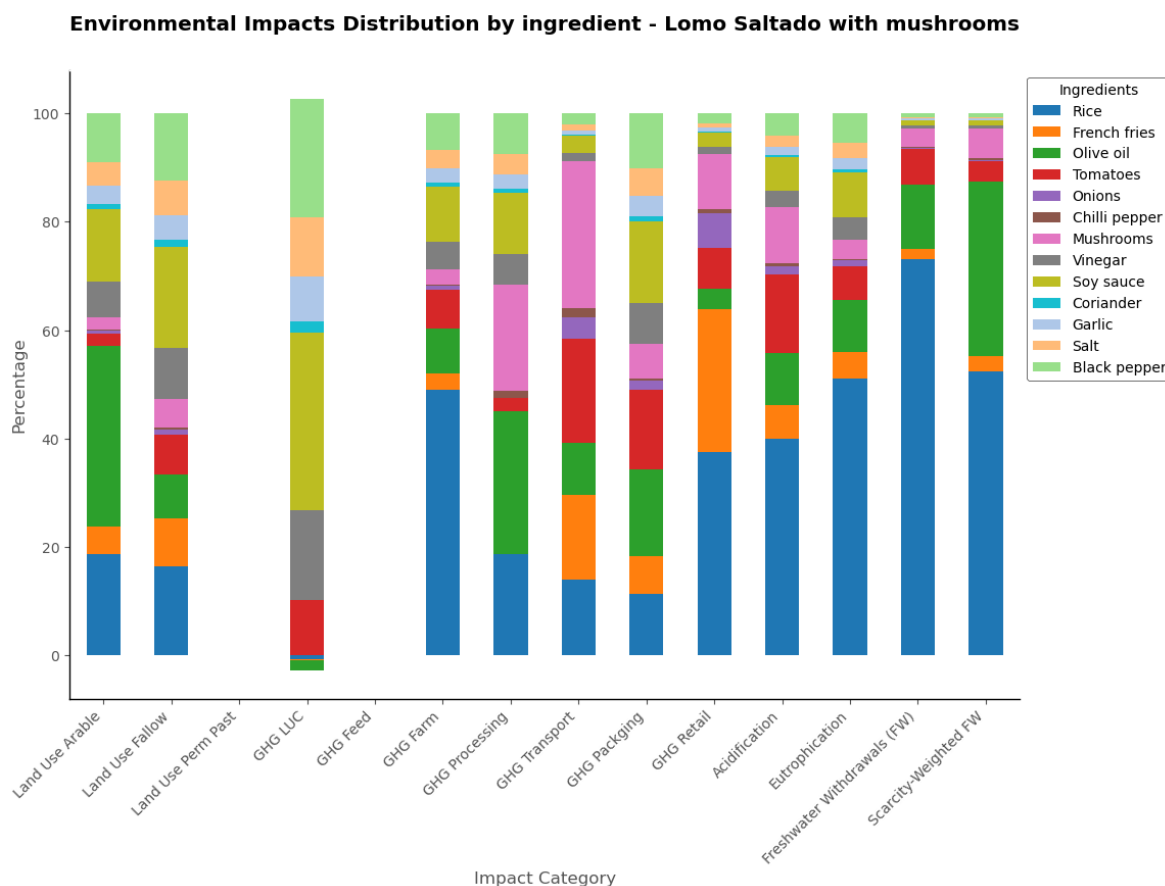


Figure 13. Stacked bar showing the environmental impacts distribution by ingredient for the Lomo Saltado with mushrooms recipe

The *Lomo Saltado* recipe with tofu, presented similar results to the other plant-based recipes. Tofu and rice are the ingredients with the highest overall environmental impact across all categories (see Figure 14). Tofu has the highest impact on land use, averaging 25%, followed by olive oil, averaging 17%. Tofu also shows the highest impact for GHG emissions across all life cycle stages, with 38% on average. However, rice is the ingredient with the highest impact in the remaining categories, contributing 39% to acidification, 48% to eutrophication, 71% to freshwater withdrawals and 52% to scarcity-weighted freshwater withdrawals. As with the other plant-based versions, there are no contributions to land use in permanent pasture or GHG emissions for feed since this recipe does not involve livestock. Again, other ingredients, such as tomatoes, vinegar, French fries, and soy sauce, still contribute to specific categories, highlighting their varied roles in the recipe's environmental footprint.

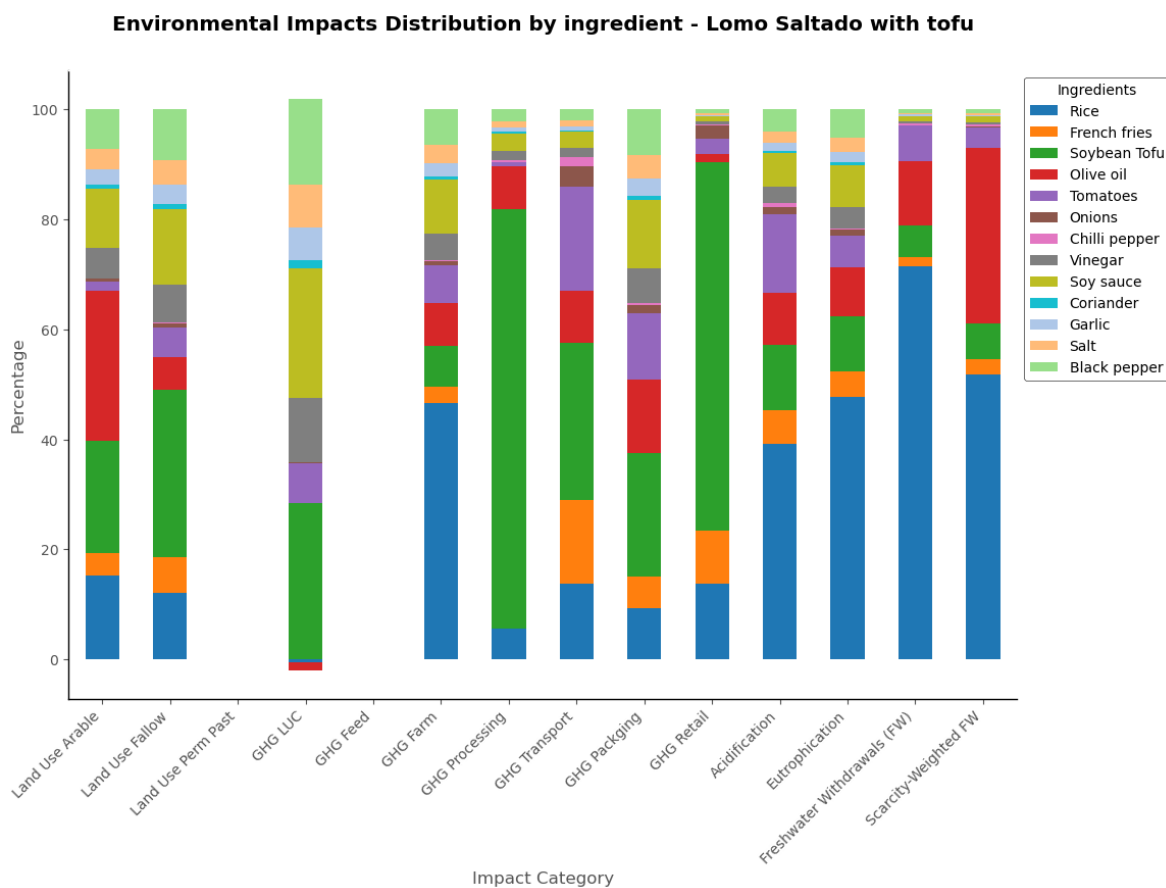


Figure 14. Stacked bar showing the environmental impacts distribution by ingredient for the *Lomo Saltado* with tofu recipe

4.1.3. Results per impact category

Figure 15 provides a comprehensive comparison of the environmental impacts across the different versions of the *Lomo Saltado* recipe. In terms of GHG emissions, the recipe with beef exhibits the highest overall emissions, particularly in the farming stage, with emissions of 20.74 kg CO₂ eq, significantly surpassing all other recipes. In comparison, the emissions

from other ingredients are below 3.4 kg CO₂ eq. The recipes with salmon and chicken follow the recipe with beef in terms of emissions, but their impacts are considerably lower. The recipe with beef is also significantly higher than other recipes for GHG emissions for land use change with 8.5 kg CO₂ eq. Plant-based recipes show much lower emissions across all life cycle stages, with minimal emissions compared to the beef version.

Regarding land use, the recipe with beef shows a very high impact in permanent pasture (76 m²), a category with no impact from other recipes. For arable and fallow land, all recipes show relatively low and similar impacts, with plant-based recipes showing values below 8.9 m².

In terms of acidification potential, the recipe with beef again leads with the highest impact at 0.11 kg SO₂ eq, followed by the version with chicken at 0.05 kg SO₂ eq. Plant-based recipes display lower impacts, all below 0.03 kg SO₂ eq. For eutrophication, the recipe with beef has the highest impact with 0.103 kg PO₄₃ eq, followed by the recipe with salmon with a moderate impact at 0.06 kg PO₄₃ eq. The plant-based and chicken versions have a lower impact, with plant-based recipes below 0.03 kg PO₄₃ eq and the recipe with chicken at 0.036 kg PO₄₃ eq.

As previously shown in the spider web chart in Figure 8, the results for freshwater withdrawals show a different picture, with the recipe with salmon requiring the most freshwater withdrawals at 1634 liters, followed closely by the recipe with beef with 1432 liters. The recipes with chicken and chickpeas have similar impacts, with 1219 liters and 1201 liters respectively. The remaining plant-based options show the lowest impacts, all below 1099 liters.

For scarcity-weighted freshwater withdrawals, the recipe with beef again shows the highest impact at 40831 liters eq. This is followed closely by the recipe with chickpeas at 39697 liters eq. The recipe with salmon follows with 37432 liters eq. The recipe versions with chicken, tofu and mushrooms have similar impacts, with values of 35220 liters eq, 33379 liters eq and 33003 liters eq respectively.

In summary and similar to the results shown in Figure 8, the recipe with beef consistently shows the highest environmental impacts across all categories, particularly in GHG emissions (land use change and farming stage), land use (permanent pasture), acidification, and eutrophication. The recipe with salmon also has significant impacts but is lower than beef in most categories, except for freshwater withdrawals. Plant-based options generally have lower impacts, with chickpeas having slightly higher impacts than mushrooms and tofu, yet still significantly lower than the beef recipe.

The details of the impact values per ingredient for the recipe with the highest impact on each impact category are given in section 8.6 of the SI.

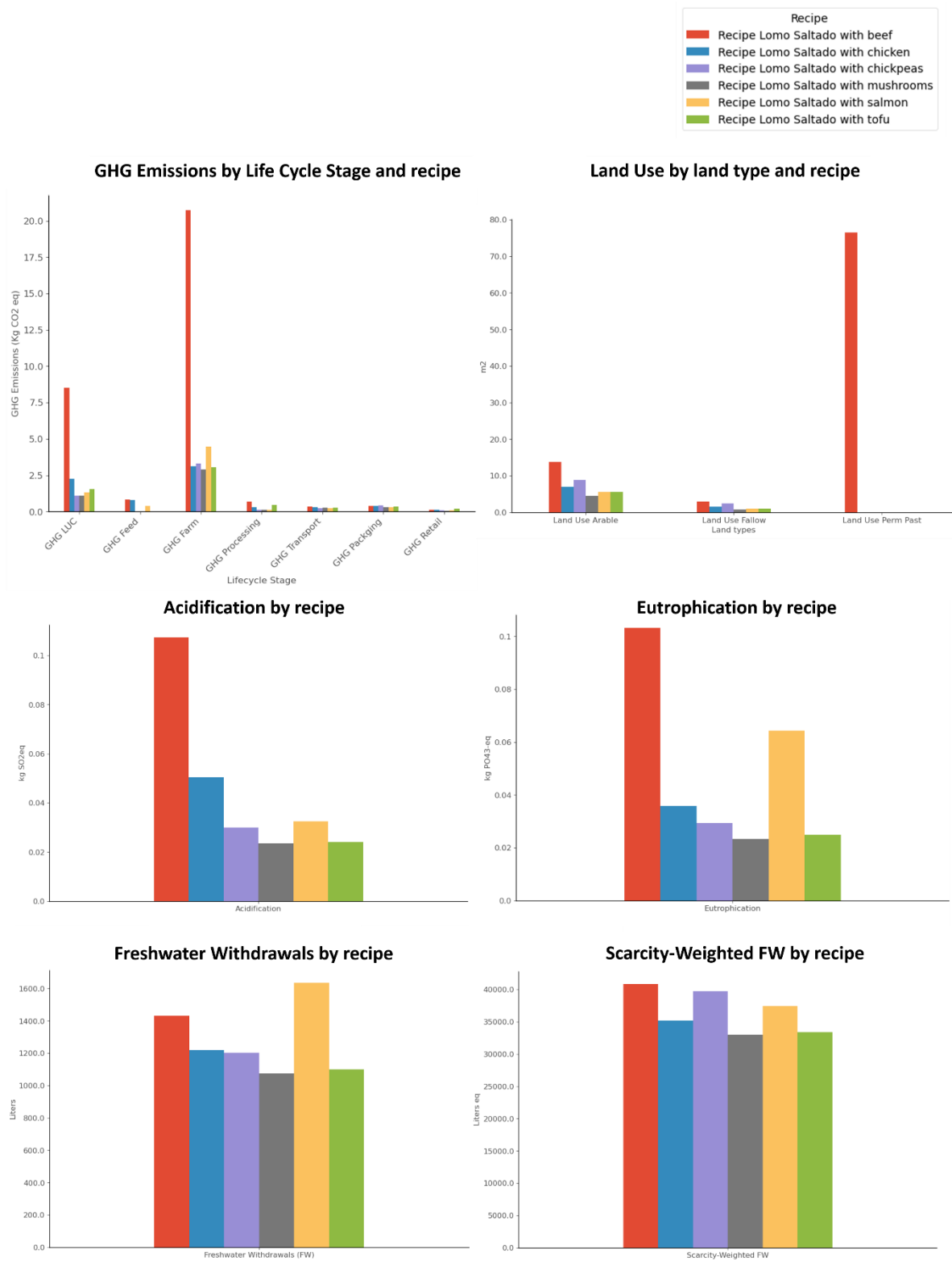


Figure 15. Comparison of the different versions of Lomo Saltado across the six impact categories

4.1.4. Sensitivity analysis

The sensitivity analysis assessed how changes in the quantity of the main ingredient – beef, chicken, salmon, chickpeas, mushrooms and tofu – and its impact values affect the results for different *Lomo Saltado* recipes. This analysis included a range of scenarios, including a 10% increase and decrease in the main ingredient's quantity (measured in grams), a substitution of 10% of meat-based ingredients with a plant-based alternative, and a 10% increase and decrease in the impact values.

Table 3 presents the original values sourced from Poore and Nemecek (2018) database for the different versions of the *Lomo Saltado* recipe, along with their respective outcomes across the six different environmental impact categories.

Table 3. Original results for the different versions of the Lomo Saltado recipe across the six impact categories using Poore and Nemecek (2018) database.

Recipe	GHG emissions (kg CO₂ eq)	Land use (m²)	Acidification (kg SO₂ eq)	Eutrophication (kg PO₄₃ eq)	Freshwater withdrawals (liters)	Scarcity-weighted freshwater withdrawals (liters eq)
<i>Lomo Saltado</i> with beef	31.67	93.21	0.11	0.10	1432.11	40831.37
<i>Lomo Saltado</i> with chicken	7.34	8.50	0.05	0.04	1218.72	35218.95
<i>Lomo Saltado</i> with salmon	6.92	6.63	0.03	0.06	1634.13	37431.61
<i>Lomo Saltado</i> with chickpeas	5.30	11.29	0.03	0.03	1201.43	39697.06
<i>Lomo Saltado</i> with mushrooms	4.77	5.24	0.02	0.02	1073.57	33003.10
<i>Lomo Saltado</i> with tofu	5.87	6.53	0.02	0.02	1099.05	33379.38

4.1.4.1. Sensitivity analysis for the amounts of the main ingredients

The quantities of the main ingredients were first decreased by 10%, reducing them from the original 455 grams to 409.5 grams. Table 4 illustrates the outcomes of the recipes with this 10% reduction in the main ingredients.

Table 4. Results for the different versions of the Lomo Saltado recipe after decreasing the amount of the main ingredient by 10%.

Recipe	GHG emissions (kg CO2 eq)	Land use (m2)	Acidification (kg SO2 eq)	Eutrophication (kg PO43 eq)	Freshwater withdrawals (liters)	Scarcity-weighted freshwater withdrawals (liters eq)
<i>Lomo Saltado</i> with beef	28.96	84.40	0.10	0.10	1392.48	39868.46
<i>Lomo Saltado</i> with chicken	7.06	8.16	0.05	0.03	1200.43	34817.27
<i>Lomo Saltado</i> with salmon	6.69	6.48	0.03	0.06	1574.30	36808.67
<i>Lomo Saltado</i> with chickpeas	5.23	10.67	0.03	0.03	1184.87	38847.57
<i>Lomo Saltado</i> with mushrooms	4.75	5.22	0.02	0.02	1069.80	32823.01
<i>Lomo Saltado</i> with tofu	5.74	6.39	0.02	0.02	1092.73	33161.67

Table 5 displays the variations across the different impact categories compared to the original values when the quantity remains at 455 grams shown in Table 3.

Table 5. Variations across the different impact categories compared to the original results from Poore and Nemecek (2018) database for the different versions of the Lomo Saltado recipe.

Recipe	Change in GHG emissions	Change in Land use	Change in Acidification	Change in Eutrophication	Change in Freshwater withdrawals	Change in Scarcity-weighted freshwater withdrawals
<i>Lomo</i>	-8.56%	-9.45%	-8.03%	-7.83%	-2.77%	-2.36%

<i>Saltado with beef</i>						
<i>Lomo Saltado with chicken</i>	-3.78%	-4.01%	-5.80%	-3.75%	-1.50%	-1.14%
<i>Lomo Saltado with salmon</i>	-3.41%	-2.32%	-3.49%	-6.51%	-3.66%	-1.66%
<i>Lomo Saltado with chickpeas</i>	-1.38%	-5.49%	-2.91%	-2.33%	-1.38%	-2.14%
<i>Lomo Saltado with mushrooms</i>	-0.43%	-0.27%	-1.02%	-0.36%	-0.35%	-0.55%
<i>Lomo Saltado with tofu</i>	-2.22%	-2.20%	-1.18%	-1.01%	-0.58%	-0.65%

Reducing the main ingredient by 10% resulted in significant environmental reductions in all the impact categories, particularly for beef. GHG emissions decreased by 8.56%, land use by 9.45%, acidification by 8.03% and eutrophication by 7.83%. Chicken showed moderate reductions, with GHG emissions decreasing by 3.78%, land use by 4.01% and eutrophication by 3.75%. However, acidification showed a higher reduction of 5.80%. Salmon exhibited similar trends, with a 3.41% reduction in GHG emissions, a 2.32% reduction in land use, a 3.49% reduction in acidification, and a significant decrease in eutrophication by 6.51%.

Plant-based options generally presented smaller reductions, with the exception of chickpeas, which, despite a minor reduction in GHG emissions of 1.38%, it showed a notable reduction in land use by 5.49%. Tofu also showed modest reductions as well, with the highest reductions being a 2.22% reduction in GHG emissions and a 2.20% reduction in land use. Mushrooms experienced very small reductions overall, with the highest being a 1.02% reduction in acidification.

The lowest reductions across all impacts were observed for freshwater withdrawals and scarcity-weighted freshwater withdrawals, with maximum reductions of 3.66% and 2.36% respectively.

Subsequently, the quantities of the main ingredients were increased by 10%, increasing them from the original 455 grams to 500.5 grams. Table 6 shows the outcomes of the recipes with this 10% increase in the main ingredients.

Table 6. Results for the different versions of the Lomo Saltado recipe after increasing the amount of the main ingredient by 10%.

Recipe	GHG emissions (kg CO2 eq)	Land use (m2)	Acidification (kg SO2 eq)	Eutrophication (kg PO43 eq)	Freshwater withdrawals (liters)	Scarcity-weighted freshwater withdrawals (liters eq)
<i>Lomo Saltado</i> with beef	34.38	102.03	0.12	0.11	1471.74	41794.29
<i>Lomo Saltado</i> with chicken	7.62	8.84	0.05	0.04	1237.01	35620.62
<i>Lomo Saltado</i> with salmon	7.16	6.79	0.03	0.07	1693.96	38054.55
<i>Lomo Saltado</i> with chickpeas	5.37	11.91	0.03	0.03	1217.99	40546.54
<i>Lomo Saltado</i> with mushrooms	4.79	5.25	0.02	0.02	1077.35	33183.19
<i>Lomo Saltado</i> with tofu	6.00	6.67	0.02	0.03	1105.38	33597.10

The variations in the impact categories compared to the original values when the quantity remains at 455 grams mirrored those displayed in Table 5, but in a positive direction.

4.1.4.2. Sensitivity analysis for reducing the quantity of the main ingredients by 10% in meat-based Lomo Saltado recipes and substituting with a plant-based option

Given the observed positive environmental implications of reducing the quantity of the main ingredients, it is interesting to explore the impact of replacing the reduced amount (45.5 grams reduction) with a plant-based alternative, particularly in the context of the *Lomo Saltado* recipe featuring meat (beef, chicken, or salmon). Mushrooms are selected due to their previously minimal environmental impact in this recipe. This study focuses on analyzing a scenario where the reduced amount of meat in the Lomo Saltado recipe is substituted with mushrooms. Specifically, a reduction of 45.5 grams (10% of 455 grams) in the meat quantity is replaced with mushrooms. The results of these substitutions are summarized Table 7.

Table 7. Results for the Lomo Saltado meat-based recipes after replacing the 10% reduction with mushrooms.

Recipe	GHG emissions (kg CO2 eq)	Land use (m2)	Acidification (kg SO2 eq)	Eutrophication (kg PO43 eq)	Freshwater withdrawals (liters)	Scarcity-weighted freshwater withdrawals (liters eq)
<i>Lomo Saltado</i> with beef	28.98	84.42	0.10	0.10	1396.26	40048.55
<i>Lomo Saltado</i> with chicken	7.08	8.18	0.05	0.03	1204.20	34997.36
<i>Lomo Saltado</i> with salmon	6.71	6.49	0.03	0.06	1578.08	36988.76

Table 8 shows the change compared to the original values when the quantities of the main from Poore and Nemecek (2018) database.

Table 8. Variations across the different impact categories compared to the original results from Poore and Nemecek (2018) database for the different versions of the meat-based Lomo Saltado recipe.

Recipe	Change in GHG emissions	Change in Land use	Change in Acidification	Change in Eutrophication	Change in Freshwater withdrawals	Change in Scarcity-weighted freshwater withdrawals
<i>Lomo Saltado</i> with beef	-8.49%	-9.44%	-7.80%	-7.74%	-2.50%	-1.92%
<i>Lomo Saltado</i> with chicken	-3.50%	-3.84%	-5.33%	-3.51%	-1.19%	-0.63%
<i>Lomo Saltado</i> with salmon	-3.11%	-2.10%	-2.75%	-6.38%	-3.43%	-1.18%

When replacing 10% of the meat with mushrooms, the recipes showed better environmental impacts than those originally containing only meat. The differences are minimal when comparing the recipes with 10% less meat to those where the 10% reduction is compensated with mushrooms. For instance, beef with mushrooms resulted in slightly worse GHG emissions (-8.49% compared to -8.56%), and land use impacts remained almost identical (-9.44% compared to -9.45%). Chicken with mushrooms showed a marginal increase in impacts across all categories, with an average increase of only 0.20% compared to the reduction-only scenario. Similarly, salmon with mushrooms demonstrated

a similar performance, except for acidification, which increased by 0.74% (-2.75% with mushrooms compared to -3.49% with just the reduction). Despite these slight increases, substituting a portion of meat with mushrooms can still maintain or improve the environmental performance of these recipes.

4.1.4.3. Sensitivity for the impact values

The impact values for all the six impact categories were first decreased by 10%. Table 9 illustrates the outcomes of the recipes with this 10% reduction in the impact values.

Table 9. Results for the different Lomo Saltado recipes after decreasing the impact values by 10%.

Recipe	GHG emissions (kg CO2 eq)	Land use (m2)	Acidification (kg SO2 eq)	Eutrophication (kg PO43 eq)	Freshwater withdrawals (liters)	Scarcity-weighted freshwater withdrawals (liters eq)
<i>Lomo Saltado</i> with beef	28.96	84.40	0.10	0.10	1392.48	39868.46
<i>Lomo Saltado</i> with chicken	7.06	8.16	0.05	0.03	1200.43	34817.27
<i>Lomo Saltado</i> with salmon	6.69	6.48	0.03	0.06	1574.30	36808.67
<i>Lomo Saltado</i> with chickpeas	5.23	10.67	0.03	0.03	1184.87	38847.57
<i>Lomo Saltado</i> with mushrooms	4.75	5.22	0.02	0.02	1069.80	32823.01
<i>Lomo Saltado</i> with tofu	5.74	6.39	0.02	0.02	1092.73	33161.67

Table 10 shows the change across the different impact categories compared to the original values, maintaining consistent impact values as derived from Poore and Nemecek (2018) database, as shown in Table 3.

Table 10. Variations across the different impact categories compared to the original values from Poore and Nemecek (2018) database for the different versions of the Lomo Saltado recipe.

Recipe	Change in GHG emissions	Change in Land use	Change in Acidification	Change in Eutrophication	Change in Freshwater withdrawals	Change in Scarcity-weighted freshwater withdrawals
<i>Lomo Saltado</i> with beef	-8.56%	-9.45%	-8.03%	-7.83%	-2.77%	-2.36%
<i>Lomo Saltado</i> with chicken	-3.78%	-4.01%	-5.80%	-3.75%	-1.50%	-1.14%
<i>Lomo Saltado</i> with salmon	-3.41%	-2.32%	-3.49%	-6.51%	-3.66%	-1.66%
<i>Lomo Saltado</i> with chickpeas	-1.38%	-5.49%	-2.91%	-2.33%	-1.38%	-2.14%
<i>Lomo Saltado</i> with mushrooms	-0.43%	-0.27%	-1.02%	-0.36%	-0.35%	-0.55%
<i>Lomo Saltado</i> with tofu	-2.22%	-2.20%	-1.18%	-1.01%	-0.58%	-0.65%

The results obtained by reducing the impact values by 10% are identical to those obtained by reducing the grams of the main ingredient by 10%, as shown in Table 5. This consistency is logical as the calculations for the results in the DELI Calculator are based on the multiplication of both the recipe ingredients quantities and the impact values. Therefore, the same interpretation applies to these results.

Subsequently, when the impact values were increased by 10%, the outcomes mirrored the results displayed in Table 6, aligning with the earlier explanation.

4.1.5. Uncertainty analysis

The results showed that beef has the highest environmental impact of the ingredients analyzed and therefore an uncertainty analysis is performed for this ingredient. Originally, the intention was to evaluate the GHG emissions for all six main ingredients. However, there are two main limitations to this analysis. Firstly, as mentioned earlier, in Poore and Nemecek (2018) database, impact values for ingredients such as salmon, chickpeas and

mushrooms were assigned from a broader food group, making them less specific. This made it challenging to mirror the exact same food items used in Poore and Nemecek's food groups due to the variability in the food items list in other databases. Secondly, there were different types and parts of the same ingredient, as well as variations such as canned, dried, or frozen forms, which made it difficult to find an appropriate value for the ingredient. To simplify this analysis, beef was chosen not only because it had the highest impact, but also because it was consistently represented across the databases. Table 11 presents the GHG emissions values for beef across 10 different databases, including Poore and Nemecek (2018) database.

Table 11. GHG emissions in kg CO₂ eq for beef across 10 databases.

Database	Beef
Poore and Nemecek (2018) database	59.6
SU-EATABLE LIFE	25.75
Database for Double Pyramid 2016	24.89
DataFIELD	33.12
UCPH - Clune 2017	28.77
AGRIBALYSE 3.0	27.09
Chinese Food Life Cycle Assessment Database (CFLCAD)	7.4
The Big Climate Database	60.25
SHARP INDICATORS DATABASE	34.04
Drew et al. 2020 Database	20.33

Table 12 shows the mean, standard deviation and variance for the GHG emissions values for beef from the 10 databases, including the value from the Poore and Nemecek (2018) database. Figure 16 shows the distribution of GHG emissions for the 10 databases.

Table 12. Statistical analysis of GHG emissions values for beef based on the values across 10 databases.

Ingredient	Mean	Median	Standard deviation	Variance	95% confidence interval
Beef	32.12	27.93	16.43	270.07	(20.37, 43.88)

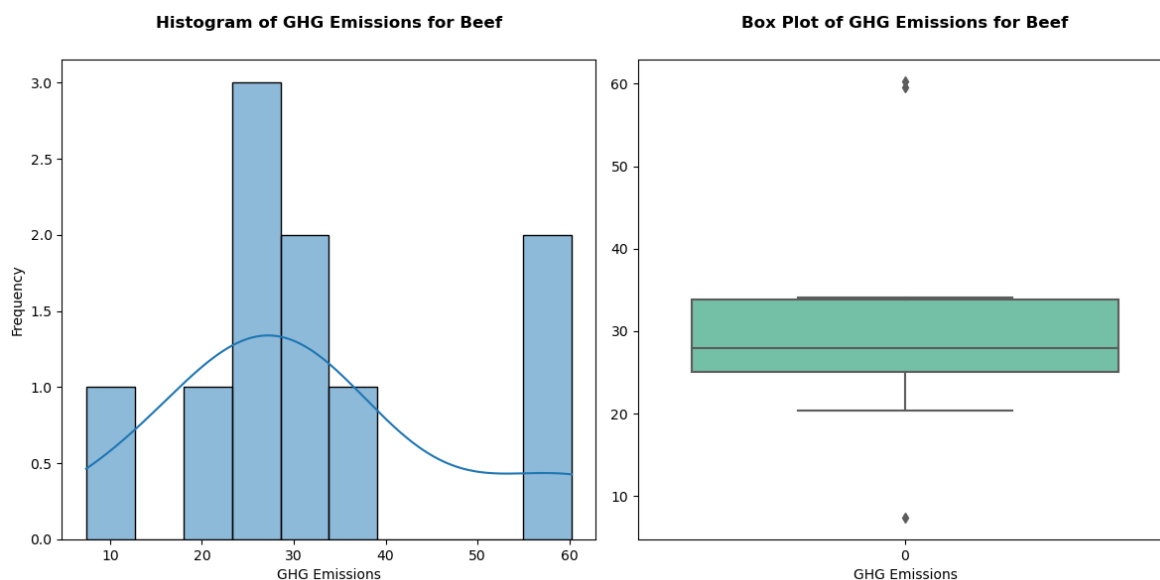


Figure 16. Distribution of the GHG emissions for beef across 10 databases.

This analysis shows the overall variability of GHG emission values across all available databases, and also shows where the Poore and Nemecek (2018) database stands in relation to the other databases. It can be seen that the Poore and Nemecek (2018) value (59.6 kg CO₂ eq) is higher than the other databases, with the exception of The Big Climate Database, which averages around 60.25 kg CO₂ eq. There is also another outlier from CFLCAD with a value of 7.4 kg CO₂ eq. These values were not excluded from the analysis, as the Poore and Nemecek (2018) database is the reference database for the DELI Calculator, and since this is already an outlier, the other outliers are included. In general terms, the results shows that the 95% confidence interval (20.37, 43.88) indicating that it can be said with 95% confidence that the true mean value of GHG emissions for beef is within this range. The average GHG emissions across the databases is 32.12 and there is a dispersion from this value of 16.43.

Given that the Poore and Nemecek (2018) database serves as the reference value for the DELI Calculator, it is also important to compare its values with those from other databases. This comparison highlights the uncertainty around the values used in the DELI Calculator. Table 13 shows the variability of the GHG emissions values compared to the value from the Poore and Nemecek (2018) database.

Table 13. Variability of GHG emissions values for beef compared to the Poore and Nemecek (2018) value.

Ingredient	Variance	% Variance
Beef	176.86	22.31%

The “variance” column presents the range of GHG emissions values for beef from the databases, excluding Poore and Nemecek (2018) database. The “% Variance” column quantifies the percentage deviation of these values from the Poore and Nemecek (2018) database value.

For beef, the variability from the databases is 176.86, which means that there is a significant range of values obtained from different databases. The % Variance is 22.31%, indicating that there is a significant variability in emissions data from different databases compared to the Poore and Nemecek (2018) database. This is consistent with the results seen earlier, where the values obtained by Poore and Nemecek (2018) are higher than those obtained by the other databases.

4.2. Test sessions

4.2.1. Results from test sessions

As detailed in the section 3.2.5., for the user experience evaluation, test sessions were conducted with eight test users over two days. On the first day, six users participated, while the remaining two were tested on the second day. During these sessions, the developer tracked the time to compare it with the duration specified in the design phase. The results from these sessions are summarized in Table 14. Notably, only 63% of the test users had a Python interpreter available on their computers. Among these users, only two were able to run the scripts successfully. This was due to two users having MacBooks where the directory path specified in the constants.py file did not match the MacOS format. Of the other three users, one had VS Code installed, but lacked the Python extension, and the other two successfully ran the scripts and got the results. Due to the tight schedule between sessions, there was insufficient time to troubleshoot these problems. As a contingency plan, the developer's computer was used to run the scripts. Nevertheless, these challenges provided valuable insights from all users into template structure and usability of the DELI Calculator.

Table 14. Summary of the results from the test sessions

Measure	User 1	User 2	User 3	User 4	User 5	User 6	User 7	User 8	Mean
Number of recipes	2	2	3	2	2	4	2	5	2.75
Average time preparing the material for running the scripts (minutes)	2.5	NA	NA	3	NA	4.5	3	3.25	3.25
Average time spent on templates (per recipe in minutes)	6	4.5	5	8	6.5	9	3.5	6	6.1
Average time saving the files (per recipe in seconds)	10	7	11	10	8	12	6	8	9.0
Average user time running the	6.5	5.5	7.5	8.5	6.5	11*	10*	9.5	8.125

scripts (developer's computer) (minutes)									
Computation time (minutes)	0.5	0.5	0.5	0.5	0.5	4*	2*	0.5	1.125
Total time assuming only one recipe (minutes)	15.7	10.6	13.2	20.2	13.6	28.7	18.6	19.4	17.5
Total time per user (including all the recipes assessed by the user)	21.7	15.1	23.2	28.2	20.1	55.7	22.1	43.4	28.7

*User who was able to run the scripts on a personal computer.

NA – represent the users that did not have a Python interpreter available on their personal computers.

Total time per user including all the recipes assessed by the user

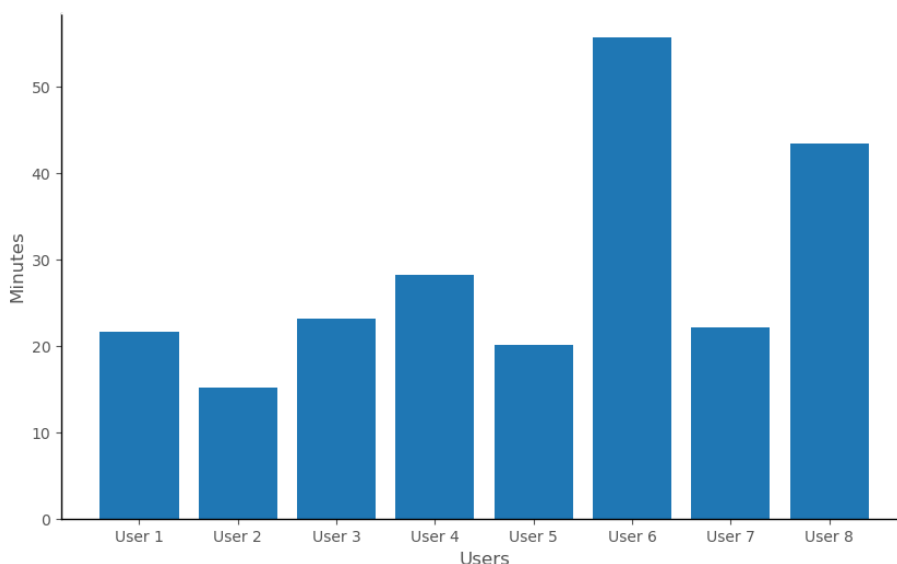


Figure 17. Total time spent per user on the test sessions including all the recipes assessed.

As detailed in Section 3.2.2.1., "Average time preparing the material for running the scripts" measures the duration required for users to download the Python source code and save it locally. Subsequently, "Average time spent on templates" quantifies the duration users invest in completing the Recipe template for each recipe, from opening the file to saving it with the appropriate name. Additionally, "Average time saving the files" captures the time users spend saving completed templates to the designated ". /recipes" folder.

"Average time running the scripts" evaluates the overall time users dedicate to executing the scripts, divided into user time and computation time. User time accounted for 88% of the total script running time, including the performance from the users who were able to run the DELI Calculator on their computers. The computation time in the developer's computer

averaged 0.5 minutes, indicating how quickly the DELI Calculator can run in future uses on the same computer once the environment is set up.

Based on the previous results, the blueprint map was updated with a new procedure for the cases where the user could not run the scripts alone. The time measures were also updated, including the values for all the test sessions. All these updates are highlighted in orange boxes (see Figure 18). Compared to the previous version, two activities were updated, and one new activity was added. The new activity involves the developer receiving filled-out Excel templates from users and saving them in the “./recipes” folder to initiate the script execution. The two updated activities address instances where the developer and user discovered that running the scripts was not possible or when the test user indicated they did not have any Python interpreter available. In these cases, the developer had to email the templates to the users and then receive them back once completed, saving them in the “./recipes” folder so the user could continue with the process.

The users who were able to run the scripts independently experienced the longest processing times, primarily due to processing demands of the various packages involved. Additionally, the initial set up was prolonged by low performance of the computers when opening VS Code. It is worth noting that the initial setup and configuration of the environment, while time-consuming for the first time, is a one-time investment. Subsequent uses of the calculator do not require this setup process.

Moreover, the time to complete the templates was originally set at 4 minutes, but users took on average 6 minutes, showing an increase of 50%. This was partly because users did not have the recipe and the amounts ready as well as requiring additional time to check details. This also varied greatly from recipe to recipe, with some having far more ingredients than others. Furthermore, the time for running the scripts increased by 363% (from 2 minutes to 9.25 minutes), as users preferred to run the visualizations.py file chart by chart and receive explanations from the developer, and this value includes the time taken by users who were able to run the scripts on their personal computers. Lastly, the test sessions were initially designed to last 20 minutes for an average of 2 to 3 recipes, but with the results obtained, there was a 45% increase in the time, from 20 to 29 minutes (see Table 14 and Figure 17).

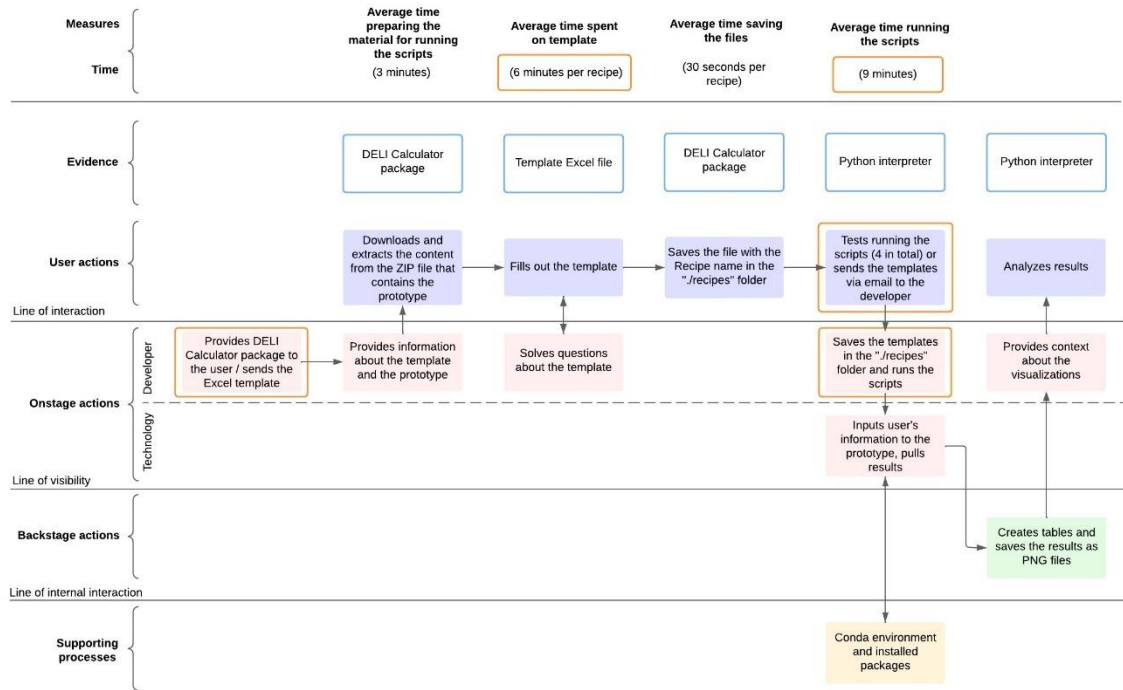


Figure 18. Updated Service Blueprint based on the test user evaluation for the Food Recipe Environmental Impact Calculator prototype compared to the original design in section 3.2.2.1.

Other findings from the test sessions revealed that the users needed certain food items that were not available in the database, such as sour cream, pumpkin seeds, salami, canned tomatoes, curry paste and others. Also, some of the users struggled with the different units of the food items, especially when it came to cups. In these cases, they preferred to use grams instead, or they used online converters to get the exact amount in the unit specified in the database. This was also a reason why the measure of time spent on filling in the template increased.

The results of each test session are shown in section 8.7. of the SI, in relation to the same graphs shown in section 4.1.

4.2.2. User experience results

After each test session, the test users completed a survey to evaluate their experience with the DELI Calculator. Figure 19 summarizes the results for the Likert scale questions.

As stated in the section 3.2.5., the survey consisted of seven questions, with the first five employing a Likert scale format, while the remaining two questions were open. The Likert scale ranged from 1 to 5, with high scores indicating more positive responses. The questions are shown below.

1. From 1 to 5, how well do you understand the purpose of this Food recipe environmental impact calculator? (1. not at all, 5. very well)
2. From 1 to 5, how useful do you think the results from the Food recipe environmental impact calculator are? (1. not useful at all, 5. very useful)
3. From 1 to 5, how easy/clear was it to enter the recipe details into the Excel template? (1. not easy/clear at all, 5. very easy/clear)
4. From 1 to 5, how clear was the procedure to use the Food recipe environmental impact calculator? (1. not clear at all, 5. very clear)
5. From 1 to 5, how would you rate the overall experience using the Food recipe environmental impact calculator? (1. not pleasant, 5. very pleasant)
6. Is there anything that you want to highlight from this prototype?
7. Are there any comments or suggestions that can help us improve the user experience and prototype?

These results show that the majority of the users (87.5%) understood the purpose of the DELI Calculator very well, indicating that the purpose of this prototype was clearly communicated. Users generally found the results of the DELI Calculator useful, with no one rating below 4 (62.5% voted for useful and 37.5% voted for very useful). Regarding the ease of entering the recipe details into the Excel template, all the users agreed that the process was easy/clear (57.1% voted for easy/clear and 42.9% voted for very easy/clear), indicating that the Excel template was considered user-friendly. Most users found the process clear, with only one user rating it as moderately clear. In terms of the overall experience, the users rated it as very pleasant, with no ratings below “4”, indicating a very positive experience.

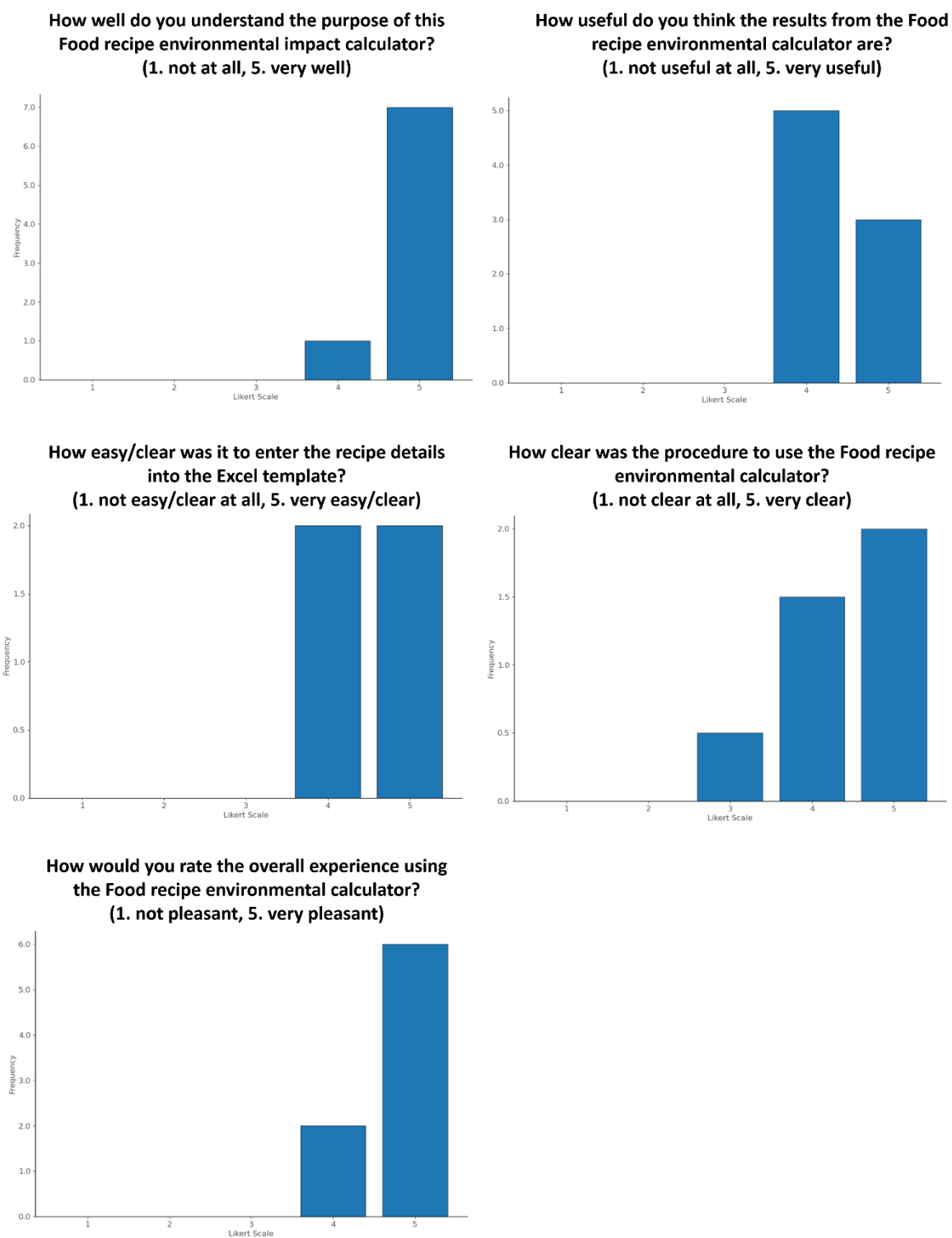


Figure 19. Results from the Likert scale questions of the user experience survey.

Feedback from the last two open questions of the survey highlighted both strengths and areas for improvement in the DELI Calculator. Users emphasized the DELI Calculator's usability, informative results and the thoroughness of the work.

The excel template was very user-friendly, although there is room for improvement based on the issues encountered by users. The spider web chart for comparing the impact of different recipes was considered useful, and the detailed per-recipe insights were appreciated. One user valued the quick resolution of an error encountered during the test session, specifically addressing duplicates in the results for items like cow milk and peas. The wide variety of ingredients, including regional items such as tapioca and cassava, and the clear presentation of their environmental impacts were well-received. The DELI Calculator's support for different measurement units was also beneficial. Users appreciated being guided through the program and having the results interpreted with assistance. However, some confusion regarding the different types of land use impact indicates a need for clearer explanations in this area.

Users also provided several suggestions to improve the DELI Calculator. They recommended simplifying the results to be understandable for a broader range of users and making comparisons to known impacts for better context. Adding explanations about the impact categories would help users understand the results more deeply. For instance, if the indicator for acidification has a value like 0.1 kg SO₂ eq, it would be helpful to include a more familiar conversion, such as equating this amount to driving a certain number of kilometers in a diesel car. Including more ingredients in the template and categorizing them (e.g. fruits and vegetables, dairy products, meats) would enhance usability. Users also suggested incorporating pictures to help differentiate similar items, like tomato sauce and paste, and adding a unit converter on the user interface for different measurements (e.g., cups to grams).

Additionally, users expressed interest in sensitivity analysis to identify which parameters affect the most each impact category per product. They also pointed out the need for improving the color scheme to distinguish ingredients in plots more clearly. Lastly, allowing users to specify the number of servings for a recipe, rather than manually scaling each ingredient, and enabling selection between multiple measurement units would further improve the user experience.

The detailed responses from users can be found in section 8.8. of the SI.

5. Discussion

This chapter aims to explore the background of the results presented in Chapter 4. The analysis revealed that the *Lomo Saltado* recipe with beef had the highest environmental impact across most indicators, while the version with mushrooms and other plant-based alternatives had the lowest impacts. The sensitivity analysis highlighted how variations in impact values and the amount in grams of the main ingredients affect the results. The uncertainty analysis showed the variability of the GHG emissions values for beef across

different databases. Additionally, the DELI Calculator demonstrated both strengths and areas of improvement based on the user experiences.

Section 5.1. focuses on understanding the details and differences between dietary patterns for all six indicators. Section 5.2. evaluates the DELI Calculator, based on the results of the test cases and user feedback.

5.1. Environmental impacts of diets

The analysis of the environmental impacts of different dietary patterns, using six versions of the *Lomo Saltado* recipe as a case study – beef, chicken, salmon, chickpeas, mushrooms and tofu – revealed significant variations across the six impact categories.

5.1.1. GHG emissions

Based on the results shown in Figure 8 to 15, the beef-based recipe consistently exhibited the highest environmental impacts across most categories, particularly in GHG emissions and land use. In contrast, tofu, chickpeas, and mushrooms versions showed much lower GHG emissions, ranging from 15% to 19% of those generated by the beef recipe. Equivalent results were obtained by Takacs et al. (2022), who found that red meat-based meals had, on average, 14 times higher emissions than their plant-based counterparts. In this thesis, red meat-based meals were only six times higher than plant-based versions, potentially due to Takacs et al. (2022) including a high amount of dairy products such as milk and cheese in their assessment, which increased the contribution of livestock to the impacts. Furthermore, Takacs et al. (2022) reported a GHG emissions of 5.78 kg CO₂ eq for a serving of beef lasagna, compared to 31.7 kg CO₂ eq for four servings of *Lomo Saltado* with beef in this study. It is important to note that even in the beef version, *Lomo Saltado* is a vegetable-heavy dish, which may contribute to the comparatively lower impacts.

Similarly, Silva et al. (2023) found that meat and dairy consumption significantly contribute to the carbon footprint in Brazil. Their study showed that vegan and no-red-meat diets have a lower average emissions, mirroring the findings of this thesis, where plant-based options and chicken and salmon versions consistently showed lower impacts than the recipe with beef. Interestingly, Silva et al. (2023) also observed that vegetarian and vegan diets (4.57 kg CO₂ eq and 3.1 kg CO₂ eq) averaged a slightly higher GHG emissions than pescatarian and no-red-meat diets (4.48 kg CO₂ eq and 3.07 kg CO₂ eq). This is similar to the results of this thesis, where the results of *Lomo Saltado* recipe with chicken and salmon (7.33 kg CO₂ eq and 6.93 kg CO₂ eq) had a slightly higher impact than the chickpeas version (5.29 kg CO₂ eq).

Pairotti et al. (2015) reported slightly different results, indicating that while vegetarian diets do show improvements, the reduction is not as drastic as observed in this study. Travassos et al. (2020) found that in Brazil, beef accounts for 41% of GHG emissions. In contrast, this study found that the *Lomo Saltado* with beef represents about 86% of emissions when

considering the recipe as a dietary representation. This discrepancy likely arises from the fact that the *Lomo Saltado* recipe does not involve the full range of ingredients found in a typical diet, thus highlighting the substantial impact of beef consumption within the recipe.

Moreover, the low contribution of the transport, packaging and retail stages to the total GHG emissions in general, averaging 12% overall and only 3% for beef, aligns with the findings of San Miguel and Ruiz (2021), where these phases were not a major contributor to the environmental impacts. This indicates that using local ingredients does not necessarily make meals more sustainable unless the transportation stage accounts for an important share of the environmental impacts.

5.1.2. Land use

Land use impacts followed the GHG emission trends, with the beef recipe again having the highest impact due to the significant land requirements of livestock farming. The other versions had considerably lower land use impacts, with chickpeas showing an 88% reduction, tofu a 93% reduction and mushrooms the highest at 94% compared to beef. These results align with the findings from Poore & Nemecek (2018), which indicate that a vegetarian diet can reduce land use by approximately 50%, while a vegan diet can achieve up to 75% reduction. Similarly, Bunge et al. (2024) reported that plant-based alternatives can reduce land use by up to 45%.

Mazac et al. (2023), found that plant-based alternatives for the same meals compared to beef presented an 81-86% reduction in land use, while other animal-sourced meal such as chicken and fish showed 73-60% less land use than the beef meal. Compared to the results from this thesis, the chicken and salmon versions showed 91 and 93% reductions, respectively, compared to the recipe with beef. The differences in the percentages are due to the specific recipe used (a chickpea patty with roasted root vegetables and a soy-based cream sauce), which had high amounts of vegetables compared to the protein source. Additionally, the database used, AGRIBALYSE 3.1. (Asselin-Balençon et al., 2022), contributed to variations in values.

As shown above, the results from this thesis reveal an even greater difference between the *Lomo Saltado* with beef and the plant-based alternatives. This is likely due to the limited range of ingredients included in the recipe and the lack of nutritional balance in the quantities used for the plant-based versions. In a real case, ensuring nutritional balance might require higher amounts of plant-based ingredients compared to the beef version. This adjustment could have impacted the results, further highlighting the substantial land use savings achieved by switching to plant-based alternatives.

5.1.3. Acidification

Acidification potential followed a similar pattern, with the beef-based recipe leading, followed by chicken and salmon at 47% and 30% of the beef's impact respectively. The

remaining versions had impacts under 28%, with mushrooms having the lowest. The significant impact of beef is attributed to ammonia emissions from cattle manure (Putman et al., 2023), which, given the quantity in the recipe, results in the highest acidification potential. These findings are consistent with Takacs et al. (2022), who reported that vegetarian meals have the lowest terrestrial acidification potential compared to meat-based options. They found that red meat-based options had up to 16 times higher acidification potential than plant-based options. In this thesis, the beef-based recipe was up to 5 times higher than plant-based versions, reflecting the different recipe compositions, as *Lomo Saltado* does not include dairy products like cheese and milk.

The results from this thesis also aligns with the findings of Mazac et al. (2023), who reported that the plant-based options for their reference recipe had 93-94% less acidification potential than the beef meal, while the other animal-sourced options had 71-87% less impact compared to beef.

5.1.4. Eutrophication

Eutrophication impacts were highest for the beef recipe, which could be represented as nitrogen leaching in different ways from cattle ranches to water bodies (Putman et al., 2023), with salmon following at 62% of the beef's impact. This high impact from salmon is attributed to nutrient release from fish waste and uneaten feed (Bailey & Eggereide, 2020). Chicken, chickpeas, tofu, and mushrooms recipes had progressively lower impacts, with mushrooms having the least at 23%. These findings align closely with previous LCA studies, such as those by Takacs et al. (2022) and Baroni et al. (2014), which also reported the highest impacts from red-meat based recipes and the lowest from plant-based options. Interestingly, Takacs et al. (2022) noted that vegetarian meals and chicken meals had similar eutrophication potential, a trend that also appears in the results of this thesis.

Furthermore, the findings of Mazac et al. (2023) also indicate that beef and fish are the ingredients with the highest impact for this category. However, the magnitude of the difference between these ingredients and the plant-based options and chicken is much greater than the results presented in Figure 15. This variance may be due to the factors mentioned earlier, including the use of different databases and recipes with different amounts and ingredients.

5.1.5. Freshwater withdrawals

For freshwater withdrawals, the salmon recipe had the highest impact, with beef close behind at 88% of salmon's impact. Salmon, particularly “Fish (farmed)”, which is the group from which salmon derives its impact values in the Poore and Nemecek (2018) database, entails water-intensive processes during the farming stage, requiring significant amounts of water to maintain optimal conditions for growth in ponds (Lekang et al., 2016; Ryan et al., 2022). In contrast, the water consumption associated with beef can be explained by the

water used for irrigation of feed crops, cattle drinking water, dust control and activities throughout the processing, distribution and consumption stages (Putman et al., 2023).

Chicken and chickpeas followed, and tofu and mushrooms had the lowest impacts in freshwater withdrawals at 67% and 66%, respectively. Takacs et al (2022) noted that animal-derived ingredients have the highest water depletion potential, with impacts 34 times higher than vegan meals. Although the results of this thesis indicate that salmon and beef have the highest freshwater withdrawal impacts, the difference between these and the plant-based options is not as pronounced. This discrepancy may be primarily due to methodological differences between databases concerning this indicator and, to a lesser extent, the composition of the recipe. As Carvalho et al. (2023) highlighted, the Poore and Nemecek database showed significant disparities compared to other databases, such as SU-EATABLE LIFE built by Petersson et al. (2021), in terms of water footprint.

In terms of the plant-based alternatives, it is noteworthy that rice emerged as the ingredient with the highest impact in this category, which can be attributed to the demands of freshwater placed on paddy fields throughout the production process, including initial land preparation and the continuous inundation of fields during plant cultivation to facilitate optimal conditions for crop growth and weed and pest control (Arif et al., 2024).

5.1.6. Scarcity-weighted freshwater withdrawals

Scarcity-weighted freshwater withdrawals presented a slightly different picture. Here, the beef recipe had the highest impact, closely followed by chickpeas at 97%. Salmon, chicken, tofu, and mushrooms recipes had progressively lower impacts, ranging from 81% to 92%. These results are consistent with the findings of Takacs et al. (2022). However, direct comparisons are challenging due to the methodological differences between LCA databases. In the Poore and Nemecek (2018) database, this indicator is calculated by multiplying the freshwater withdrawals values by the corresponding scarcity index.

According to a study by Porto Costa et al. (2023), the most effective strategy for reducing water scarcity potential linked to beef production would be to replace this ingredient with plant-based options, saving approximately 13 m³ of water scarcity per portion of 100 grams. Scaling this value up to the 455 grams of beef used in the *Lomo Saltado* recipe would yield a reduction of 59000 liters, significantly higher than the value obtained in this thesis, which was 40831 liters. Such discrepancies could also be related to the differences in the data sources, as the study by Porto Costa et al. (2023) is based on ecoinvent 3.7.1., whereas the Poore and Nemecek (2018) database is based on global values derived from studies across 119 countries.

On the other hand, a study conducted by Iversen et al. (2020) showed that salmon is predominantly farmed in northern countries such as Norway, Canada, the Faroe Islands, and the United Kingdom, with Chile as the second-largest producer after Norway. These five countries account for 98% of the total salmon production in the world. The study's results indicate that Norway has been the main producer from 1990 to 2018. Norway

typically experiences low water scarcity, a regional difference that may explain the variations observed in the impact values from the Poore and Nemecek (2018) database. This geographical context, where water scarcity is not a pressing issue, could explain the lower water scarcity impacts associated with salmon, contrasting with its higher impact in the freshwater withdrawals category due to the significant water requirements for maintaining optimal conditions in aquaculture.

Moreover, according to Rosi et al. (2017), vegetarian and vegan diets typically use around 1000 liters of water per day less than animal-based diets like omnivorous diets, which aligns with the values shown in the results section. The recipe with beef had 40831 liters eq, while the recipe with chickpeas had 39697 Liters eq. The recipe with salmon followed with 37432 liters eq, and the versions with chicken, tofu and mushrooms had similar impacts, with values of 35220 Liters eq, 33379 liters eq and 33003 liters eq respectively.

5.1.7. Breakdown into ingredients

Further breakdowns of the environmental impacts by ingredient per recipe, as shown in Figures 9-14, provide deeper insights. For example, in the beef recipe (Figure 9), beef contributed most to land use, GHG emissions, acidification, and eutrophication. However, rice contributed significantly to freshwater withdrawals and scarcity-weighted freshwater withdrawals. Similar patterns were observed for other recipes, with the main ingredient typically dominating the environmental impacts, while other ingredients like rice and olive oil also contributed significantly across various categories. Based on the previous discussion per impact category, these results align with the results showed in previous LCA studies (Baroni et al., 2014; Bunge et al., 2024; Pairotti et al., 2015; Rosi et al., 2017; Silva et al., 2023; Takacs et al., 2022)

The results underscore the substantial environmental benefits of plant-based versions of the recipe, with tofu and mushrooms being the most sustainable choices. These findings highlight the potential for significant reductions in environmental impacts through dietary changes, particularly by substituting beef and salmon with more sustainable options like tofu and mushrooms. Lastly, one significant limitation of these results is the inability to compare the environmental impact values of *Lomo Saltado* with other studies due to its highly regional nature. This limitation highlights the need for more comprehensive databases that include a wider variety of regional recipes and ingredients to facilitate better comparisons and broader applicability.

5.1.8. Sensitivity and uncertainty analysis

The sensitivity analysis revealed that the environmental impacts for the *Lomo Saltado* recipes are sensitive to changes in both the quantity of the main ingredients – beef, chicken, salmon, chickpeas, mushrooms and tofu – and the impact values. The results showed that a 10% reduction in beef produced an 8.56% reduction in GHG emissions. Similar findings were reported by Takacs et al. (2022), where a 20% reduction in the beef content in the beef lasagna recipe resulted in a 17% decrease in GHG emissions. If the quantities of the

main ingredients had been reduced by 20%, a 17% reduction would have been expected. Additionally, Takacs et al. (2022) showed that a 20% reduction in the GHG emission factors for beef led to a 16.8% decrease in GHG emissions. This outcome is consistent with the results of this thesis, as both parameters showed a proportional reduction magnitude with a 10% decrease.

On the other hand, the small reductions observed in the categories of freshwater withdrawals and scarcity-weighted freshwater withdrawals indicated that these categories were not highly sensitive to changes in the amount of ingredients like beef. This is understandable since rice is a major contributor to these impacts. As previously explained, rice cultivation demands significant water usage throughout its growth process.

The uncertainty analysis showed a high variability in the GHG emission values for beef across databases. This may be caused by several factors related to the LCA data used, such as differences in the environmental impact assessments methodologies, variations in beef origin, data compilation years, beef types, and other aspects encountered during the uncertainty analysis. According to Carvalho et al. (2023), different databases for environmental impacts of food lead to diverse estimates for common indicators. Their study, which included the Poore and Nemecek (2018) database, SU-EATABLE LIFE and SHARP-ID, found that Poore and Nemecek (2018) database had the highest median values for the impact. However, the study found that most indicators, except for water footprints, were highly correlated with each other.

5.2. *Evaluation of the prototype*

The DELI Calculator prototype demonstrated notable strengths in performance, usability, and reliability during the test sessions, but also revealed several limitations and areas for improvement. The DELI Calculator's reliance on the Poore and Nemecek database posed challenges, particularly in terms of food items resolution and the omission of certain food items, such as beverages. The fact that impact values were replicated for all the food items within the same food group made the values less precise. However, it was confirmed that the general trends were accurate based on reference studies, except for the indicators of freshwater withdrawals and scarcity-weighted freshwater withdrawals.

Improving the resolution of the food item list proved useful, though the test sessions revealed the difficulty of including the broad spectrum of culinary knowledge. The users were from different nationalities, which impacted the way they selected recipes to assess, frequently searching for regional food items that were not included in the list of 528 items. Additionally, users from different regions had varying understandings of recipe measurements, with people from Latin America accustomed to using cups and Europeans preferring grams.

Only 63% of users had VS Code installed, and even among these, only two successfully ran the scripts. The issues highlighted a significant usability barrier for users with different operating systems or incomplete software setups, necessitating the use of the developer's

computer. Users also took longer than anticipated (6 minutes instead of 4) to complete the Excel templates. This delay was partly due to the need to check recipe details and varied ingredient lists, indicating a potential benefit from a more streamlined or guided input method.

The time to run the scripts increased significantly, from an expected 2 minutes to an average of 9.25 minutes. Users ran visualizations step-by-step and sought explanations, suggesting that the availability of useful information alongside the charts might be needed. Improving user-friendliness is essential, as the impacts presented are at a midpoint level, focusing on single environmental problems rather than showing direct impacts on human health, biodiversity or resource scarcity. Providing relatable information, such as translating environmental impacts into common equivalents, could enhance understanding. For example, O'Malley et al. (2023) reported that vegetarian and omnivorous diets differed by 1.07 kg CO₂-eq/1000 kcal on average. If a third of omnivores switched to a vegetarian diet of 2000 calories per day, the savings would be equivalent to removing 340 million passenger vehicle miles from the road.

The DELI Calculator's purpose was clearly communicated and well understood by users, with a high percentage finding it very useful. Users appreciated the control they had over input data, a feature often lacking in existing online tools, which typically offer limited, default options. This control allowed users to tailor the calculator to their specific needs, enhancing its usefulness. The ability to dive deep into the specifics of each recipe's environmental impact was highly valued by the users, noting the DELI Calculator's capacity to handle a high number of recipes with detailed analysis. The DELI Calculator also impressed users with its extensive database of ingredients, although some were missing, it was valued for the inclusion of regional items. In addition, the Excel template was generally considered easy to use, with positive feedback on its clarity and usability. The overall user experience was rated highly, with most users finding the process pleasant and straightforward.

While the survey results showed an overall user experience with the DELI Calculator prototype, it is important to acknowledge potential biases that may have influenced these perceptions. The test users were not strangers to the author of this thesis, which could have impacted their feedback and the real experience with the tool. This familiarity might have led to a more positive outlook on certain aspects of the DELI Calculator's performance. Despite this, the feedback provided valuable insights into areas for improvement, particularly in terms of technical usability and database coverage.

Overall, the DELI Calculator prototype shows promise in assessing the environmental impacts of different dietary patterns, but it requires refinements to enhance usability, performance, and reliability. Addressing the technical barriers, improving time efficiency, and expanding database coverage are crucial steps toward creating a more robust and user-friendly tool. Lastly, while the current user experience with the DELI Calculator is positive, it is expected that users would have an even better perception once a dedicated user interface is built. A user interface would enhance flexibility in terms of units and ingredients, making the tool even more accessible and convenient.

5.3. *Limitations and future research*

It was not possible to directly compare the environmental impact values of *Lomo Saltado* with those from other studies since it is a very region-specific recipe. Selecting another recipe that has been previously studied would help to corroborate the environmental impacts more accurately.

The DELI Calculator, while promising, has several limitations that must be addressed in future research. Future research should focus on integrating more detailed and diverse databases to capture a more accurate and comprehensive picture of environmental impacts. Combining the Poore and Nemecek database with other databases could help address these gaps and improve the reliability of the results.

Additionally, the current analysis did not consider the variability in environmental impacts due to sourcing ingredients from different countries or whether they were organic or conventionally grown. These factors can significantly influence the overall environmental impact, and future research should include a more granular approach to ingredient sourcing.

Moreover, the results are further influenced by the absence of data on cooking methods, preparation times, waste generated during cooking and disposal processes – key stages in calculating environmental impacts for food. Improving the database to include impact values for various cooking processes, energy use, and disposal methods would provide a more comprehensive and accurate assessment of environmental impacts.

The inclusion of additional impact categories and indicators at endpoint level, such as those related to biodiversity, which is currently missing, and land use is currently used as proxy, should also be considered. This would provide a more holistic understanding of the environmental footprint of different recipes.

Future work should focus on developing a more user-friendly interface, possibly web-based, that does not require users to install specific software or possess advanced technical skills. Such an interface could streamline the process, reduce the time required to complete tasks, add useful features and improve overall user experience. For example, incorporating flexible unit conversion features within the tool to accommodate diverse user preferences seamlessly. As mentioned in the beginning of this thesis, the scope of the thesis did not cover the development of a user interface, which significantly impacted user experience.

Lastly, performing a sensitivity analysis to understand the impact of ingredient variations would be a valuable addition to the tool. This could help users comprehend how changes in ingredient amounts affect the overall environmental impact, providing more actionable insights for making sustainable dietary choices.

6. Conclusions

This thesis developed the first prototype of the DELI Calculator, designed to estimate the environmental impact for food recipes with a comprehensive set of environmental indicators. Adjustments were made to the Poore and Nemecek (2018) database to enhance its utility, such as improving the food resolution, ensuring it met the thesis's objectives effectively. To evaluate this tool, the environmental impacts of different versions of the *Lomo Saltado* recipe were assessed as representative of different dietary patterns. The results showed that the plant-based meals had substantially lower environmental impacts across all the indicators compared to the meat-based versions. The versions with beef had the highest overall environmental impact. The results also showed that both the impact values and amount of the ingredients are important parameters influencing the results. The sensitivity analysis further revealed the substantial impact reductions can be achieved by minimizing meat content, particularly red meat.

The development of DELI Calculator represents a significant step forward in empowering users to make informed decisions about their dietary choices. The tool was valued by the test users as a user-friendly and useful tool. The development process encountered challenges related to database resolution, software compatibility, and user interface design. These challenges highlighted the need for more comprehensive databases, improved software compatibility, and enhanced user guidance. Additionally, the experience underscored the importance of considering users' diverse culinary knowledge and preferences when designing such tools.

Moving forward, it is imperative to address these challenges by enhancing database coverage, improving software compatibility, and optimizing user guidance within the tool. Additionally, future iterations should consider the diverse culinary knowledge and preferences of users to create a more inclusive and user-friendly platform.

Overall, the development of the DELI Calculator highlights the importance of leveraging technology to promote sustainable food consumption practices. By providing individuals with the means to assess the environmental impact of their dietary choices, the DELI Calculator holds the potential to drive meaningful change towards a more sustainable food system. Through ongoing refinement and innovation, the DELI Calculator can continue to serve as a valuable tool in fostering informed decision-making and mitigating environmental impacts associated with food production and consumption.

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8. Supplementary Information

8.1. *DELI Calculator – TUTORIAL*





Welcome to the **DELI Calculator** tutorial!

In this guide, you will learn how to use the DELI Calculator to estimate the environmental impacts of your recipes.

Whether you're a researcher, a chef, or an eco-conscious consumer, this tool will help you make more sustainable food choices.

Before you begin...

Make sure that you have a Python interpreter and an Integrated Development Environment, such as Visual Studio Code (VS Code), available in your computer.



1. Installation

- Download the DELI Calculator package from the GitHub repository:
https://github.com/PaulaBarco/DELI_Calculator
- Open your Integrated Development Environment, such as VS Code.
- Load the DELI Calculator folder.
- Make sure you have the following Python functions and these versions available:
 - Pandas (version 1.4.2 or later)
 - Pathlib (part of the standard library in Python 3.4 and later)
 - Numpy (version 1.22.4 or later)
 - Mathplotlib.pyplot (version 3.5.3 or later)



2. Preparing the recipes



Ingredient	Amount	Unit
Bread wheat		cup
Oye		cup
Soft wheat		cup
Wheat		cup
Wheat and rye bread		cup
Wheat bread		cup
Wheat flour		cup



- **Open the Excel file named “Recipe_template”.** You will see three columns: Ingredient, Amount and Unit.
- **Search for the ingredient you want.** You can either use the shortcut Ctrl + F to find it quickly or use the Filter function to navigate through the column.
- **Write the amount in the Amount column.** **Important!** Please follow the unit defined in the Unit column. If the unit is different from “grams” and you know the exact amount in grams for that ingredient, you can delete the existing unit and enter “grams” instead. Do not convert from grams to another unit, as this will cause errors when running the scripts. Be careful while doing this to avoid any typos.



DELI Calculator
Better choices in your kitchen table

2. Preparing the recipes (continuation)

- **Repeat the same procedure for all the ingredients in your recipe.** Ensure each ingredient has the correct amount and unit.
- **Once all the amounts are filled out, save the file with the name of the recipe.** Replace the template name with a recognizable name for your recipe.
- **If you want to input more than one recipe, repeat the previous steps.** This means creating a new file for each recipe you want to assess.
- **Lastly, save all the recipe files in the folder called “recipes”.** You can find this folder at the following path: “../data/recipes”



3. Running the DELI Calculator

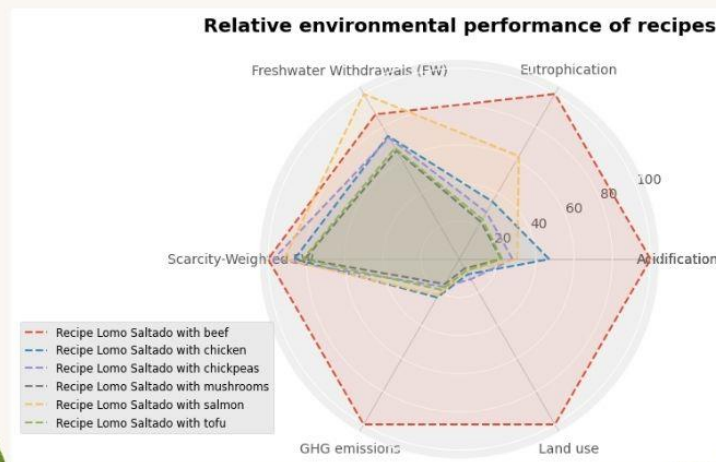
- On your Integrated Development Environment, such as VS Code, open the file named 'cleaning.py' and run the script.
- Once 'cleaning.py' has finished executing, open the Python file named 'harmonized_datasources.py' and run the script.
- After 'harmonized_datasources.py' has completed, open the Python file named 'analyzed_data.py' and run the script.
- Finally, open the 'visualizations.py' file. Run the code for each chart individually to gain a better understanding of the results.



4. Understanding the results

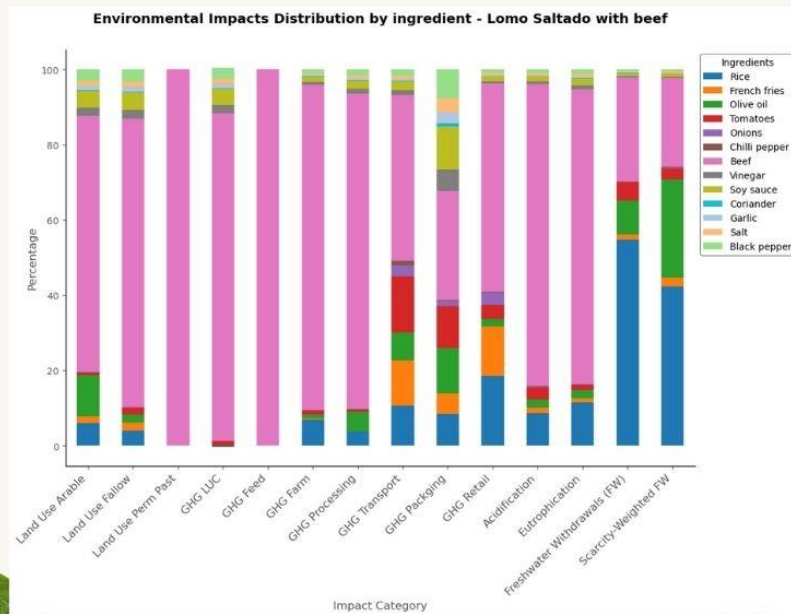
In the results, we have both visualizations and tables with the data. There are four main visualizations:

- **Spider web chart to compare recipes:** this chart shows the relative environmental performance by scaling values relative to the recipe with the highest impact for each category.



4. Understanding the results (continuation)

- **100% Stacked bar chart per recipe broken down into ingredients:** this chart visualizes the distribution of impacts for each ingredient for within recipe.

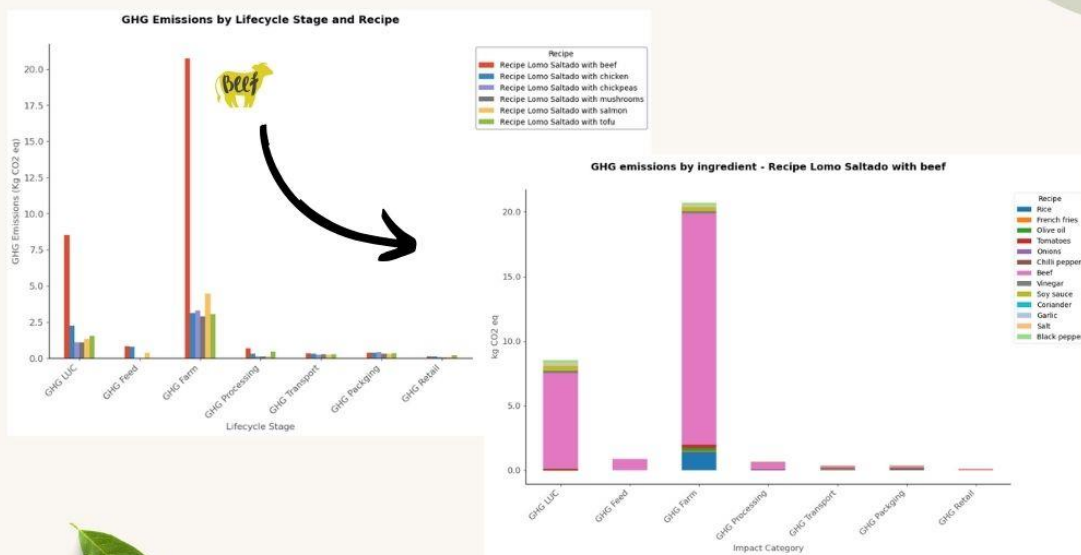


Important!
GHG emissions are now shown per life cycle stage, and land use is displayed per land type.



4. Understanding the results (continuation)

- **Bar chart and Stacked bar chart per impact category:** this compares the recipes per impact category and then shows the details into ingredients for the recipe with the highest overall impact.



5. Final remarks

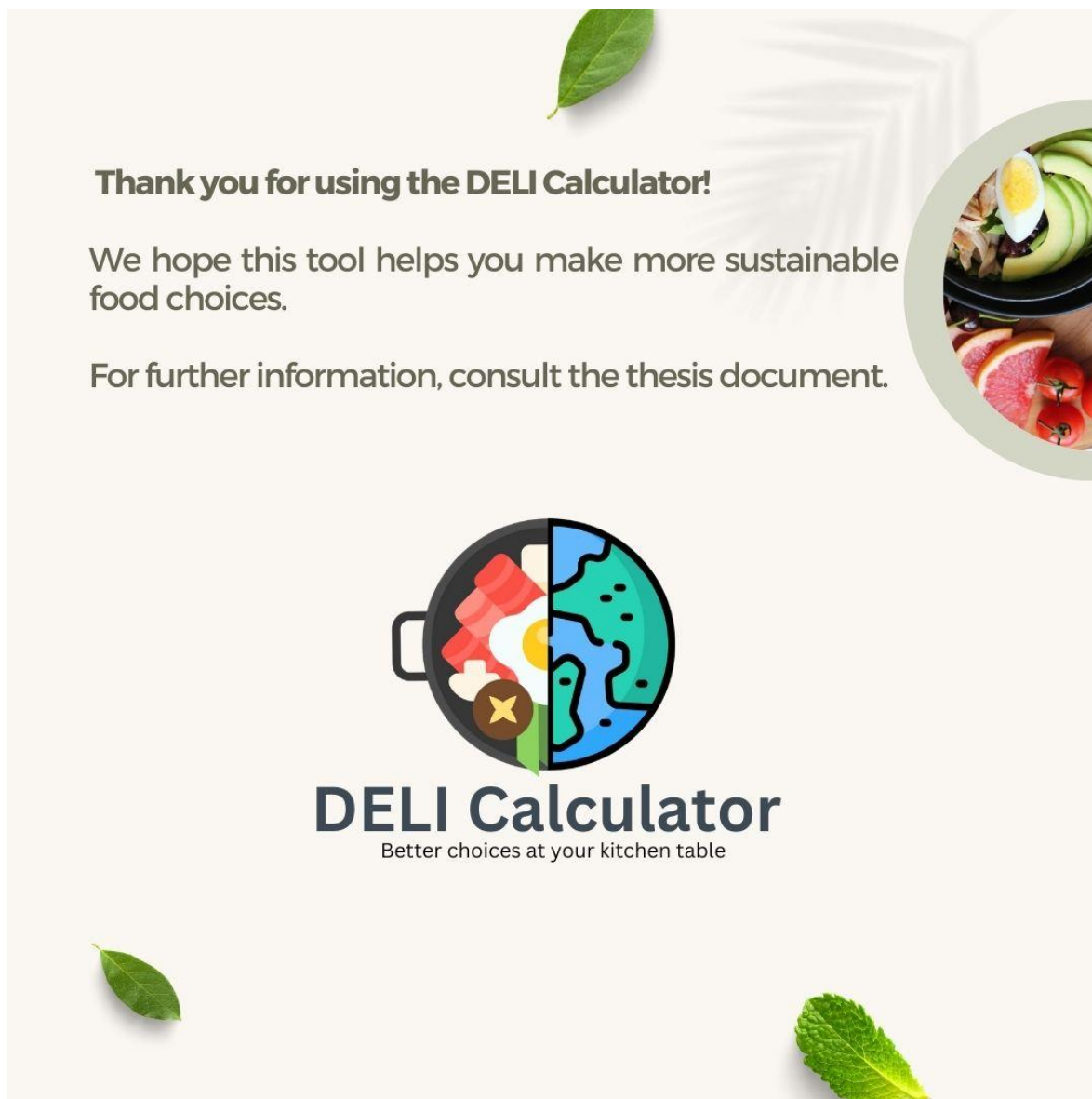
All the results can be found in './impacts'.

If there is any error when running the scripts, it is likely related to the Recipe template. Please check for the following:

- typos
- blank cells in the Ingredient or Unit column
- use of something different than period (.) as a decimal separator
- amounts should be a numeric value, no fractions
- the columns should have the labels: Ingredients, Amounts, Units.

Otherwise, ensure all dependencies are installed correctly and check for compatibility issues with your operation system.





8.2. Project work results – Autumn 2023

This section outlines the comparison methodology used to select the database, detailing the criteria employed and the resulting section. The content of this section is presented verbatim from the project report submitted in Autumn 2023.

8.2.1. Comparison methodology for open-access databases

To enable a more quantitative and visual comparison between the databases, a Pedigree matrix was utilized, following the approach proposed by the International Reference Life Cycle Data System (ILCD) handbook (JRC-IES, 2010).

According to Chen & Lee, (2021), the Pedigree matrix is a valuable tool for characterizing data quality aspects and quantifying quality ratings. The authors mention that this method is widely employed in LCA to assess input data quality during the LCI phase, as demonstrated by its implementation in the LCA database, Ecoinvent. The Pedigree matrix methods translates descriptive assessments into numerical values using Data Quality Indicators (DQIs) originally defined by (Weidema & Wesnaes, 1996). Based on the ILCD version, there are six DQIs, including completeness, precision/uncertainty, methodological appropriateness and consistency, temporal, geographical and technological representativeness, each assigned scaling scores.

ILCD handbook describe each DQI as follows:

- Completeness evaluates the proportion of elementary flows that are quantitatively included in the inventory.
- Precision/uncertainty measures the variability of the data values.
- Methodological appropriateness and consistency measures that the methodological choices align with the goal and scope of the application data and decision support context.
- Temporal representativeness examines the extent to which the data reflects the population of interest concerning the time / age of the data.
- Geographical representativeness considers whether the reference data is geographically congruent with the underlying study.
- Technological representativeness assesses whether the reference data corresponds to the technology applied in the underlying study.

To facilitate database comparisons in this research, the DQIs were adjusted to align with the study's objectives, as detailed.

In the ILCD framework, an overarching performance indicator, known as the Overall Data Quality Rating (DQR), is derived from the previously mentioned DQIs. The calculation involves aggregating all the DQIs through a weighted average, as indicated in the formula below. Notably, the rating of the weakest quality level is counted five-fold (once in the addition of all indicators and four additional times) to emphasize the impact of the lowest quality aspect on the DQR. Consequently, the denominator is increased by four.

$$DQR = \frac{C + P + M + TeR + GR + TiR + X_W * 4}{i + 4}$$

Where:

DQR: Overall data quality rating

C: Completeness

P: Precision/Uncertainty

M: Methodological appropriateness and consistency

TiR: Temporal representativeness

GR: Geographical representativeness

TeR: Technological representativeness

X_w : The weakest quality level among the DQIs (highest numerical value)

i : Number of applicable DQI.

The Data Quality Rating result is used to identify the corresponding quality level as shown in Table 15.

Table 15. Adapted overall quality level of a database according to the achieved DQR from JRC-IES, (2010)

Overall data quality rating (DQR)	Overall data quality level
≤ 1.6	“Excellent quality”
$1.6 < \text{to} \leq 2.0$	“Very good quality”
$2.0 < \text{to} \leq 3.0$	“Good quality”
$> 3.0 \text{ to} \leq 4.0$	“Fair quality”
> 4.0	“Data estimate”

8.2.2. Adjustment of the Pedigree matrix components

The criteria within the DQIs were adjusted for the purpose of comparing the “available for download” databases in this research. Specifically, the criteria were adjusted as follows:

- **Completeness:** this indicator, as defined in this study, relates to the number of food items, the stages of the life cycle, or system boundaries, and the impact categories covered by the database. Consequently, this indicator has three sub-indicators. All three sub-indicators are important; however, for this study the coverage of impact categories has a higher weight due to the necessity of incorporating impact categories to assess the impacts on biodiversity. Therefore, to calculate the final performance, a weighted average was chosen to be computed. The weights can be found next to the name of the sub-indicator.
 - Food items (20%): this refers to the number of food items covered by the database. A higher number of food items indicates better completeness.
 - Stages of life cycle or system boundaries (20%): the life cycle stages for a diet analysis should cover, at minimum, primary production, processing, transport, retail, consumption (food preparations). More life cycle stages result in a better performance.
 - Inventory categories (60%): this means that the more inventory categories covered, the better to describe the impact on biodiversity.
- **Precision/uncertainty:** this indicator evaluates the quality of the studies considered, the sources utilized in constructing the database, and the availability of the uncertainty analysis applied to the data. Consequently, this indicator has three sub-indicators. All three sub-indicators are important; however, for this study the quality of the source has a higher weight due to the necessity of building upon a foundation of high-quality, reliable data. Therefore, to calculate the final

performance, a weighted average was chosen to be computed. The weights can be found next to the name of the sub-indicator.

- Quality of the source (60%): this assesses the reliability of the sources employed in creating the database. As per Chen & Lee, (2021), reliability refers to the extent to which data is generated from sources of robust scientific foundation, such as acceptable theorems or verified observations. The following categories show the rating system associated with data quality.
 - Very good: the data sourced directly from a primary source, verified data based on measurements (experimental data).
 - Good: the data is based on secondary sources, including Environmental Product Declarations (EPDs), peer-reviewed articles, conference papers, grey literature, among others.
 - Fair: the data is based on generic datasets for LCA for the product material/product being studied.
 - Poor: the data lacks reliability and originates from sources with limited credibility. It may involve unverified claims, unsupported assumptions, or insufficiently substantiated estimates.
 - Very poor: non-qualified estimates or uneducated guesses.

It is crucial to note that the subsequent sub-indicators are not applicable to databases utilizing generic datasets for LCA as a source. Even though these generic datasets rely on studies, it is not a study that the authors directly conducted, making it difficult to track down the sources and calculations. Usually, these generic datasets require the use of LCA software to access the full details. In such cases, the overall score for the “Precision/uncertainty” data quality indicator will be contingent on qualifying as “Fair”.

- Number of studies (20%): this assesses the quantity of studies considered in the database. A higher number of studies leads to more robust results.
- Uncertainty (20%): this aspect evaluates whether authors conducted an uncertainty analysis for the values in the database and the extent of approximations made. It is important to note that performing descriptive statistics is set as a basic requirement for uncertainty analysis due to the great variance in possible methods to use. The levels are organized based on how well the uncertainty analysis was documented. The following rating scales delineated the criteria for assessing this indicator:
 - Very good: descriptive statistics are provided with calculation and relevant data, including the number of data points for each value in the database.
 - Good: descriptive statistics are available for each data value, but direct calculations are not provided.
 - Fair: descriptive statistics are not conducted for all data values in the database.

- Poor: authors mention that an uncertainty analysis was performed, but there is no documentation or values available in the database.
 - Very poor: no uncertainty analysis was performed.
- **Methodological appropriateness and consistency:** this indicator evaluates the ease of reproducing the results based on the thorough documentation of each database. The more comprehensive the documentation of the calculation procedures for different values in the database, the easier it is to reproduce those values.
 - Very good: methods are extensively described, and values, along with calculations, are available in the database. Additionally, sources for each data value are provided.
 - Good: methods are extensively described, and values for every part of the method described are available in the database. However, direct calculations are not provided.
 - Fair: methods are not thoroughly described, and values for different steps of the methods are absent, with only final values present. Sources are available for most data values.
 - Poor: direct calculations are not available in the database, and supporting material is not very explanatory. Nevertheless, sources are available for some of the values.
 - Very poor: Supporting material about calculations is unavailable, and no direct calculations are provided in the database. Moreover, there are no sources for each value. This rating also applies for the availability of full calculations upon request.
- **Temporal representativeness:** this indicator assesses the time difference between the data in the database and the current study, as well as the time frame considered for the literature review. More recent data contributes to a more accurate time frame to represent the current study.
- **Geographical representativeness:** this indicator evaluates the geographical coverage of the data, providing insights into the representation of biodiversity impacts across different regions. A higher number of covered countries and well-represented continents contributes to a more comprehensive understanding of biodiversity impacts.
- **Technological representativeness:** this indicator measures the level of detail in grouping and aggregating the data. The higher the level of detail, the better. It includes five different levels based on the typologies defined by Petersson et al., (2021) explained as follows:
 - General Categories: This includes major categories like fruits, vegetables, grains, dairy, and protein.
 - Intermediate Categories: This includes subcategories within each group such as citrus fruits, leafy greens, whole grains, poultry, cheese, olive oil.

- Detailed Categories: this further categorizes into individual food types within subcategories, like oranges, spinach, quinoa, cheddar cheese, chicken breast, etc.
- Food items and Preparations: This includes variations within specific types, considering factors like cooking methods, varieties like different apple varieties.
- Recipes or product brands: this represents each specific food item separately such as a particular brand of yogurt, a unique recipe preparation, etc.

To define quality levels and/or ranges for these indicators, a combination between the ILCD framework and the work by Chen & Lee, (2021) was performed. Subsequently, an adjustment to the intention used for this study was conducted. Finally, Table 16 shows the DQIs, the associated numerical values (ranging from 1 for very good to 5 for very poor), and quality level definitions.

Table 16. Quality levels and quality rating for the DQIs (ranging from 1 for very good to 5 for very poor), and the corresponding definition modified from JRC-IES, (2010) and Chen & Lee, (2021)

DQI / Rating		1	2	3	4	5
Completeness	Food Items	≥ 500 food items	< 400 food items	< 300 food items	< 200 food items	< 100 food items
	Stages - System Boundary	≥ 5 stages	4 stages	3 stages	2 stages	1 stage
	Impact Categories	≥ 5 impact categories	4 impact categories	3 impact categories	2 impact categories	1 impact category
Precision/ Uncertainty	Quality of the Sources	Very good	Good	Fair	Poor	Very poor
	Number of Studies	More than 800 studies	Less than 800 studies	Less than 600 studies	Less than 400 studies	Less than 200 studies
	Uncertainty	Very good	Good	Fair	Poor	Very poor
Methodological Appropriateness and Consistency		Very good	Good	Fair	Poor	Very poor
Temporal Representativeness		Less than 3 years of difference	Less than 6 years of difference	Less than 10 years of difference	Less than 15 years of difference	More than 15 years
Geographical Representativeness		≥ 20 countries and spans all continents	Between 11 and 20 countries and covers at least 4 continents	Between 5 and 10 countries and includes representation from 3 continents	Between 2 and 4 countries and covers 1 or 2 continents	1 country or region

Technological Representativeness	Recipes or product brands	Food items and Preparations	Detailed categories	Intermediate categories	General categories
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8.2.3. Pedigree Matrix evaluation

The modified Pedigree matrix enhances clarity and objectivity in selecting the most suitable database for the current research, with its application limited to open-access databases available for download at this point in time.

The Data Quality Score serves as a tool for comparing databases, where lower scores indicate higher quality in alignment with the study's objectives. Table 17; **Error! No se encuentra el origen de la referencia.** details the ranking of databases based on their Data Quality Scores, and specific values for the subcategories can be found in Table 41.

Table 17. Ranking of databases based on their Data Quality Score

	Database	Data Quality Score
1	SU-EATABLE LIFE	2.26
2	Database for Double Pyramid 2016	2.81
3	Poore & Nemecek 2018	2.98
4	DataFIELD	3.22
5	UCPH – Clune 2017	3.28
6	AGRIBALYSE	3.7
7	AGRIBALYSE + Biodiversity	3.7
8	Clark et al. 2022 Database	3.7
9	CFLCAD	3.72
10	The Cool Food Pledge	3.86
11	The Big Climate Database	3.94
12	SHARP INDICATORS DATABASE	3.98
13	Drew et al. 2020 Database	4.06

As a result of this comparison, the Data Quality Score for all the databases ranged from 2.26 to 4.06 and indicates substantial difference in data quality among them. All databases, except of the New Zealand-specific LCA database, fall under the categories of “Good quality” and “Fair quality” according to Table 15 adapted from JRC-IES, (2010).

Regarding Completeness, more specifically system boundaries, 92% of the databases included five or more stages in their analyzed systems (dataFIELD covers only two stages). Therefore, most databases covered primary production, processing, transport, packaging, and retail phases. However, in terms of impact categories, 53% of the databases (seven out of thirteen) considered only one or two impact categories, primarily focusing on GHG emissions and Land Use or Water Footprint, with four exclusively analyzing GHG emissions. When including three impacts categories, the impact categories were GHG emissions, Water Footprint and Land Use or Ecological Footprint. Only four out of thirteen databases covered four or more impact categories.

From a Precision/Uncertainty perspective, seven out of thirteen databases did not conduct a systematic literature review and instead relied on existing databases from prior literature reviews or well-known LCA databases like Ecoinvent. The databases that performed literature reviews exhibited varied performance in terms of the number of studies covered and the extent of uncertainty analysis. Regarding Methodological Appropriateness and Consistency, performance was also diverse, with four databases providing calculations and details upon request or through LCA software. The remaining nine databases demonstrated performance levels ranging from Fair to Good and Very Good, with Fair being the most frequent.

Concerning criteria measuring data suitability for the underlying study in terms of representativeness, it was observed that 70% of the databases (nine out of thirteen) had data older than six years for temporal representativeness. In terms of geographical representativeness, 54% (seven out of thirteen databases) were based on only one country or continent. The remaining databases covered more than 60 countries covered, but the authors acknowledged that these data predominantly represented Europe, North America and Australia, with lesser representation for continents and regions such as Africa, Asia, Oceania (distinct from Australia) and South America. Lastly, for technological representativeness, 80% of the databases (ten out of thirteen) had food items as the finest level of aggregation, while the remaining databases detailed categories or food groups.

It is important to note that sub-indicators such as Number of studies and Uncertainty were not applicable for the databases utilizing well-known LCA databases such as Ecoinvent and EXIOBASE. This is why their scores are empty, and the box has a dark grey background.

8.2.4. Database selection

Although SU-EATABLE LIFE received the highest score, its limitations in terms of impact categories led to the selection of the Poore & Nemecek 2018 database. This decision aligns with the research's purpose, as SU-EATABLE LIFE does not fulfill the primary need of assessing impacts on biodiversity caused by food. Nevertheless, the Poore & Nemecek database still ranked within the top 3 with a “Good quality” performance, indicating its suitability for this study. This database presents several advantages but also comes with drawbacks. It provides comprehensive coverage of environmental impacts associated with various food products, encompassing six inventory categories.

The data is recent and relevant, reflecting current production methods, supply chain dynamics, and technological advancements. The database's compilation methodology is rigorous, supported by thorough documentation. Covering 119 countries, with an initial focus on studies from less-represented countries, it offers a global perspective. However, its resolution at the food level is suboptimal, as it categorizes foods into groups, presenting challenges in understanding data specific to individual food items.

8.3. AGRIBALYSE 3.0

The information about AGRIBALYSE 3.0 is taken verbatim from the project report submitted in Autumn 2023.

AGRIBALYSE 3.1 is a reference database of environmental impact indicators for agricultural and food products in France. It was developed as part of the AGRIBALYSE program, led by ADEME and INRAE since 2009, with support from various organizations and experts. This extensive database covers 2500 food items listed in CIQUAL and maintains consistent connections between nutritional and environmental data. It includes 13 environmental impact indicators, such as Climate Change, Particulate matter, Water Use, Resource use (fossils & minerals, and metals), Land Use, Ozone Depletion, Acidification, Ionizing radiation, effect on human health, Photochemical Ozone formation, effect on human health, Eutrophication (terrestrial, marine & freshwater). The chosen impact assessment method is Environmental Footprint midpoints and single score (Asselin-Balençon A. et al., 2022).

AGRIBALYSE 3 builds upon earlier versions (mainly for raw materials production processes), incorporating additional data related to food processing, logistics, packaging from various sources. The database comprises 2517 datasets. Food products within the database are modeled on a per 1 kilogram basis for prepared product. It comprehensively covers the value chain, encompassing raw material production, processing, assembly, packaging, distribution, retail, storage, preparation at the consumer level, and disposal of packaging. Transport between each stage is included, except for transport between retail and the consumers' home. Waste and food losses are accounted for at various stages, except at consumers' home. The allocation method used is mainly economic allocation, and biophysical and mass allocation applied to specific products like dairy husbandry, and cheese production respectively.

Despite its utility, AGRIBALYSE 3 has limitations due to its compilation from various sources, resulting in sometimes inconsistent boundaries, data collection, and methods. Capital goods may not always be included, and certain inventory items related to methodological questions remain inconsistently implemented, particularly in agricultural inventories concerning carbon uptake, land use change data, and crop protection.

AGRIBALYSE® data is freely available in two formats: a full version, which requires proficiency in LCA software like SIMAPRO and OpenLCA, enabling users to access detailed production stages and modify production hypothesis; and a simplified version, accessible to everyone, consisting of spreadsheets for the LCA indicators for raw conventional agricultural products, organic products, feed, and ready-to-eat food products. Additionally, simplified versions for ready-to-eat food products are available in CSV format, comprising summary results, and detailed breakdowns by life cycle stage and ingredient. It is particularly suitable for hotspot analysis, environmental information and eco-scores and can be utilized by non-experts. The database can be downloaded by following this link: <https://doc.agribalyse.fr/documentation-en/agribalyse-data/data-access>, and download the non-expert version.

While this data is primarily tailored to the French context and food market, it can serve as an initial proxy for European countries but may not be suitable for southern region countries. Nevertheless, it should be noted that using disaggregated data outside France requires an Ecoinvent license.

AGRIBALYSE® data is accessible under the conditions of Etalab's "Open License", which can be found in its entirety here: <https://www.etalab.gouv.fr/wp-content/uploads/2018/11/open-licence.pdf>. According to this license, users have the right to adapt the AGRIBALYSE® data to create "Derived Information", products or services. Commercial use is also allowed, such as combining it with other information or including it in a service or application. However, users must mention the authorship of the information, including source and date of the last update.

Lastly, unauthorized representation or reproduction without the author's consent is prohibited under the French Intellectual Property Code (art. L122-4) and is punishable by law (art. L122-5), with exceptions for private use, individual analysis, and short quotations for critical, educational, or informational purposes.

8.4. Instructions for completing the Recipe_template

1. **Open the Excel file named "Recipe_template".** You will see three columns: Ingredient, Amount and Unit.
2. **Search for the ingredient you want.** You can either use the shortcut Ctrl + F to find it quickly or use the Filter function to navigate through the column.
3. **Write the amount in the Amount column.** Important! Please follow the unit defined in the Unit column. If the unit is different from "grams" and you know the exact amount in grams for that ingredient, you can delete the existing unit and enter "grams" instead. Do not convert from grams to another unit, as this will cause errors when running the scripts. Be careful while doing this to avoid any typos.
4. **Repeat the same procedure for all the ingredients in your recipe.** Ensure each ingredient has the correct amount and unit.
5. **Once all the amounts are filled out, save the file with the name of the recipe.** Replace the template name with a recognizable name for your recipe.
6. **If you want to input more than one recipe, repeat the previous steps.** This means creating a new file for each recipe you want to assess.
7. **Lastly, save all the recipe files in the folder called "recipes".** You can find this folder at the following path: "../data/recipes"

8.5. Instructions to replicate the results from the sensitivity analysis

Since the DELI Calculator does not include a sensitivity analysis in the source code, a manual analysis becomes necessary. The following instructions delineate the steps for adjusting both the quantities of the main ingredients in different versions of the *Lomo*

Saltado recipe and the impact values. In order to perform a sensitivity analysis, the following steps should be taken:

8.5.1. Instructions for sensitivity analysis on quantities of the main ingredients

1. Duplicate the Recipe_template files where the original recipe was saved.
2. Introduce desired changes; for instance, decrease the quantity of the beef ingredient by 10%.
3. Save the file "Sensitivity" prefixed to the original recipe name, such as "Sensitivity Lomo Saltado with beef". Alternatively, if both an increase and decrease by 10% are desired, use a recognizable name to denote the change. The file name must commence with "Sensitivity".
4. Save the modified Recipe_template files in the "./recipe" directory.
5. Note that the sensitivity analysis code is reliant on the source code; thus, ensure that the tool has executed the existing packages and scripts with the original recipes.
6. Open the Python file named "sensitivity_uncertainty.py" in your integrated environment.
7. Follow the designated sections and execute the code beneath each section. For example, the "Sensitivity for amounts of the main ingredients" section contains instructions for both increase and decrease, as altering the Recipe_template suffices. The resultant values are stored based on the recipe name. Find the results in both "./impacts" and "./interim".

8.5.2. Instructions for sensitivity analysis on impact values

1. Duplicate the Excel file called "food_item_poore_and_nemecek".
2. Adjust the values for the desired ingredient, performing increases or decreases for each impact category.
3. Save the file with a recognizable name, preferably following the suggested format:
 - a. For increases: "food_item_poore_and_nemecek_sensitivity_plus"
 - b. For decreases: "food_item_poore_and_nemecek_sensitivity_minus"
4. Open the Python file "sensitivity_uncertainty.py"
5. Run the corresponding section. If you are assessing an increase, run the "Increase" section under the "Sensitivity for impact values".
6. Find the results in both "./impacts" and "./interim" directories.

8.6. Impact values per ingredient for the Lomo Saltado recipe with the highest impact on each impact category

As mentioned in 3.2.3.3., the 'visualizations.py' file also generates results for the recipe with the highest overall impact in each impact category. For these recipes, a stacked bar chart is created to show the breakdown of impacts by ingredients. This process is repeated for all six impact categories. Figure 20 - 25 displays the results for the recipe with the highest impact value in each impact category.

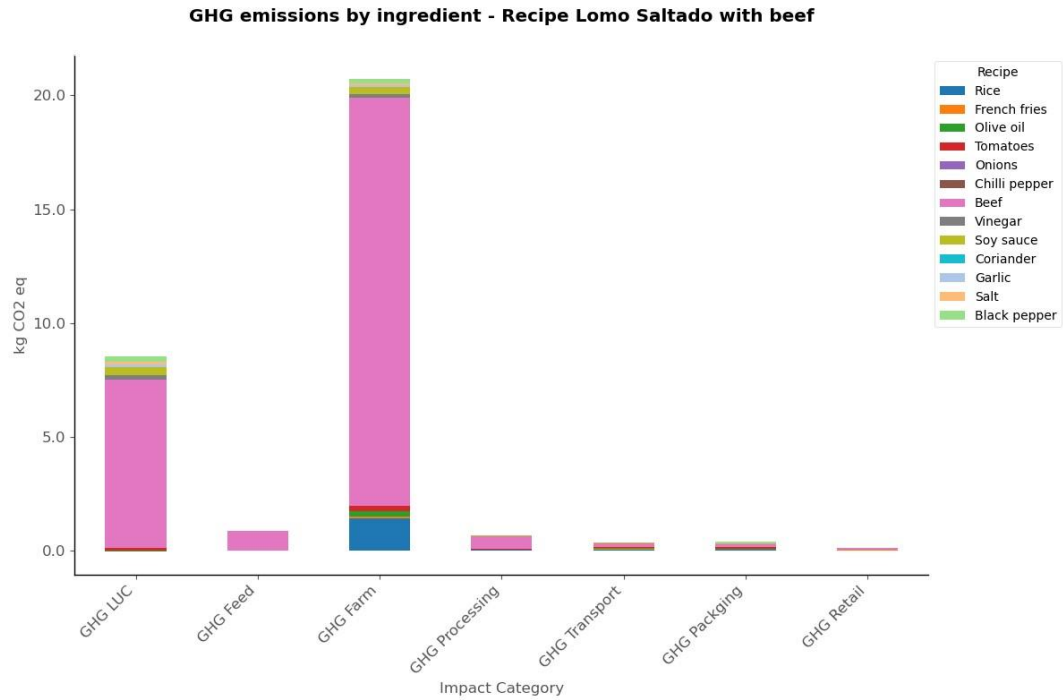


Figure 20. GHG emissions by ingredient for the recipe with the highest GHG emissions: Lomo Saltado with beef.

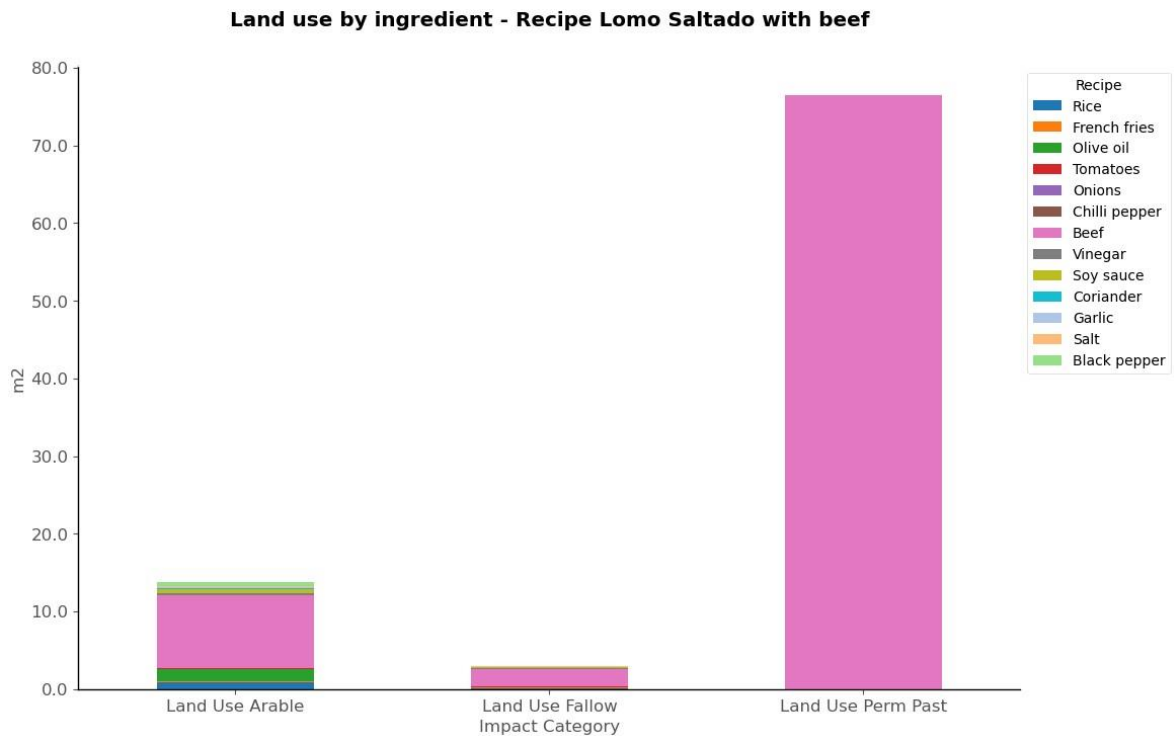


Figure 21. Land use by ingredient for the recipe with the highest Land use impact: Lomo Saltado with beef.

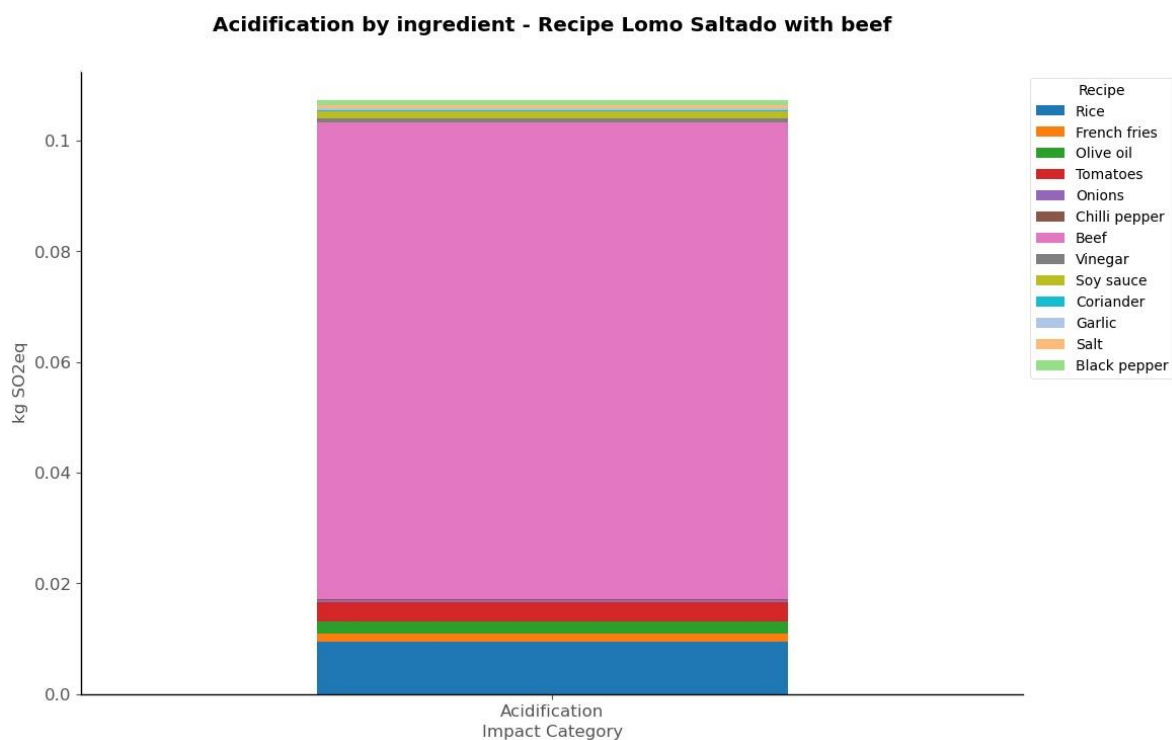


Figure 22. Acidification by ingredient for the recipe with the highest Acidification potential: Lomo Saltado with beef.

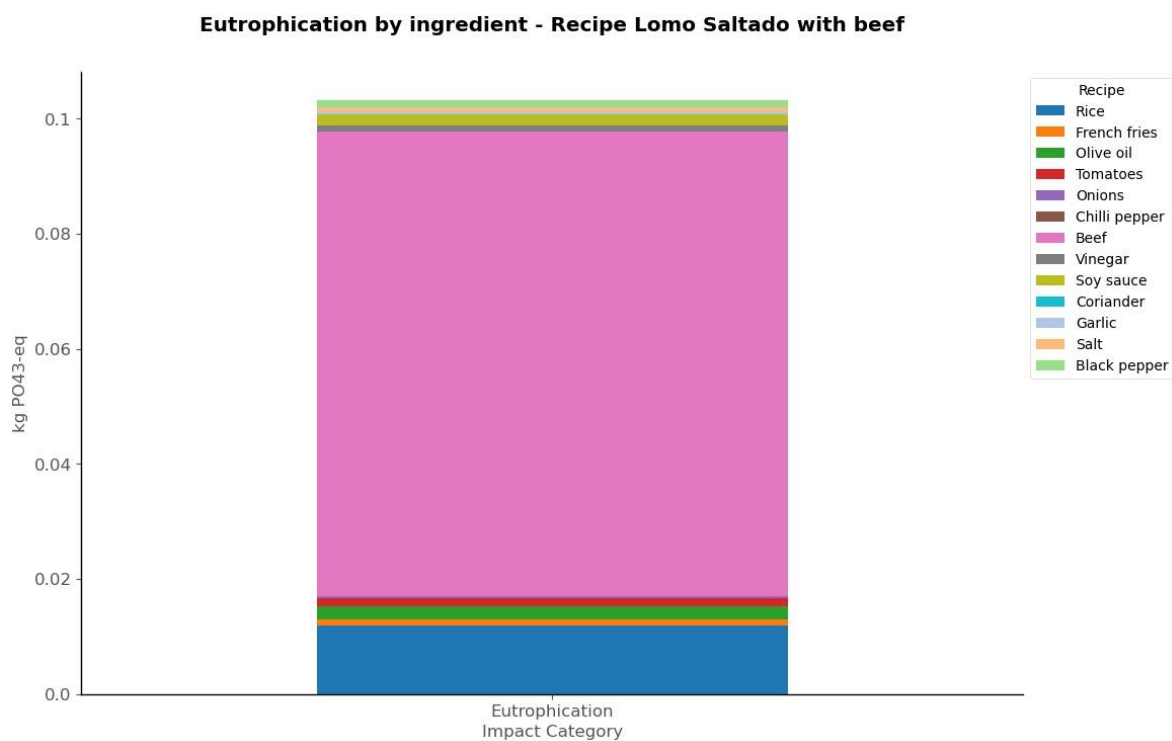


Figure 23. Eutrophication by ingredient for the recipe with the highest Eutrophication potential: Lomo Saltado with beef.

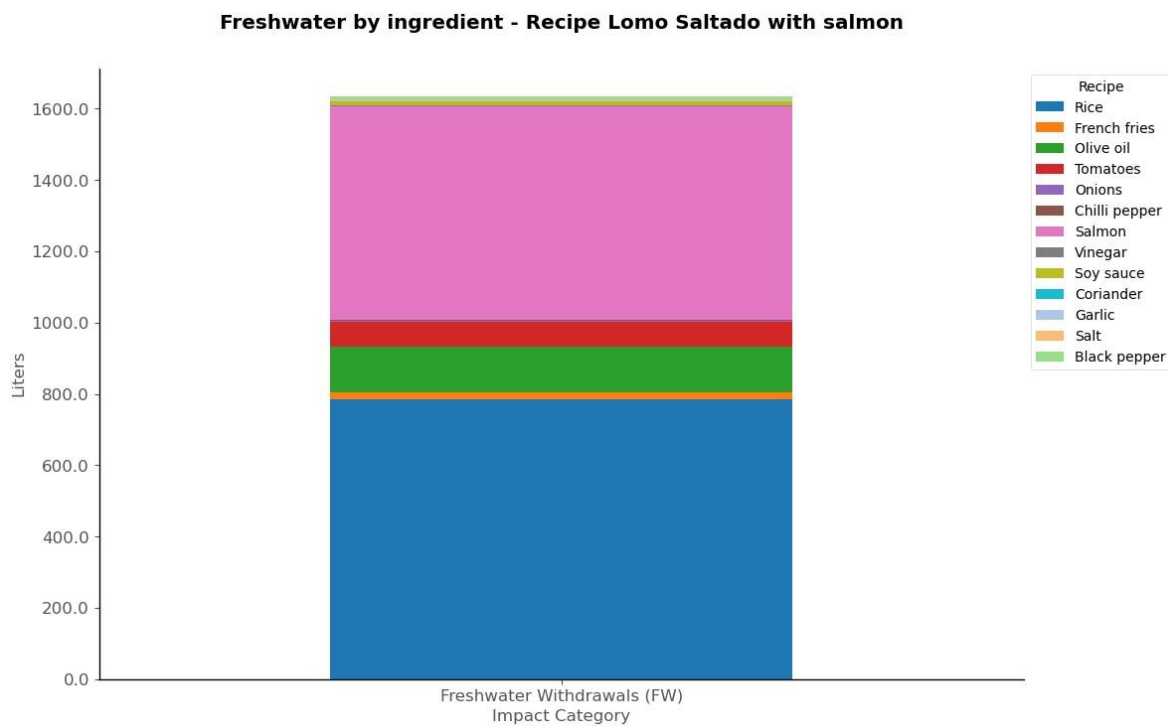


Figure 24. Freshwater withdrawals by ingredient for the recipe with the highest Freshwater withdrawals: Lomo Saltado with salmon.

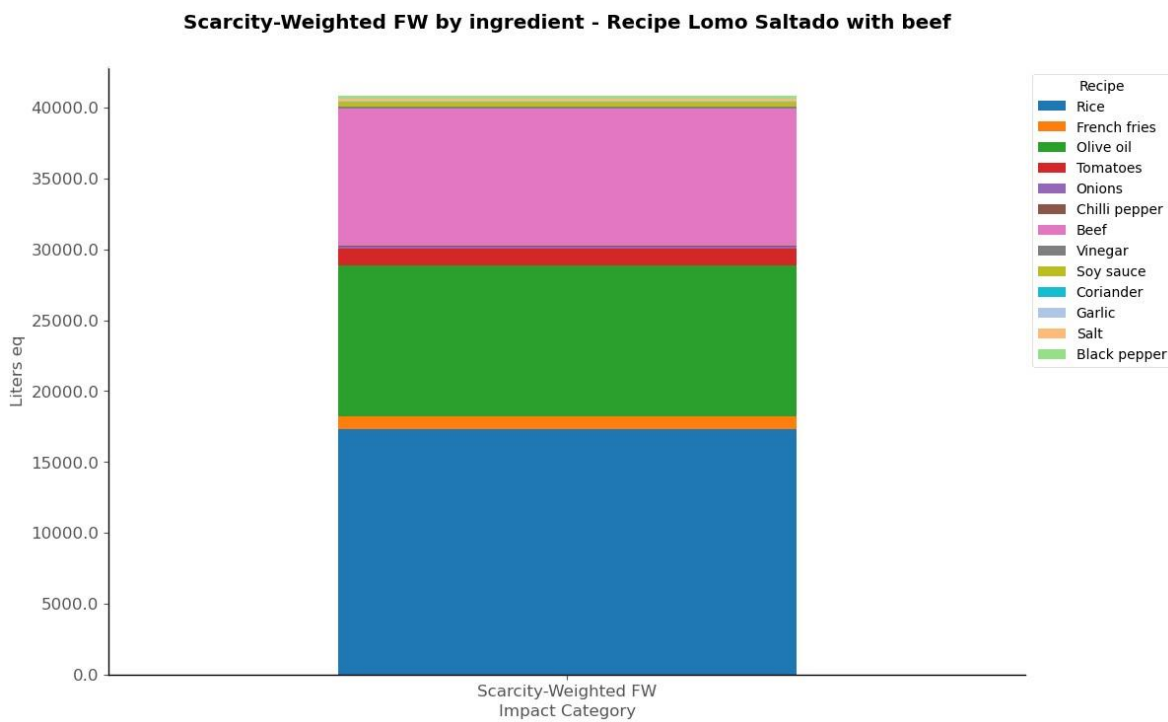


Figure 25. Freshwater withdrawal by ingredient for the recipe with the highest Scarcity-weighted freshwater withdrawal: Lomo Saltado with beef.

8.7. Results from the test sessions

In the following subsections, the results from the different test users are presented. Only the visualizations are included here. For more details, please refer to the “Test sessions” folder located within the DELI Calculator package.

8.7.1. Results from User 1

8.7.1.1. Recipes

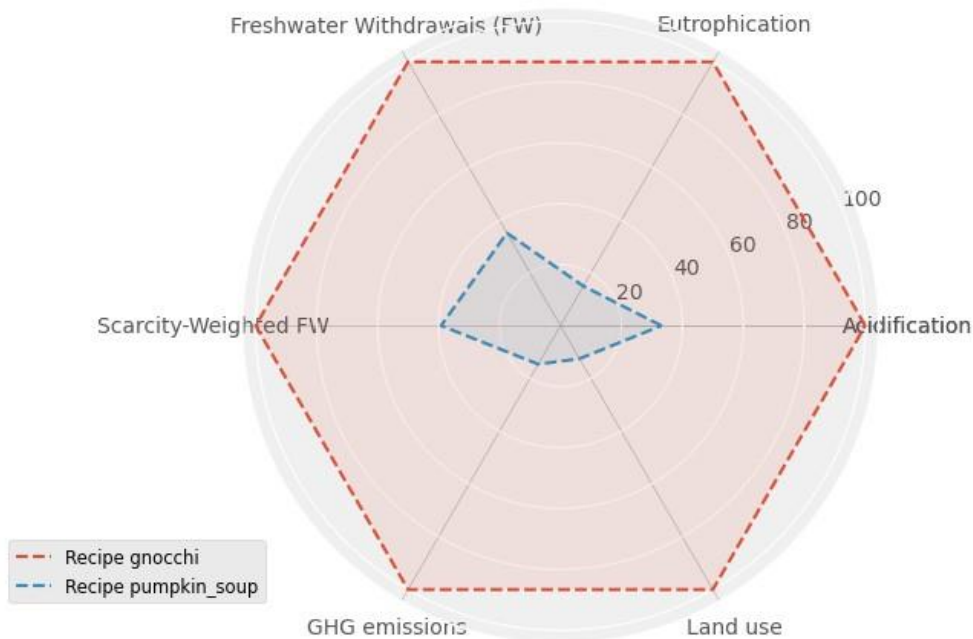
Table 18. Composition of the recipe: Pumpkin soup

Ingredient	Amount	Unit
Soymilk	500	grams
Onions	1	unit
Carrots	6	unit
Hokkaido pumpkins	1	unit
Lemons	1	unit
Spices Misc.	2	tbsp
Butter	50	grams

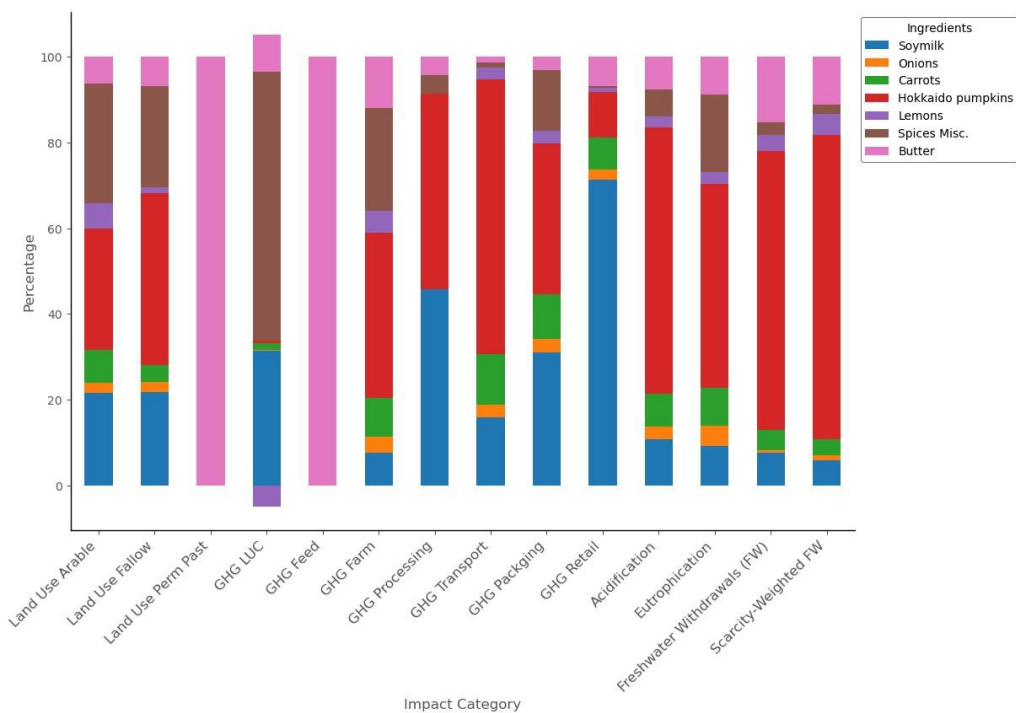
Table 19. Composition of the recipe: Gnocchi

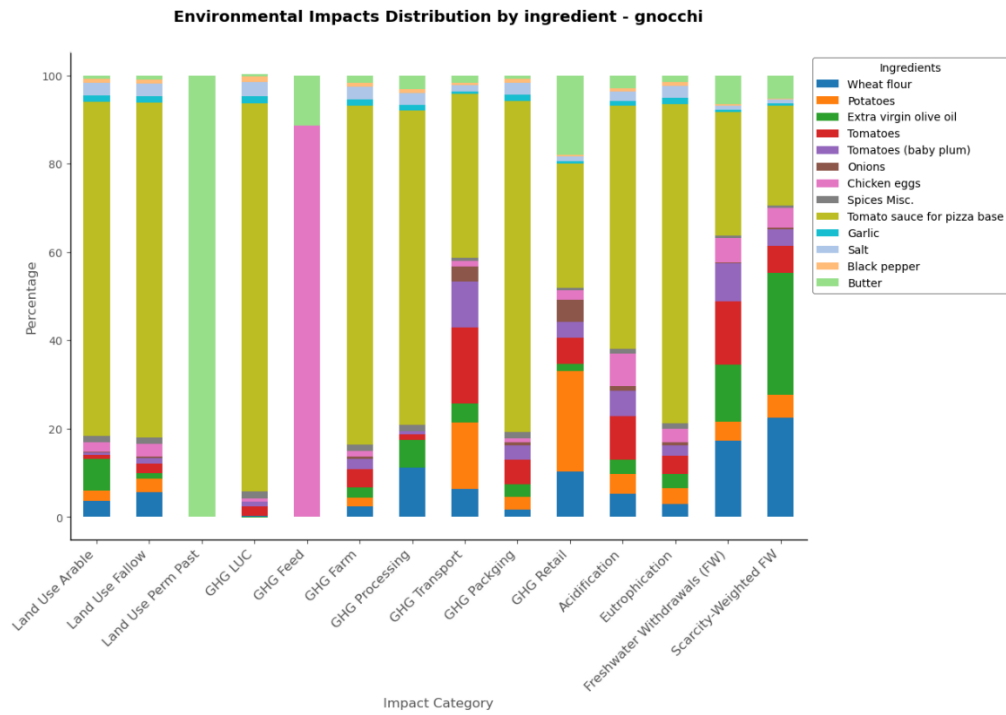
Ingredient	Amount	Unit
Wheat flour	150	grams
Potatoes	500	grams
Extra virgin olive oil	2	tbsp
Tomatoes	2	unit
Tomatoes (baby plum)	12	unit
Onions	1	unit
Chicken eggs	1	unit
Spices Misc.	1	tbsp
Tomato sauce for pizza base	800	grams
Garlic	3	clove
Salt	1.5	tbsp
Black pepper	0.5	tbsp
Butter	60	grams

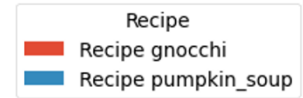
Relative environmental performance of recipes



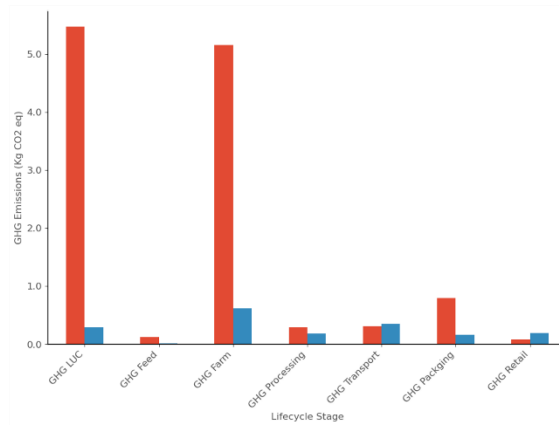
Environmental Impacts Distribution by ingredient - pumpkin_soup



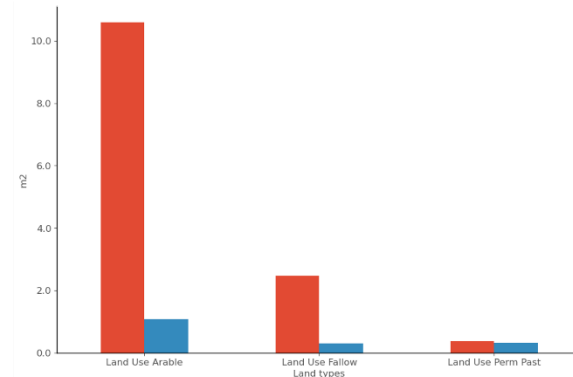




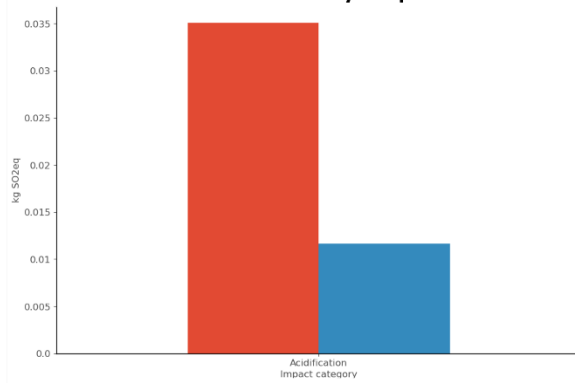
GHG Emissions by Life Cycle Stage and recipe



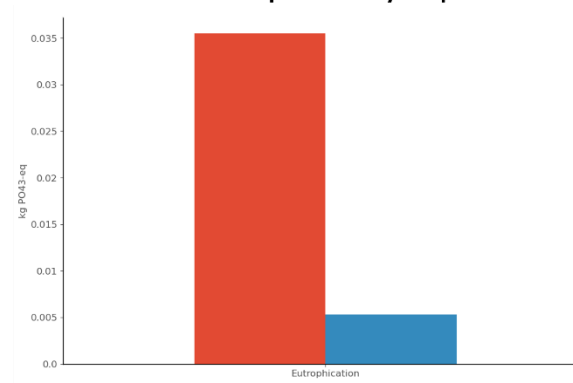
Land Use by land type and recipe



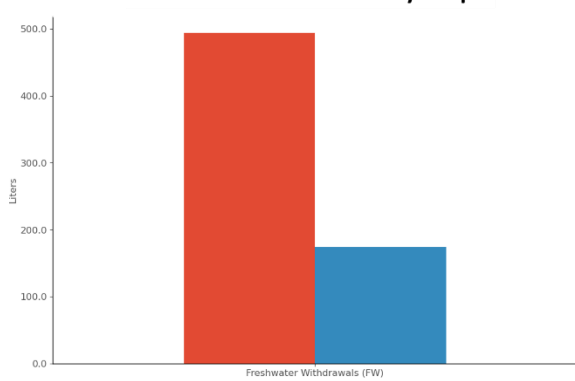
Acidification by recipe



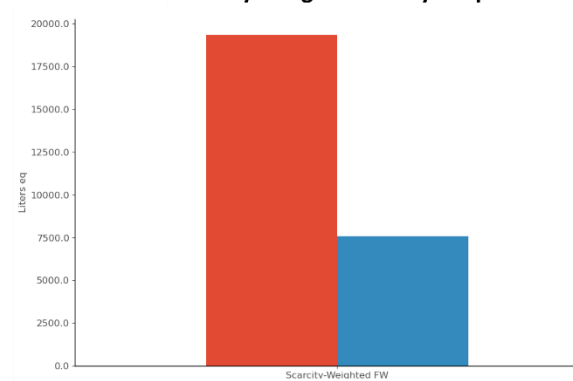
Eutrophication by recipe



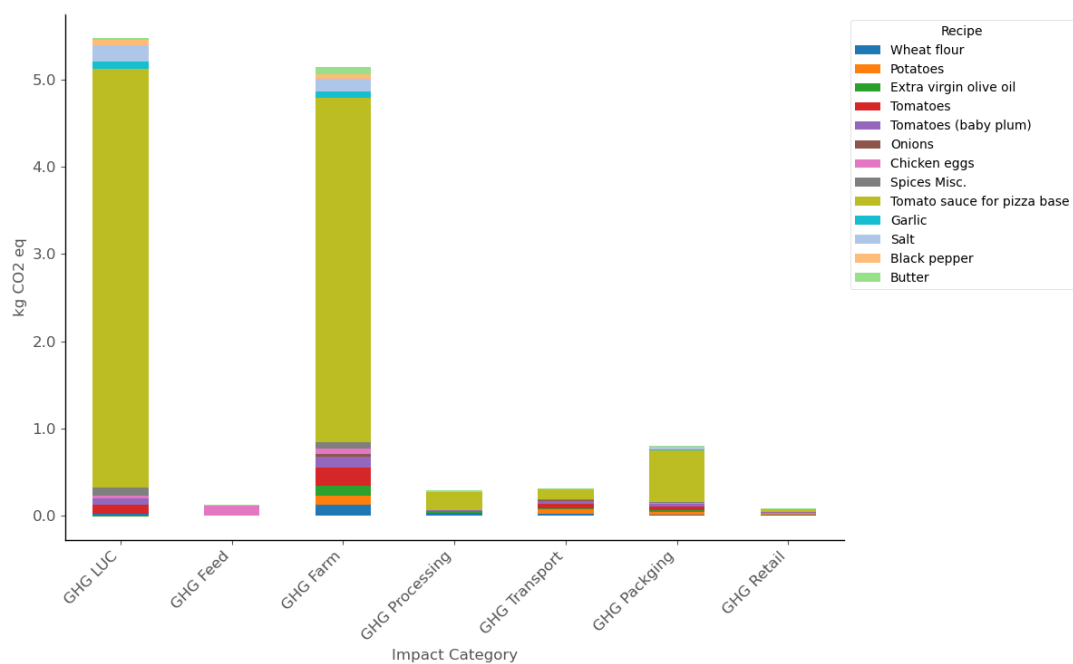
Freshwater Withdrawals by recipe



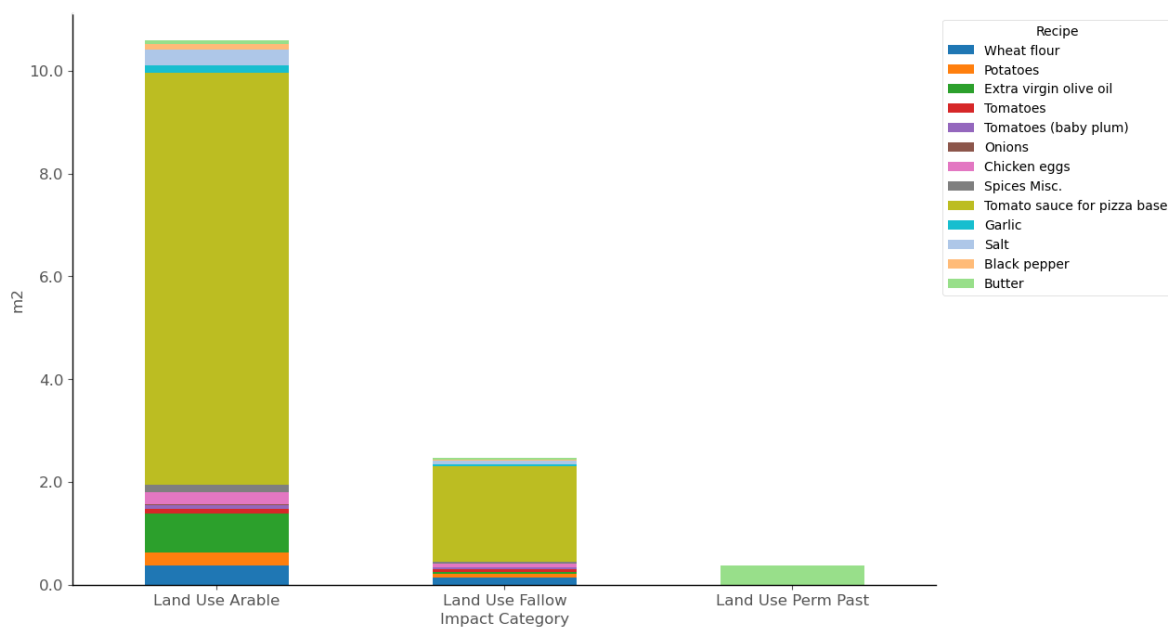
Scarcity-Weighted FW by recipe



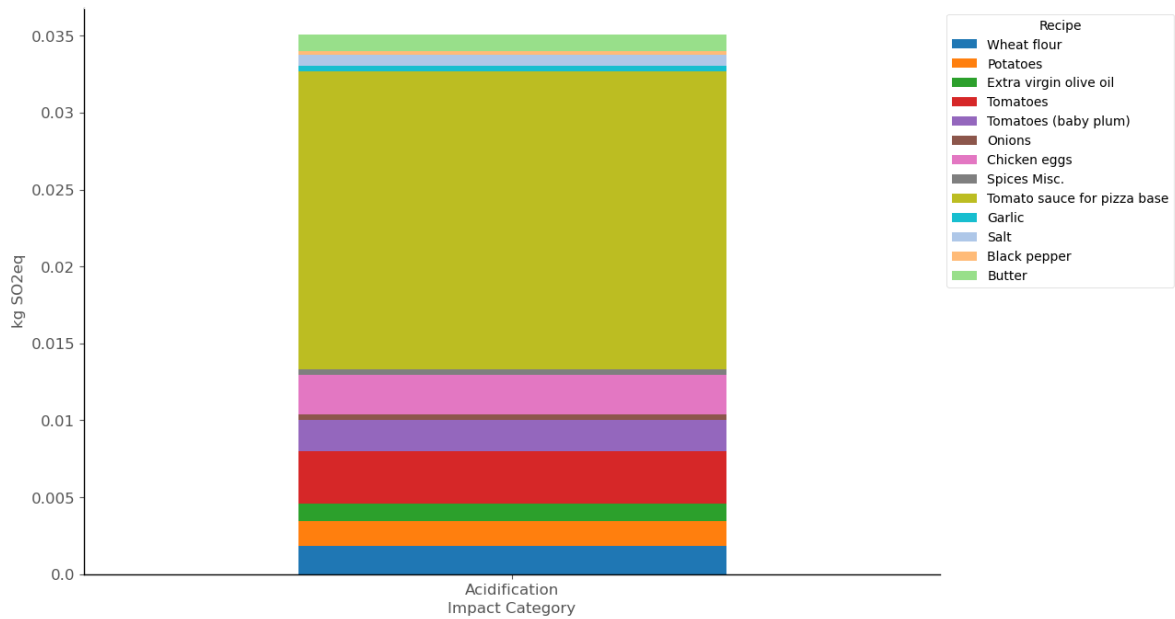
GHG emissions by ingredient - Recipe gnocchi



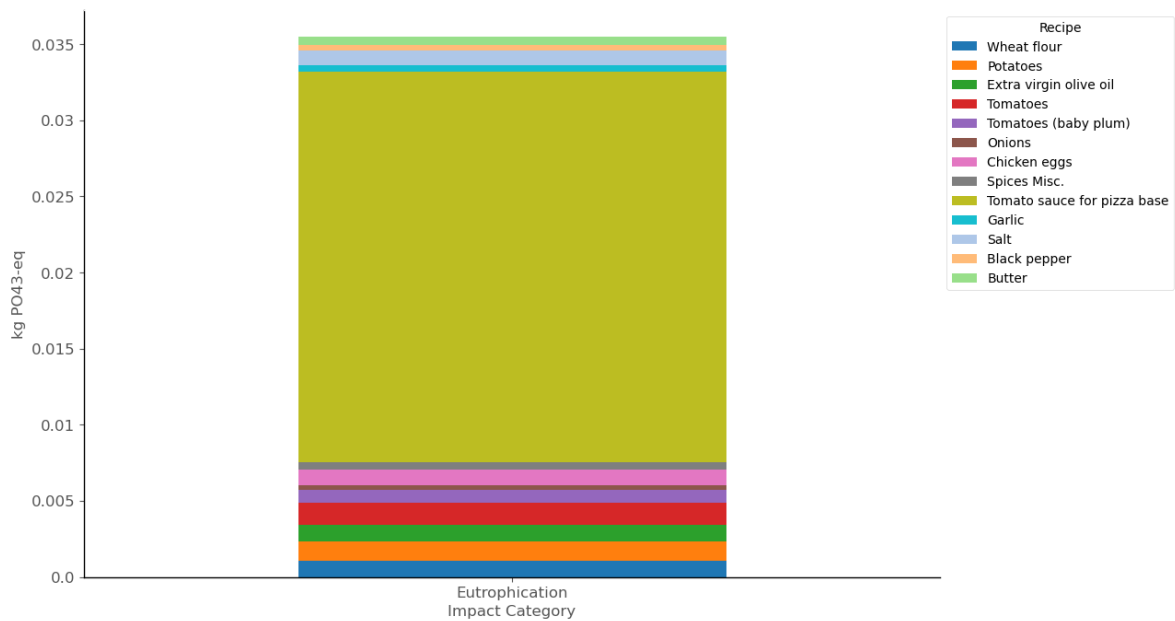
Land use by ingredient - Recipe gnocchi

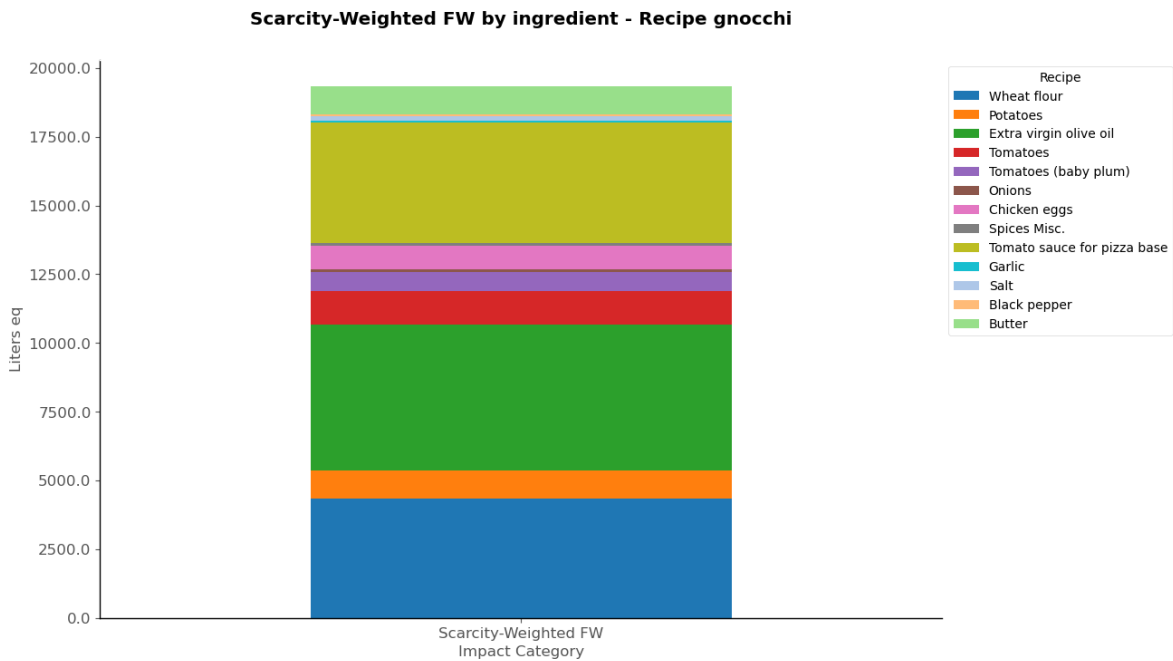
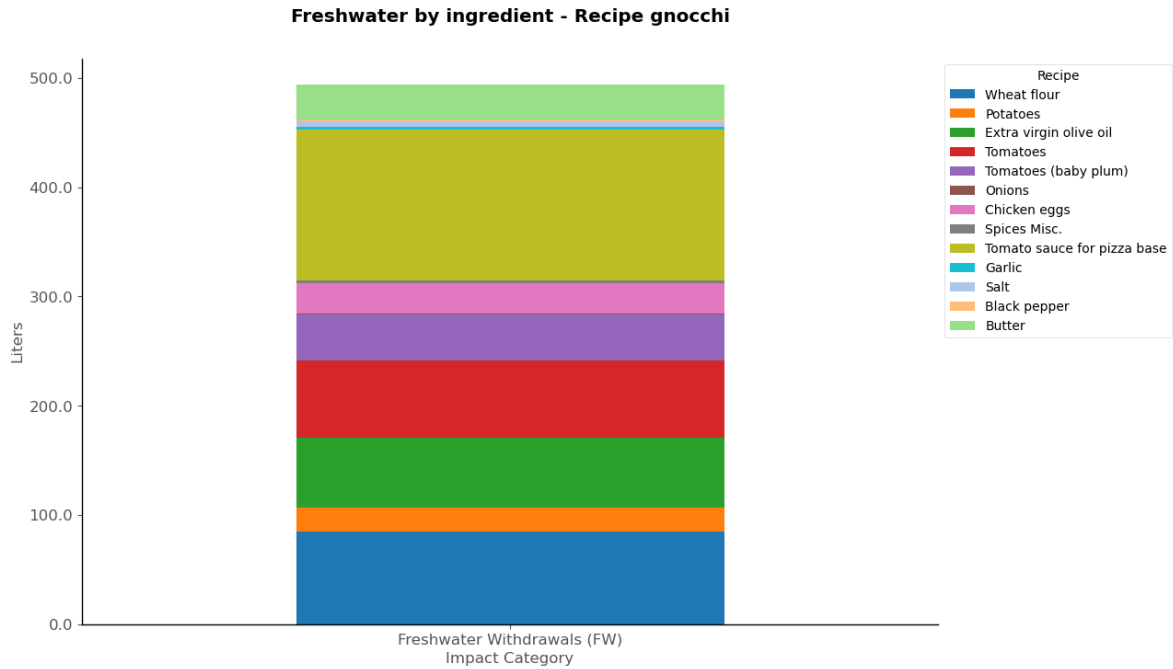


Acidification by ingredient - Recipe gnocchi



Eutrophication by ingredient - Recipe gnocchi





8.7.2. Results from User 2

8.7.2.1. Recipes

Table 20.Composition of the recipe: Aubergine vegetarian

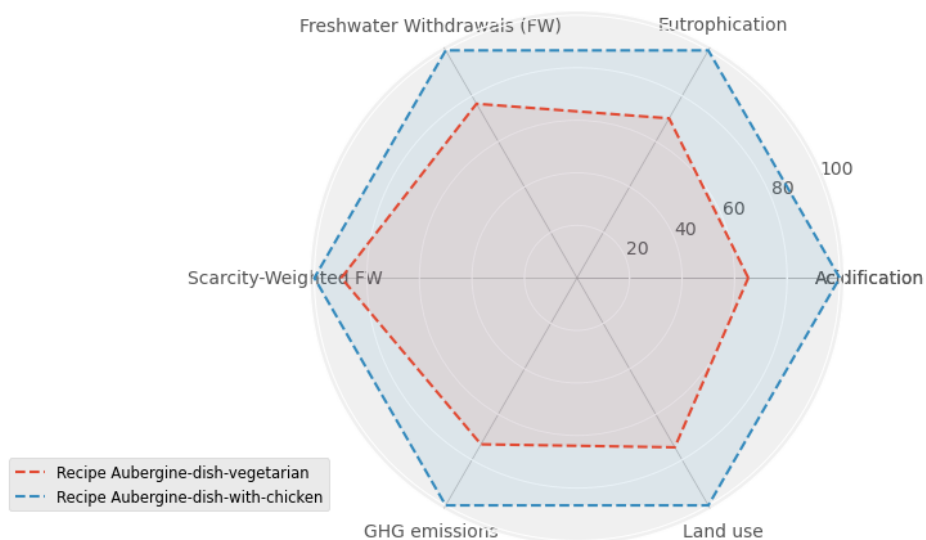
Ingredient	Amount	Unit
Chickpeas	400	grams
Extra virgin olive oil	8	tbsp

Tomatoes	300	grams
Onions	2	unit
Zucchini	3	unit
Apricot	250	unit
Tomato sauce	5	tbsp
Coriander	1	tbsp
Garlic	6	clove
Sea salt	1	tbsp
Cinnamon	0.5	tbsp
Cumin	1	tbsp

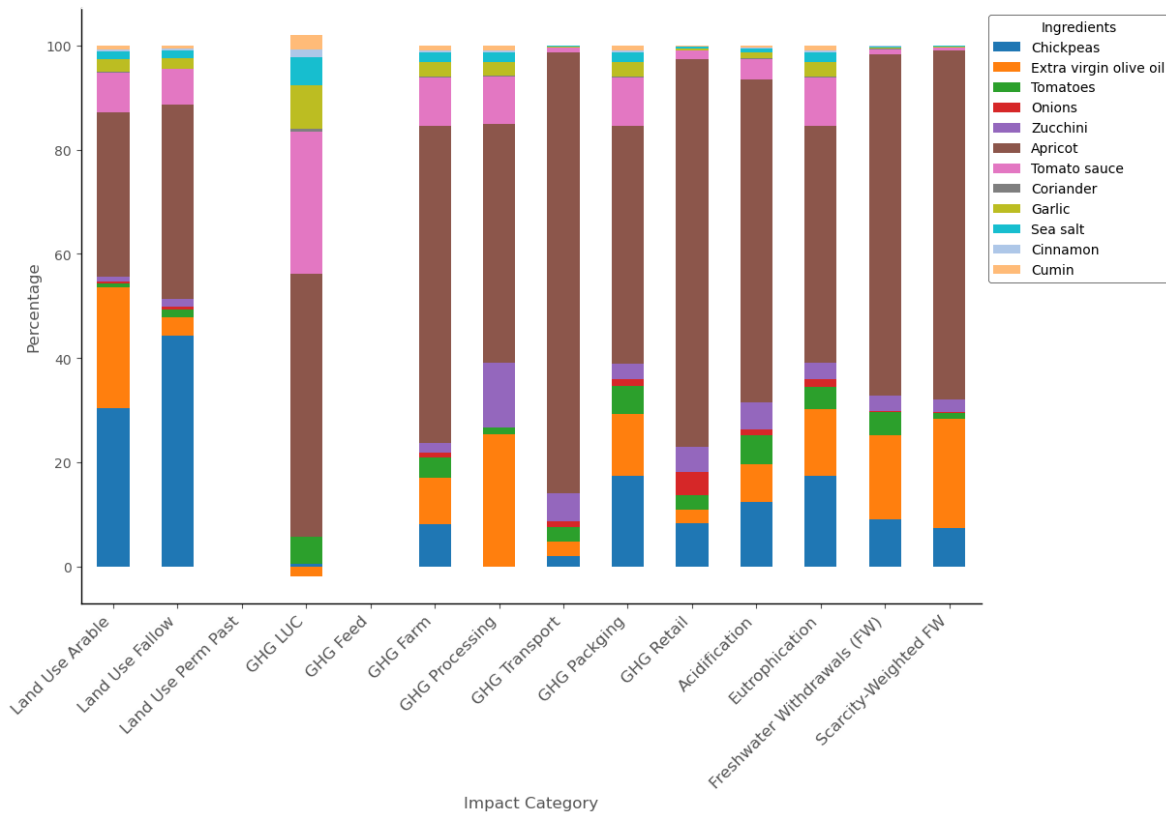
Table 21. Composition of the recipe: Aubergine with chicken

Ingredient	Amount	Unit
Extra virgin olive oil	8	tbsp
Tomatoes	300	grams
Onions	2	unit
Zucchini	3	unit
Apricot	250	unit
Chicken	400	grams
Grana Padano cheese, from cow's milk	100	grams
Tomato sauce	5	tbsp
Coriander	1	tbsp
Garlic	6	clove
Sea salt	1	tbsp
Cinnamon	0.5	tbsp

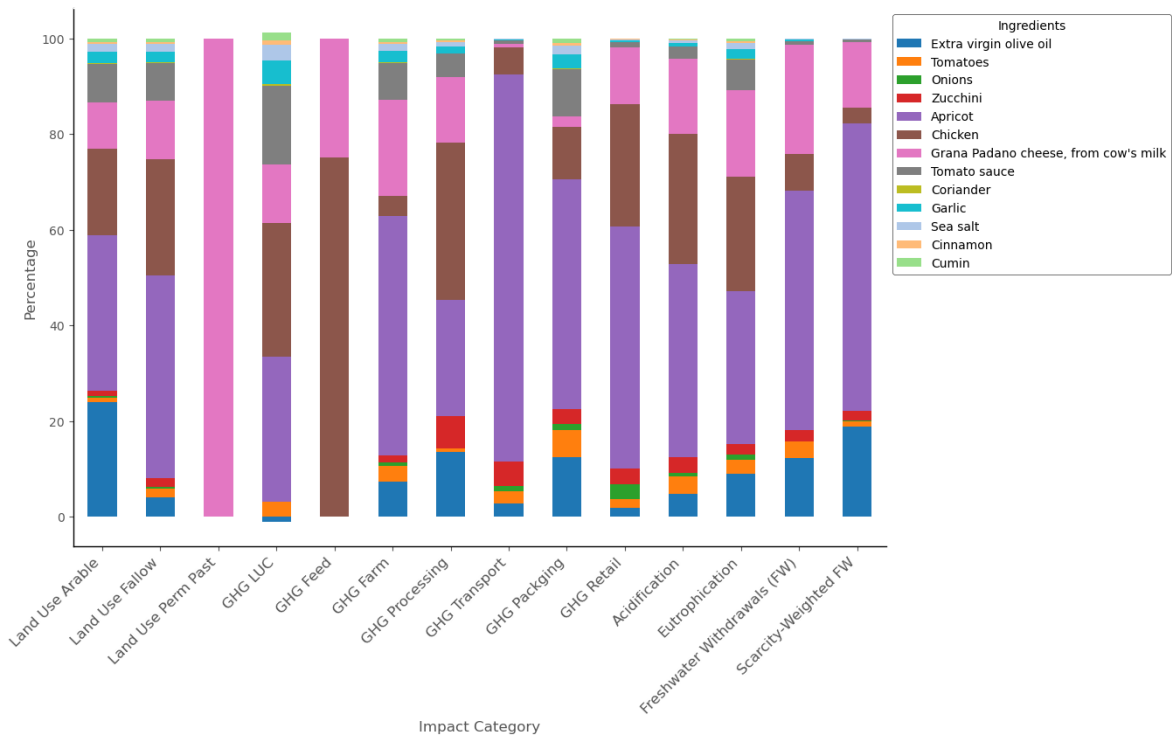
Relative environmental performance of recipes

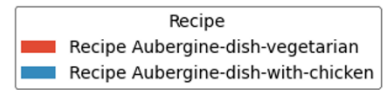


Environmental Impacts Distribution by ingredient - Aubergine-dish-vegetarian

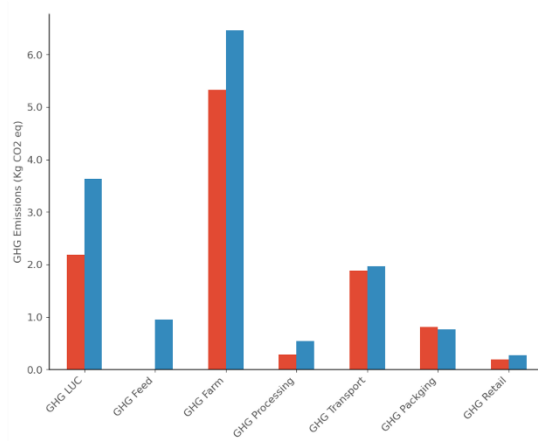


Environmental Impacts Distribution by ingredient - Aubergine-dish-with-chicken

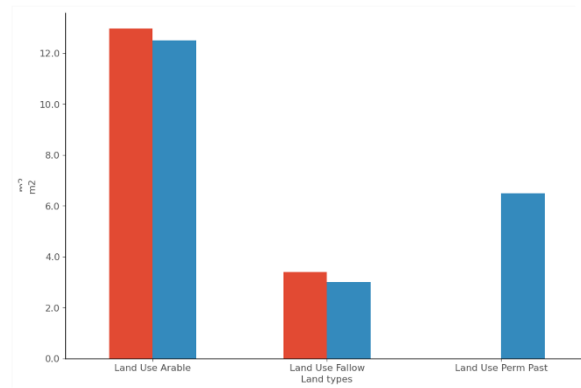




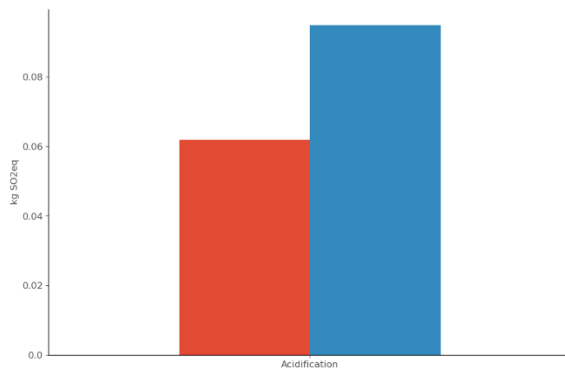
GHG Emissions by Life Cycle Stage and recipe



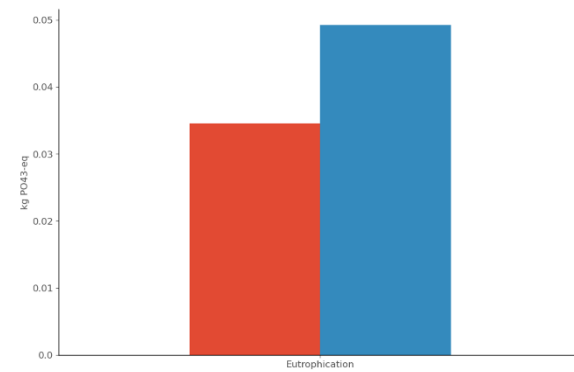
Land Use by land type and recipe



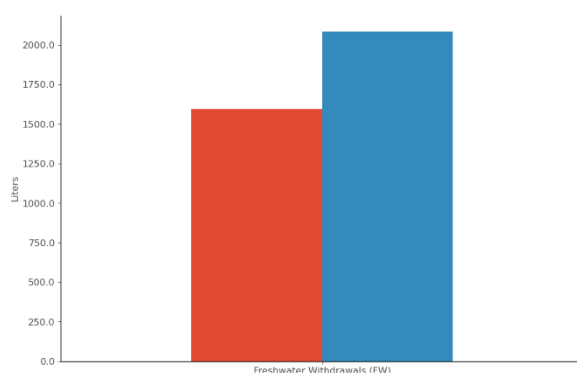
Acidification by recipe



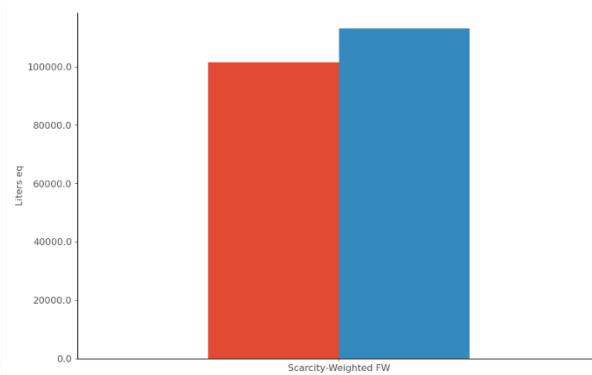
Eutrophication by recipe



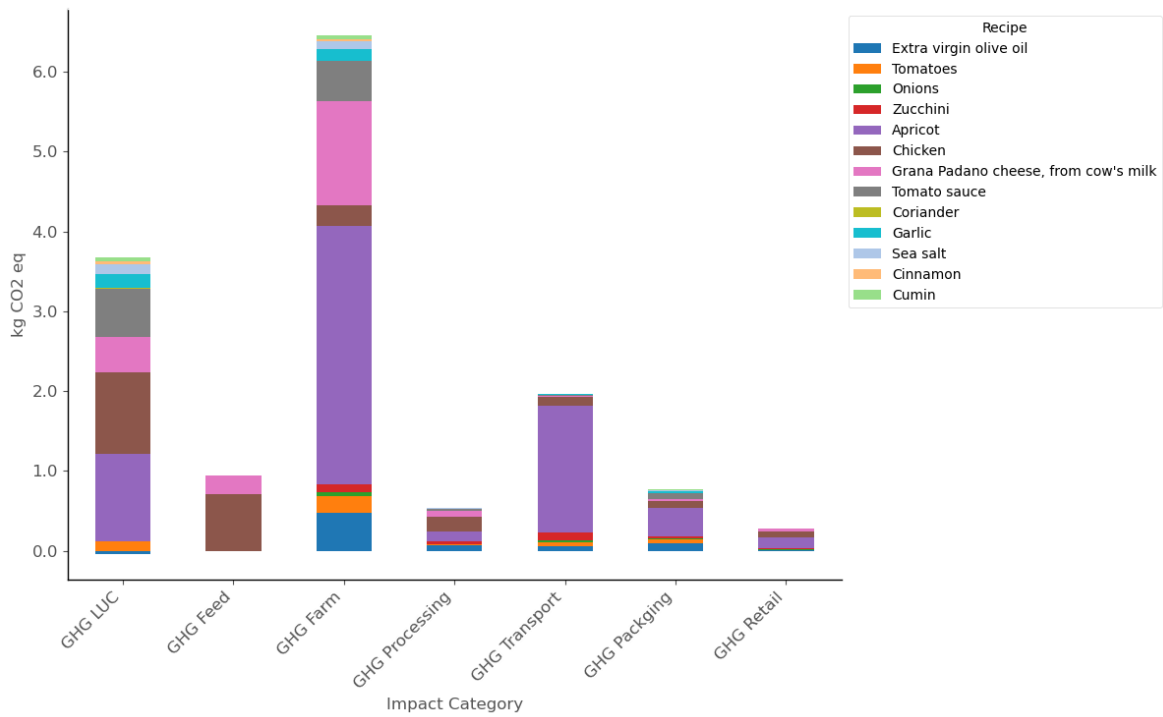
Freshwater Withdrawals by recipe



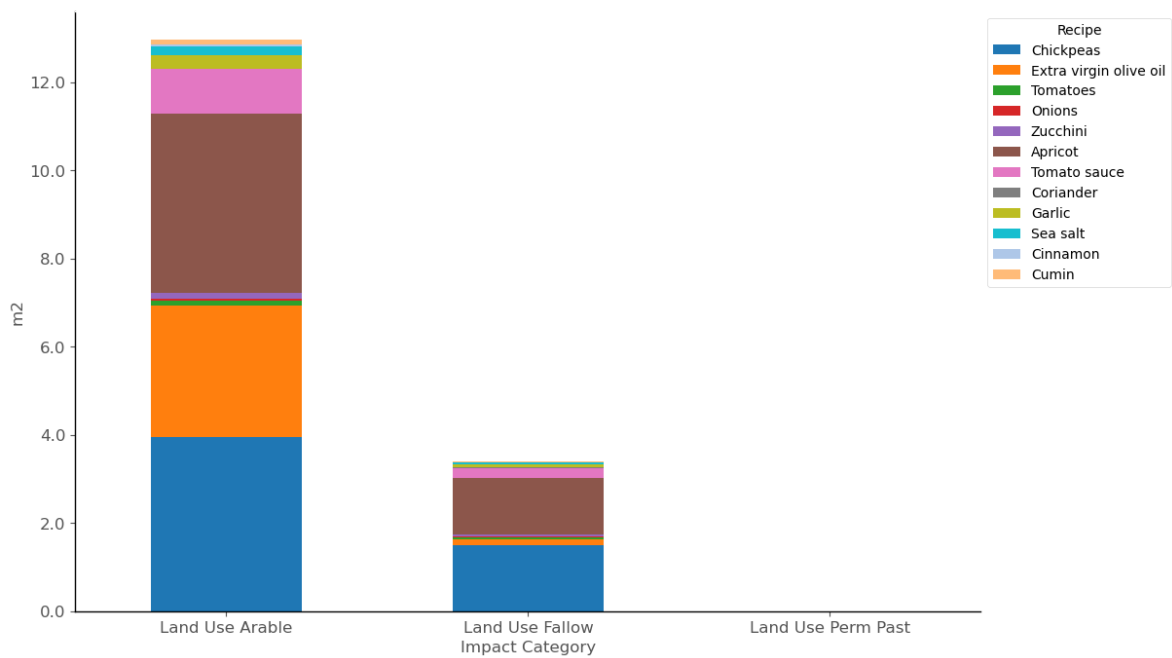
Scarcity-Weighted FW by recipe



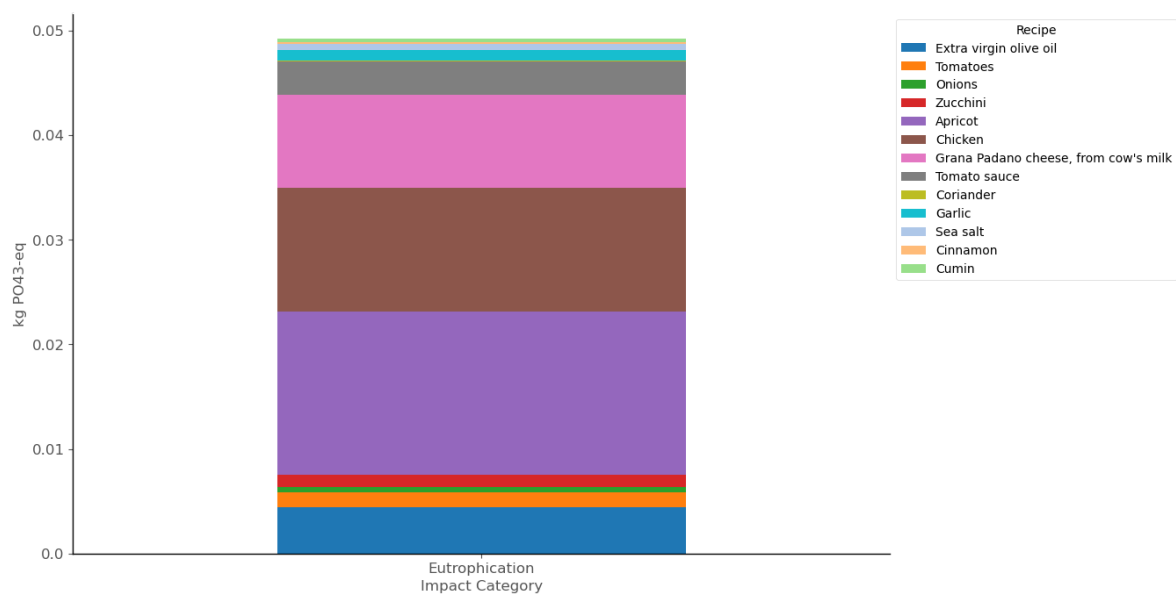
GHG emissions by ingredient - Recipe Aubergine-dish-with-chicken



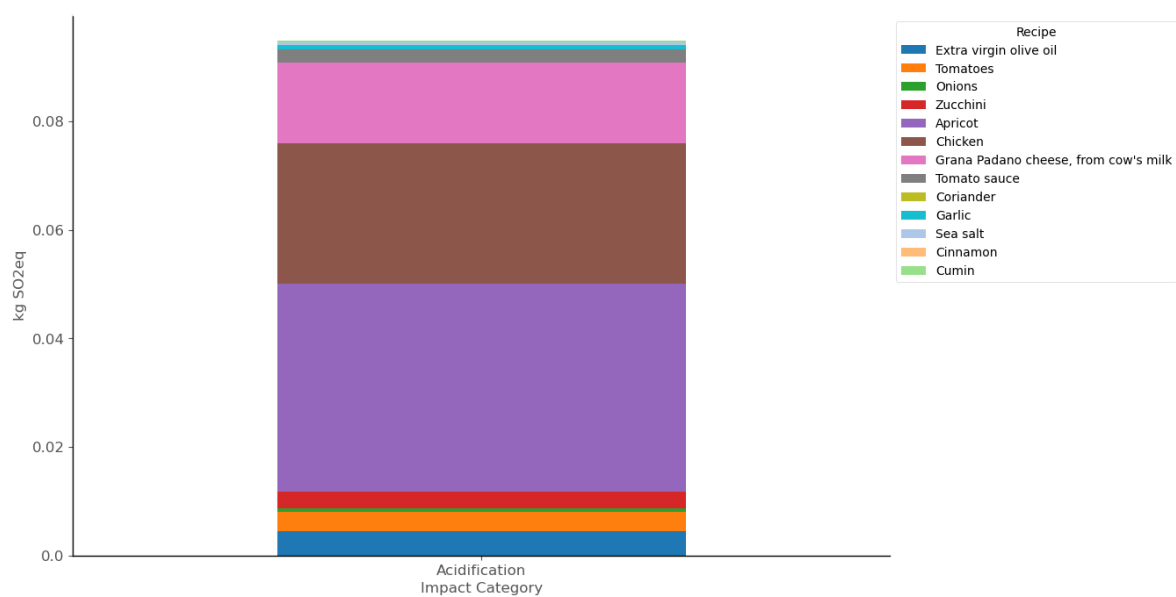
Land use by ingredient - Recipe Aubergine-dish-vegetarian

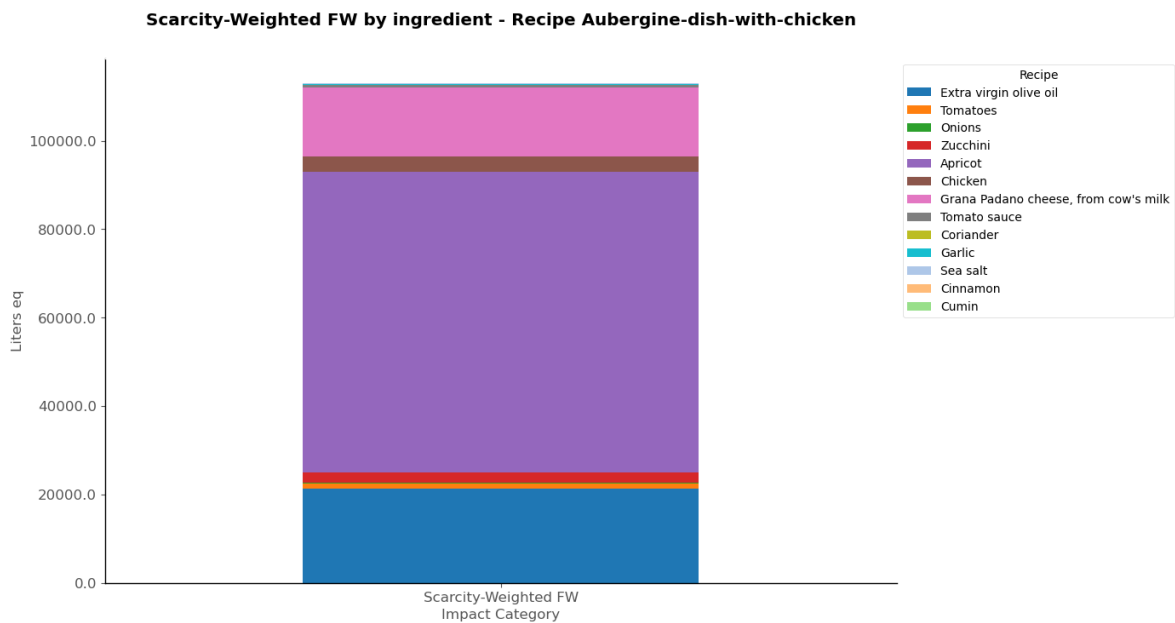
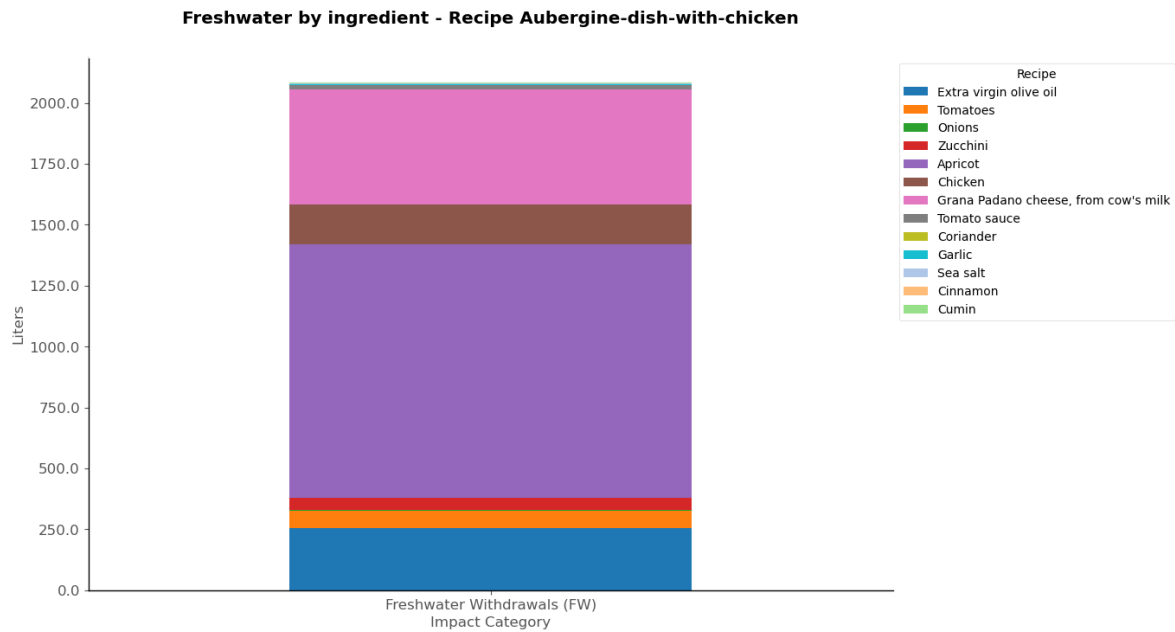


Eutrophication by ingredient - Recipe Aubergine-dish-with-chicken



Acidification by ingredient - Recipe Aubergine-dish-with-chicken





8.7.3. Results from User 3

8.7.3.1. Recipes

Table 22. Composition of the recipe: One pot

Ingredient	Amount	Unit
Potatoes	600	grams
Sugar	1	tbsp

Onions	1	unit
Mushrooms	200	grams
Beef sausages	100	grams
Chicken	600	grams
Tomato sauce	400	grams
Thyme	2	tbsp
Broth poultry	2	dl

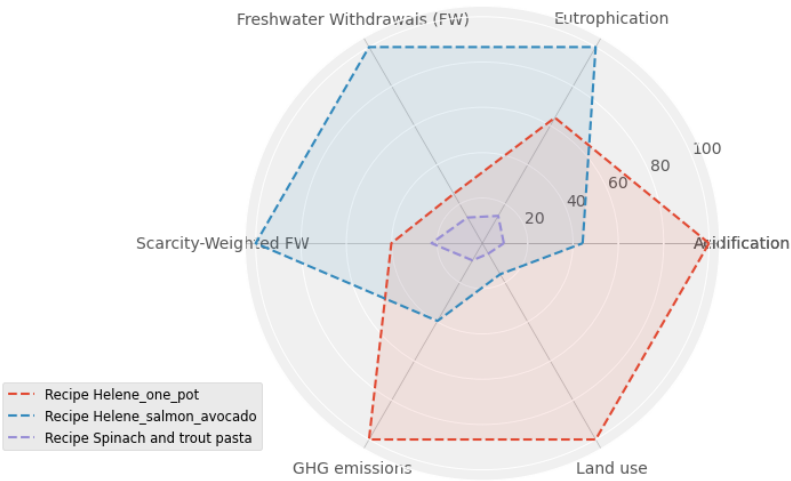
Table 23. Composition of the recipe: Salmon and avocado

Ingredient	Amount	Unit
Rice	2	cup
Sunflower oil	1	tbsp
Onions	0.5	unit
Chilli pepper	0.5	unit
Lemons	0.7	unit
Avocado	1	unit
Mango	1	unit
Salmon	800	grams
Soy sauce	2	tbsp
Coriander	3	tbsp
Salt	0.5	tbsp
Black pepper	0.5	tbsp

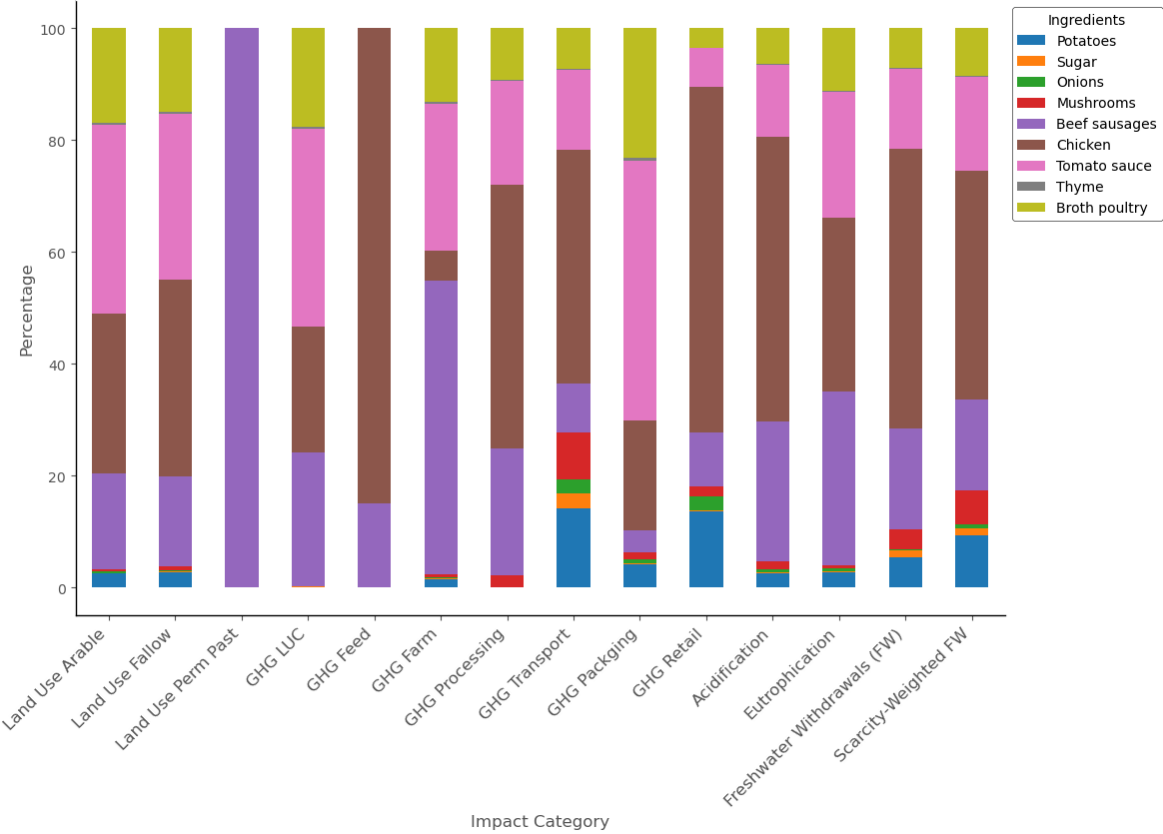
Table 24. Composition of the recipe: Spinach and trout pasta

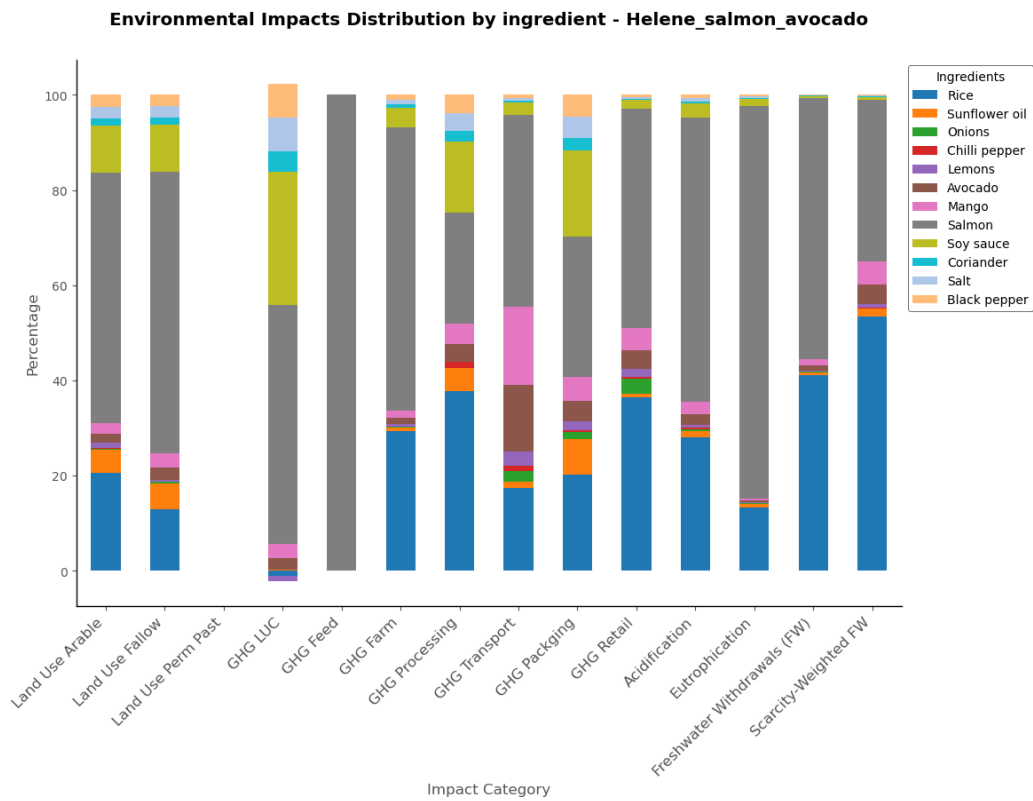
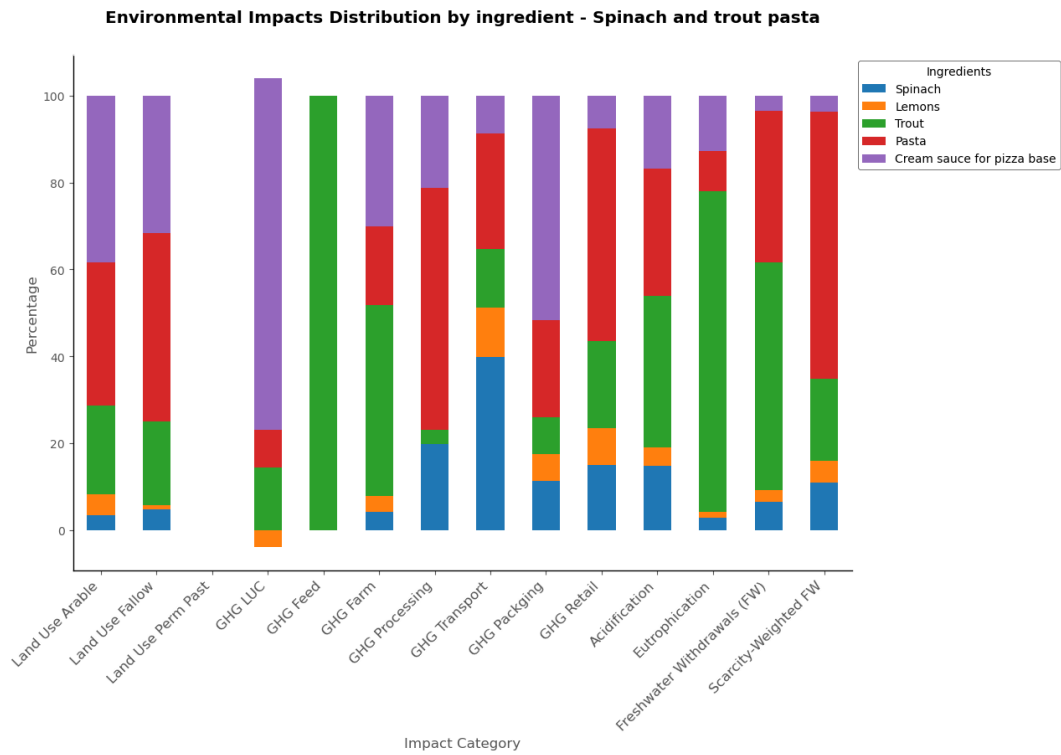
Ingredient	Amount	Unit
Spinach	200	grams
Lemons	1	unit
Trout	100	grams
Pasta	200	grams
Cream sauce for pizza base	50	grams

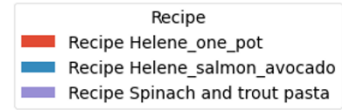
Relative environmental performance of recipes



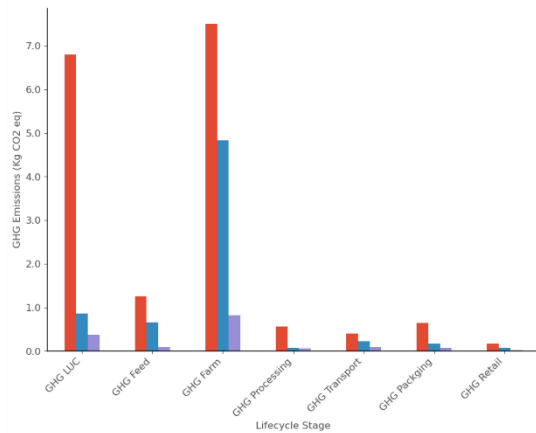
Environmental Impacts Distribution by ingredient - Helene_one_pot



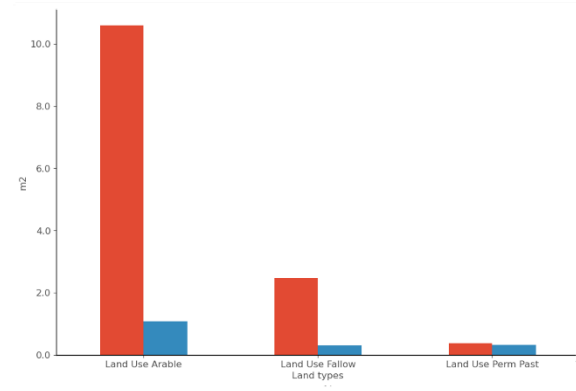




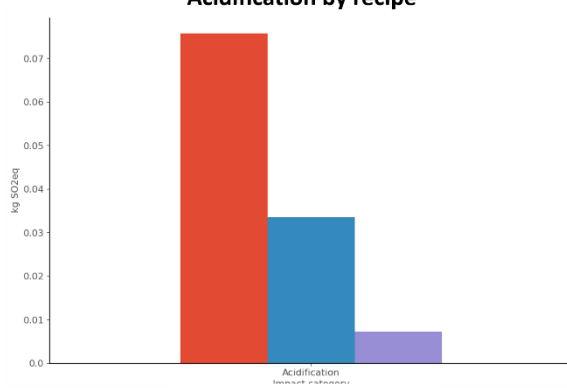
GHG Emissions by Life Cycle Stage and recipe



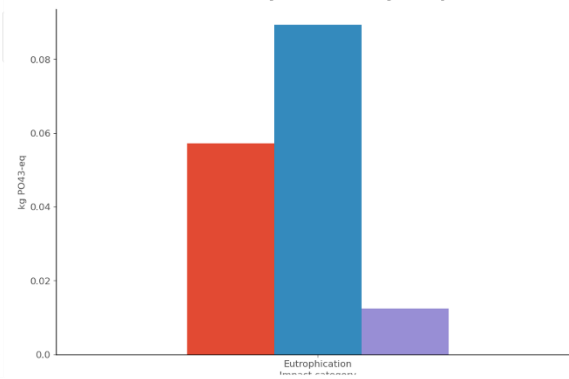
Land Use by land type and recipe



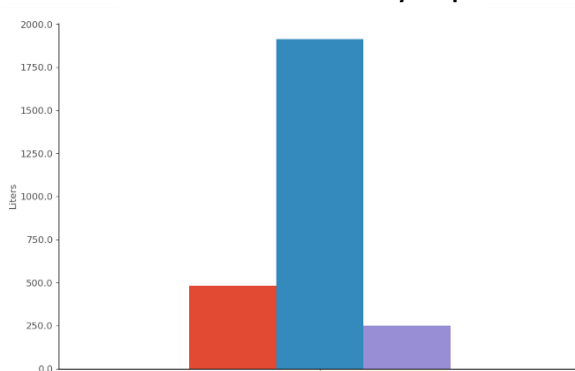
Acidification by recipe



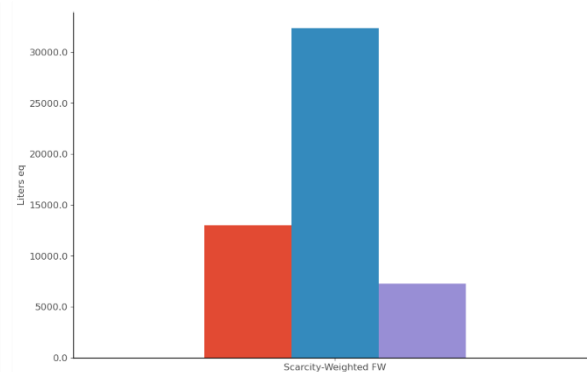
Eutrophication by recipe

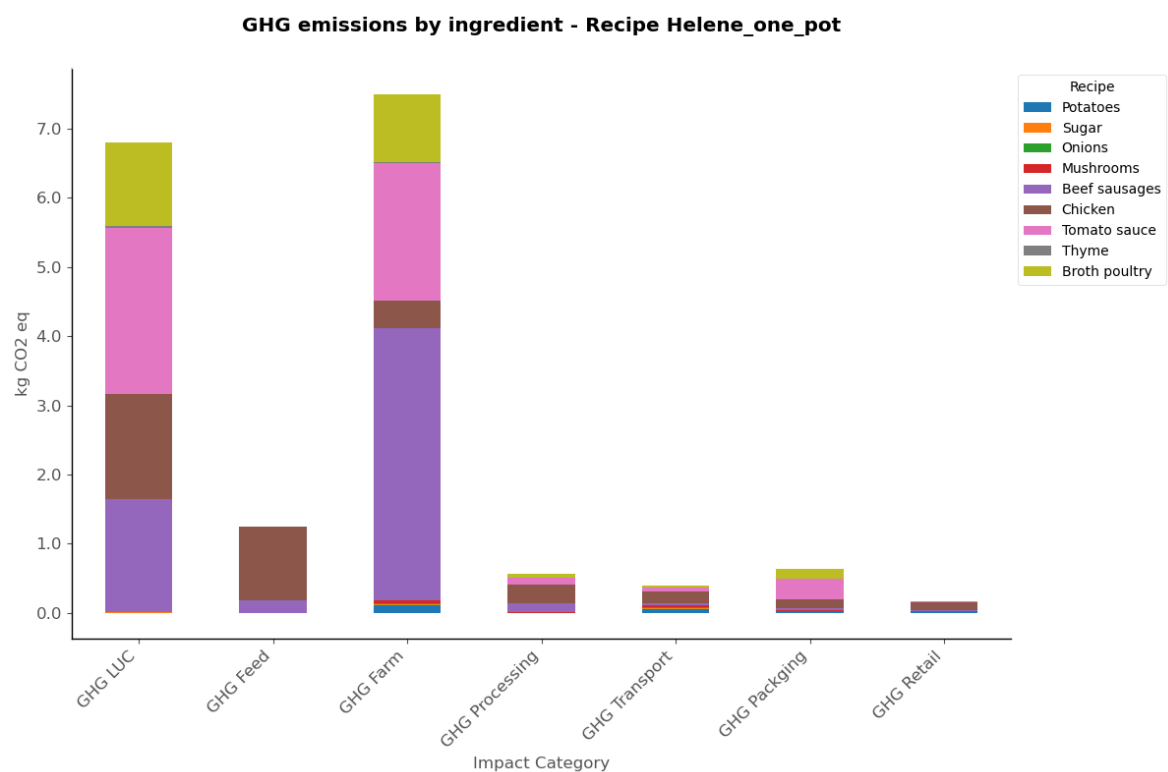
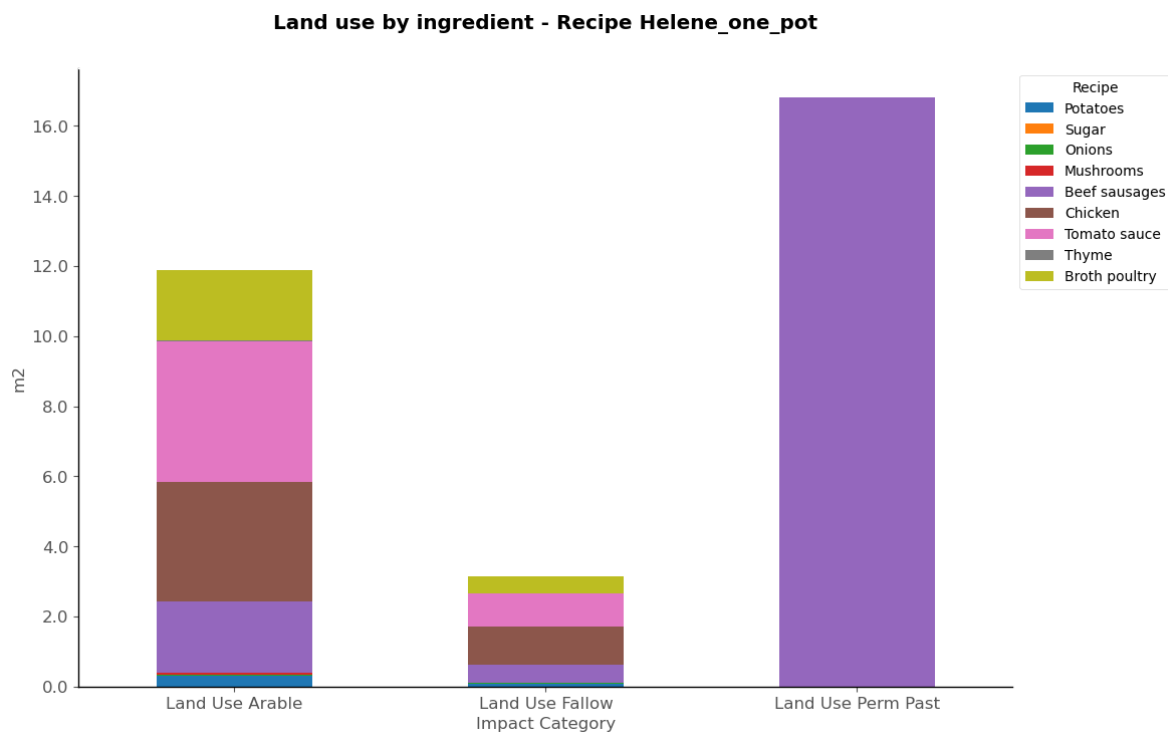


Freshwater Withdrawals by recipe

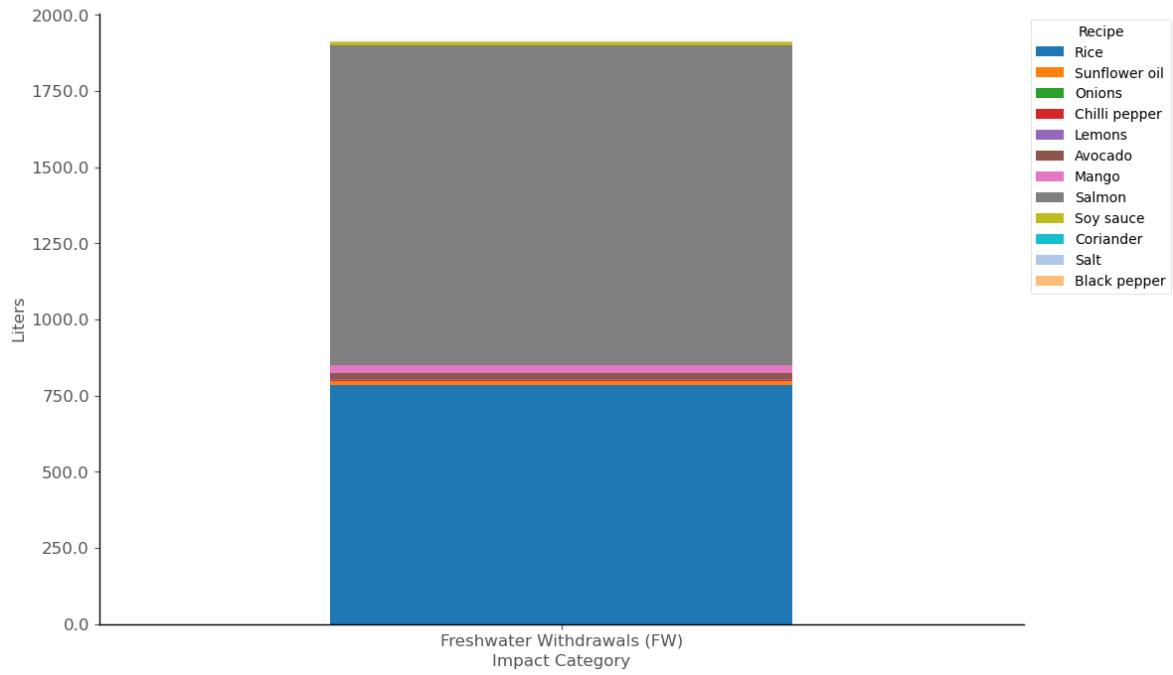


Scarcity-Weighted FW by recipe

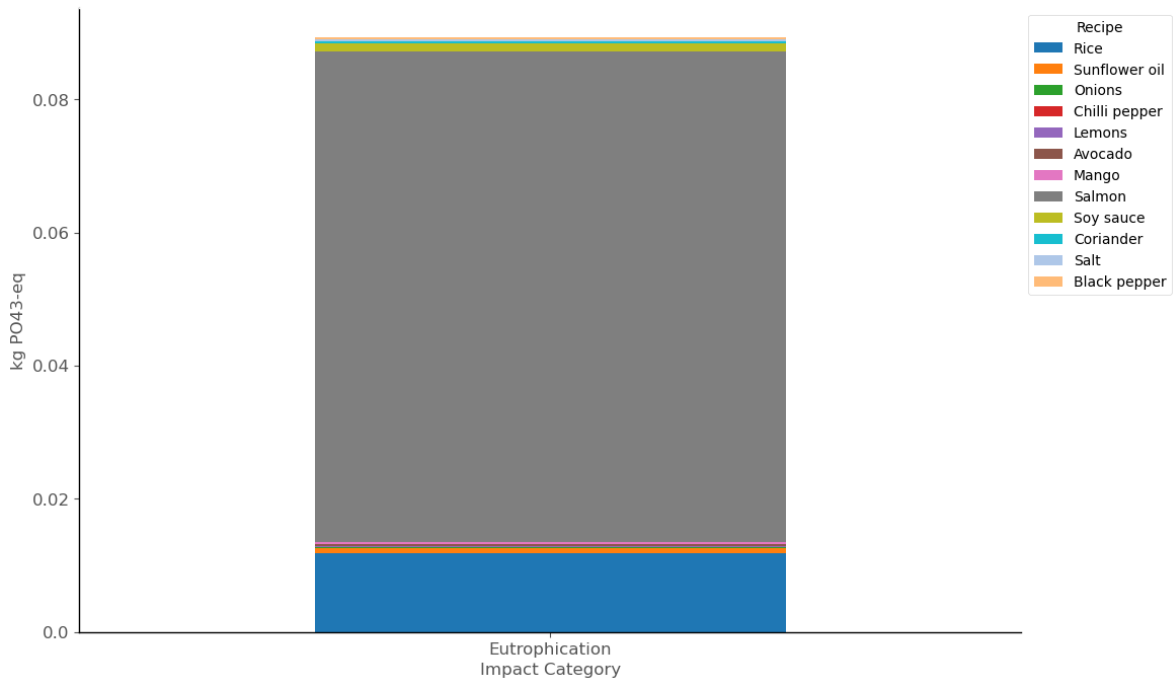


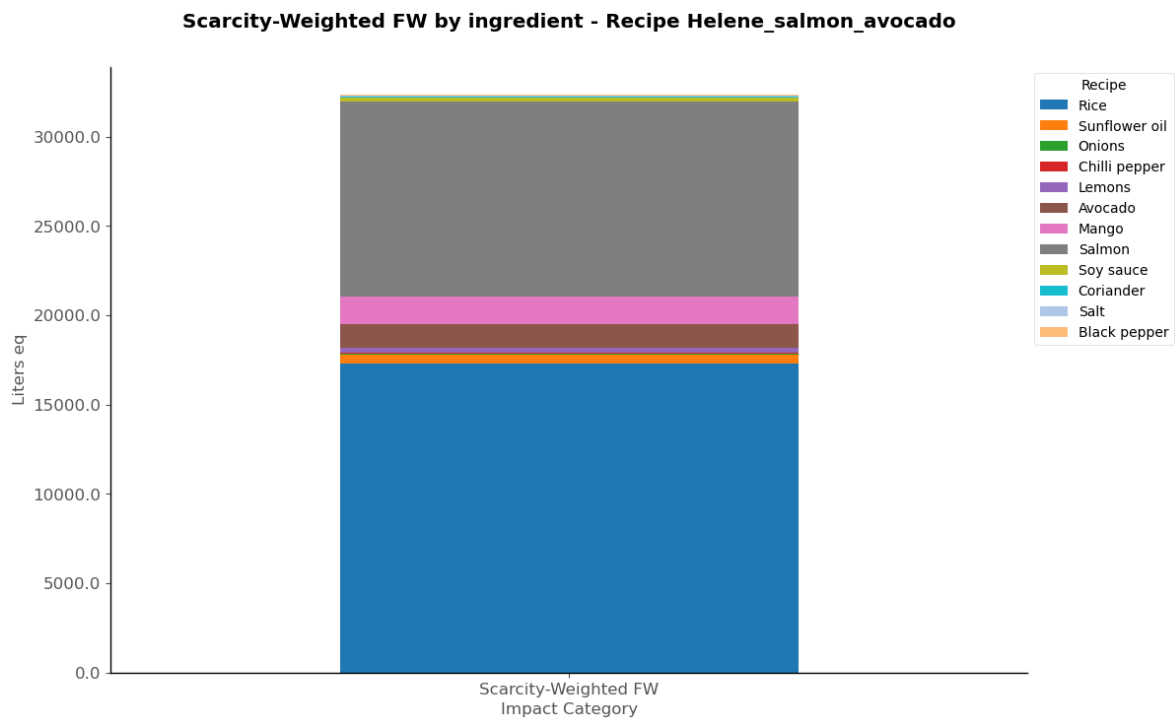
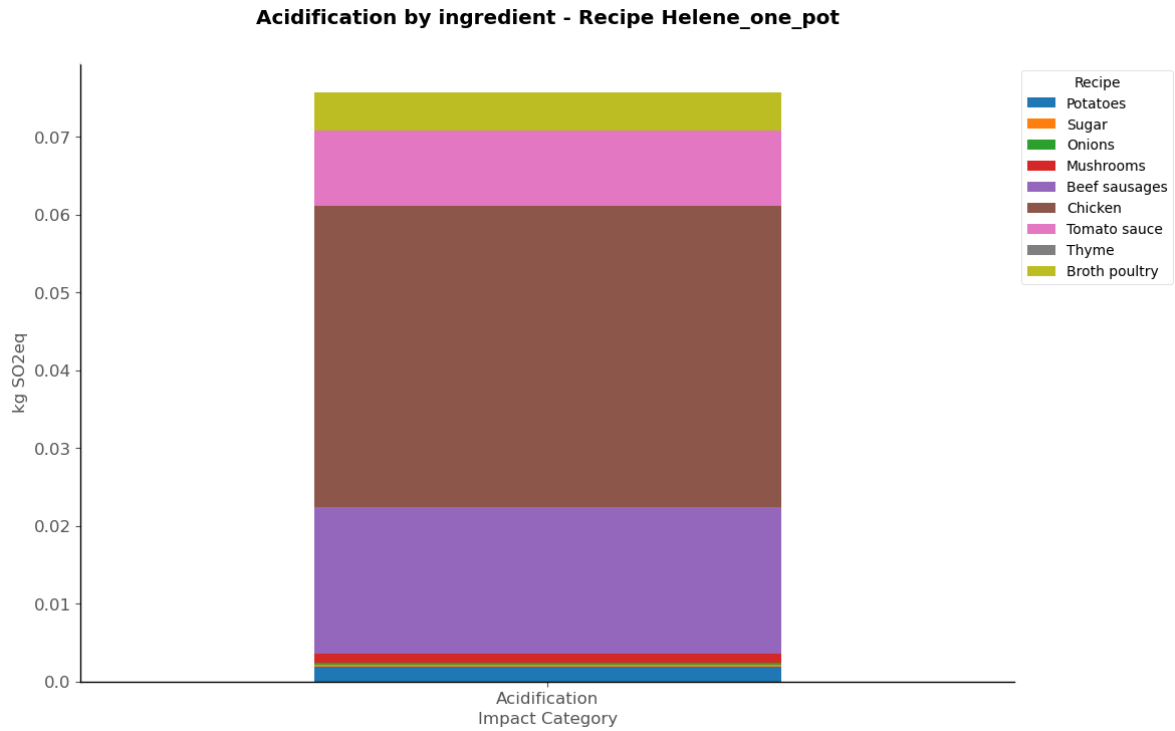


Freshwater by ingredient - Recipe Helene_salmon_avocado



Eutrophication by ingredient - Recipe Helene_salmon_avocado





8.7.4. Results from User 4

8.7.4.1. Recipes

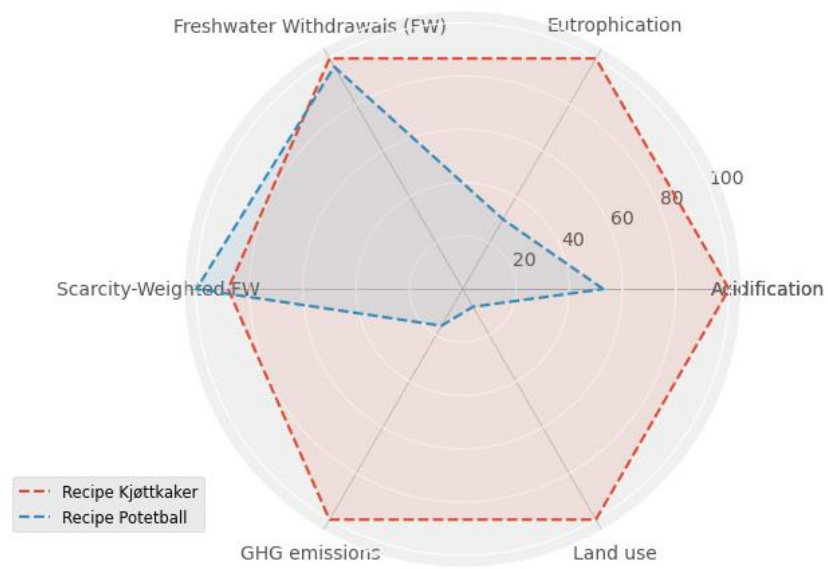
Table 25. Composition of the recipe: Kjøttkaker

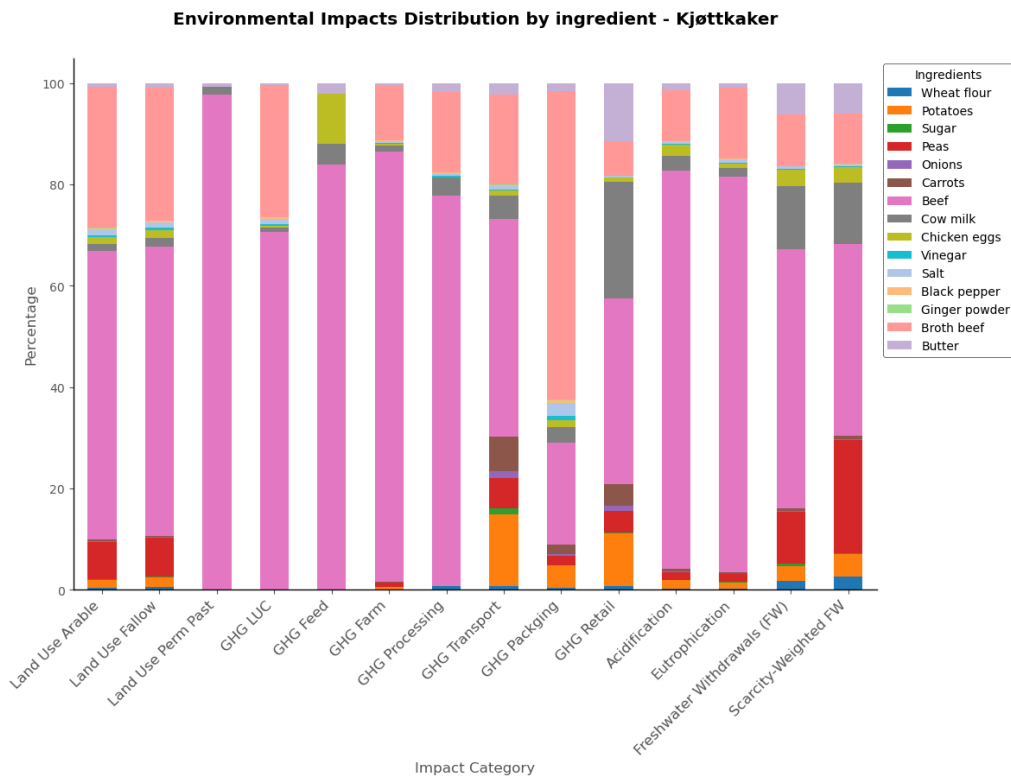
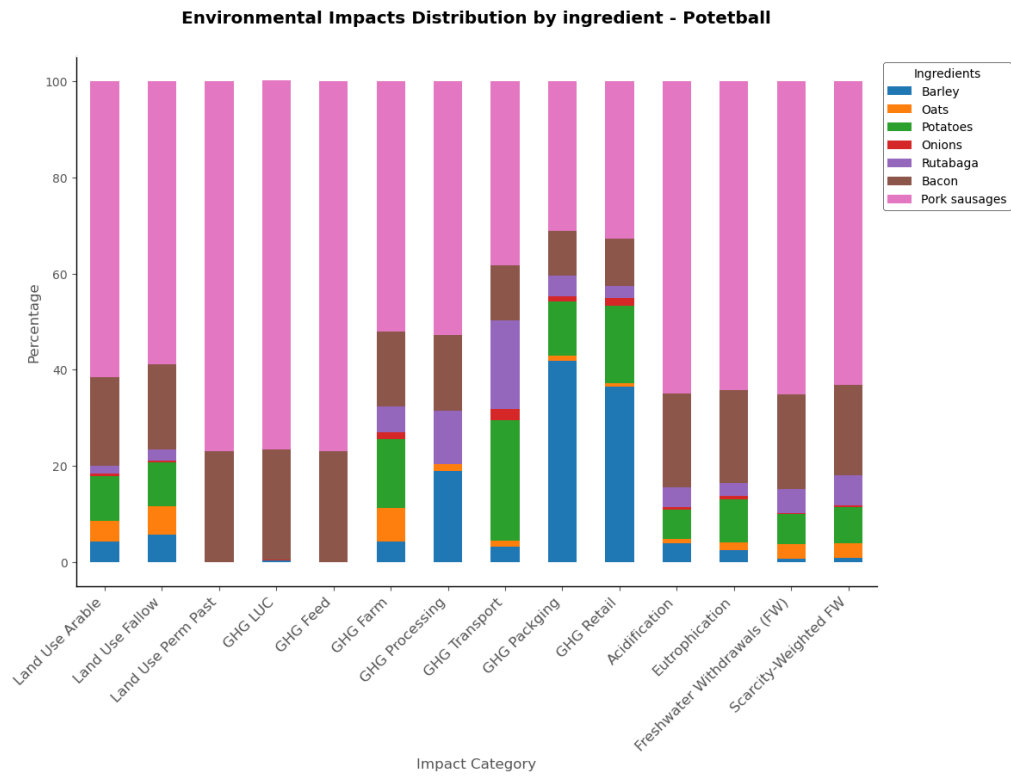
Ingredient	Amount	Unit
Wheat flour	3	tbsp
Potatoes	600	grams
Sugar	0.5	tbsp
Peas	250	grams
Onions	0.5	unit
Carrots	4	unit
Beef	500	grams
Cow milk	2	dl
Chicken eggs	1	unit
Vinegar	0.5	tbsp
Salt	1	tbsp
Black pepper	0.25	tbsp
Ginger powder	0.1	tbsp
Broth beef	5	dl
Butter	5	tbsp

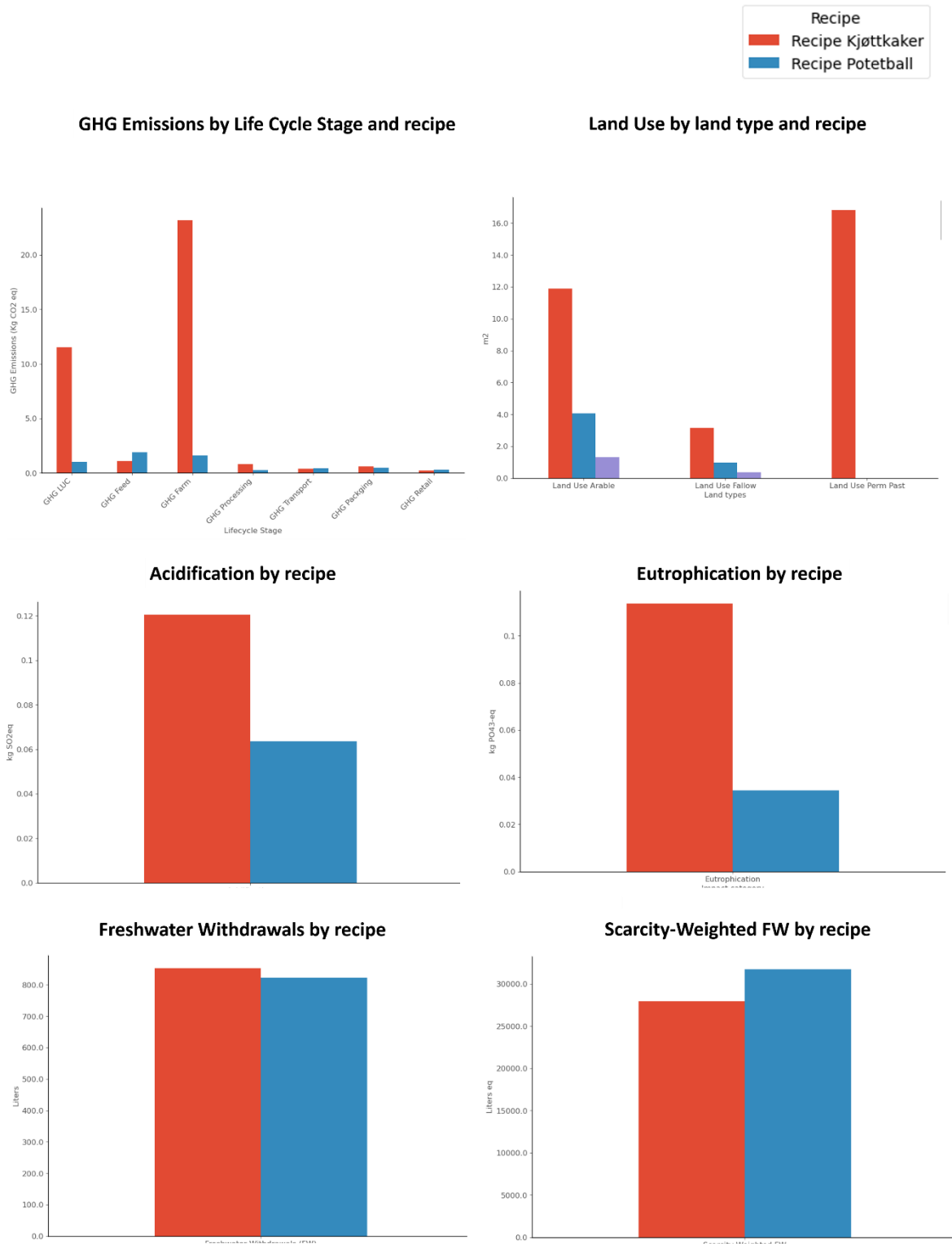
Table 26. Composition of the recipe: Potetball

Ingredient	Amount	Unit
Barley	4	dl
Oats	0.5	cup
Potatoes	1200	grams
Onions	1	unit
Rutabaga	500	grams
Bacon	150	grams
Pork sausages	500	grams

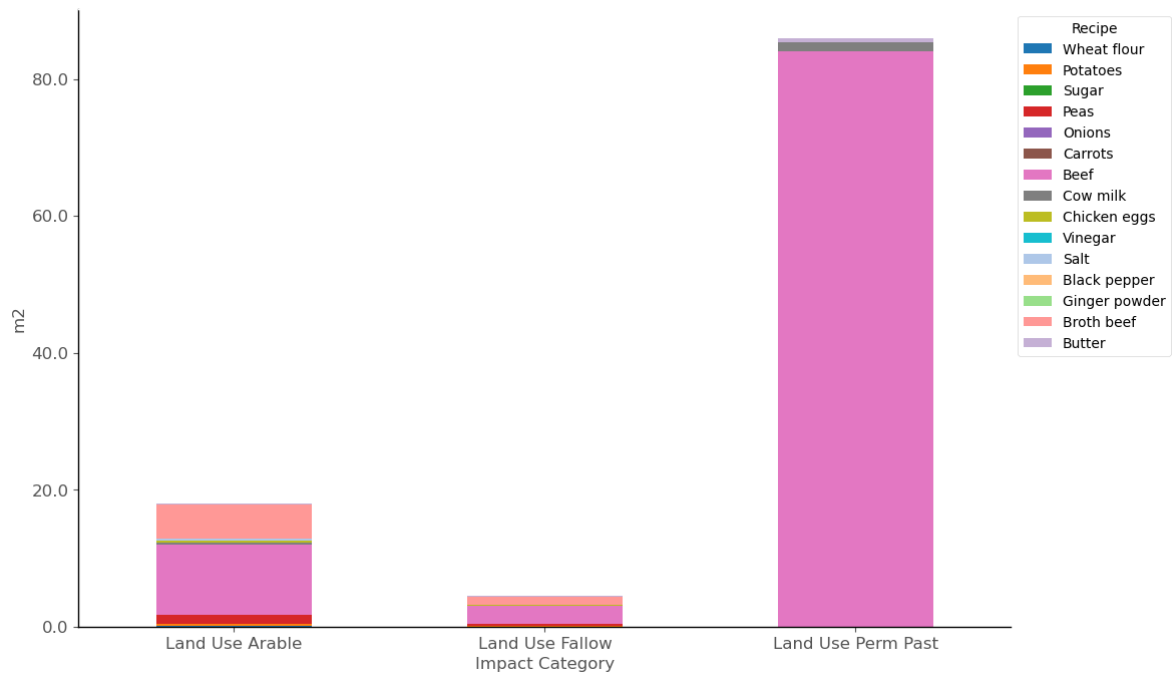
Relative environmental performance of recipes



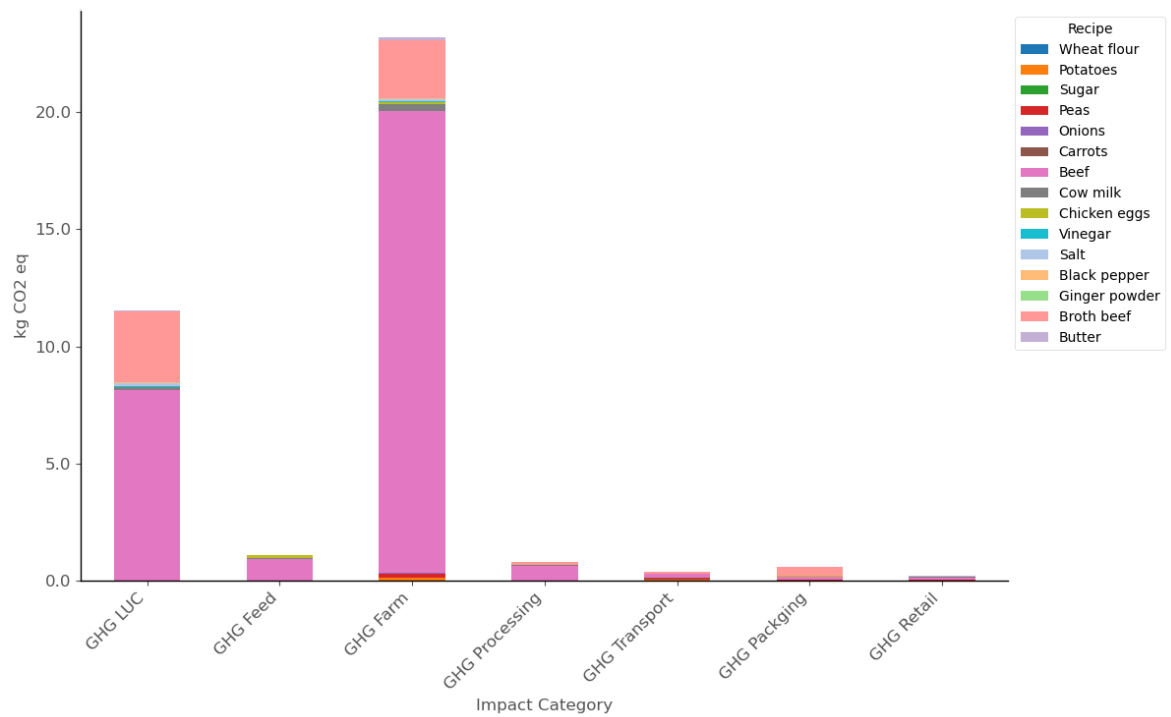




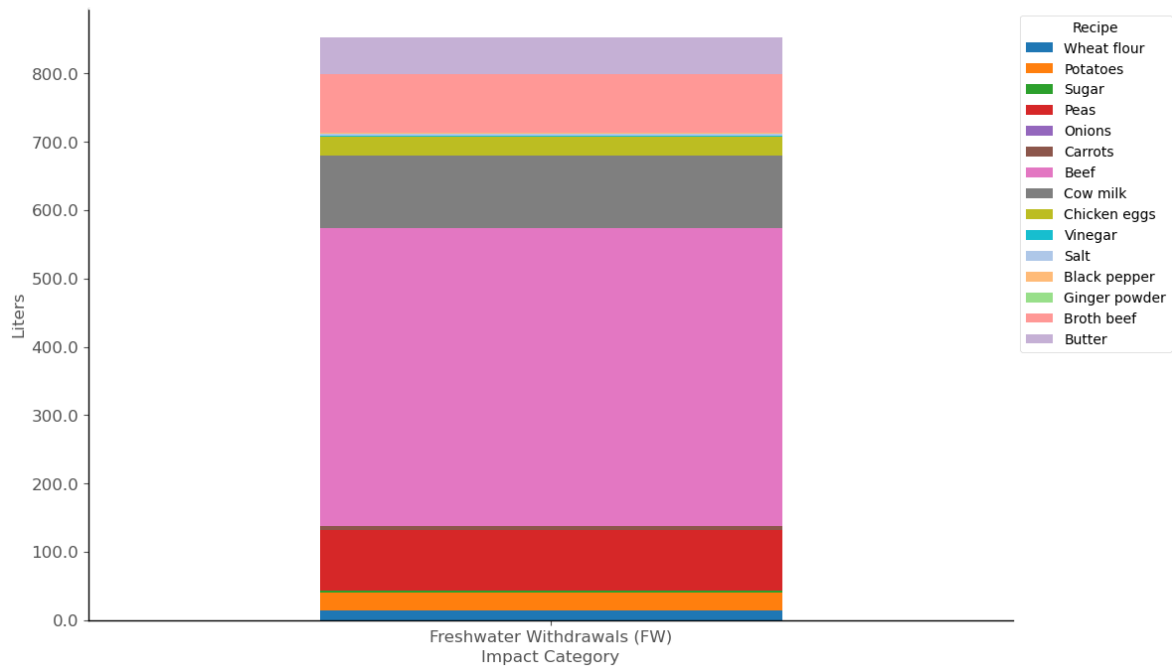
Land use by ingredient - Recipe Kjøttkaker



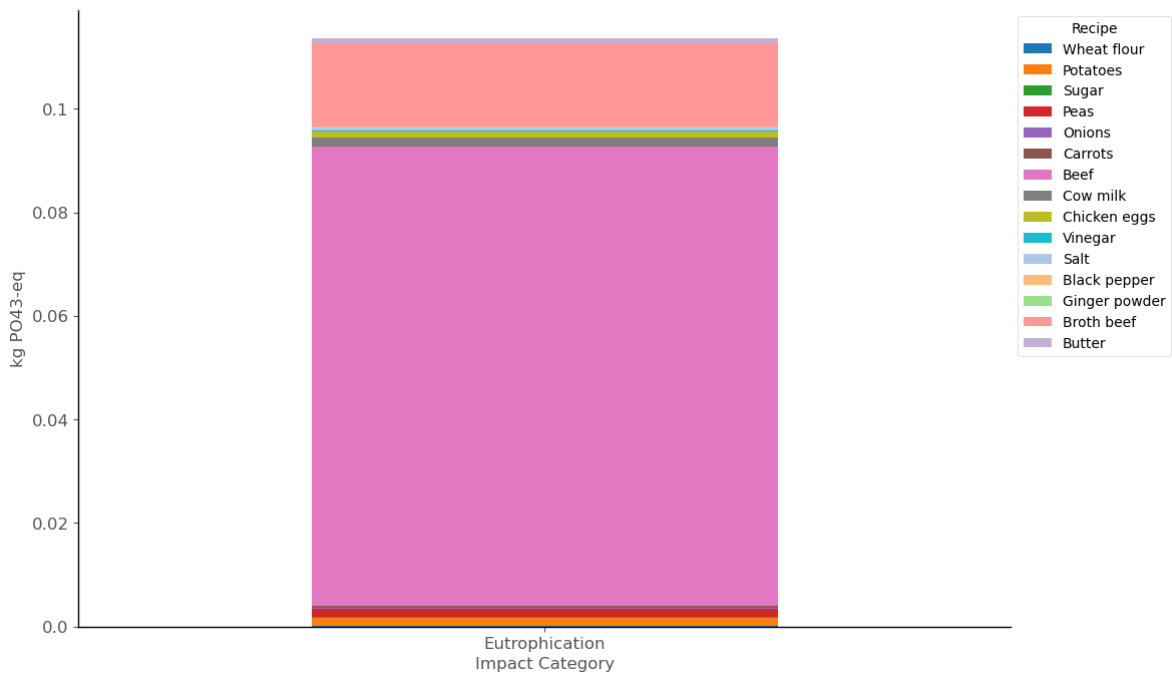
GHG emissions by ingredient - Recipe Kjøttkaker

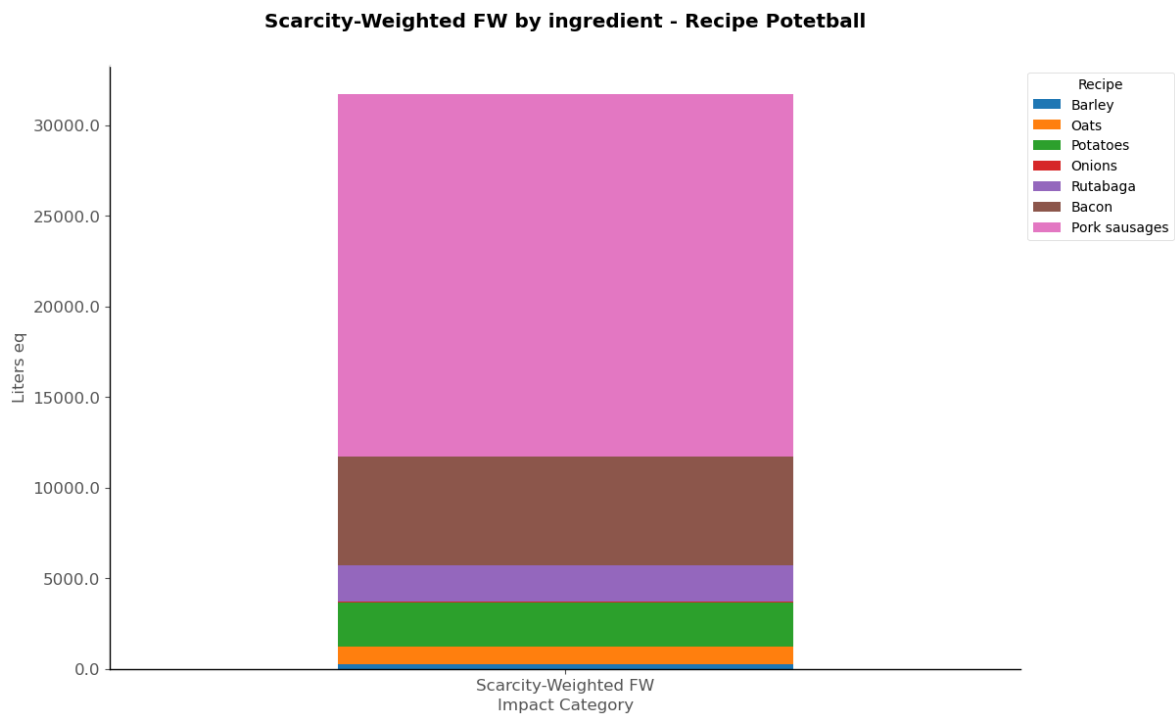
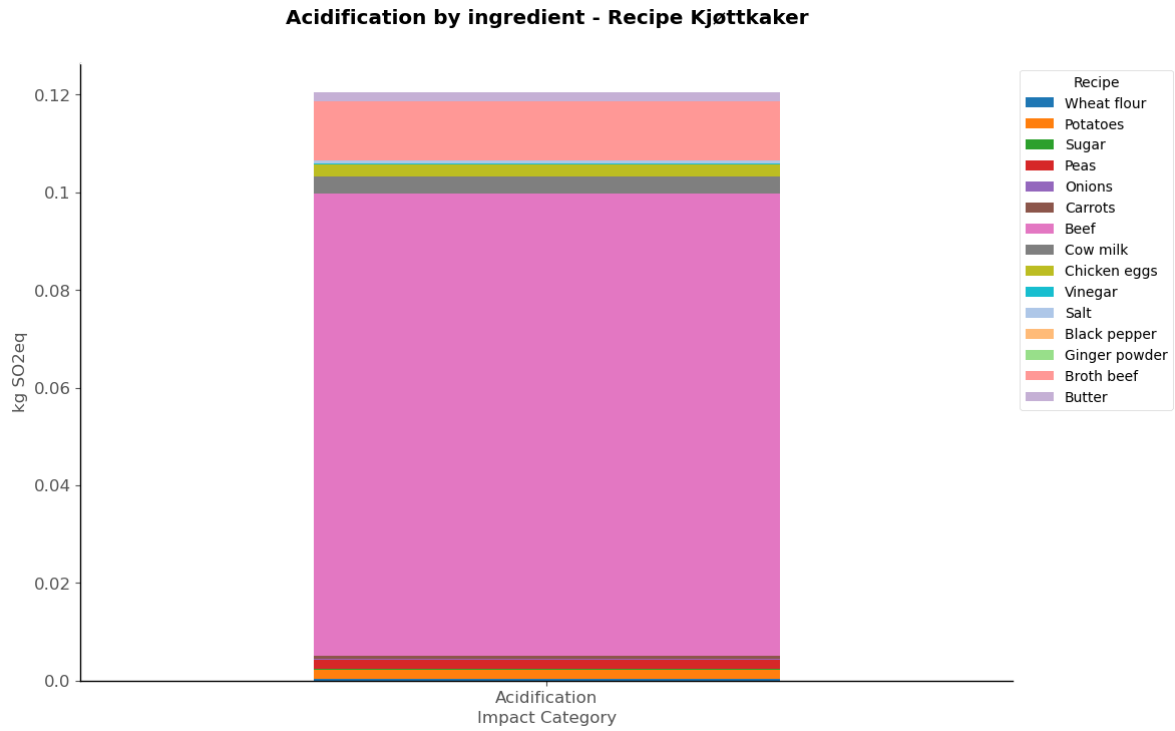


Freshwater by ingredient - Recipe Kjøttkaker



Eutrophication by ingredient - Recipe Kjøttkaker





8.7.5. Results from User 5

8.7.5.1. Recipes

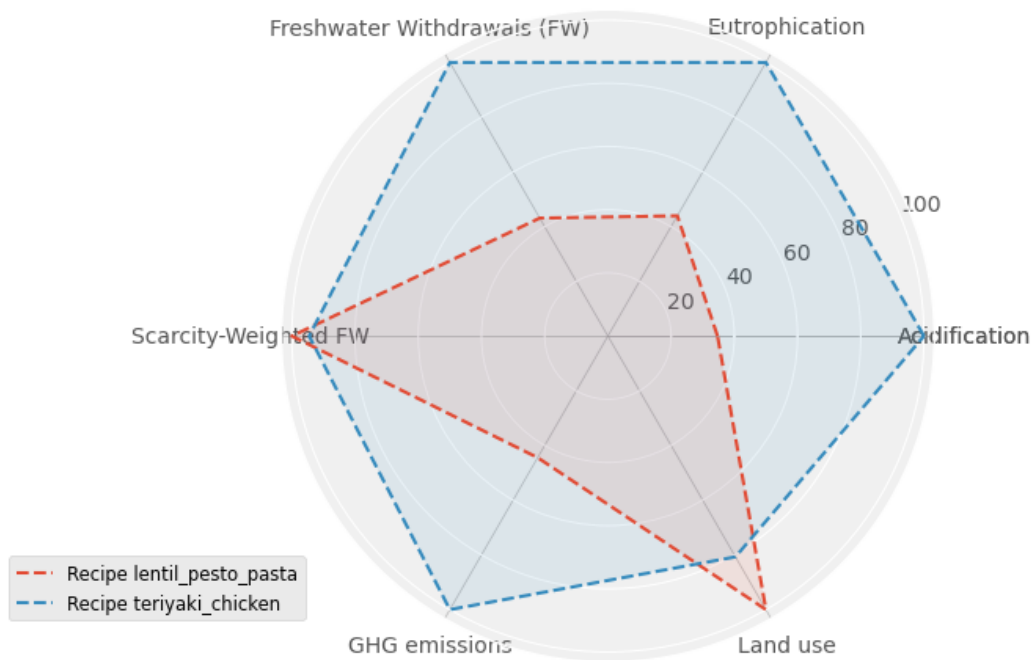
Table 27. Composition of the recipe: Lentil pesto pasta

Ingredient	Amount	Unit
Lentils	380	grams
Extra virgin olive oil	2	tbsp
Onions	1	unit
Carrots	1	unit
Pasta	300	grams
Vinegar	1	tbsp
Pesto	100	grams
Garlic	2	clove
Salt	0.5	tbsp
Black pepper	0.5	tbsp

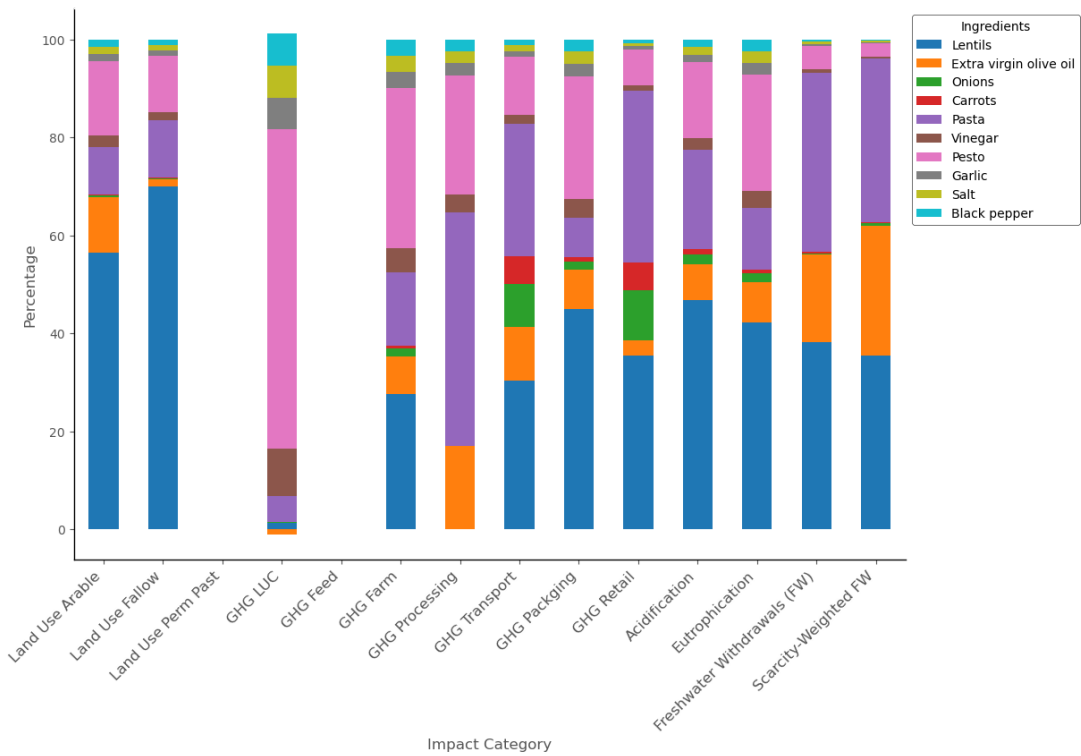
Table 28. Composition of the recipe: Teriyaki chicken

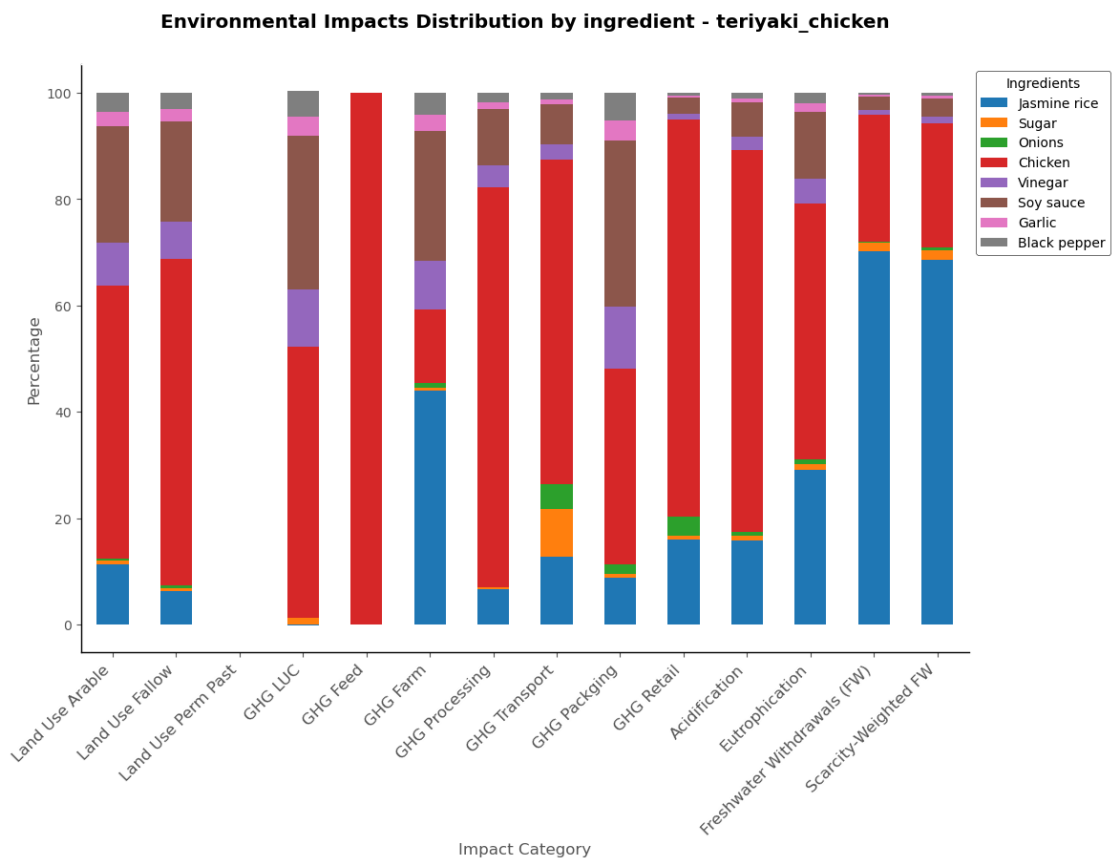
Ingredient	Amount	Unit
Jasmine rice	1.5	cup
Sugar	2	tbsp
Onions	1	unit
Chicken	500	grams
Vinegar	3	tbsp
Soy sauce	6	tbsp
Garlic	3	clove
Black pepper	1	tbsp

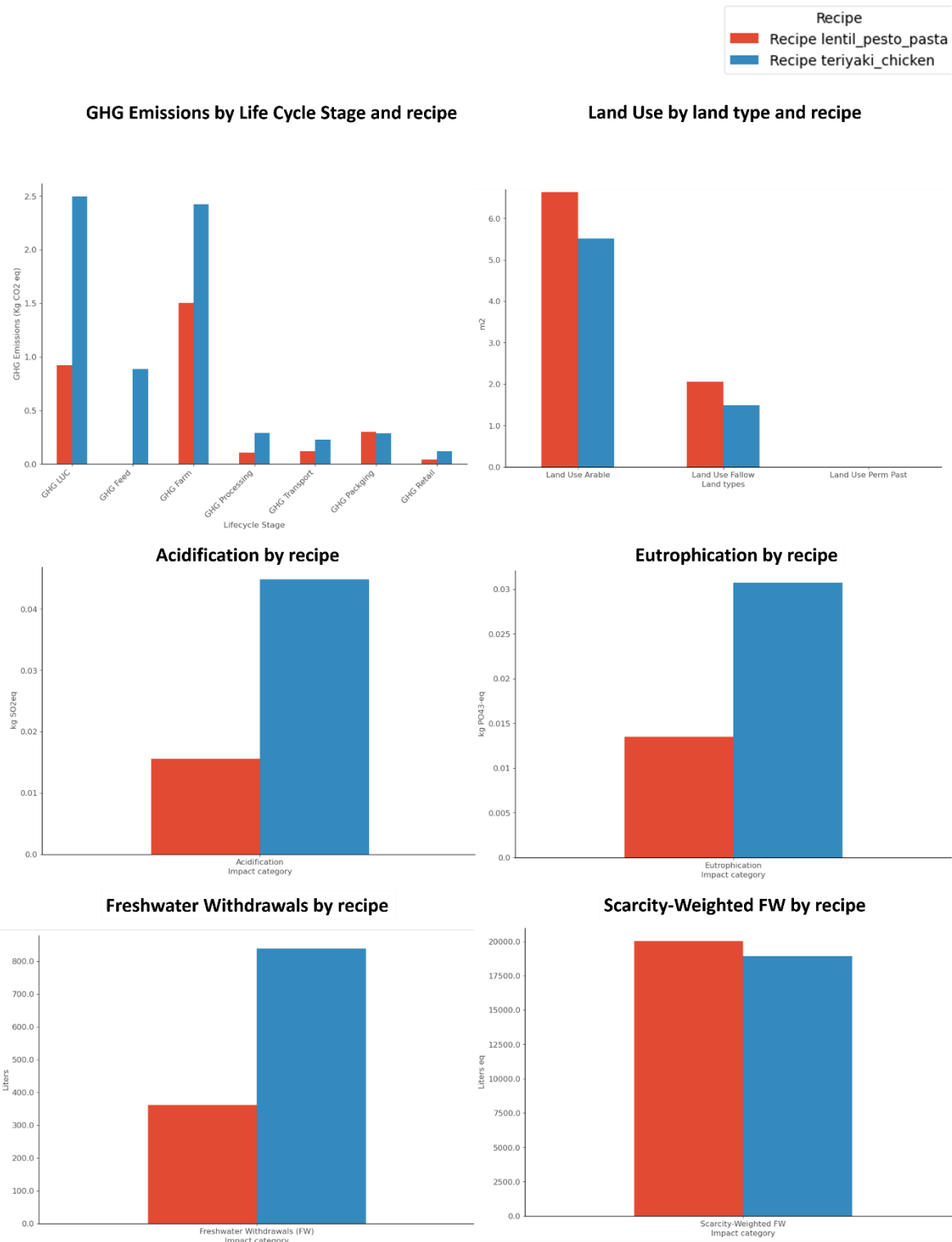
Relative environmental performance of recipes

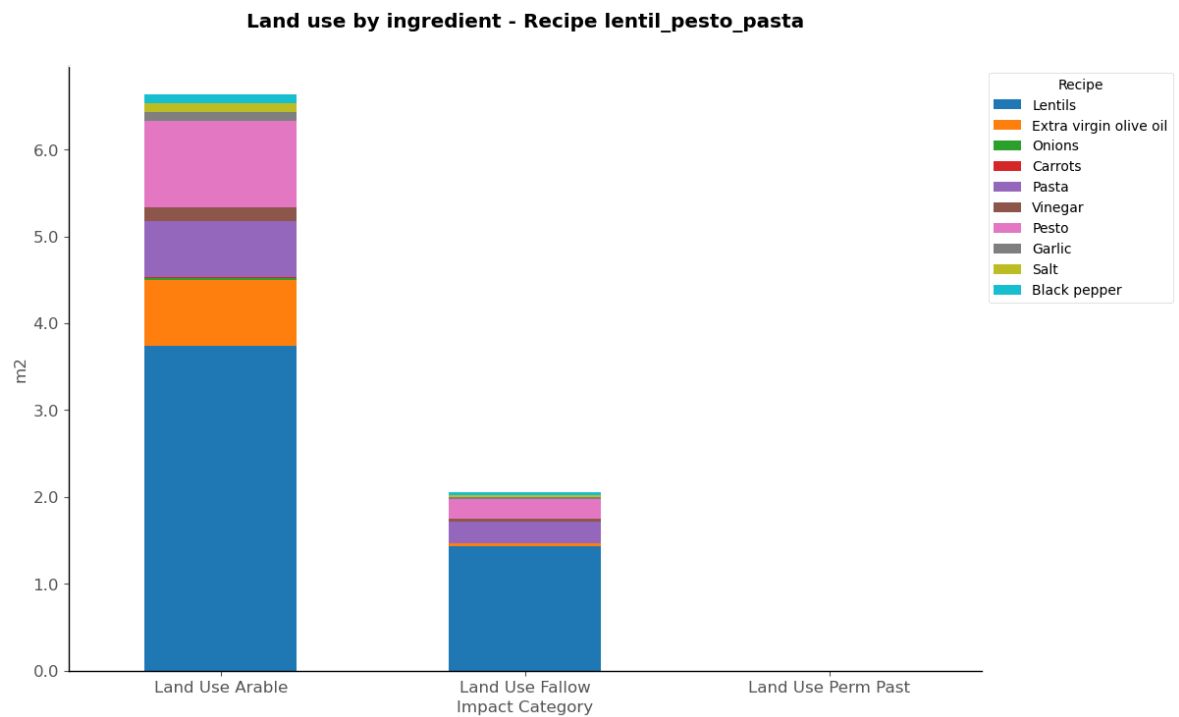
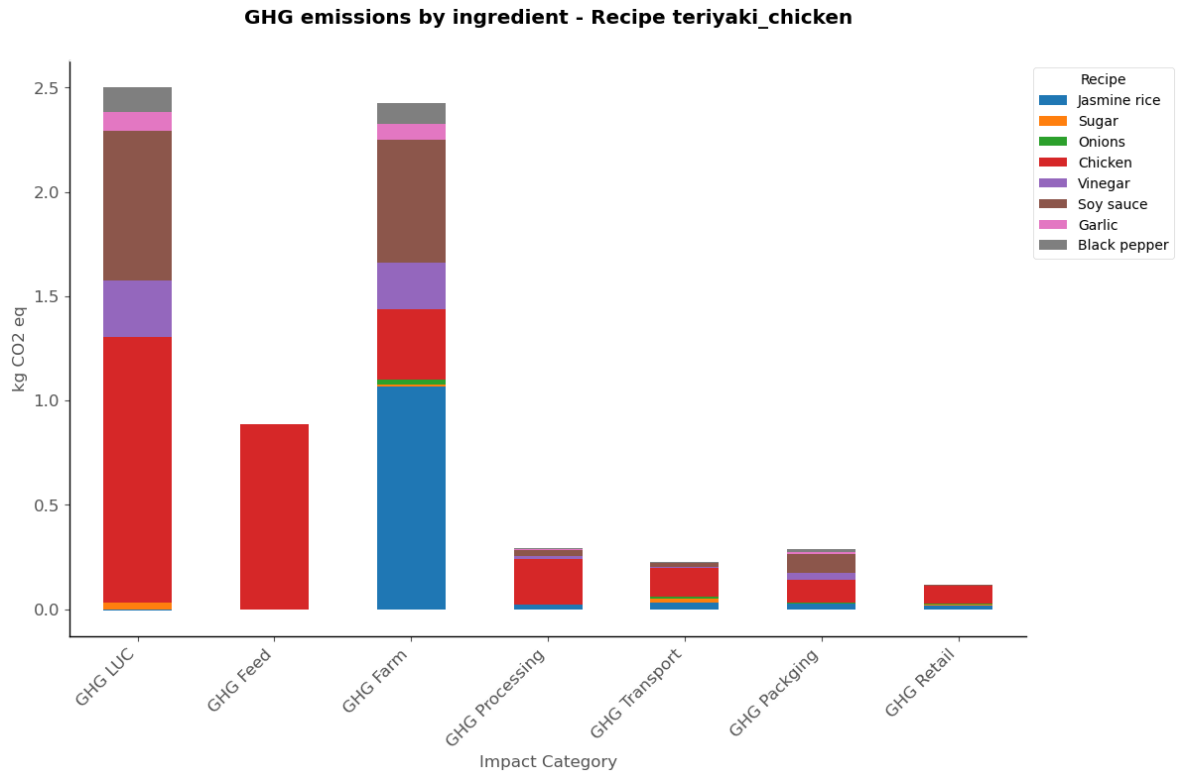


Environmental Impacts Distribution by ingredient - lentil_pesto_pasta

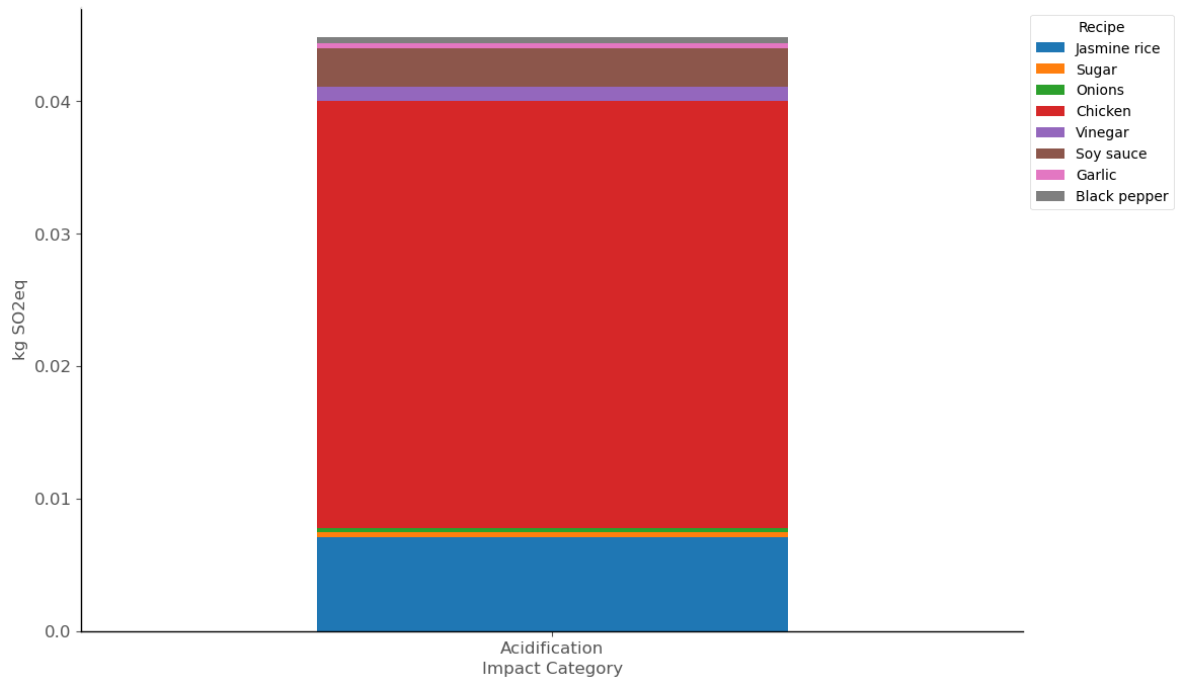




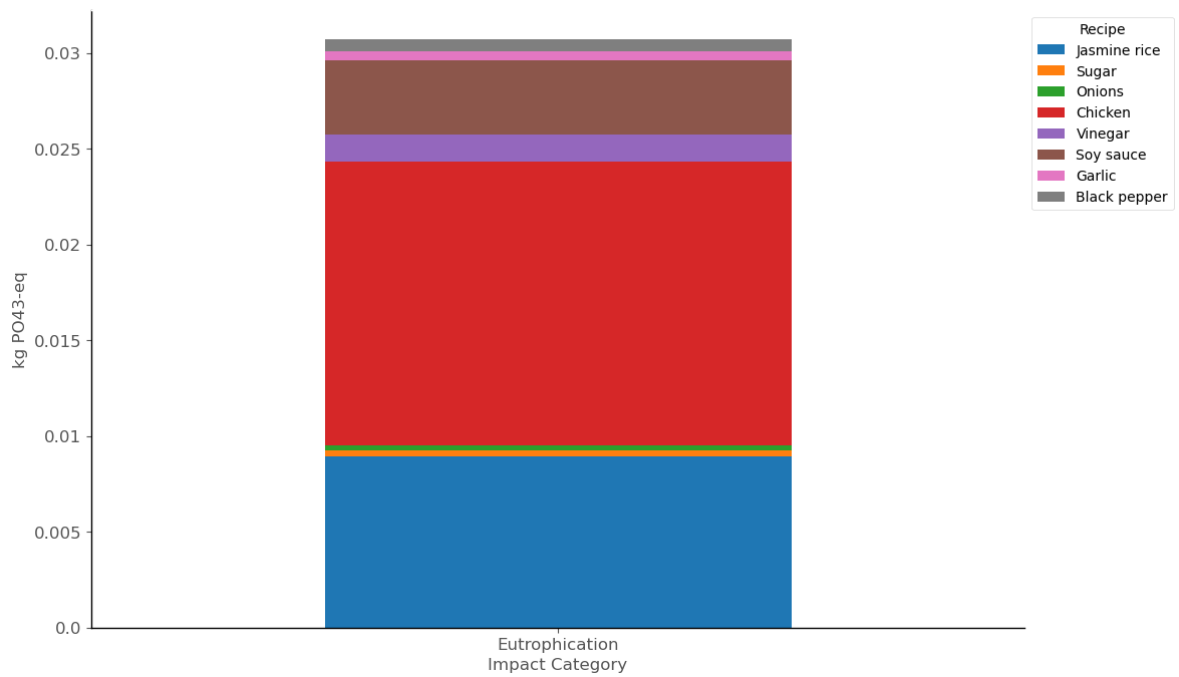


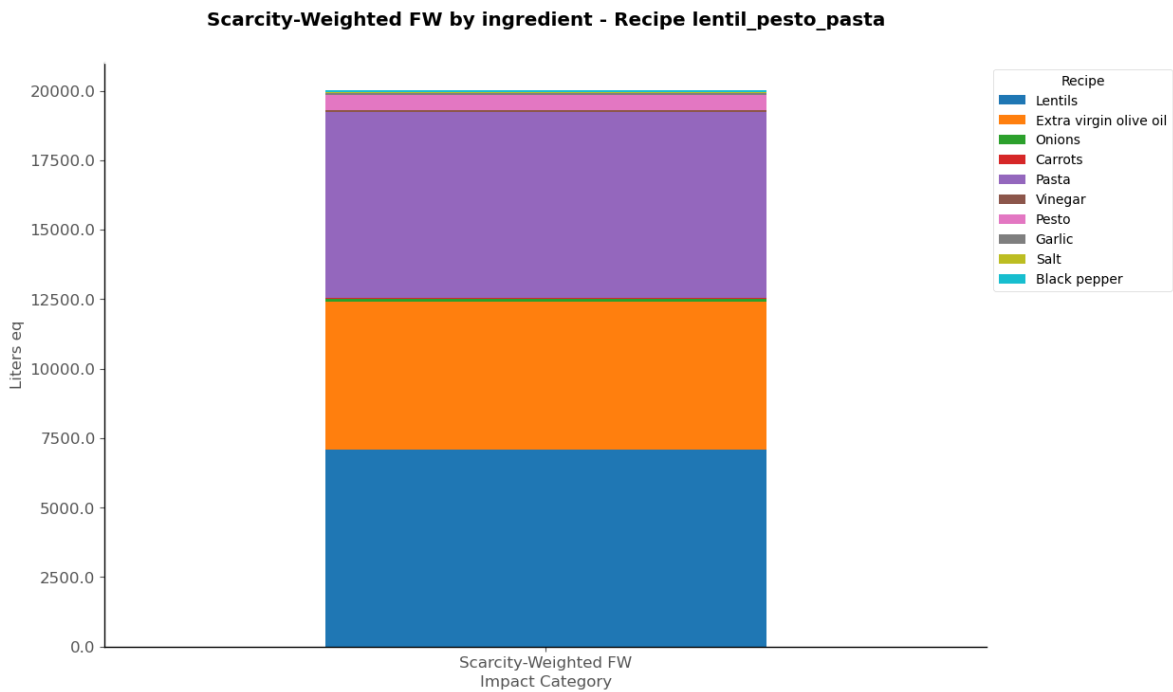
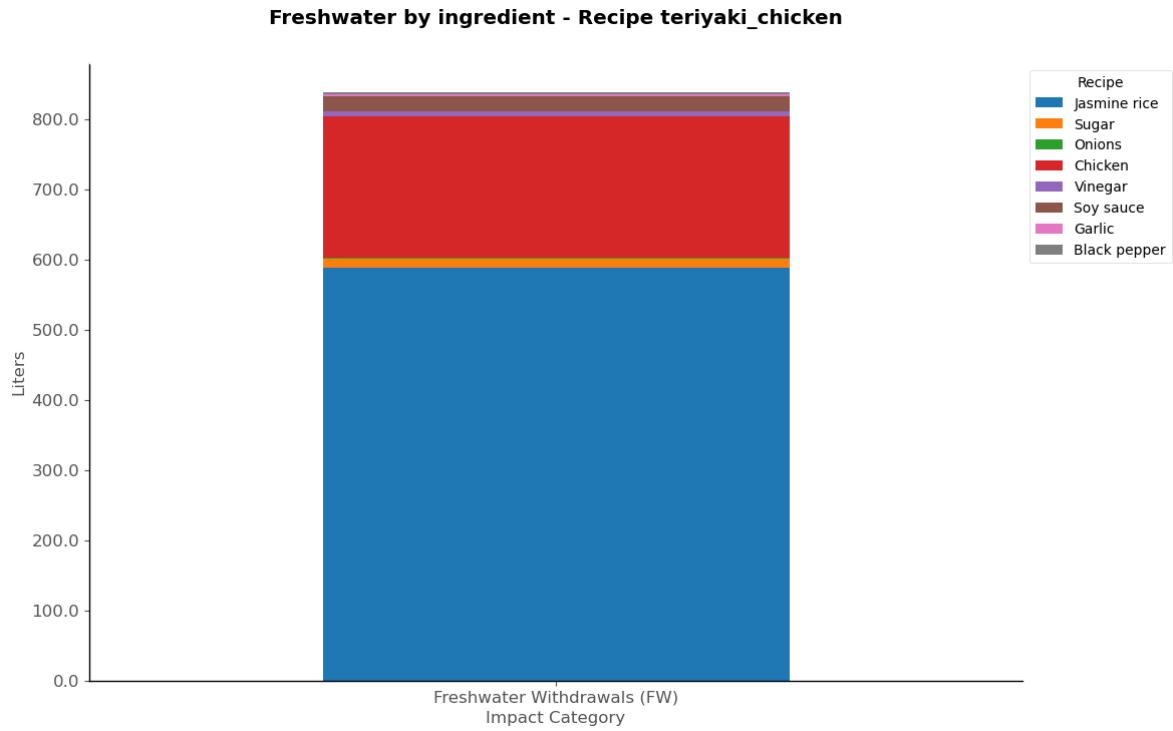


Acidification by ingredient - Recipe teriyaki_chicken



Eutrophication by ingredient - Recipe teriyaki_chicken





8.7.6. Results from User 6

8.7.6.1. Recipes

Table 29. Composition of the recipe: cookies

Ingredient	Amount	Unit
Sugar	5	tbsp

Peanuts	200	grams
Bananas	1	unit
Chocolate	50	grams
Nutritional yeast	1	tbsp

Table 30. Composition of the recipe: Salad

Ingredient	Amount	Unit
Sunflower seeds	2	grams
Tomatoes (baby plum)	4	unit
Carrots	0.5	unit
Cucumber	0.25	unit
Lettuce	1	unit
Beetroot	1	unit
Spinach	30	grams
Avocado	0.5	unit
Salad dressing	2	tbsp

Table 31. Composition of the recipe: Stroganoff with chickpeas

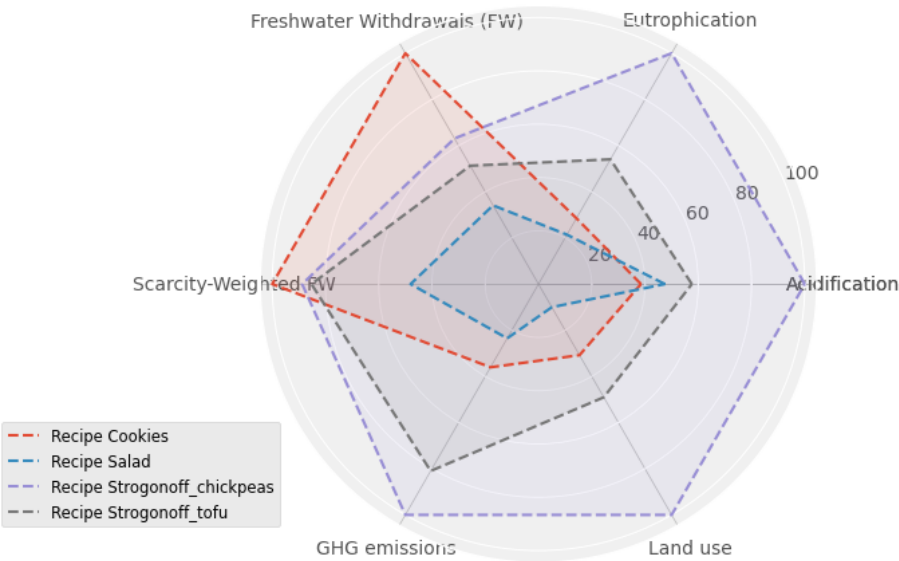
Ingredient	Amount	Unit
Wheat flour	1	tbsp
Chickpeas	340	grams
Cononut milk	258	grams
Sunflower oil	2	tbsp
Onions	0.5	unit
Mushrooms	35	grams
Mustard	1	tbsp
Tomato sauce	266	grams
Salt	1	tbsp
Black pepper	1	tbsp
Paprika	1	tbsp

Table 32. Composition of the recipe: Stroganoff with tofu

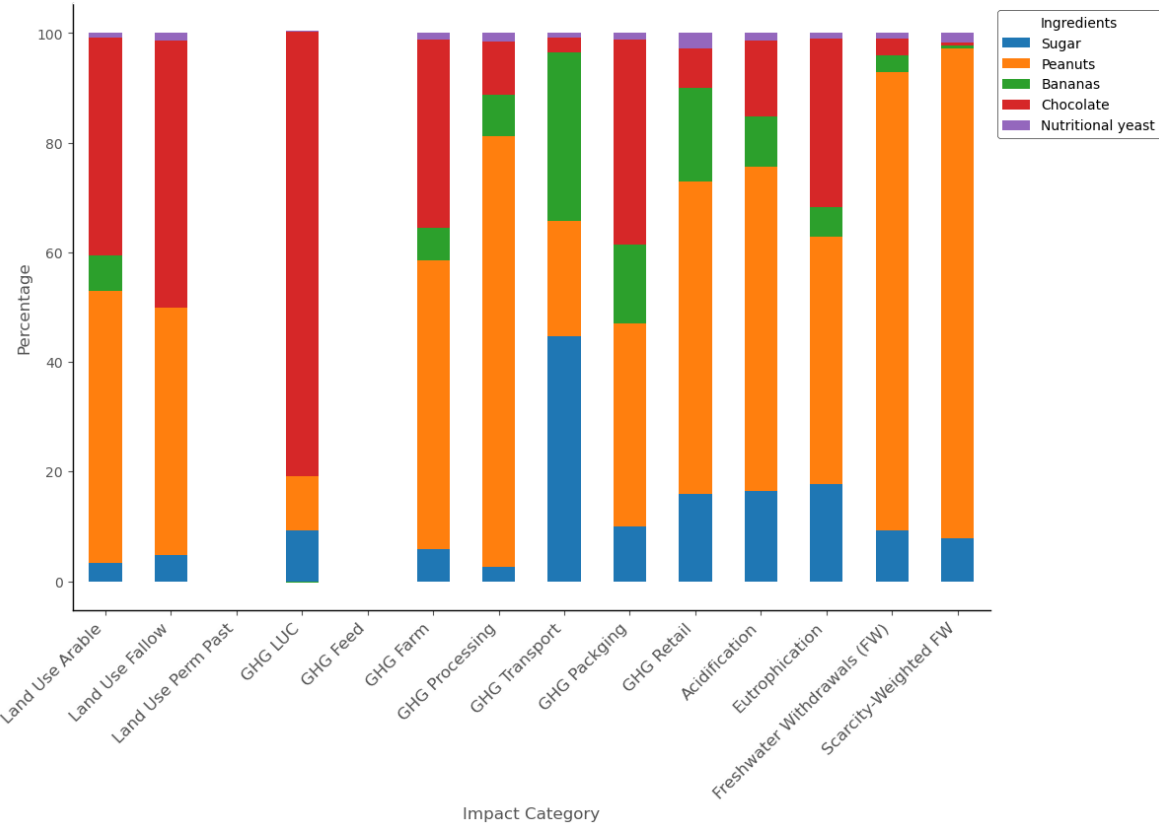
Ingredient	Amount	Unit
Wheat flour	1	tbsp
Soymilk	258	grams
Soybean Tofu	524	grams
Extra virgin olive oil	2	tbsp
Onions	0.5	unit
Mustard	1	tbsp

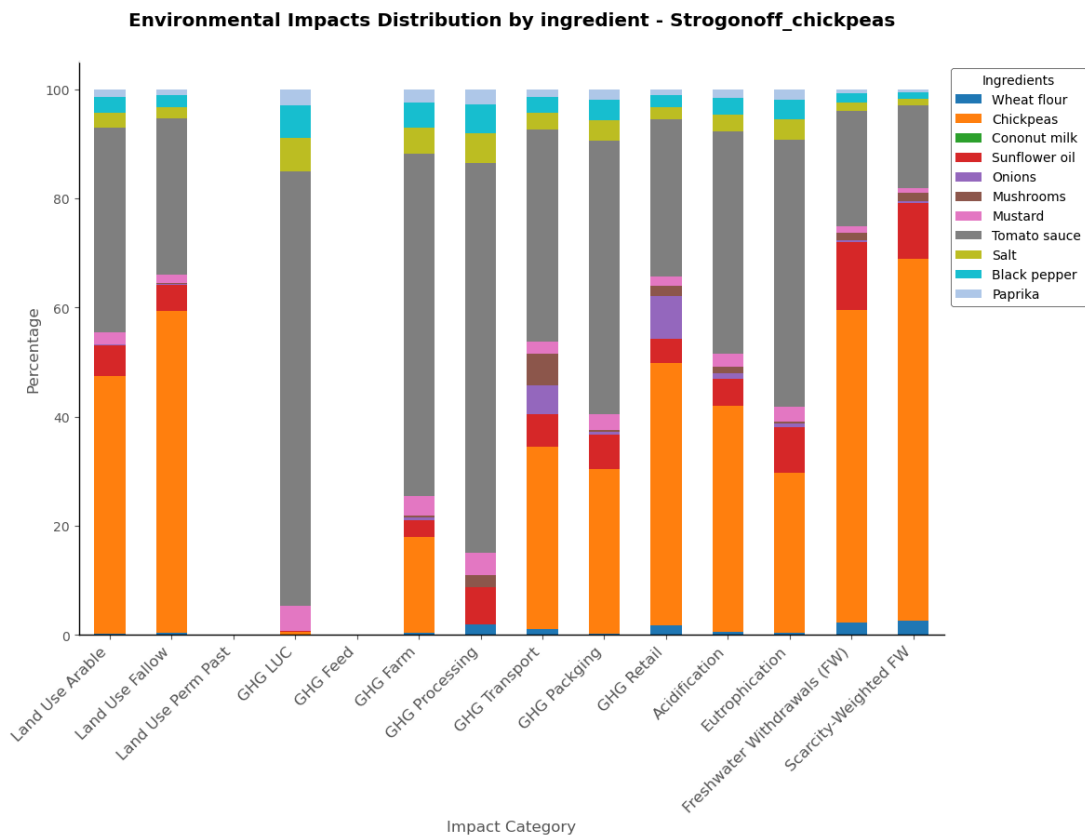
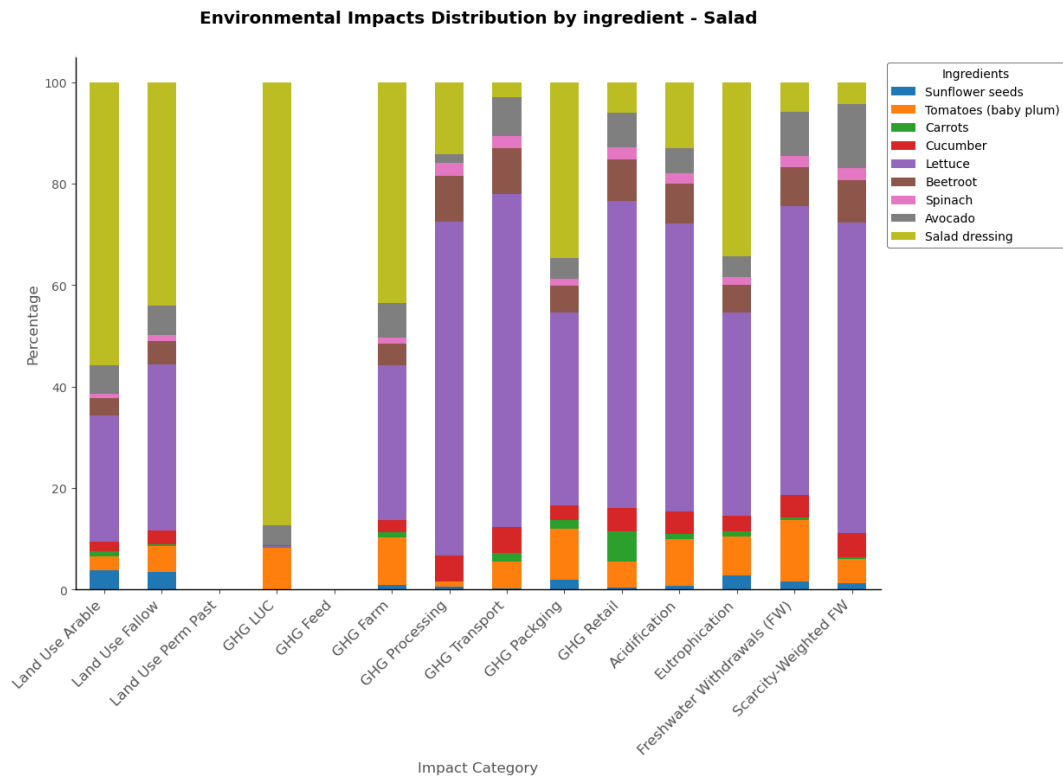
Ketchup	5	tbsp
Salt	1	tbsp
Black pepper	1	tbsp

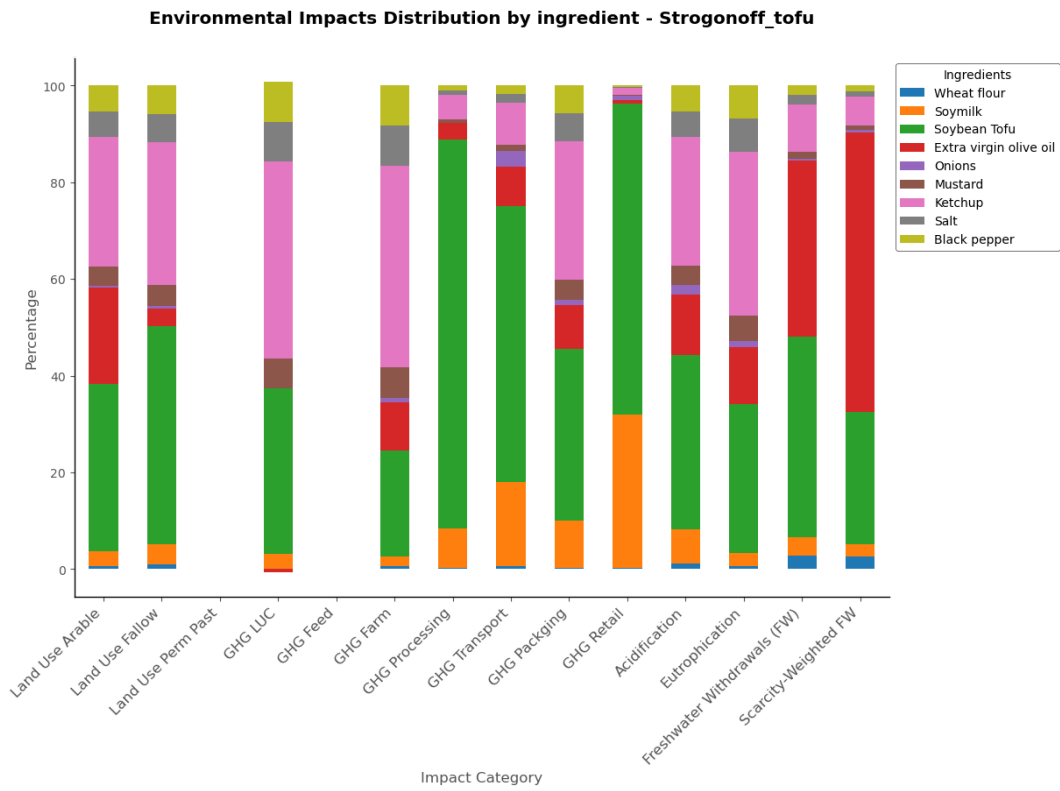
Relative environmental performance of recipes

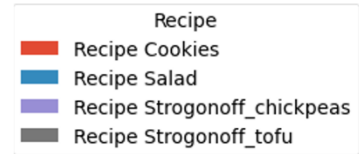


Environmental Impacts Distribution by ingredient - Cookies

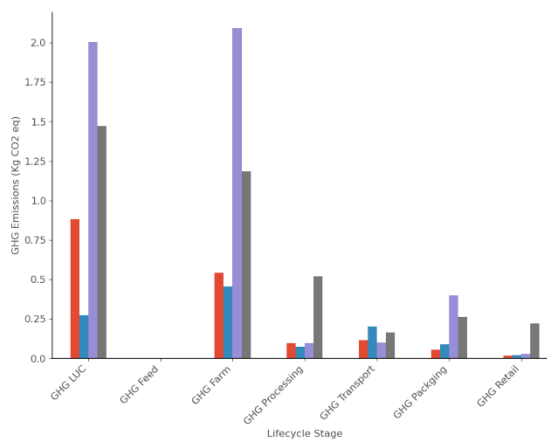




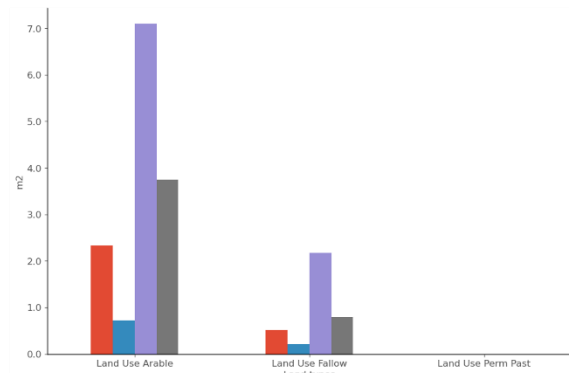




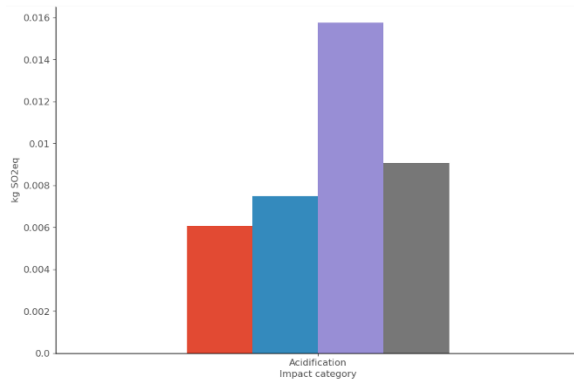
GHG Emissions by Life Cycle Stage and recipe



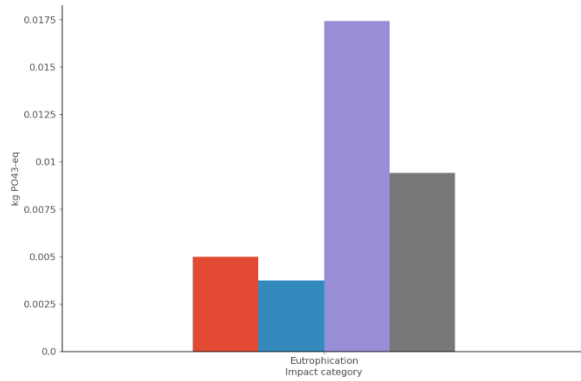
Land Use by land type and recipe



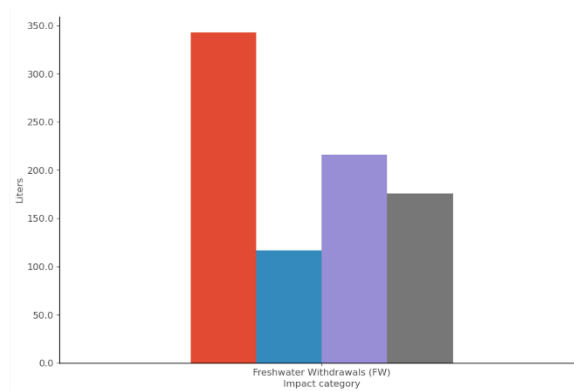
Acidification by recipe



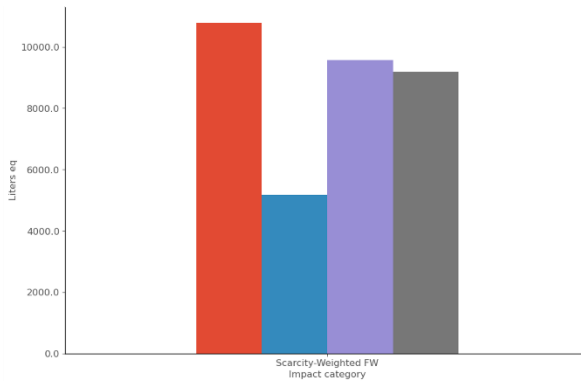
Eutrophication by recipe



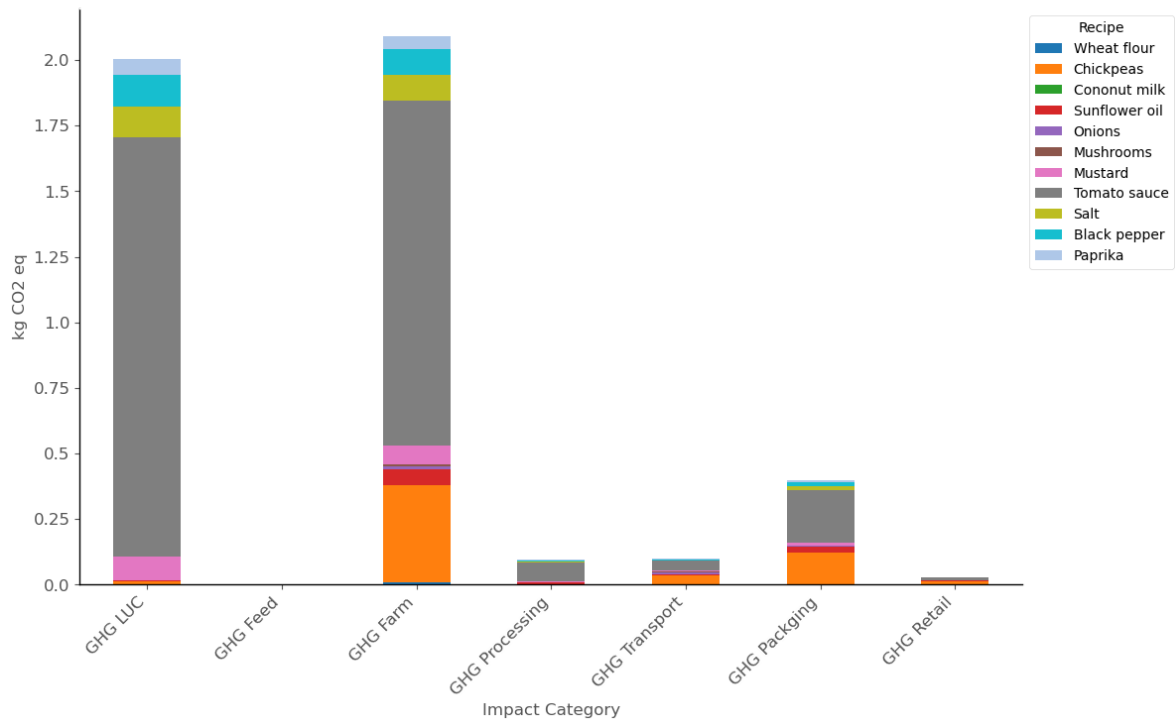
Freshwater Withdrawals by recipe



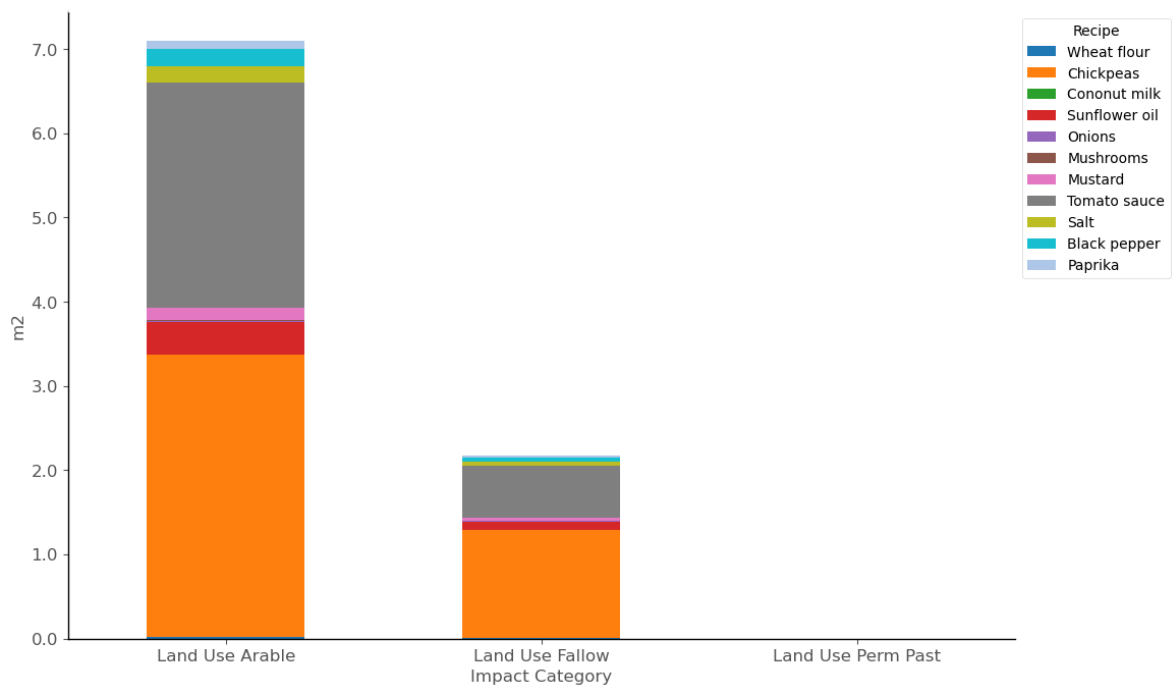
Scarcity-Weighted FW by recipe



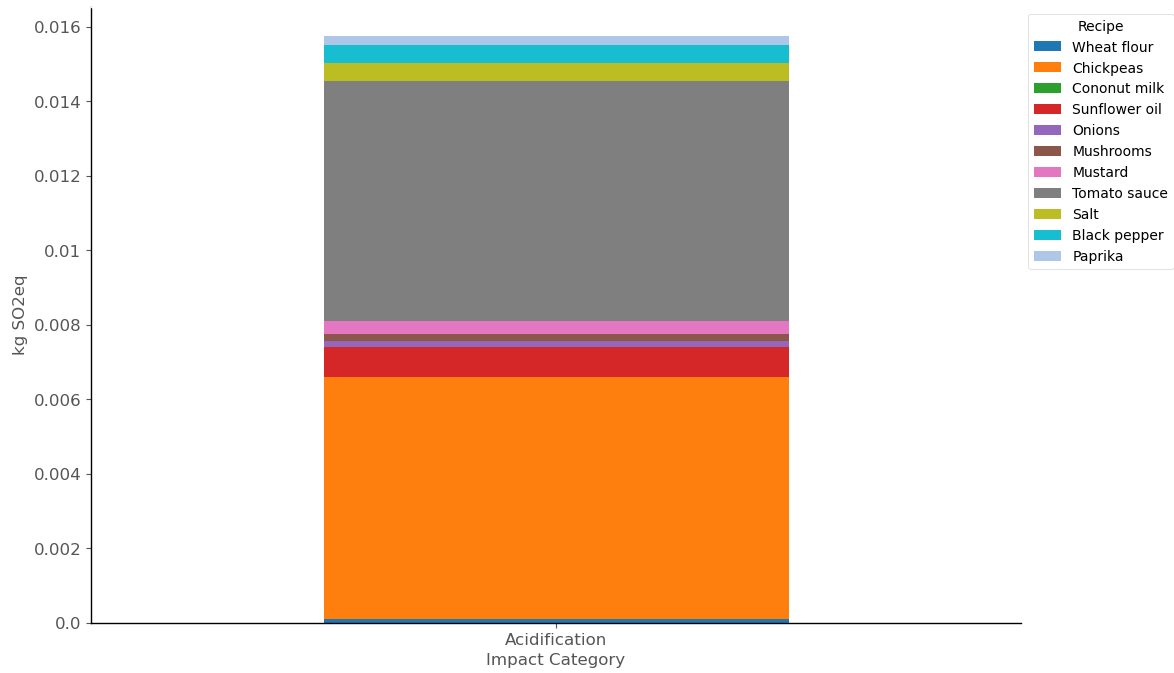
GHG emissions by ingredient - Recipe Strogonoff_chickpeas



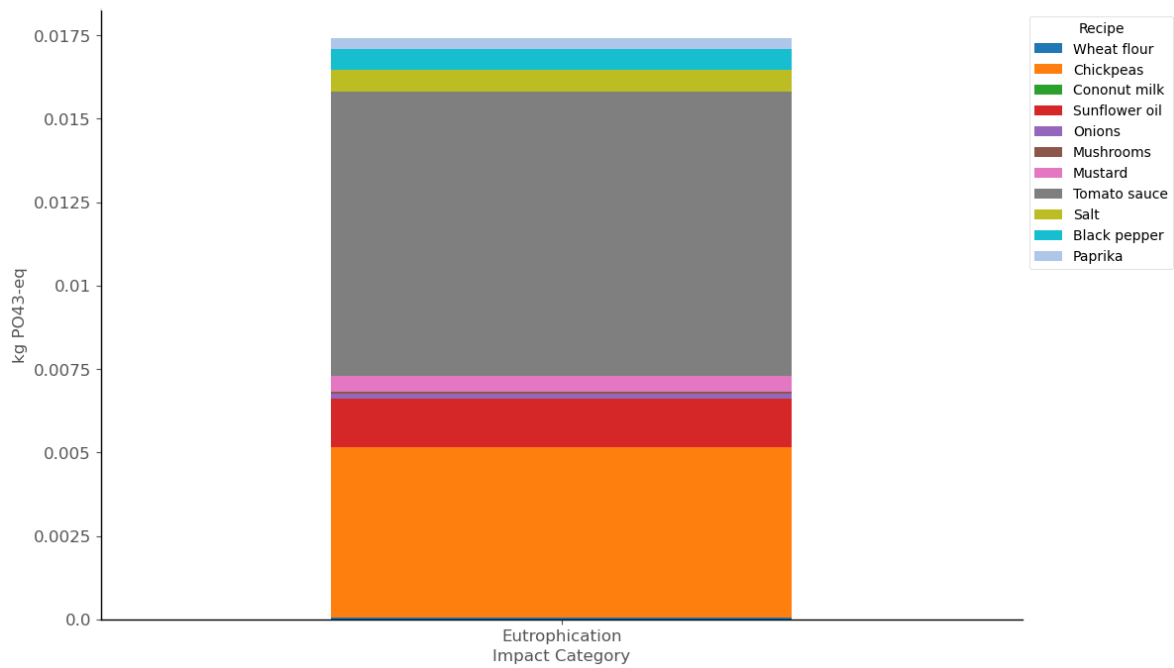
Land use by ingredient - Recipe Strogonoff_chickpeas

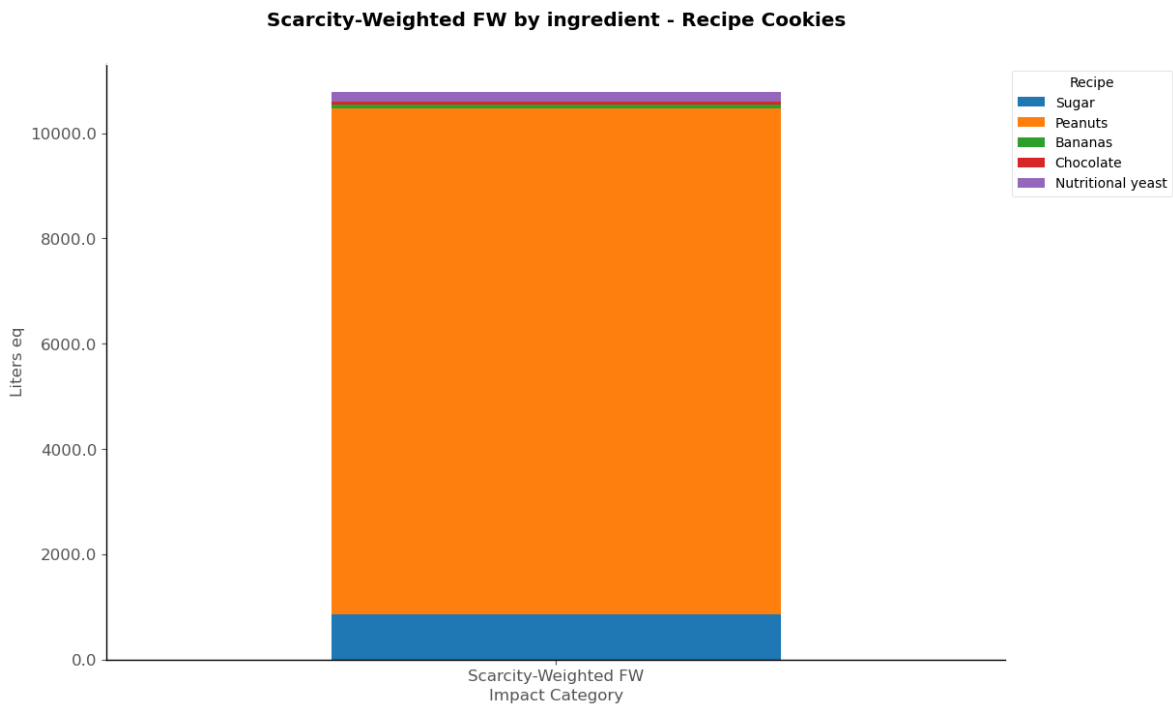
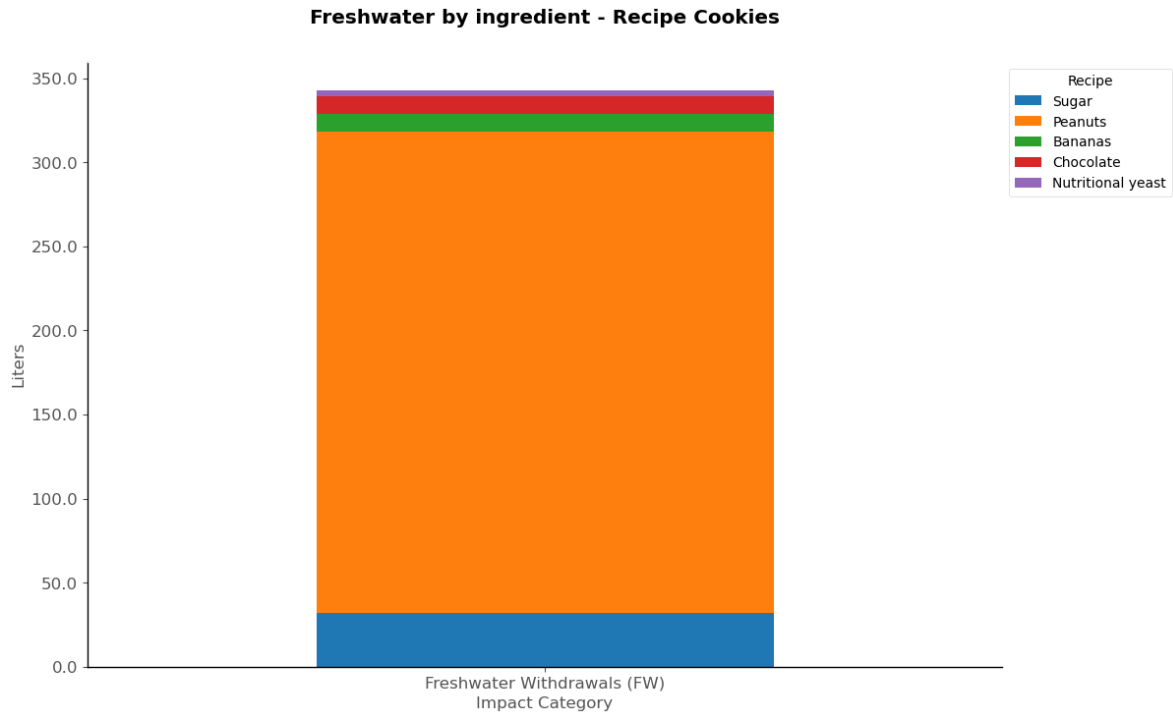


Acidification by ingredient - Recipe Stroganoff_chickpeas



Eutrophication by ingredient - Recipe Stroganoff_chickpeas





8.7.7. Results from User 7

8.7.7.1. Recipes

Table 33. Composition of the recipe: Asian soup

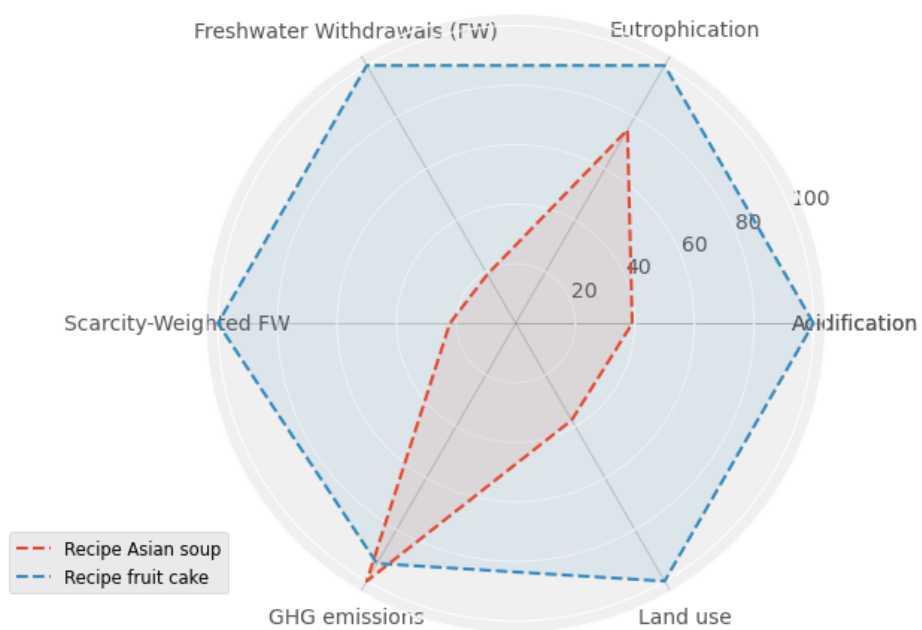
Ingredient	Amount	Unit
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Sugar	3	tblsp
Tomatoes (baby plum)	5	unit
Onions	1	unit
Carrots	5	unit
Cucumber	0.3	unit
Asian noodles	300	grams
Sesame oil	3	tblsp
Soy sauce	3	tblsp
Coriander	3	tblsp
Horseradish	150	grams
Broth vegetables	4	dl

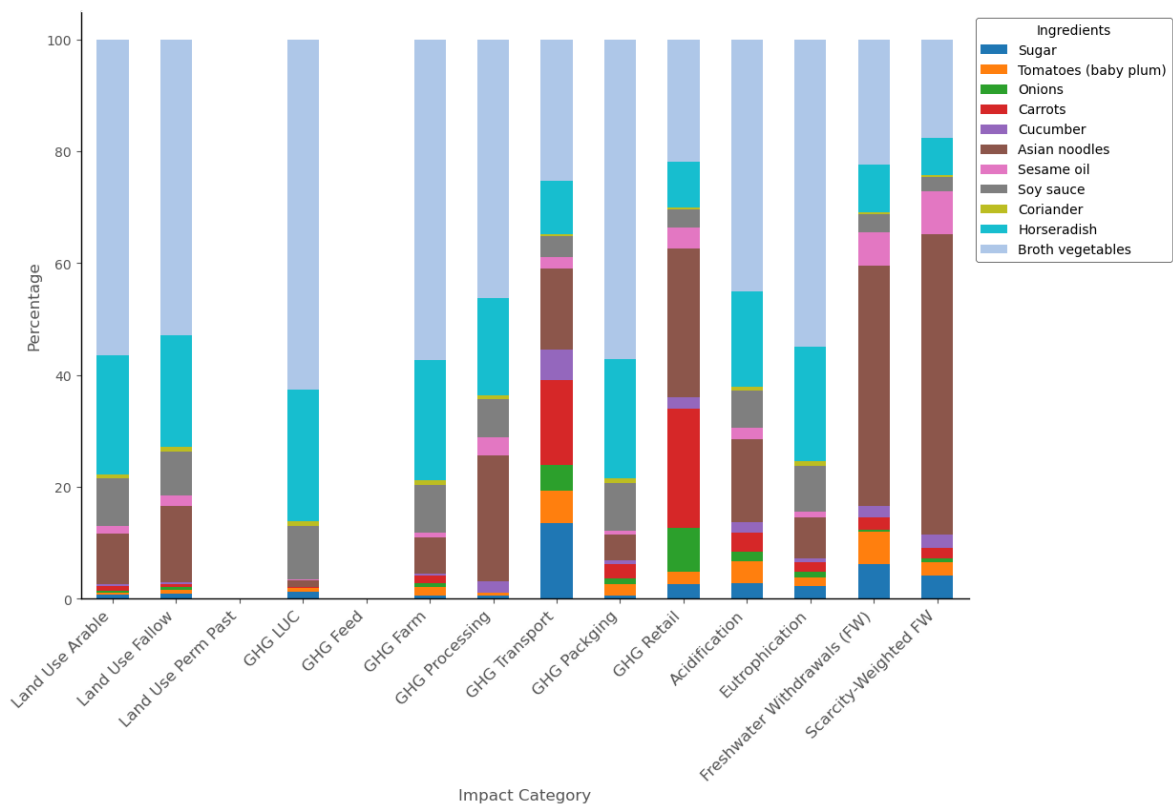
Table 34. Composition of the recipe: Fruit cake

Ingredient	Amount	Unit
Wheat flour	2	cup
Sugar	8	tblsp
Blueberries	200	grams
Cream cheese	200	grams
Chicken eggs	2	unit
Cookies	10	unit
Vanilla	1	tblsp
Butter	250	grams
Cream	400	grams

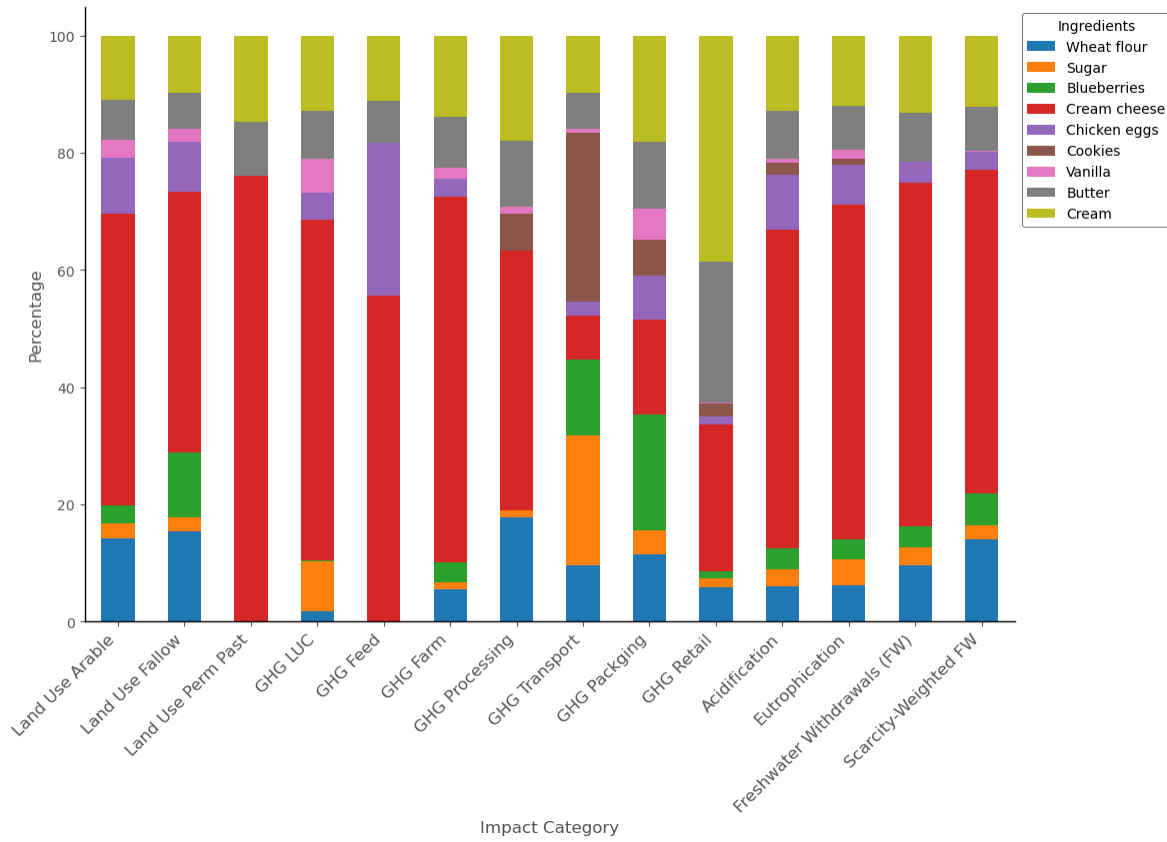
Relative environmental performance of recipes

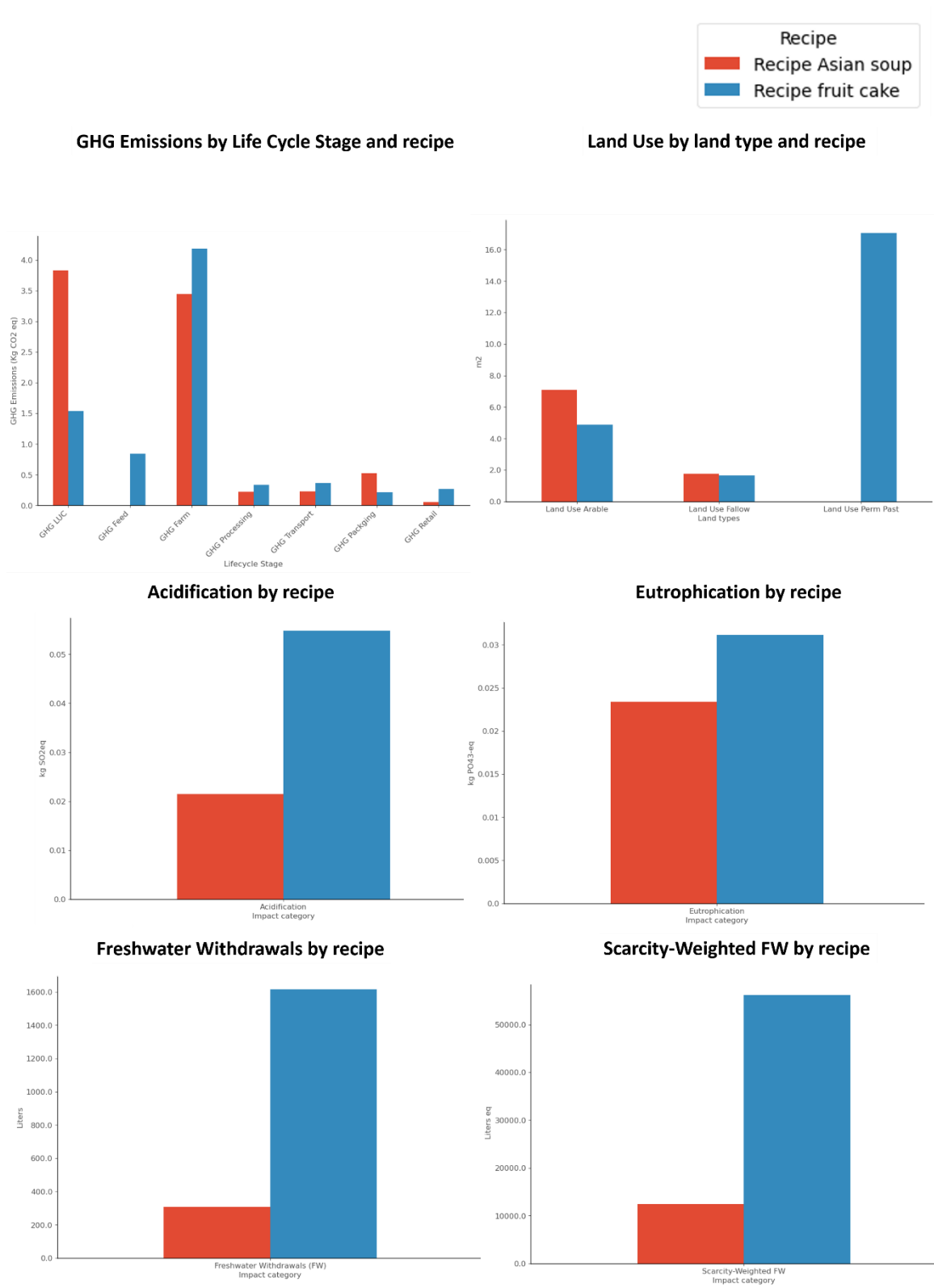


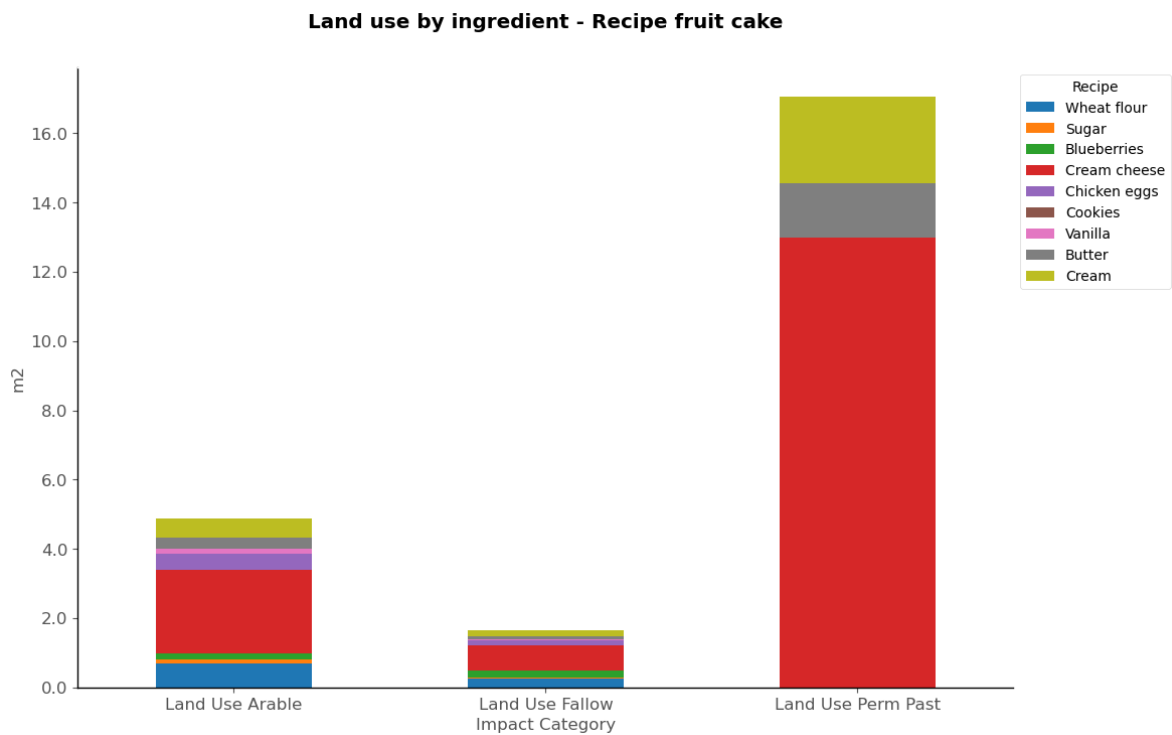
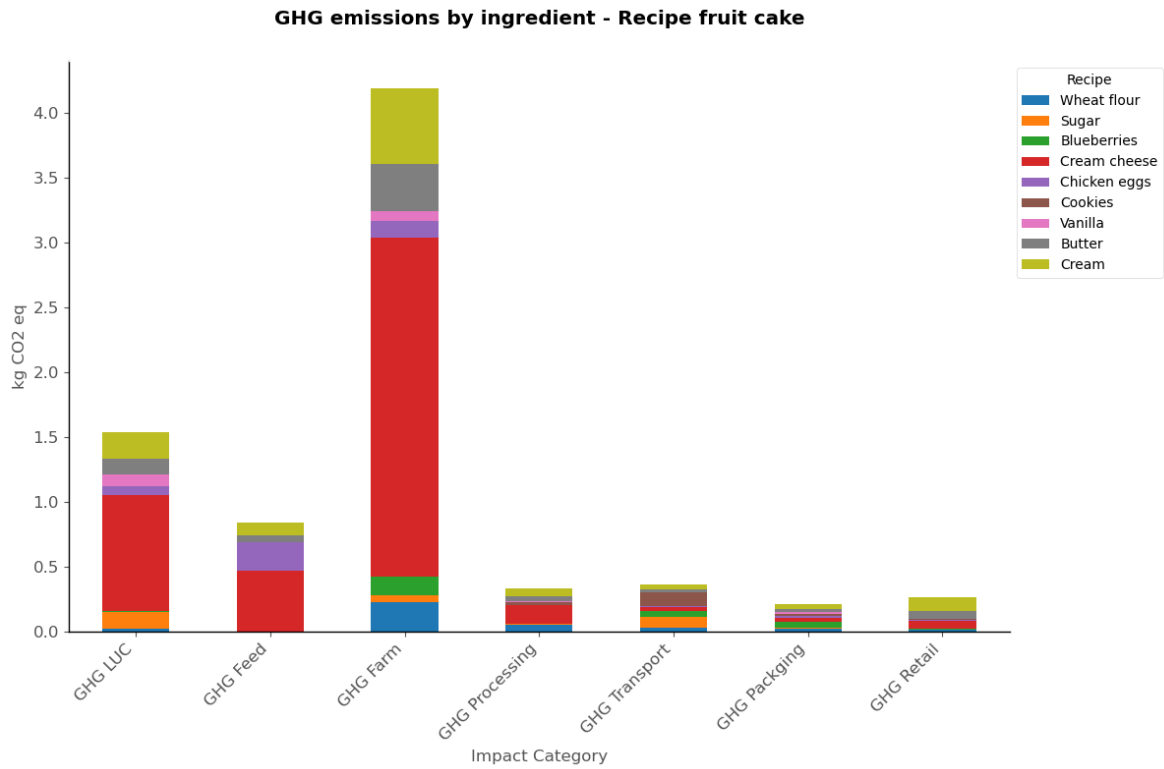
Environmental Impacts Distribution by ingredient - Asian soup



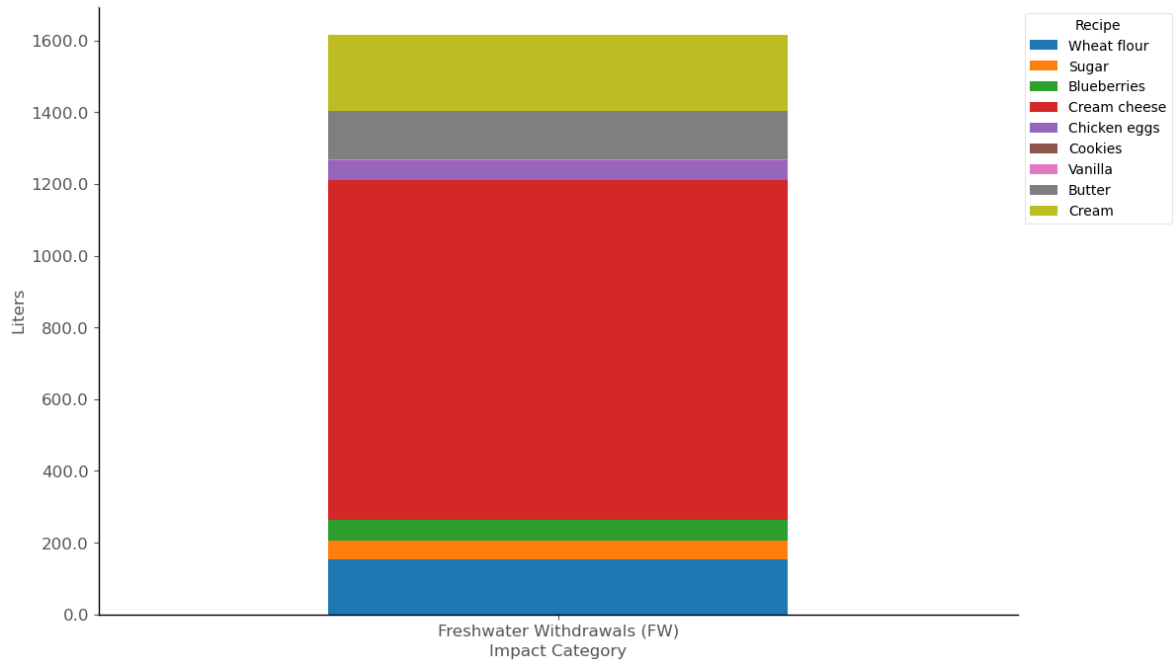
Environmental Impacts Distribution by ingredient - fruit cake



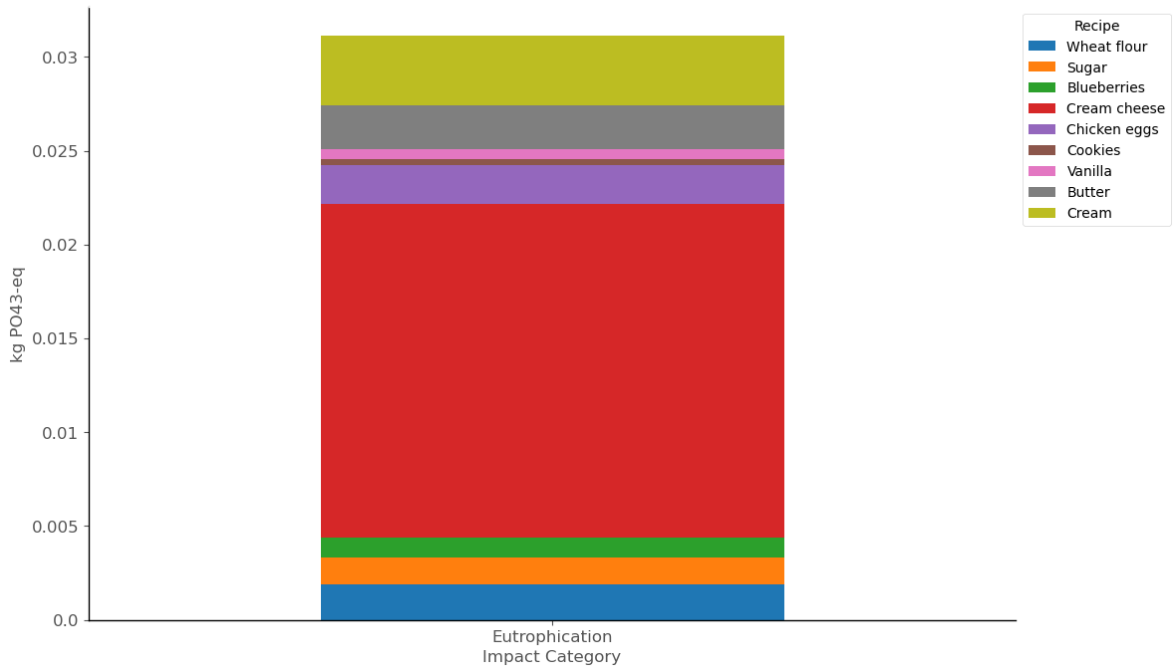




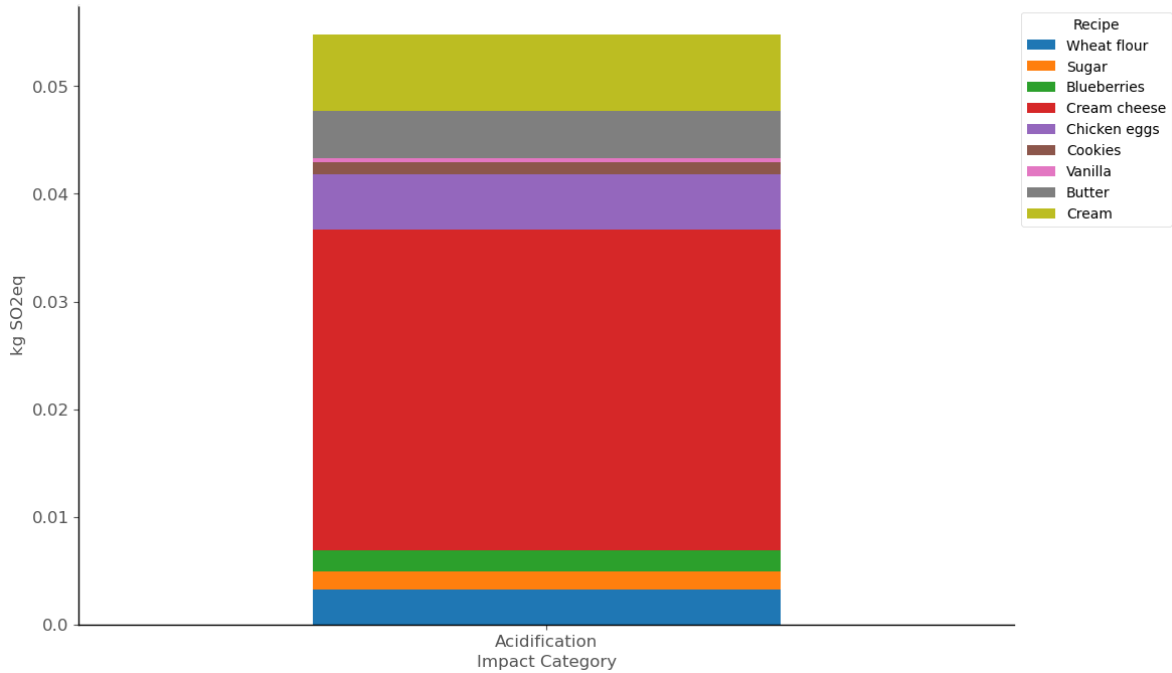
Freshwater by ingredient - Recipe fruit cake



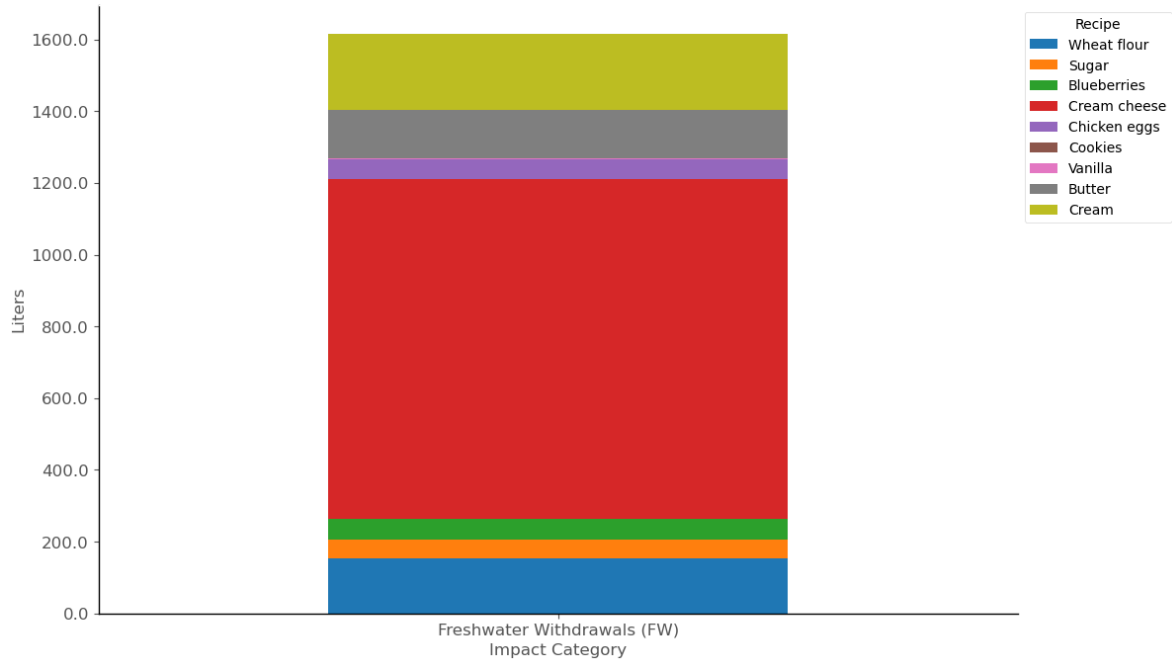
Eutrophication by ingredient - Recipe fruit cake

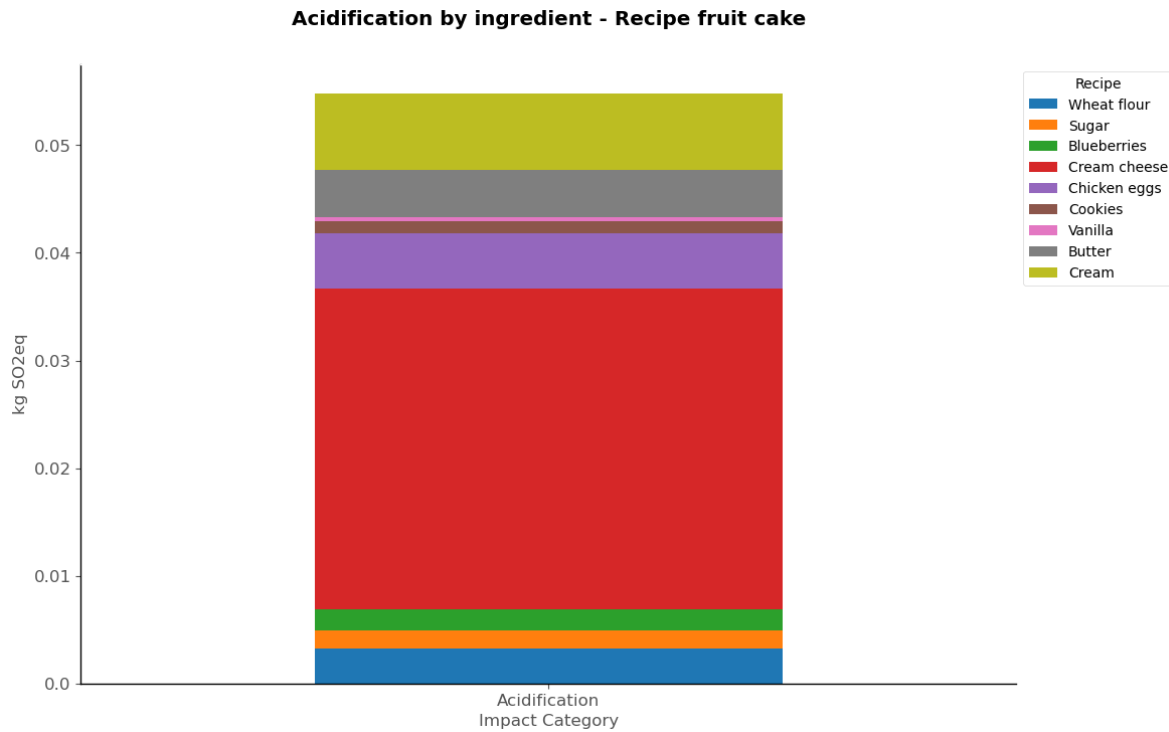
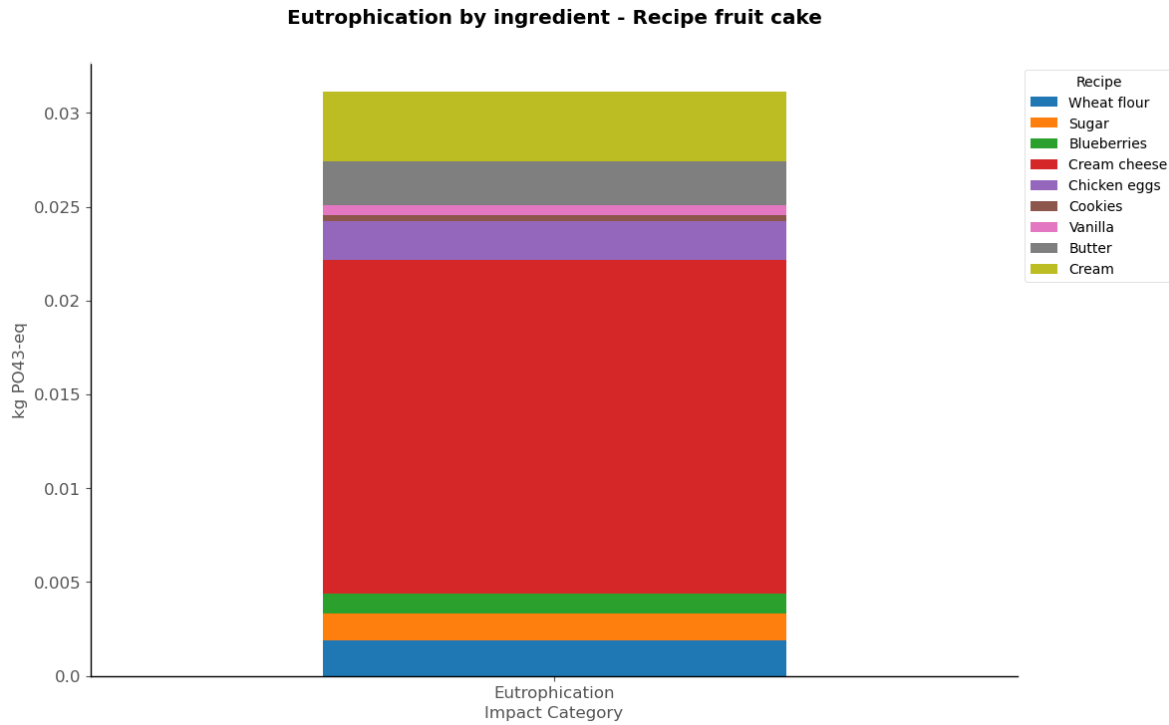


Acidification by ingredient - Recipe fruit cake



Freshwater by ingredient - Recipe fruit cake





8.7.8. Results from User 8

8.7.8.1. Recipes

Table 35. Composition of the recipe: Chicken in creme freice

Ingredient	Amount	Unit
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Sunflower oil	3	tbsp
Onions	3	unit
Sweet pepper	2	unit
Mushrooms	500	grams
Chicken	800	grams
Pasta	500	grams
Spices Misc.	2	tbsp
Tomato paste	20	tbsp
Salt	1	tbsp
Black pepper	1	tbsp
Cream	340	grams

Table 36. Composition of the recipe: Chilli con carne

Ingredient	Amount	Unit
Jasmine rice	1.5	cup
Red kidney beans	400	grams
Sunflower oil	3	tbsp
Onions	3	unit
Chilli pepper	1	unit
Sweet corn	200	grams
Sweet pepper	2	unit
Mushrooms	400	grams
Chicken	800	grams
Tomato sauce	800	grams
Salt	1	tbsp
Black pepper	1	tbsp

Table 37. Composition of the recipe: Kulling og søtpotet

Ingredient	Amount	Unit
Sweet potatoes	1200	grams
Onions	4	unit
Broccoli	2	unit
Chicken	800	grams
Salt	1	tbsp
Black pepper	1	tbsp

Table 38. Composition of the recipe: Meeeeeat

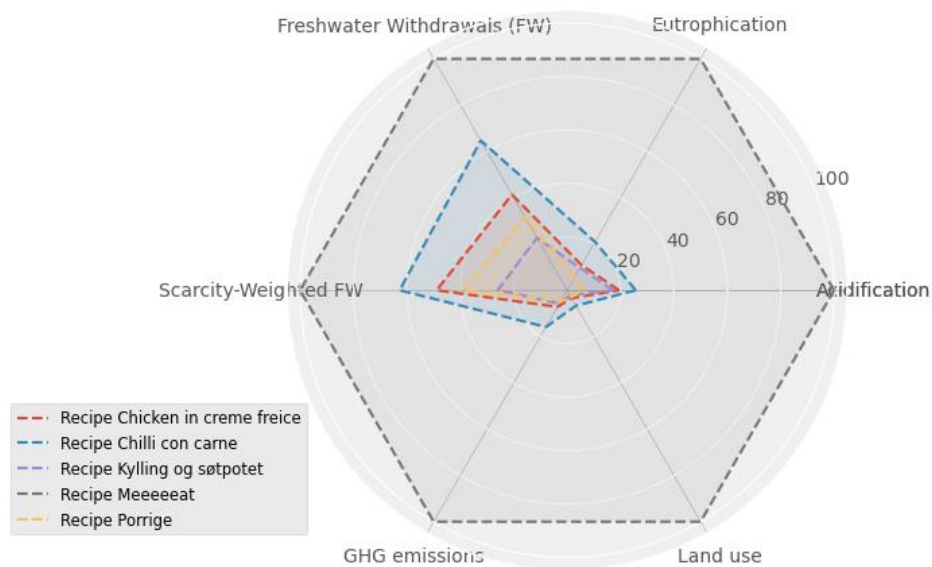
Ingredient	Amount	Unit
Potatoes	1600	grams
Onions	2	unit
Broccoli	2	unit

Cranberries	400	grams
Beef	1800	grams
Salt	2	tbsp
Black pepper	2	tbsp
Butter	30	grams
Cream	340	grams

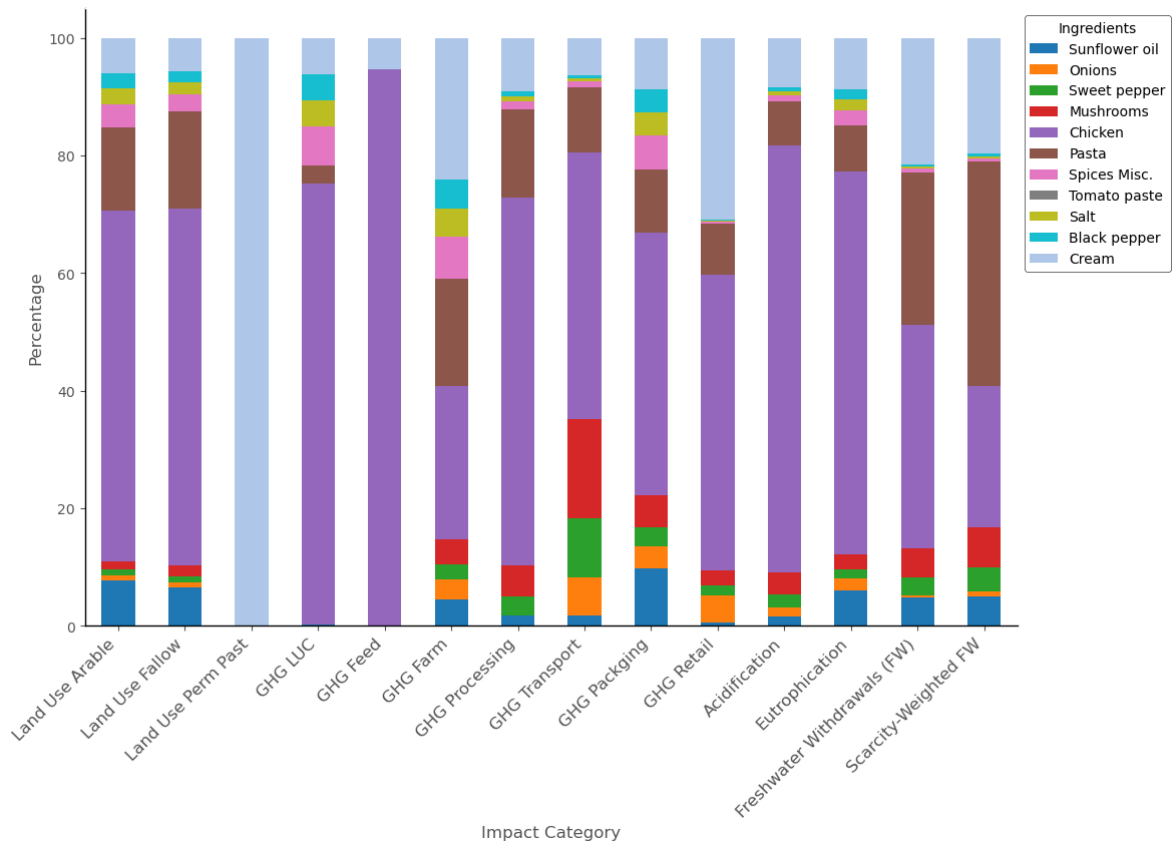
Table 39. Composition of the recipe: Porridge

Ingredient	Amount	Unit
Oats	9	cup
Soymilk	1400	grams
Sunflower seeds	50	grams
Bananas	6	unit
Raisin	150	grams
Salt	2	tbsp
Cinnamon	1	tbsp

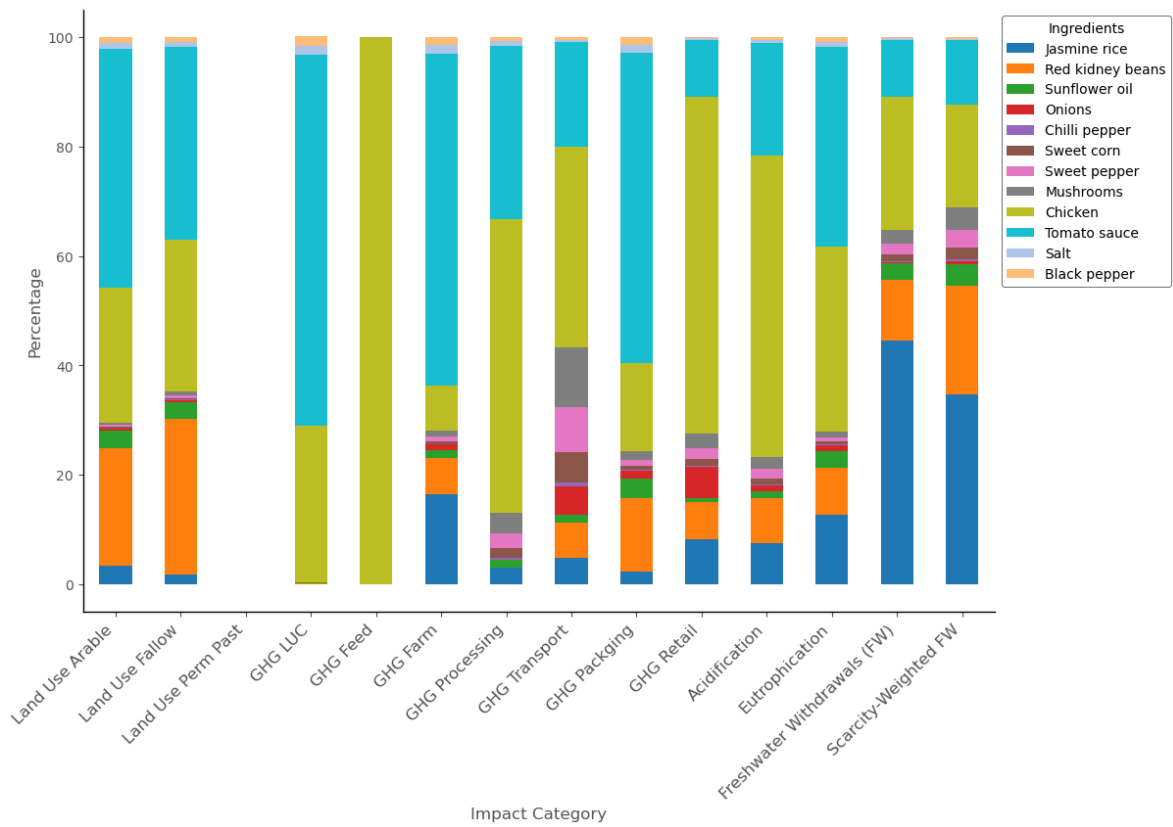
Relative environmental performance of recipes



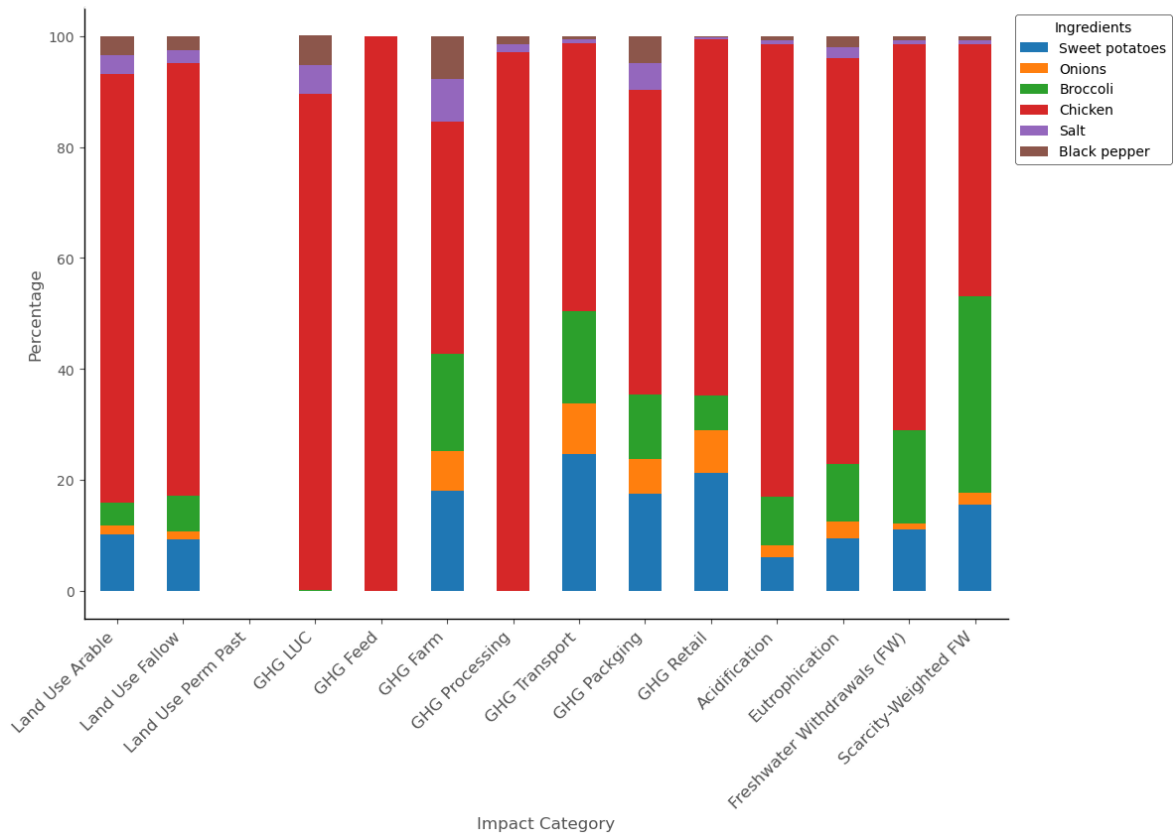
Environmental Impacts Distribution by ingredient - Chicken in creme freice



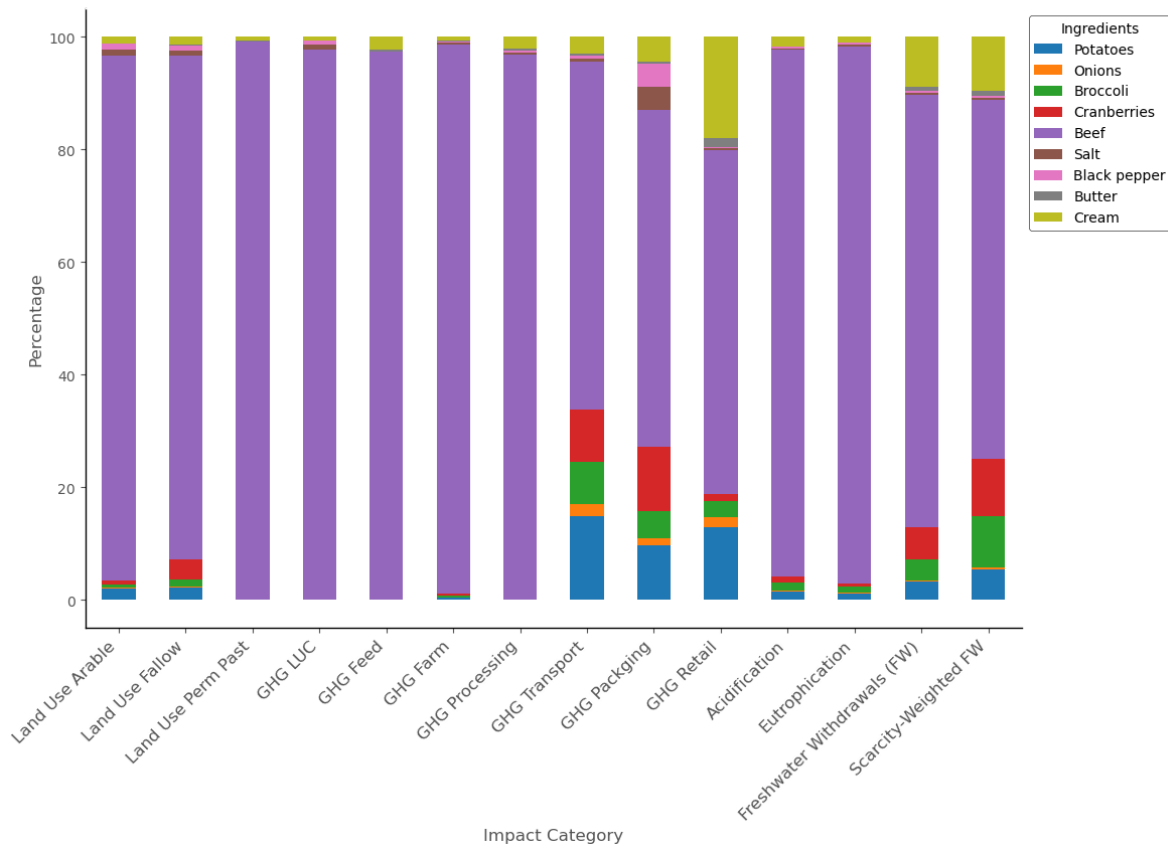
Environmental Impacts Distribution by ingredient - Chilli con carne

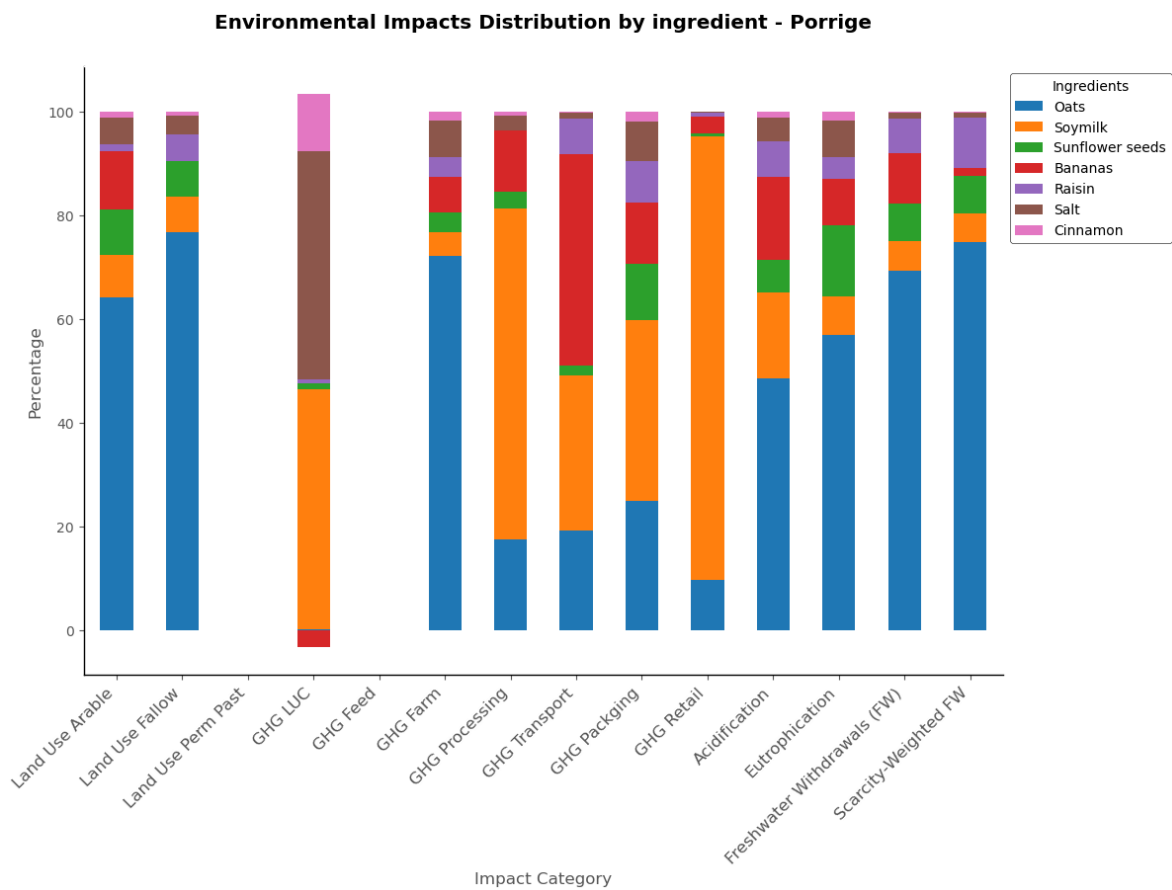


Environmental Impacts Distribution by ingredient - Kylling og søtpotet



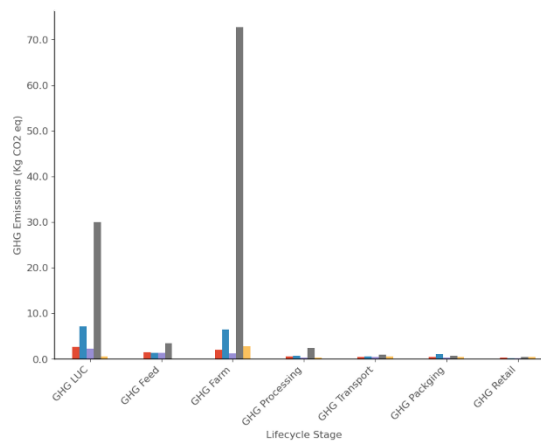
Environmental Impacts Distribution by ingredient - Meeeeeat



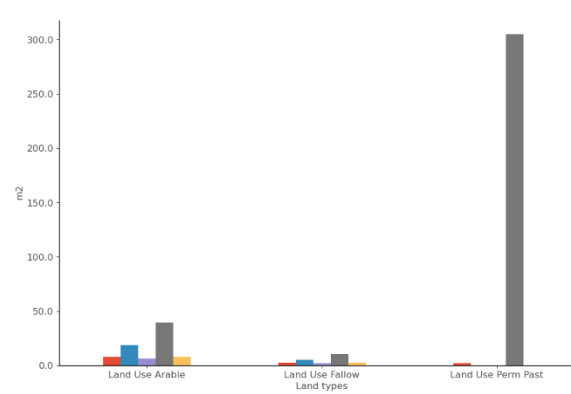




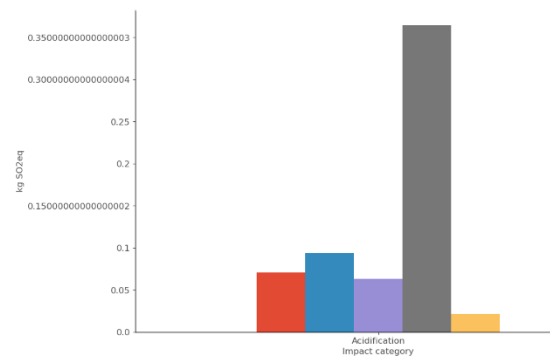
GHG Emissions by Life Cycle Stage and recipe



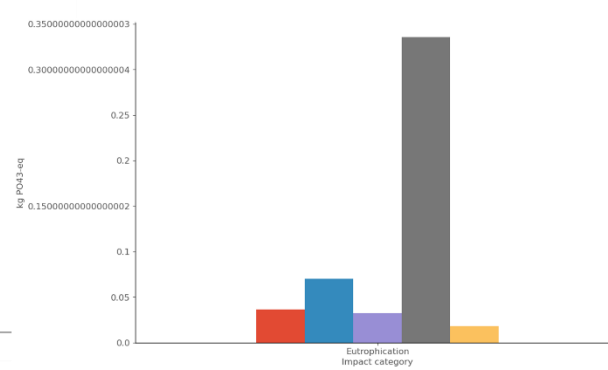
Land Use by land type and recipe



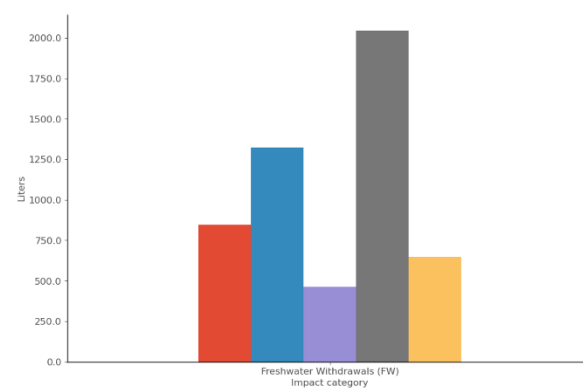
Acidification by recipe



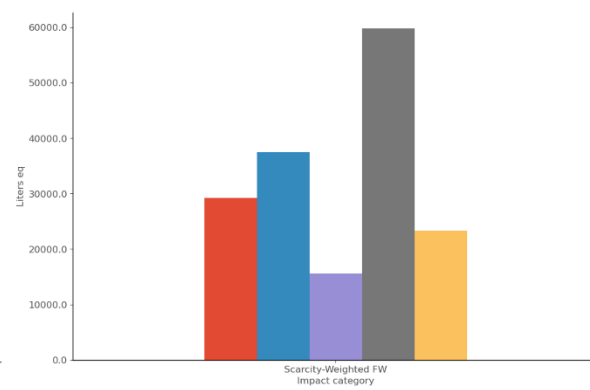
Eutrophication by recipe

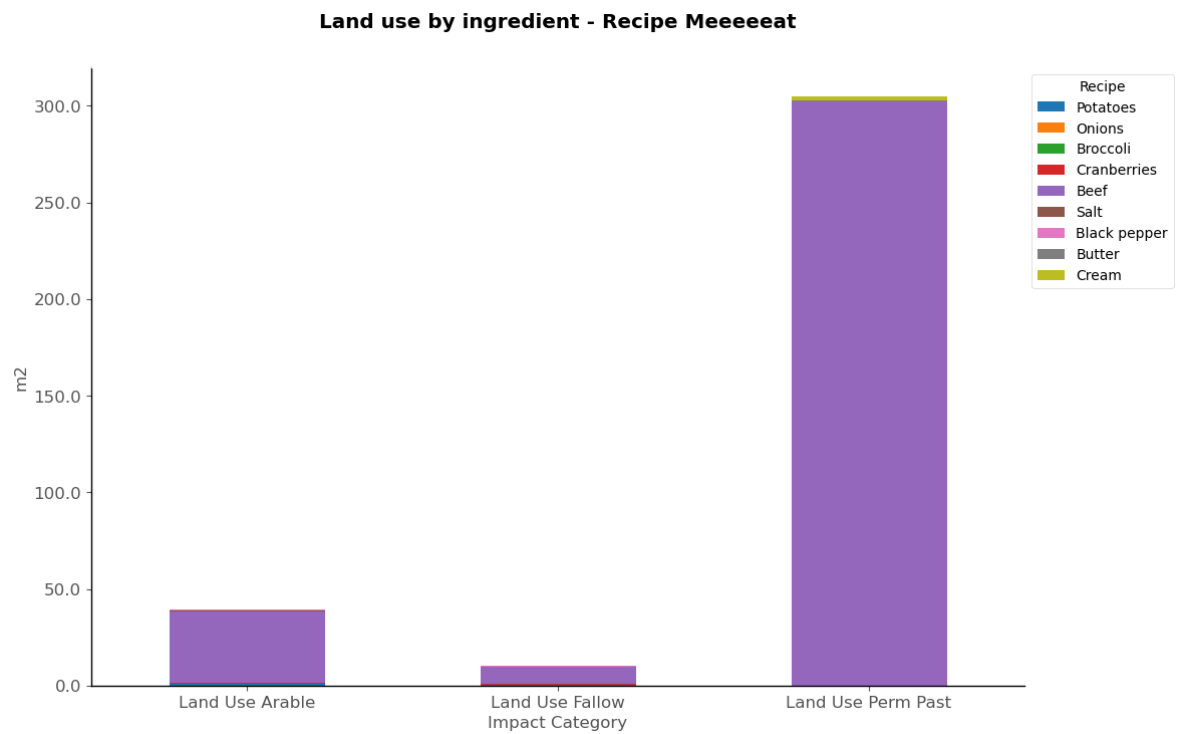
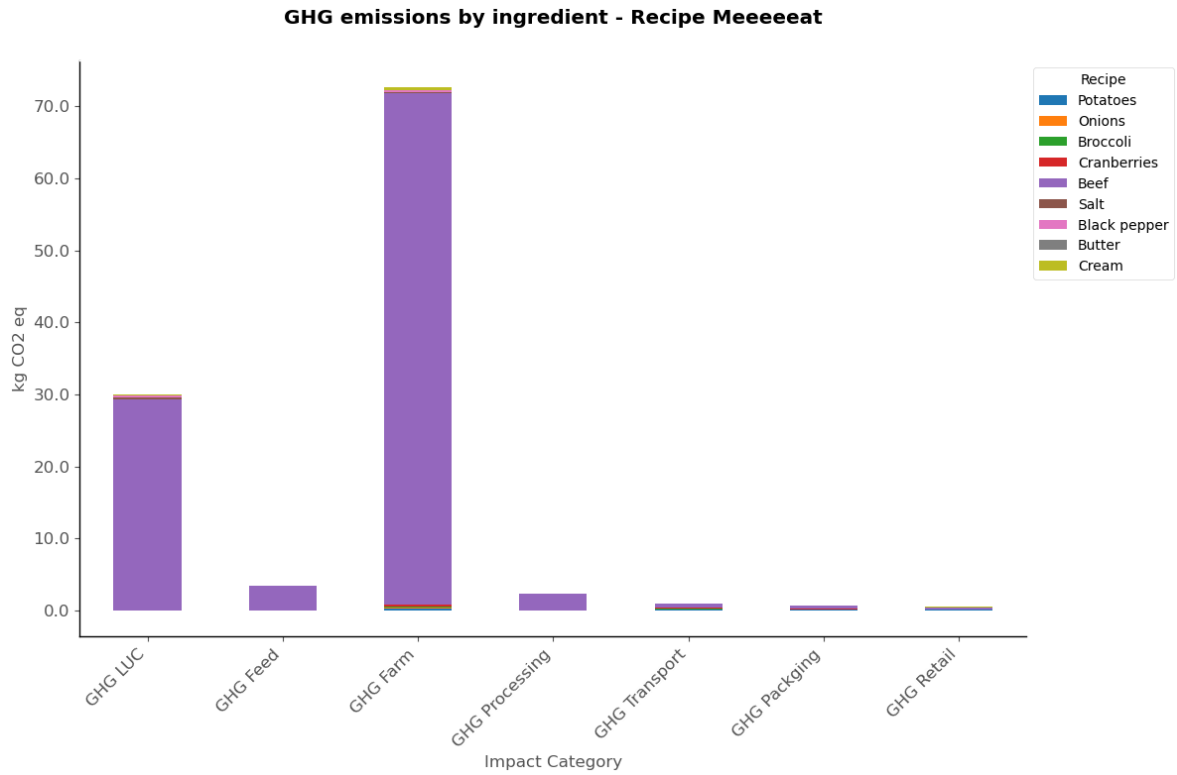


Freshwater Withdrawals by recipe

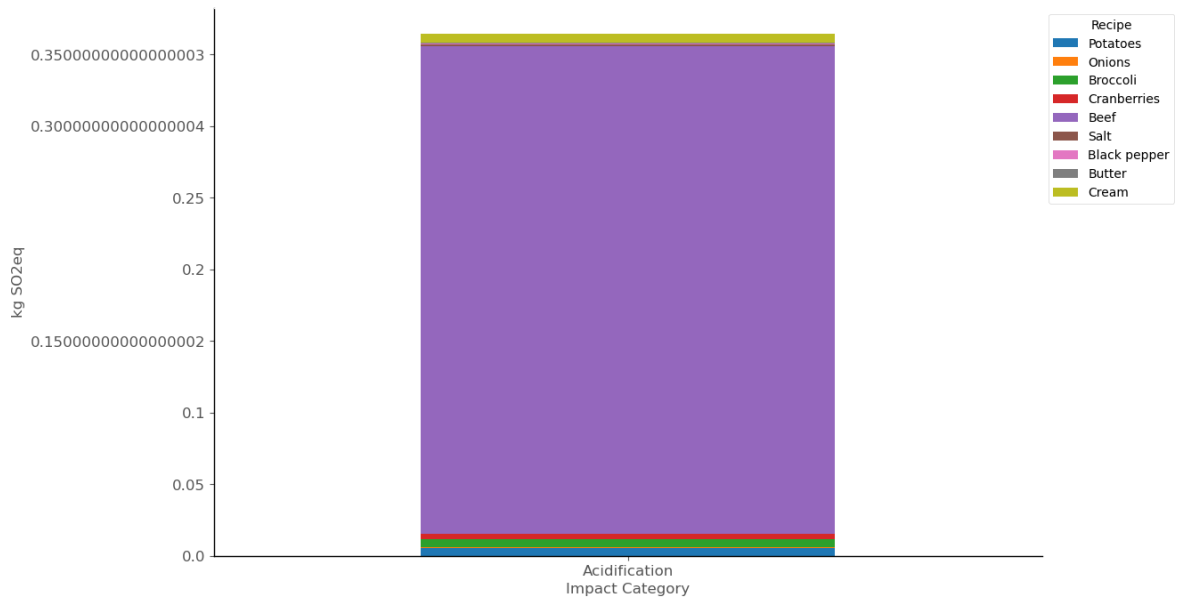


Scarcity-Weighted FW by recipe

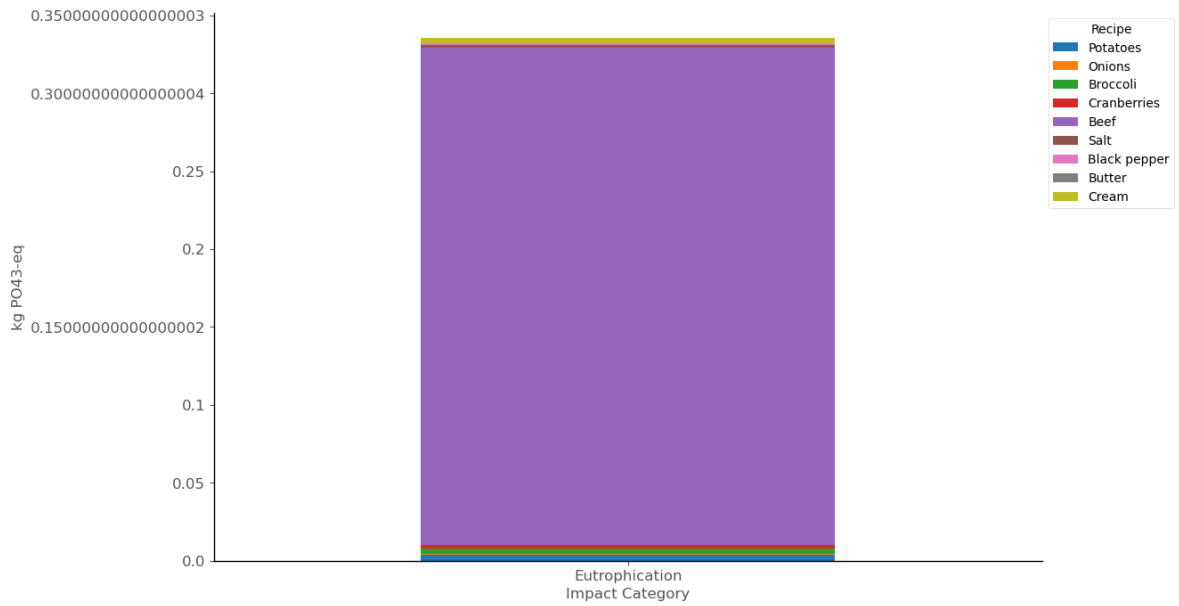


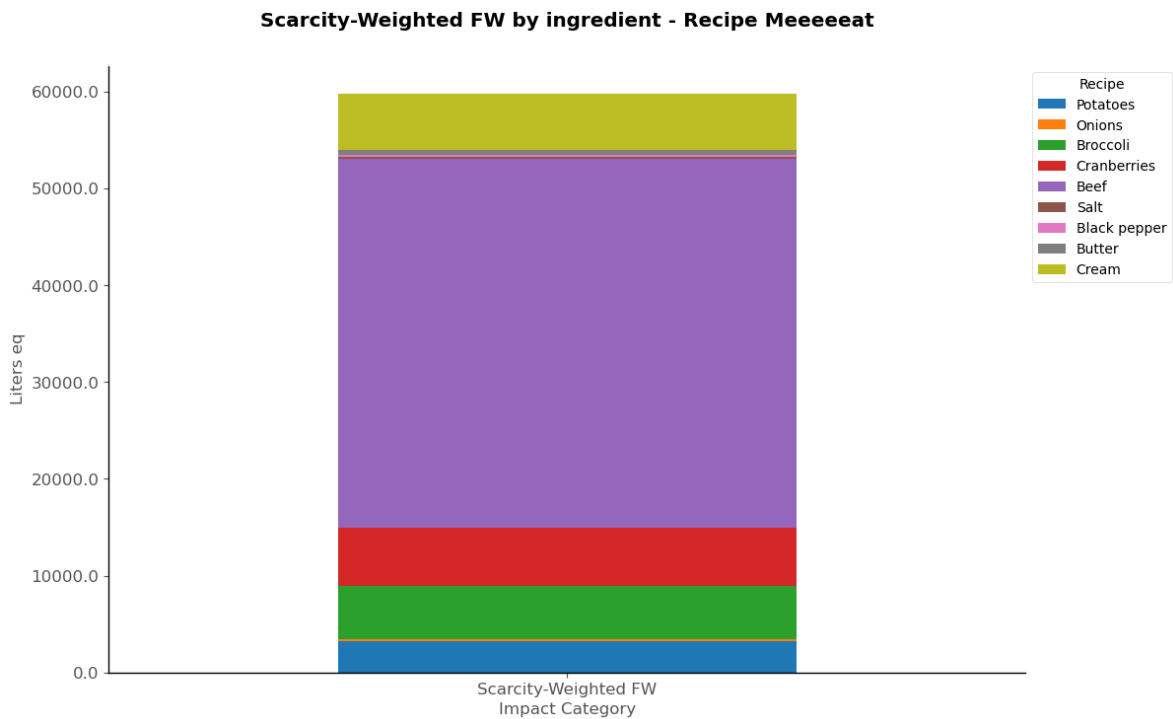
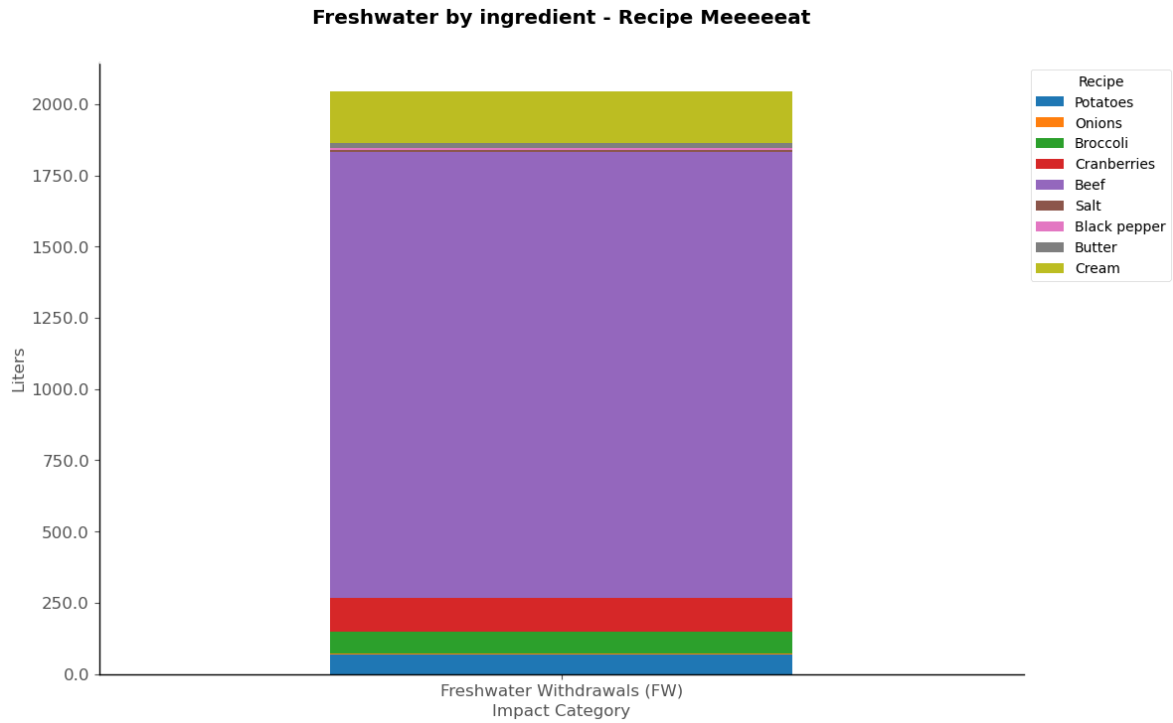


Acidification by ingredient - Recipe Meeeeeat



Eutrophication by ingredient - Recipe Meeeeeat





8.8. Results from the user experience survey

Table 40 shows the full answers from the test users once the test sessions was completed.

Table 40. Results from the survey after the test sessions

User	From 1 to 5, how well do you understand the purpose of this Food recipe environmental impact calculator? (1. not at all, 5. very well)	From 1 to 5, how useful do you think the results from the Food recipe environmental impact calculator are? (1. not useful at all, 5. very useful)	From 1 to 5, how easy/clear was it to enter the recipe details into the Excel template? (1. not easy/clear at all, 5. very easy/clear)	From 1 to 5, how clear was the procedure to use the Food recipe environmental impact calculator? (1. not clear at all, 5. very clear)	From 1 to 5, how would you rate the overall experience using the Food recipe environmental impact calculator? (1. not pleasant, 5. very pleasant)	Is there anything that you want to highlight from this prototype?	Are there any comments or suggestions that can help us improve the user experience and prototype?
User 1	5	4	5	4	4	The following error showed up when running the cleaning.py script on my computer: FileNotFoundError: [Errno 2] No such file or directory: './data/raw/aaq0216_datas2.xls'	The results could be simplified, that they are understandable for a large range of users and maybe make some comparisons to known or more impacts for better understanding. Explanations about the impact categories could also be useful, of people want to know more.
User 2	5	4	5	5	5	The excel template was very easy to use, and the code was easy to run. I liked the "spider web" diagram used to compare the impact of different recipes. I also liked that you included the impact of the different life cycle stages. It was a bit unclear with the different types of land use impact, what they represent, and how they are modeled.	Adding more ingredients, and also including a little description of each impact category and the "scarcity" so that the user understands more what the results mean (but maybe that is also a part of the user interface, which is out of scope).
User 3	5	5	4	5	5	We found some duplicates when analysing the results (for	Maybe add some more types of unit for different ingredients

						cow milk, peas, and something more). This was however a something Paula fixed quickly!	
						Impressive work as well!	
User 4	5	4	5	5	5	A good variety of ingredients! And also cool to see the impacts so easily. Paula was very good at explaining everything :)	Some missing ingredients: sour cream/creme fraiche, lime, salami. You could categorize some ingredients, for example "fruits and vegetables", "dairy products", "spices and herbs", "meats"
User 5	5	5	5	5	5	The prototype has a wide variety of food items from different types of diets. The graphics were well presented: first, with the main comparison, and later, with more details per recipe. Good regional items also such as tapioca, cassava.	Pictures could help people to choose (between tomato sauce or paste, for example). It would also be good to have a converter on the website for different units (cups to grams). Separate the food items by category for easier research.
User 6	5	5	4	3	5	It is good that one is talked through using the program and that the results are interpreted with someone else.	It would be interesting to obtain information on the sensitivity of the parameters in order to see which parameter would most impact the GHG emissions per product in relation to their quantity.
User 7	4	4	4	4	5	Super interesting! Nice to see the differences between recipes and also the difference in impact when water scarcity is considered for the different ingredients. Think it is super cool, and I was extremely impressed with the amount of work that has gone in. The	Some missing food items - even though there are sooo many already which is wonderful! A lot of popular supermarket items in Norway include sauces, curry pastes etc, that are missing from the list. For example it could be interesting to see if there are 'most popular supermarket

						result is very commendable, I think the prototype is excellent. There are of course limitations to the prototype, but for one main person's work in a limited timeframe, it has come a very long way and it's something to be very proud of!	items' in Norway and see if the calculator covers them, if Norway is a focus.
User 8	5	4	4	4	4	It was very interesting and enlightening to be able to compare multiple recipes and see how they impacted the environment in different aspects. Another great feature was the amount of different ingredients that was supported, and that all measurement units could be given in grams instead of the default unit.	It could be useful to have a larger variety in the color scheme to easier identify the ingrediences in the plots. It would be easier to make certain the different recipies was scaled correctly if one could specify how many servings the recipies are for instead of 'manually' having to scale each ingredient. To be able to select between multiple measurement units would also be a plus.

Table 41. Full version of the Pedigree Matrix for comparing the "available for download" databases

Database / DQIs	Completeness (C)				Precision / Uncertainty (P)				Methodological Appropriateness and Consistency (M)	Temporal Representativeness (TiR)	Geographical Representativeness (GR)	Technological Representativeness (TeR)	Data Quality Score
	Food items	System boundary	Impact categories	Score	Quality of the sources	Number of studies	Uncertainty	Score					
	20	20	60		60	20	20						
SU-EATABLE LIFE	323 food items for GHG emissions and 320 food items for Water footprint	Primary production, processing, packaging, transport, and retail stages	GHG emissions & Water Footprint		Peer-reviewed papers, conference proceedings, public reports, studies with described data collection methods, and EPDs.	841 publications for GHG emissions and 88 publications for Water footprint	Uncertainty Analysis conducted using the three quality flags method. Findings from all aspects were integrated to determine the degree of uncertainty, and recommendations for High Uncertainty		The three-step framework ensures both methodological robustness and usability for future research. Also, all the calculations and relevant information to reproduce the values are available for download	GHG emissions - (1998 - 2019) Water footprint - (2005-2018)	GHG emissions - 85 countries in Africa, Asia, America, Europe, Oceania Water Footprint - 41 countries in Africa, Asia, America, Europe, Oceania	Food items	
	2	1	4	3,00	2	1	1	1,60	1	2	1	2	2,26
Poore and Nemecek (2018)	40 food groups	Primary production, processing, transport, packaging, and retail phases.	GHG emissions, land use, freshwater withdrawals, freshwater withdrawals weighted by local water scarcity, acidifying emissions, and eutrophying emissions		Studies included in peer-reviewed journals, or are PhD theses, ISO compliant reports, LCA databases, or conference proceedings with clear data and methods	570 publications	Methodology adopted to handle variance in studies, emphasizing the importance of addressing various sources of variability and describing the process of generating random numbers to create a more comprehensive dataset		Well established method for building and standardizing the inputs and management inventory. The calculations are explained in the supplementary material but are not provided in the database. Sources for each data point are not described in the database	2000 - 2016	119 countries in Europe, North America, Oceania, Brazil, and China, but limited in Africa and Central Asia	Food groups	
	5	1	1	1,80	2	3	1	2,00	2	3	1	4	2,98

SHARP INDICATORS DATABASE	944 food items	Primary production, use of primary packaging, transport, food losses and waste, and food preparations	GHG emissions & Land use		Life cycle inventory data were retrieved from Agri-Footprint 2.0, Ecoinvent 3.3, CAPRI, and supplemented by recent literature and technical reports	Not applicable	Not applicable	The full calculations are available upon request for scientific purposes	The Agri-footprint 2.0 was released in December 2015 and updated in 2017. The data age ranges from 4 to 7 years Ecoinvent 3.3 was released in August 2016, then the data is at least 7 years old The CAPRI model is based on 2004.	Denmark, Czech Republic, Italy and France (Europe)	Food items	
	1	1	4	2,80	3		3,00	5	3	4	2	3,98
dataFIELD	322 food items	Primary production, and processing	GHG emissions		Peer reviewed journals articles and NGO reports, databases, industry-based reports, government reports, and conference proceedings	326 publications	Systematic approach to handling variability in impact factors for food production, considering data points, proxy assignments, and adjustments for foods with fewer observations or specific conditions. The calculation of confidence intervals adds a statistical measure to the assessment of diet-level variability	Four-step process with clear guidance. Calculations are available in the database. Part of the data is not completely clean, for example, having different ways to refer to the same country. Also, some of the fields have errors in the values	2005 - 2016	Between 60 to 70 countries in mainly Europe, North America and Oceania (Australia). Few studies in Central America, Africa and Asia.	Food items	
	2	4	5	4,20	2	4	1	2,20	2	3	2	2

AGRIBALYSE	2500 food items	Raw material production, processing, assembly, packaging, distribution, retail, storage, preparation at the consumer level, disposal of packaging.	Climate Change, Particulate matter, Water Use, Resource use (fossils & minerals, and metals), Land use, Ozone Depletion, Acidification, Ionizing radiation, effect on human health, Photochemical Ozone formation, effect on human health, Eutrophication (terrestrial, marine & freshwater)	The database comprises 2517 datasets from previous versions of Agribalyse, ACYVIA (Life Cycle Analysis in Food Processing Industries), Ecoinvent 3.88, World Food LCA database v3.5			Not applicable	Not applicable	The format that includes all the details and allows for data adjustments requires expert users utilizing SimaPro and OpenLCA. The format freely available on the website lacks the necessary detail for emissions sources, and adjustments to the detail data cannot be made	The latest version of the database, Agribalyse 3.1, was released in October 2022. The data range from 2013 to 2023.	France (Europe)	Food items	
				3		3,00				1	5	2	3,70
AGRIBALYSE + Biodiversity	2497 food items	Raw material production, processing, assembly, packaging, distribution, retail, storage, preparation at the consumer level, disposal of packaging.	Biodiversity, Land competition, Freshwater Eutrophication, Marine Eutrophication, Acidification, Cumulative Energy Demand, Water Consumption, Climate Change	This database uses Agribalyse 3.0 and as a reference for biodiversity impacts, the compilation of characterization factors proposed by Nitschelm, L., et al 2021			Not applicable	Not applicable	There is not enough documentation to track down the values in the database also the sources used. The paper was built to support research that is still no published.	The paper was published in 2023. Therefore, the data in the database is at least from 2013 to 2022	France (Europe)	Food items	
				3		3,00				1	5	2	3,70

Chinese Food Life Cycle Assessment Database (CFLCAD)	80 food items for GHG emissions, 93 food items for Water use, and 50 food items for Land use	GHG emissions: Raw material production, storage, processing, packaging, transportation, preparation at home, food losses along the food supply chain	GHG emissions, Water use and Land use	Literature review using the China National Knowledge Infrastructure (CNKI) for journals in Chinese and Google Scholar databases for journals in English	125 publications for GHG emissions, 66 publications for water use and 11 publications for land use	The paper presents summary statistics, such as mean and standard deviation, for GHG emissions, water use, and land use across various food groups based on literature. However, it lacks information on confidence intervals and the documentation does not elaborate on the impact of these statistics on the different food categories.	The original paper provides a clear description of the process and includes various conversion factors for each food group. However, the actual calculations are not presented, and the food items lack clear names. In attempting to reproduce mean values for food categories, observed discrepancies arise as the reproduced values slightly differ from those presented in the paper.	2005-2020	China (Asia)	Food items		
	5	3	3	3,40	2	5	3	2,80	3	1	5	2
The Big Climate Database	500 food items	Agriculture, food processing, packaging, transportation, retail	GHG emissions	EXIOBASE v3.316b2 FAOSTAT	Some publications, reports, websites used to complete data. No summary of number of these documents.	Not applicable	The procedure is extensively described in the supplementary material and the sources are described for each value. Additionally, the data behind the values in the database can be tracked in both the online and the excel version. However, there is no access to direct calculations in the database.	Crop cultivation: 2011 and 2018 Animal production: 2011 and 2018 Markets: 2018 Food processing: No temporal consistent data Remaining activities: 2011	Denmark for retail stage, and 43 countries for food processing not located in Denmark (Europe)	Food items		

	1	1	5	3,40	4		4,00	2	3	5	2	3,94
Clark et al. 2022 Database	57185 food items	Raw material production, food processing, packaging, transportation, retail	GHG emissions, Water Scarcity, Land Use, Eutrophication Potential		HESTIA Blue Food Assessment	HESTIA gives access to 395 peer reviewed studies and reports from a wide time frame	The authors used Monte Carlo analysis to estimate the average environmental impact and its variability. This involved randomly selecting and pairing producer-level impact estimates with the composition of each ingredient in the products.	The documentation provides a clear explanation of the procedure, and an uncertainty analysis was conducted. However, the database file's tabs are challenging to follow without the supplementary material. Additionally, understanding the details of food items in the "Department" column in the "Figure S1-S5" tab is not straightforward	HESTIA gives access to 395 peer reviewed studies and reports from a wide time frame. HESTIA is based on Poore & Nemecek study.	United Kingdom (Europe)	Food items	
	1	1	2	1,60	2	4	2	3	3	5	2	3,70
New Zealand-specific LCA Database	346 food items	Farming and processing, transportation, transit packaging, consumer packaging, warehouse and distribution, refrigeration, and supermarket overheads	GHG emissions		Database of Hoolohan et al. (2013)	Approximately 20 publications	Due to the absence of uncertainty estimates in the LCA studies and a lack of reliable proxy estimates, the uncertainty within the database was uniformly set at 20% for all foods.	The process is clearly outlined in the main text, but unfortunately, the supplementary documentation lacks the necessary details about the methods. Additionally, the database itself does not provide direct calculations for the procedures described.	2003 - 2011	United Kingdom (Europe)	Food items	
	2	1	5	3,60	2	5	4	3	4	5	2	4,06

The University of Copenhagen database of food emissions factors - UCPH - Clune 2017	168 detailed food categories	Raw material production, food processing, packaging, transportation, retail	GHG emissions	Peer reviewed scholarly journal papers; conference proceedings; EPDs, and grey literature	369 publications	The database presents information for each food category, encompassing the number of studies, GWP scenarios, and descriptive statistics, while also conducting a thorough analysis of variability, including correlations between GWP values and factors such as geographic locations, farming methods, or species. The values are not listed in the database, but in the paper.	The procedure is well described. The variability analysis is not part of the database, and this makes it difficult to validate the values because there are different names for the food categories on each file. Also, part of the data is not completely clean, for example, having different ways to refer to the same country. No direct calculations are shown in the database	2000 - 2015	Approximately 78 countries. Predominantly Europe (68%), followed by North America and Oceania. Asia, Africa and South America were less represented	Detailed categories		
	4	1	5	4,00	1	4	2	1,80	3	3	2	3
The Cool Food Pledge (CFP)	54 food groups	Primary production, processing, transport, packaging, and retail phases.	GHG emissions & Land use	Based on Poore and Nemecek (2018)	570 publications	Uncertainty analysis conducted by Poore & Nemecek	Excel sheet with calculations and scenario development can be obtained from the authors upon request.	2000 - 2016	119 countries in Europe, North America, Oceania, Brazil, and China, but limited in Africa and Central Asia	Food groups		
	5	1	4	3,60	2	3	1	2,00	5	3	1	4

Database for Double Pyramid 2016	1300 food items			Crop, farm process, industrial process, transport and storage, consumption and final disposal	GHG emissions, Water Footprint, and Ecological Footprint	LCA databases such as Ecoinvent, LCA food, Water Footprint Network, and Ecological Footprint Network. Most of the sources were taken from verified publications like EPDs and Climate Declaration, and documents associated with scientific publication, executive summary, working papers and presentations.		566 documents associated with scientific publication, executive summary, working papers and presentations.	No uncertainty analysis was performed	The procedure is well described in the supplementary material and the sources are described for each value. However, there is no access to direct calculations in the database, and there is no methodology to extrapolate the values missing for each food category.	Ecoinvent 3.31 was released in July 2014, then the data is at least 9 years old LCA Food Database was accessed in May 2016. Then the data is at least 7 years old Water Footprint Network data was accessed in 2013, which makes the data at least 10 years old. Ecological Footprint Network data was viewed in 2013, which makes the data at least 10 years old. Publications from 1996 to 2015	66 countries, mainly from Northern Europe and Mediterranean, followed by North America, and Asia. Less represented: Oceania, South America, Central Europe, and Africa	Food items	
	1	1	3	2,14	2	3	5	3,00	3	3	3	3	2	2,81

